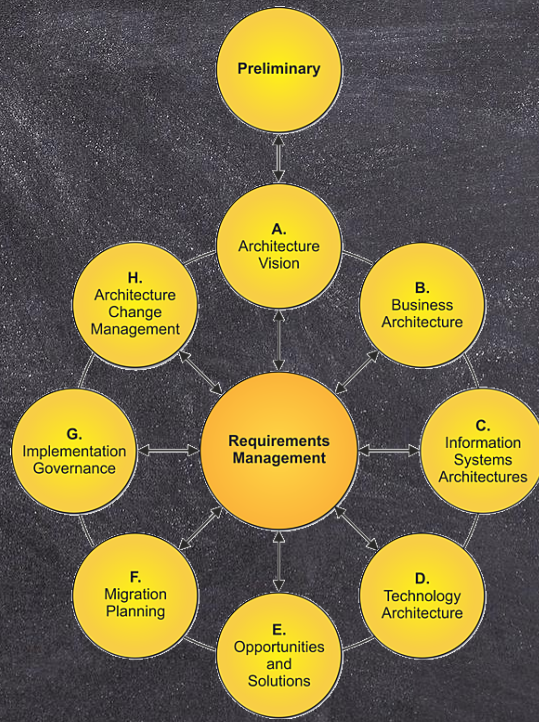




Enterprise Architecture (Foundation) with the TOGAF Standard, 10th Edition

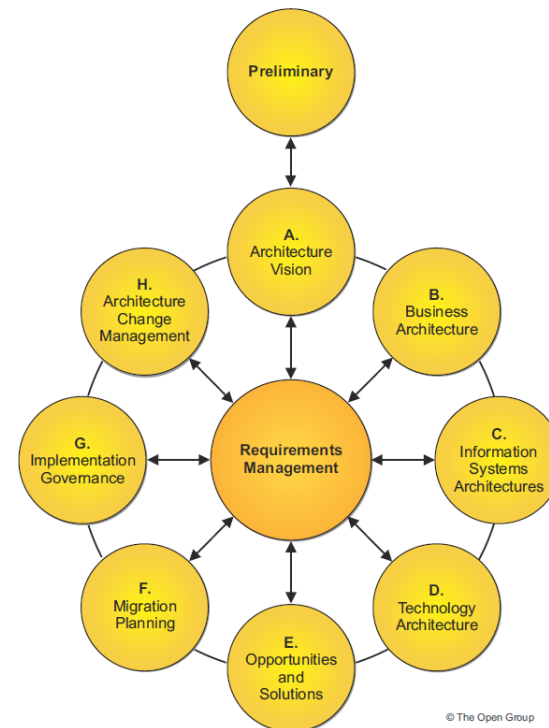
Biggest and best course for TOGAF and Enterprise Architecture | All-in-one carefree package to pass the part 1 exam | Learn from 2 EA experts

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Document outline (1/3)

Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition

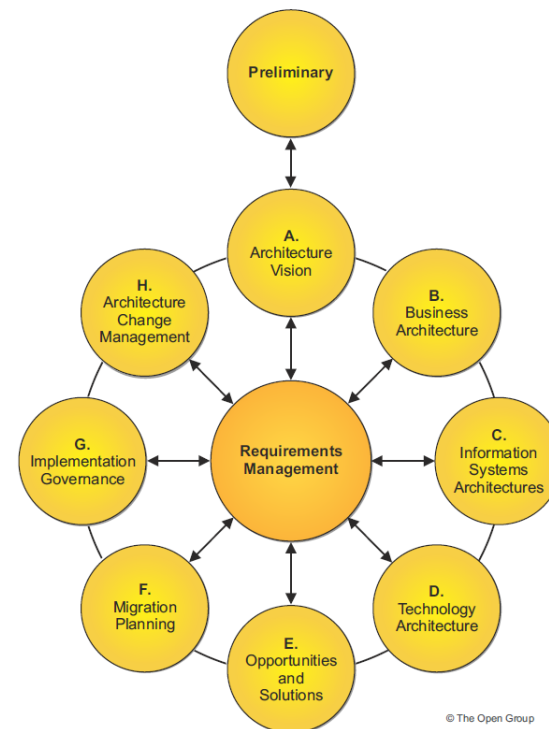
1	<u>Welcome</u>
2	<u>Introduction to Enterprise Architecture</u>
3	<u>Architecture Development Method (ADM)</u>
4	<u>Preliminary Phase</u>
5	<u>Phase A: Architecture Vision</u>
6	<u>Phase B: Business Architecture</u>
7	<u>Phase C: Information Systems Architectures</u>
8	<u>Phase D: Technology Architecture</u>



Document outline (2/3)

Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition

9	<u>Phase E: Opportunities and Solutions</u>
10	<u>Phase F: Migration Planning</u>
11	<u>Phase G: Implementation Governance</u>
12	<u>Phase H: Architecture Change Management</u>
13	<u>Requirements Management Phase</u>
14	<u>Applying the ADM</u>
15	<u>Architecture Content & Core Concepts</u>
16	<u>Architecture Governance & Architecture Capability</u>



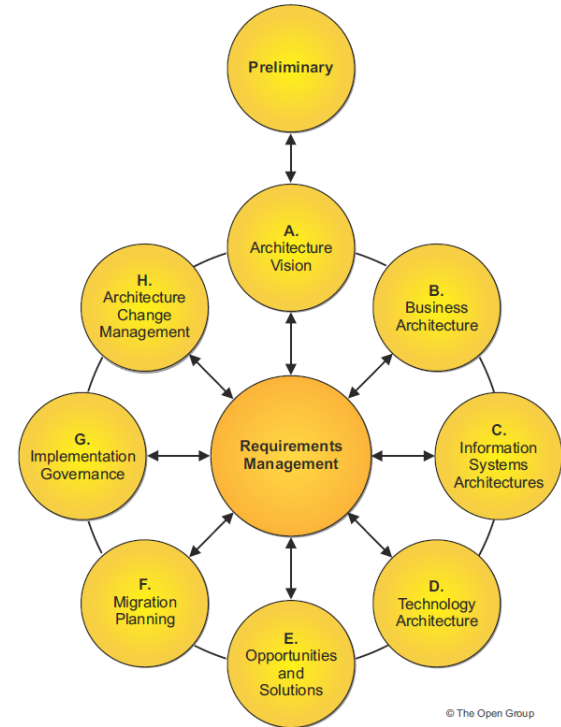
Document outline (3/3)

Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition

17 Exam Preparation & Definitions

18 Practice Test

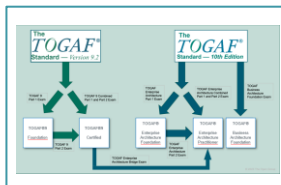
19 Wrap-up



Course outline (1/5)

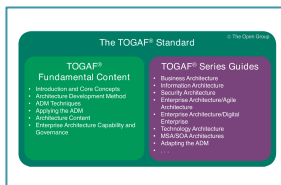
Enterprise Architecture Foundation with the TOGAF® Standard, 10th Edition

1. Welcome



- Course outline
- Course instructors
- TOGAF Certification Program

2. Introduction to Enterprise Architecture



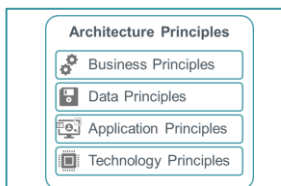
- Enterprise Architecture Definition, Purpose & Benefit
- Domains, Levels, Partitions & Abstractions
- The TOGAF Standard, 10th Edition

3. Architecture Development Method (ADM)



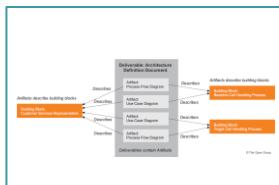
- Overview of the ADM Phases
- Artifacts & Deliverables, Building Blocks
- Stakeholders, concerns, viewpoints and views

4. Preliminary Phase



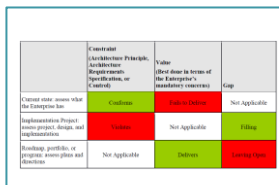
- Purpose, Objectives, Steps & Approach
- Architecture Principles

5. Phase A: Architecture Vision



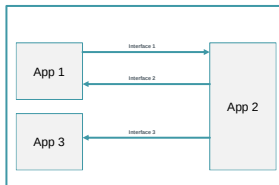
- Statement of Architecture Work
- Business Transformation Readiness Assessment
- Architecture Vision & Definition Document

6. Phase B: Business Architecture



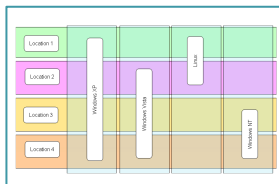
- Purpose, Objectives, Steps & Approach
- Gap Analysis

7. Phase C: Information Systems Arch.



- Purpose, Objectives, Steps & Approach
- Application & Data Architecture Artifacts

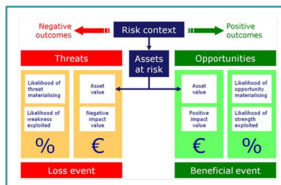
8. Phase D: Technology Architecture



- Purpose, Objectives, Steps & Approach
- Technology Architecture Artifacts



9. Phase E: Opportunities and Solutions



- Interoperability & Risk Management
- Architecture Roadmap
- Implementation and Migration Plan

10. Phase F: Migration Planning

8 Governance Checkpoints and Success/Failure Criteria

5.1 Governance Checkpoints

5.2 Metrics and KPIs

In addition, the following metrics will be used to determine the success of this architecture work:

Metric	Measurement Technique	Target Value	Estimated Further Value

<<Back to the Request for Architecture Work and restate index >>

5.3 Success/Failure Criteria

5.4 Acceptance Procedure

<<Describe the process to be used, the acceptance item list >>

- Purpose, Objectives, Steps & Approach
- Implementation Governance Model

11. Phase G: Implementation Governance

Portfolio	Project	Solution Delivery
- Identify the products needed by the enterprise - Identify the boundaries of the products - Set constraints for the product owner - Define the enterprise's backlog	- Identify the products needed by the enterprise - Identify the boundaries of the products - Set constraints for the product owner - Set constraints for the agile team	Set constraints for the agile team

➢ The Enterprise Architect serves the stakeholders guarding the mission, vision, goals and investment roadmap, hence guarding enterprise value

- Architecture Contracts
- Compliance Assessment
- Support of Agile Software Development

12. Phase H: Arch. Change Management

4 Change Rationale and Impacts

4.1 Drivers for Change

The following drivers for change are relevant (provide notes on all that apply).

Business Drivers	
Reduces as used developments	
Reduces exceptions	
Reduces materials	
Reduces technology innovations	
Strategic change	
Technology Drivers	
New technology requests	
Asset management cost reduction	
Technology withdrawal	
Standardisation	
Operational Drivers	
Improvements based on operational experience	
Capacity planning	

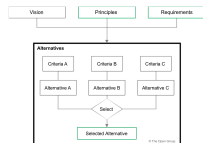
- Purpose, Objectives, Steps & Approach
- Change Request

13. Requirements Mgmt. Phase



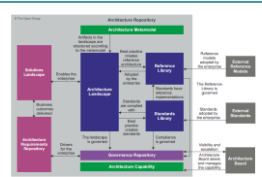
- Purpose, Objectives, Steps & Approach
- Requirements Impact Assessment

14. Applying the ADM



- ADM Techniques & Iterations
- Information Flow
- Architecture Alternatives & Trade-off Method

15. Content & Core Concepts



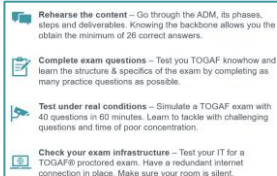
- Content Framework & Enterprise Metamodel
- Enterprise Continuum
- Architecture Repository

16. Architecture Governance & EA Capability



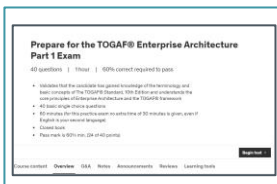
- Corporate & EA Governance
- Architecture Board & Capability

17. Exam Preparation & Definitions



- Key Definitions
- Exam Preparation & Time Management
- Question Analysis & Answer selection

18. Practice Test



- TOGAF EA Part 1 Sample Exam
- 40 Questions, Answers & Explanations

19. Wrap-up



- Next Steps & Good Luck
- Course Materials Download



Professional

- Expert in Enterprise Architecture Management
- Master degree in mechanical engineering
- Managing partner at palladio Consulting
- Certified in TOGAF, Business Analysis & SCRUM Master
- Consulting experience in Enterprise Architecture Management since 2016 (> 20 business/IT projects)



Personal

- Father of two daughters
- Board game geek
- Hiking in the mountains

About your Instructors

Dr. Christopher Schulz



Professional

- PhD in Computer Science focusing on Enterprise Architecture
- Managing partner at [palladio Consulting](#)
- Certified in TOGAF, ArchiMate, COBIT and ISO/IEC 20000
- 15 years consulting experience in Enterprise Architecture Management (> 50 business/IT projects)
- Professional trainer since 2008 (TOGAF and IREB)



Personal

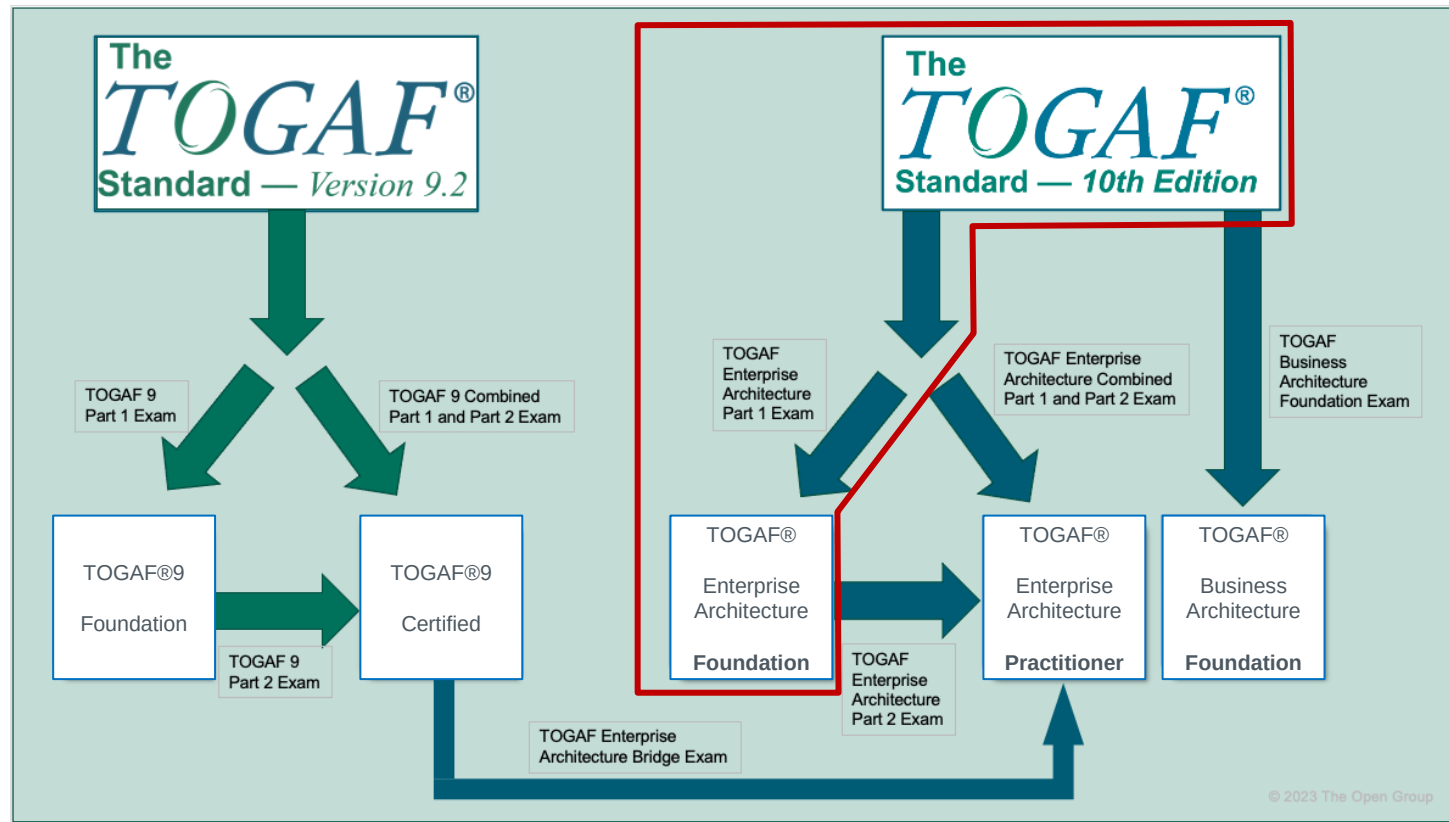
- Two young kids
- Sports & outdoors
- Blogging & writing (15 books, > 500 posts)

TOGAF Certification Program – Certification Portfolio

This course focuses on the Enterprise Architecture Foundation certification



This
Course



Setup

- Validates that the candidate has gained **knowledge of the terminology** and **basic concepts** of **The TOGAF® Standard, 10th Edition** and understands the **core principles of Enterprise Architecture** and the **TOGAF® framework**
- **40 basic single choice questions**
- **60 minutes** + 30 minutes for non-native speakers (English)
- **Closed book**
- **Pass mark: 60% (24 out of 40 questions)**

Info

- [TOGAF® Enterprise Architecture Part 1 Exam \(OGEA-101, English\)](#)



Conditions

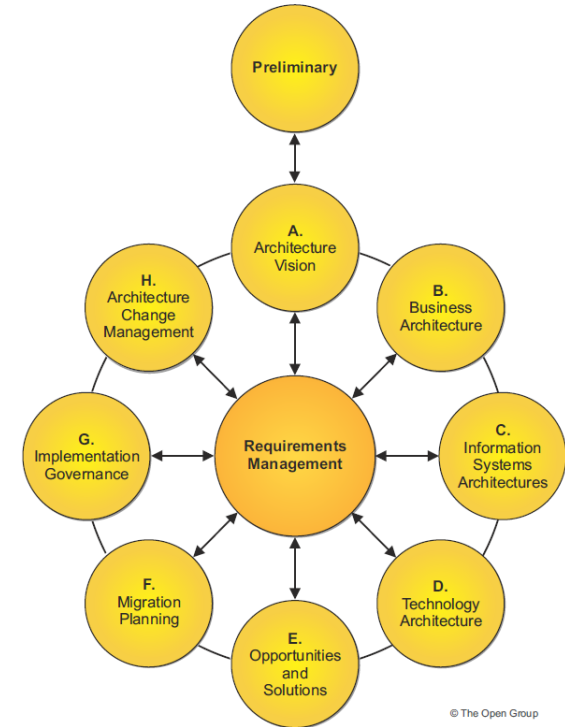
- Booked at & conducted by **Pearson Vue** on behalf of The Open Group
- **Test Center** or **remote** under controlled conditions (video surveillance)
- **Extra time**, if English is 2nd language <https://certification.opengroup.org/ESL>
- **Translation dictionary** may be used and must be presented for inspection

Document outline

Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition



1	Welcome
2	Introduction to Enterprise Architecture
3	Architecture Development Method (ADM)
4	Preliminary Phase
5	Phase A: Architecture Vision
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Enterprise – Definition

An enterprise is any collection of organizations that have common goals

- **Entire Enterprise** or **one or more specific areas** of interest
- May comprise **multiple enterprises**
- May include **partners, suppliers, customers**
- Considered as a **system**
- May develop and maintain **several independent Enterprise Architectures**
- **Examples:** Corporation vs. division of a corporation, government agency vs. single government department, partnerships and alliances of businesses



Image source: [Pixabay.com](https://pixabay.com/) / Pixabay

Architecture – Definition

The TOGAF Standard enhances the ISO/IEC/IEEE 42010:2011 terminology

1. The fundamental **concepts or properties of a system** in its environment embodied in its **elements, relationships** and in the **principles of its design & evolution**.
(Source: ISO/IEC/IEEE 42010: 2011)
2. The structure of **components**, their **interrelationships** & the **principles** and **guidelines** governing their **design & evolution** over time.
(Source: The Open Group TOGAF® Standard — Introduction and Core Concepts)

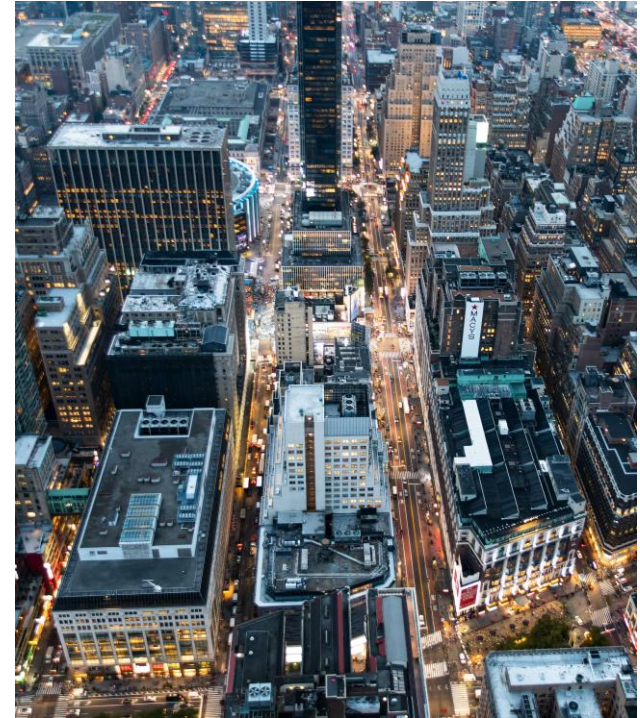


Image source: [Pexels.com](https://www.pexels.com/) / Pexels

Architecture Domains – Introduction (1/2)

The BDAT Architecture Domains divide an Enterprise Architecture into 4 subsets

Business Architecture

Defines the business strategy, governance, organization and key business processes

Data Architecture

Describes the Structure of an organization's conceptual, logical & physical data assets, data management resources

Application Architecture

Provides a blueprint for the individual applications to be deployed, their interactions and their relationships to the core business processes of the organization

Technology Architecture

Describes the digital architecture and the logical software and hardware infrastructure capabilities and standards that are required to support the deployment of business, data and applications services

IT Architecture

Specific domains as **combination** (e.g., digital architecture, risk & security architectures)

➤ **Single or holistic consideration** of architecture domains

Architecture Domains – Introduction (2/2)

Each Architecture Domain describes different “things”

Business Architecture



- Business Capabilities
- Business Processes
- Value Streams
- Information Concepts
- Organization Units
- ...

Data Architecture



- Information Objects
- Data Objects
- Data management resources
- ...

Application Architecture



- Applications
- Interfaces
- Application Domains
- Application Functions
- Application Services
- ...

Technology Architecture



- IT infrastructure
- Middleware
- Network elements
- Technology platforms
- Cloud services
- Runtime environments

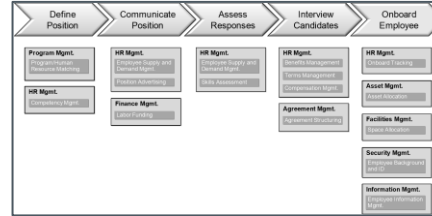
Architecture Domains – Visualization (1/2)

Each Architecture Domain can be described & visualized by several artifacts

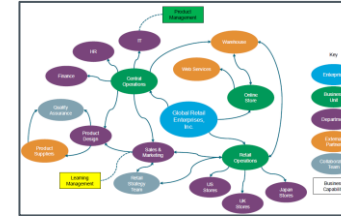
Business Architecture



Business Capability Map

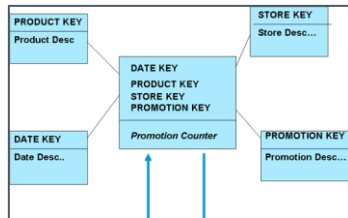


Value Stream Map



Organization Map

Data Architecture



Data Dissemination Diagram

		Business Function	
		Sales	HR
Data Entity	Product Margin Data	X	
	Customer Data	X	
	Employee Data		X
	Payroll Data		X
	Insurance Data		X
	Revenue Data	X	

Data Entity/Business Function Matrix

		Application	
		Sales-System	HR-System
Data Entity	Product Margin Data	X	
	Customer Data	X	
	Employee Data		X
	Payroll Data		X
	Insurance Data		X
	Revenue Data	X	

Application / Data Matrix

Architecture Domains – Visualization (2/2)

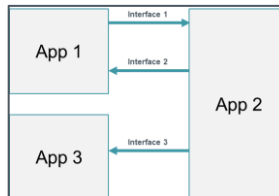
Each Architecture Domain can be described & visualized by several artifacts

Application Architecture



ID	Name	Category
L_APP_01	Enterprise Resource Planning	Platform
L_APP_02	Customer Relationship Management	Platform
L_APP_03	Data Warehouse	Platform

Application Portfolio Catalog



Application Communication Diagram

		Organization	
		Sales	HR
Application	CRM	X	
	ERP		X
	DWH	X	X
	Videocall	X	X
	DMS	X	X

Application / Organization Matrix

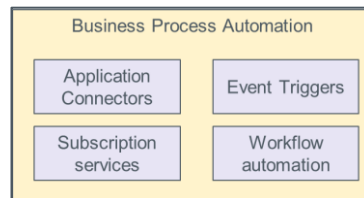
...

Technology Architecture

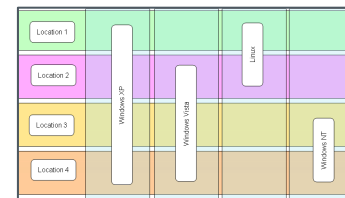


ID	Name	Category
L_TEC_01	Workflow	Automation
L_TEC_02	Event trigger	Automation
L_TEC_03	App Connector	Automation
L_TEC_04	ETL	BI

Technology Portfolio Catalog



Platform Decomposition Diagram

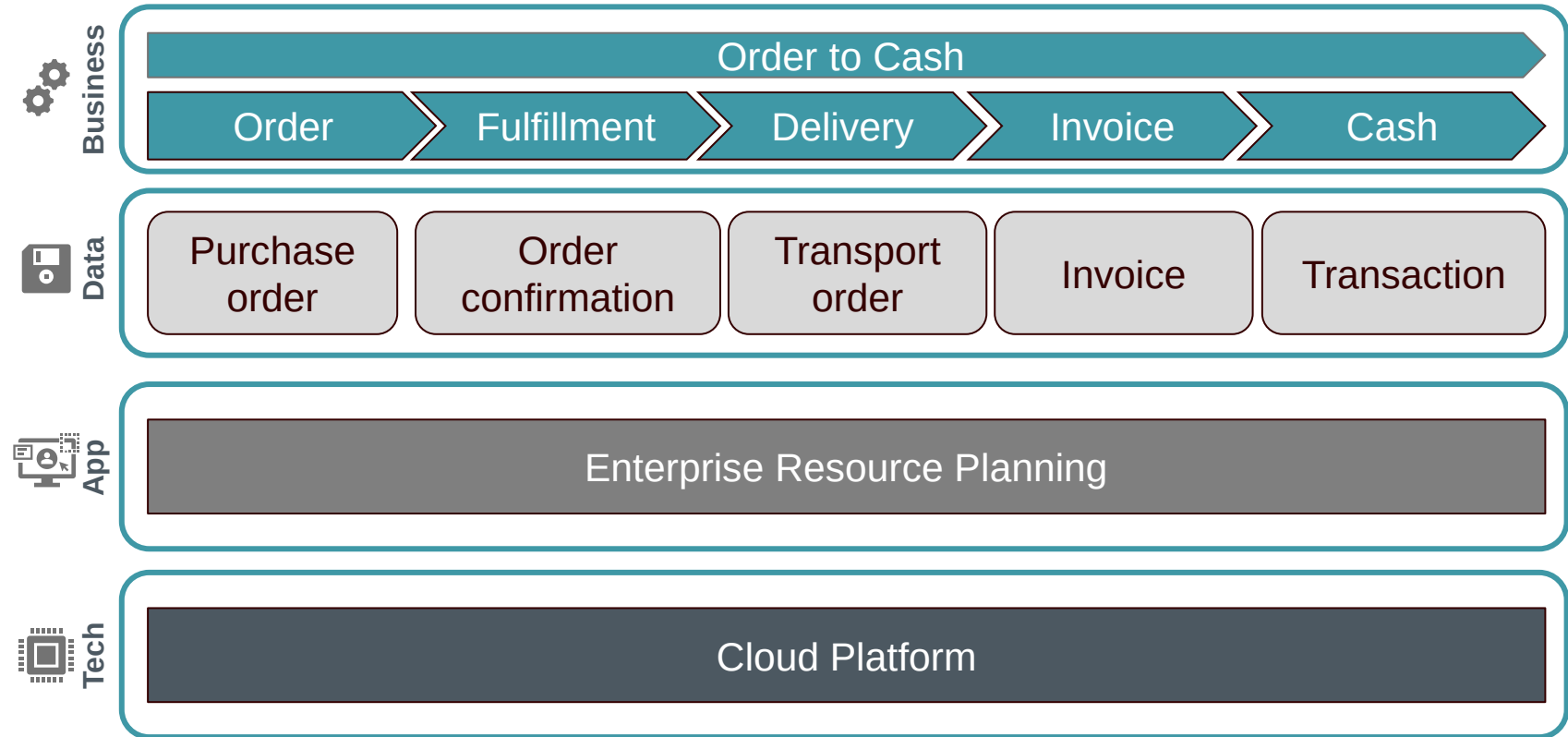


Environments & Location Diagram

...

Architecture Domains – Example

Simplified Enterprise Architecture for the Order to Cash process



Architecture States – Overview

Enterprise Architects manage multiple architecture states



Baseline Architecture
Current state acting as
reference for all change



Resting Architecture
State where the enterprise receives
value if all change activity is suspended



Candidate Architecture
Future state that stakeholders
have not approved yet



Transition Architecture
Fully functional future state that partially
realizes targets with a specific time and
target conformance



Target Architecture
Future state that
stakeholders have approved

Today  Future

Scoping Architectures – Scope, Detail, Domains & Time

Four dimensions define and limit the scope of an architecture



Enterprise Scope (breadth)

What is the full extent of the enterprise and what part of that extent will this architecting effort deal with?

- Organizations
- Business unit
- Departments
- Processes



Level of Detail (depth)

To what level of detail should the architecting effort go?

How much architecture is "enough" (effort between architecture and system design & development)?



Architecture Domains

Which domains should be looked at?

- Business
- Data
- Application
- Technology



Time Period (planning horizon)

What is the time period that needs to be articulated for the Architecture Vision?

Does it make sense (practicality, resources) to be covered in a detailed Architecture Description?

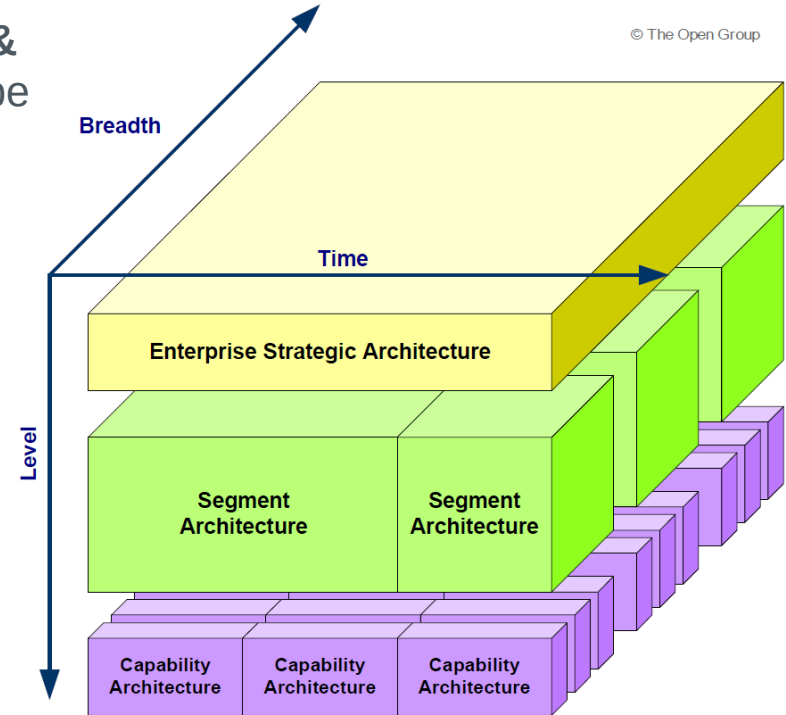
Scope of the architectural activity is mostly **limited by people, finance, objectives, stakeholder concerns and the organisational authority of the EA team**

Architecture Levels – Strategic, Segment & Capability Architecture

The concept divides the Architecture Landscape into three levels of granularity

Provide an **organizing framework** for **change & operations** describing & classifying the landscape

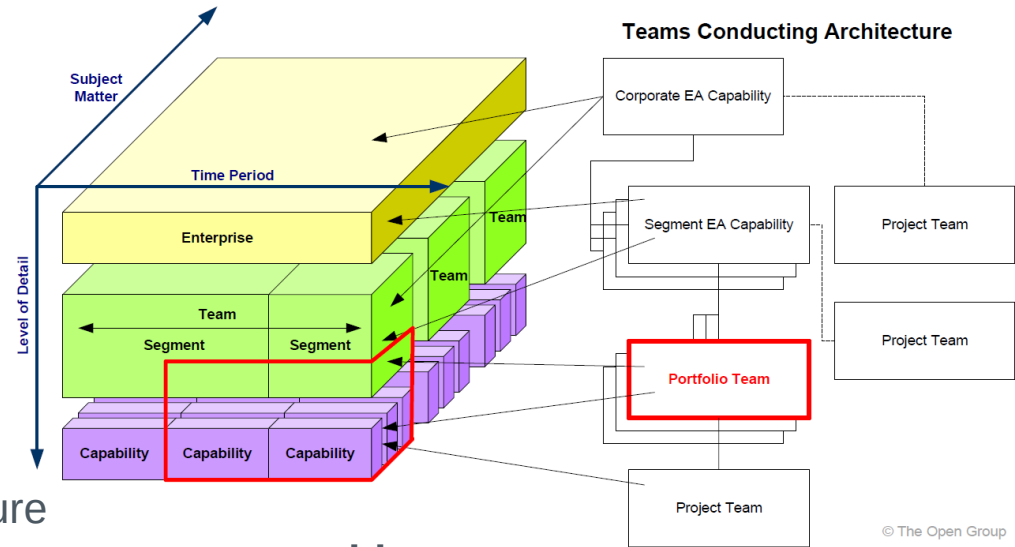
1. **Strategic Architecture** supports direction setting at an executive level
2. **Segment Architecture** supports direction setting and the development of architecture roadmaps at a program or portfolio level
3. **Capability Architecture** supports the development of effective architecture roadmaps realizing capability increments



Architecture Partitioning – Approach

A divided subset of an architecture facilitates its development & management

1. Establish several architecture partitions, providing **defined boundaries, governance and ownership**
2. Apply partitioning to architecture until **each architecture has one owning team**
3. Each **team** carrying out architecture activity within the enterprise **owns one or more architecture partitions** and will execute the ADM to define, govern and realize their architectures



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➤ **Adopt a partitioning model that reflects the enterprise' operating model**

Architecture Partitioning – Benefits

Partitioning helps to simplify the development & management of Architectures



Conflict Management

Organizational unit architectures conflict with one another



Parallelization

Partitions allow for specific groups of architects to own and develop specific elements of the architecture at the same time



Re-use

Effective architecture re-use requires modular architecture segments that can be taken and incorporated into broader architectures and solutions



Manageable Complexity & Governance

Partitioning divides an Architecture into subsets to reduce complexity, ensure effective governance and simplify the development & management

Architecture Abstraction Levels – Introduction

The concept motivates to ask structured questions about an architecture

Why (Contextual)

Why is the architecture needed?

How (Logical)

How do we structure the functionality?

What (Conceptual)

What functionality and other requirements need to be met by the architecture?

With what (Physical)

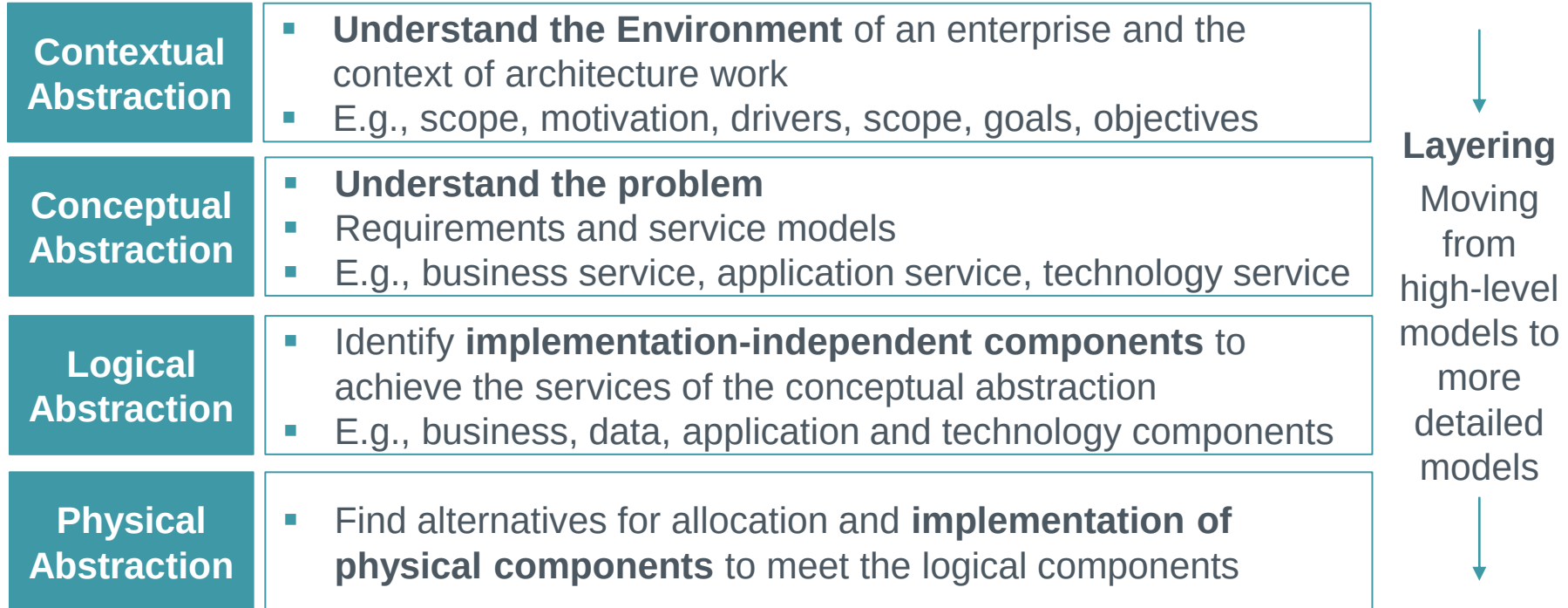
With what assets shall we implement this structure?

➤ Dividing a **problem area** into **smaller problem areas** easier to model & to solve

Note: Why, what and how have no connection to Zachman® Enterprise Architecture Framework

Architecture Abstraction Levels – Definition

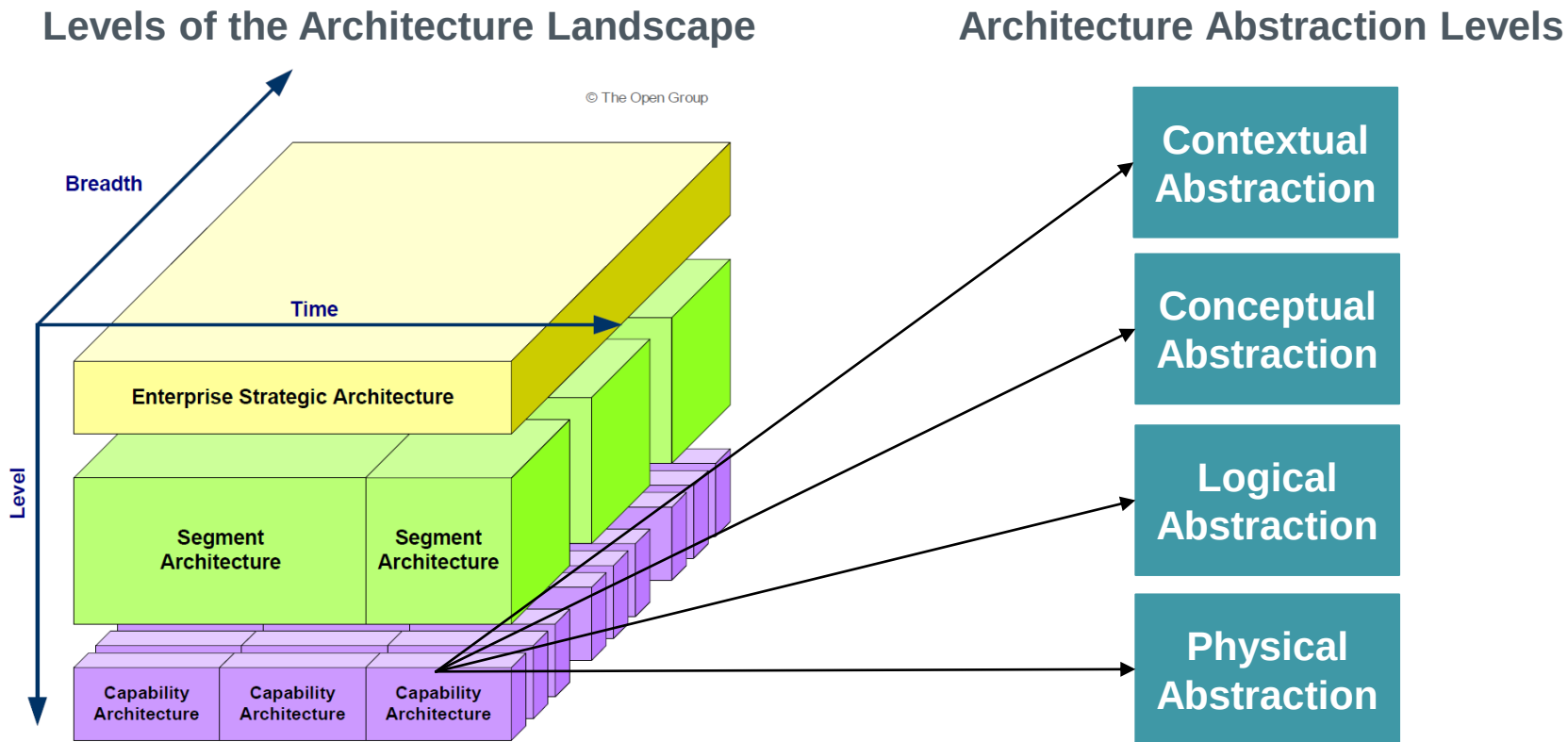
The concept divides an architecture effort into four distinct levels



➤ **Abstraction levels** cross the **Architecture Domains**

Abstraction Levels vs. Levels of the Architecture Landscape

One or more Abstraction Levels describe the Levels of the Architecture Landscape



Building Block – Definition

The potentially re-usable component delivers architectures and solutions

- Is a **package of functionality** defined to **meet the business needs** across an organization (generally **recognizable** as "a thing" by domain experts)
 - Has normally a **type that corresponds to the Enterprise Metamodel** (e.g., actor, business service, application, data entity)
 - Can be defined at **various levels of detail**, depending on the objectives of the Enterprise Architecture and the architecture development stage
 - Can lead to **improvements in legacy system integration, interoperability** and flexibility in the **creation of new systems and applications**
- An **organization** must decide what **arrangement of Building Blocks** works best

Building Block – Characteristics

A good Building Block meets several criteria



Considers **implementation & usage** and evolves to **exploit technology & standards**



Is **re-usable** and **replaceable** and **well specified**



May be **assembled** from and **subassembly** of other **Building Blocks**



May **interoperate** with other, **inter-dependent Building Blocks** based on a **published and stable interface**



Should have **defined boundaries** and **specification** which are **loosely coupled** to its **implementation** (i.e., possible to realize a Building Block in several ways)

Building Block – Architecture & Solution Building Blocks

ABBs and SBBs are special types of Building Blocks



Building Blocks (BBs)

Potentially re-usable component

Meet the business needs

Defined at various levels of detail

Architecture Building Blocks (ABBs)

Architectural components
Describe the required capability
Logical or supplier-independent

Implement the
capability

Shape the
specification

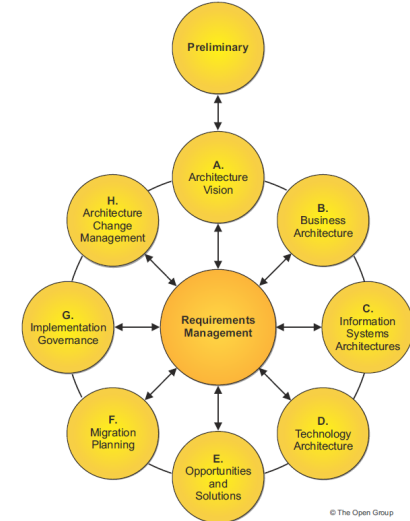
Solution Building Blocks (SBBs)

Solution components
Realize the required capability
Physical or implementation-specific

The TOGAF Standard – Introduction

TOGAF is a best practice framework for Enterprise Architecture

- **Enterprise Architecture Framework** to develop **any kind of architecture** in **any context**
- Developed through the **collaborative efforts** of the community
- Can be applied for a **range of use-cases** (e.g., agile enterprise, digital transformation)
- Describes a **standard cycle of change**, used to plan, develop, implement, govern, change and sustain an architecture
- Describes the **Building Blocks** in an enterprise used to **deliver business services & information systems**



The TOGAF Standard – Suitability

TOGAF is a suitable framework for Enterprise Architecture

- Enables organizations to **operate in an efficient & effective way** using a proven and recognized set of best practices to **address business & technology trends**
- Enables the organization to build **workable & economic solutions**
- Adds value, standardizes & de-risks **architecture development**
- Results in an Enterprise Architecture that is...
 - consistent
 - reflects the needs of stakeholders
 - employs best practice
 - **considers current and future needs** of the business



The TOGAF Standard – Tailoring & Integration

The generic framework can be tailored & integrated with other frameworks

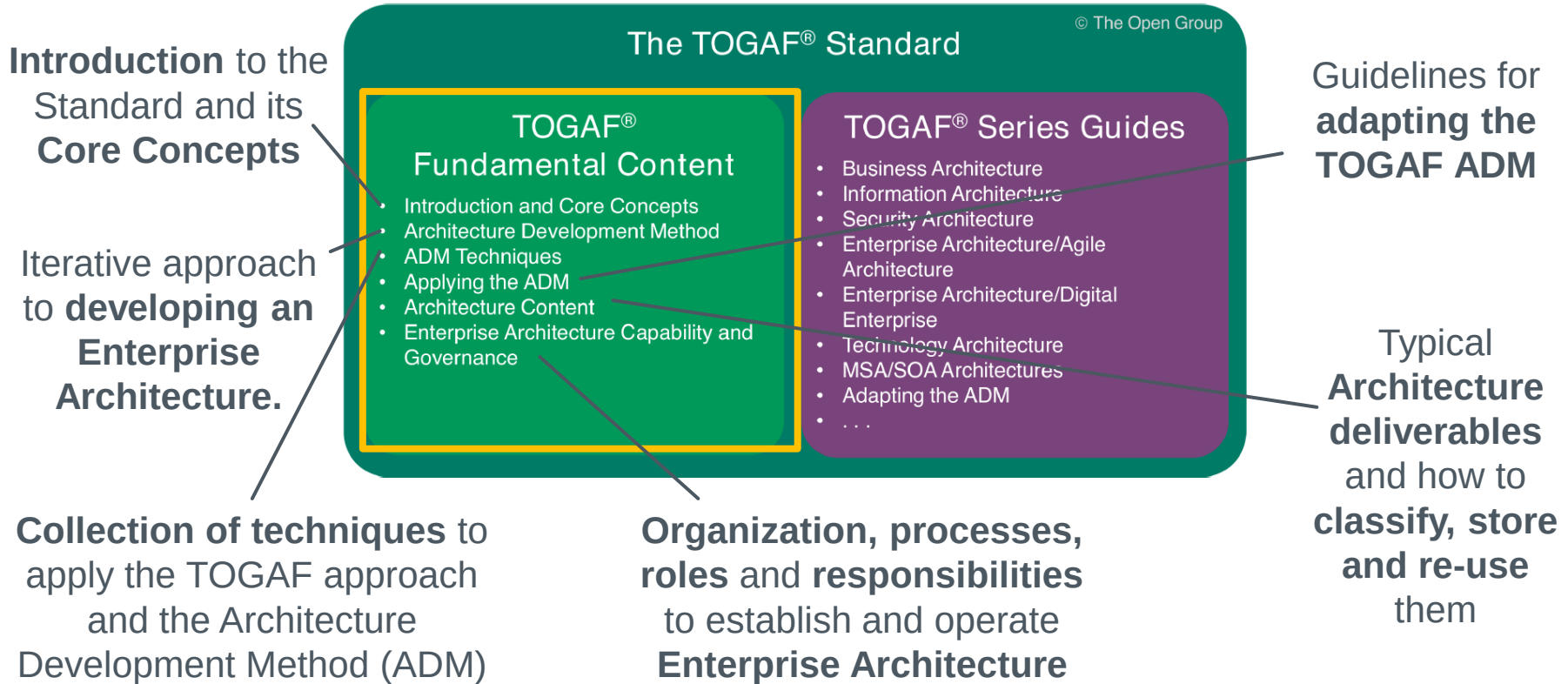
- May **adopt elements** from **other frameworks**
- Allows the **replacement or extension of its deliverables** by a more specific set (e.g., defined by other frameworks)
- Allows the **integration of TOGAF methods to other standard frameworks** or best practices (e.g. ITIL®, COBIT®, PRINCE®)
- **Should be tailored and integrated** into the **processes** and **organization structures**
- May be used as a **standalone framework**



Image source: [Pexels.com](https://www.pexels.com/) / Pexels

The TOGAF Standard – Structure & Documents

The Fundamental Content is covered by six documents



The TOGAF Standard – ADM, Supporting Guidelines & Techniques

The TOGAF ADM is supported by an extended set of resources



The Open Group Standard

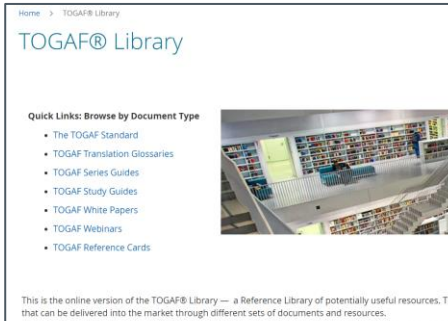
TOGAF® Standard — ADM Techniques

The Open Group

TOGAF® Series Guide

**A Practitioners' Approach to Developing
Enterprise Architecture Following the TOGAF® ADM**

Prepared by The Open Group Architecture Forum



TOGAF Fundamental Content Documents

- ADM
- ADM Techniques
- Applying the ADM

TOGAF Series Guides

Guidance on how to use and
adapt the ADM for specific
needs

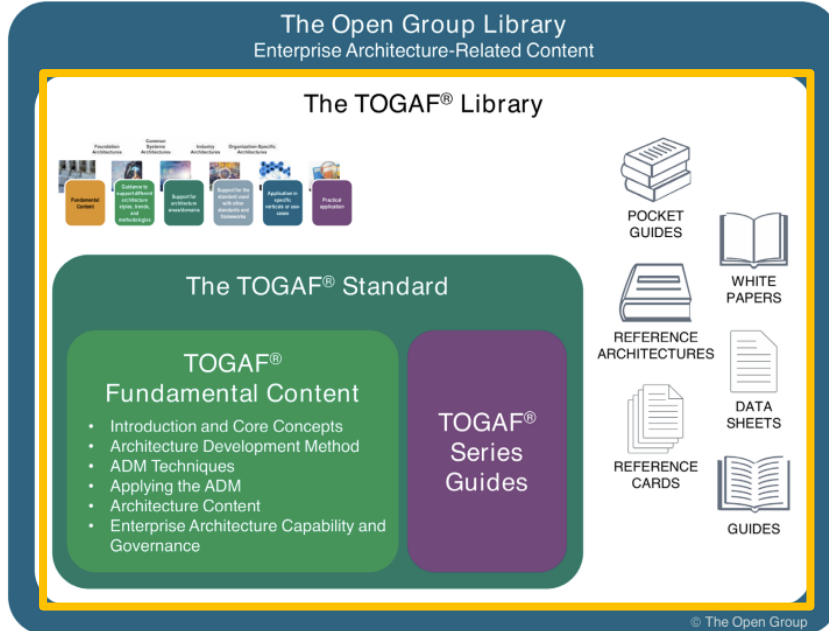
White Papers & Guides

Classified and referenced in
the TOGAF Library

➤ Individual guidelines & techniques are **described separately** so that they can be **referenced from the relevant points in the Architecture Development Method (ADM)**

The TOGAF Library – Overview

The portfolio of additional guidance material supports the practical application



- Accompanies the **TOGAF Standard**
- Offers **guidance material** to support the **practical application of TOGAF**
- Contains **guidelines, templates, patterns and reference material**
- Accelerates the **creation of new architectures** for the enterprise
- Maintained under the governance of **The Open Group Architecture Forum**

<https://Publications.OpenGroup.org/TOGAF-Library>

Enterprise Architecture – Purpose (1/2)

A framework for continuous change links strategic direction & business value

- **Manages complexity & risks and supports Change**
- **Optimizes** processes into an integrated environment that is **responsive to change** and **supportive of the delivery of the business strategy & mission**
- **Provides enterprises a strategic context for the evolution and reach of digital capability** in response to the changing needs of the business environment
- Achieves a **balance** between **business transformation** and **operational efficiency**
- Allows **business units** to innovate for **business goals** and **competitive advantage**
- Enables an **integrated strategy** with **synergies** across the enterprise and beyond

Enterprise Architecture – Purpose (2/2)

The business capability guides effective change to improve an enterprise

- **Governs (directs and controls) the change activity to realize the expected value**
- Describes **current and future state** of an enterprise as well as the **gap**
- Documents **processes around personal data** that can be easily understood
- Addresses the **end state**, performs **preference trade-off and value realization for big and little questions**



Image source: [Pexels.com](https://www.pexels.com/) / Pexels

Enterprise Architecture – Benefits

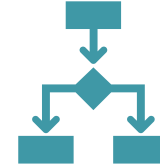
Enterprise Architecture has several key benefits for an organization



Better **return on existing investment & reduced risk** for future investment



More **effective and efficient digital transformation and operations**



More **effective strategic decision-making** by C-Level executives & business leaders



More **effective and efficient business operations**



Faster, simpler and cheaper **procurement**

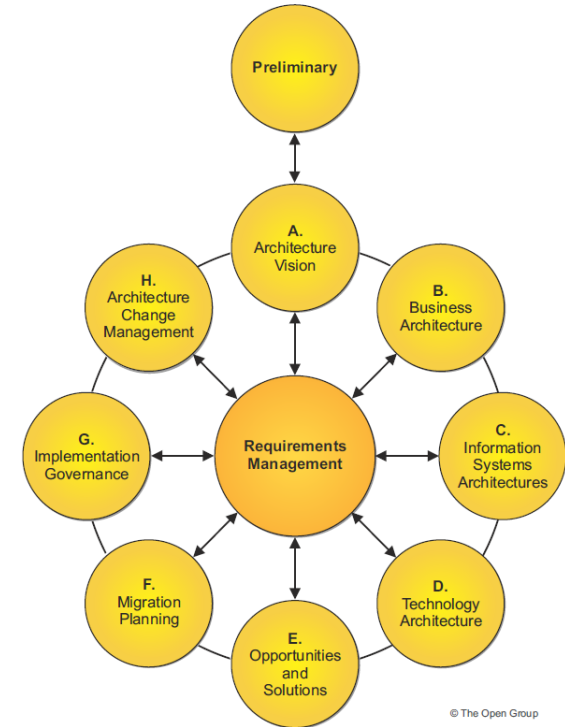


Right **balance across conflicting demands**

Document outline

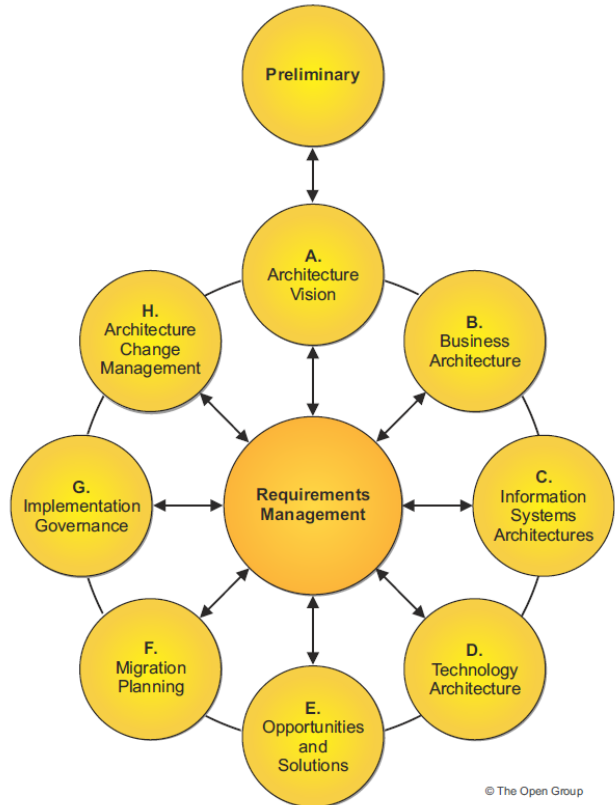
Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition

1	Welcome
2	Introduction to Enterprise Architecture
3	Architecture Development Method (ADM)
4	Preliminary Phase
5	Phase A: Architecture Vision
6	Phase B: Business Architecture
7	Phase C: Information Systems Architectures
8	Phase D: Technology Architecture



Architecture Development Method (ADM) – Overview (1/9)

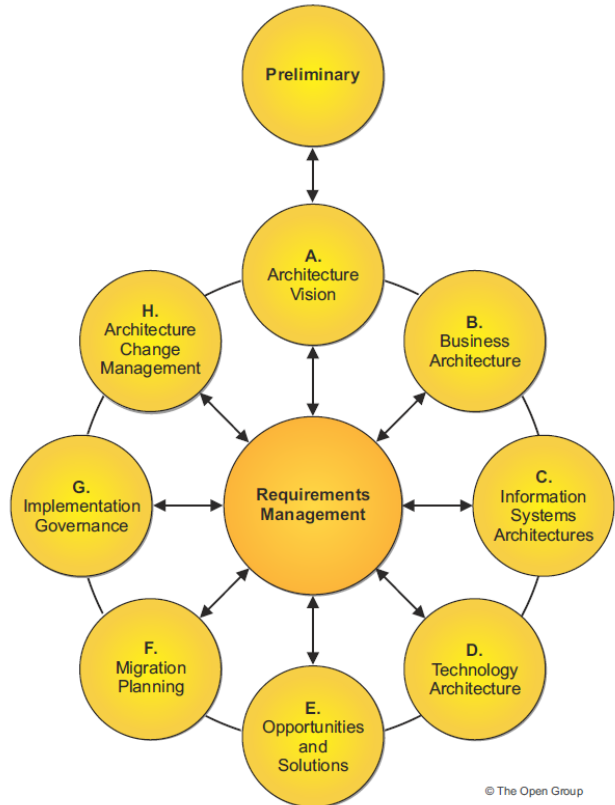
The method develops & manages the lifecycle of an Enterprise Architecture



- **Core** of the TOGAF Standard
- **Tested & repeatable** process for developing architectures
- **Establishes** an **architecture framework**
- **Develops architectures** and **architecture content**
- **Governs** the **realization & transition** of architectures
- **Iterative cycle** of continuous **architecture definition** and **realization**
- **Transforms enterprises** in a **controlled** manner in response to **business goals** and **opportunities**

Architecture Development Method (ADM) – Overview (2/9)

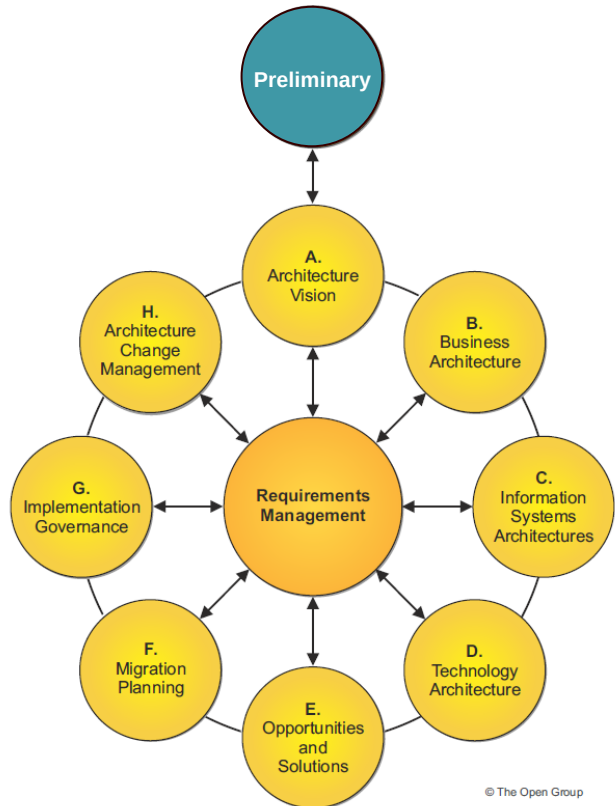
The method develops & manages the lifecycle of an Enterprise Architecture



- Step-by-step approach with 10 phases (8 arranged in a **cycle**)
- **Each phase** is divided into **steps**
- **Recommendations** to **define and deploy an EA**
- Should be **adapted** to the needs of the enterprise and to **support different architecture styles**
- Is **not a „waterfall“** method and **does not mandate a specific sequence (Phases & Steps)**
- Reference model:
 - **Deliver solutions in an architected way**
 - **Identify interacting components and relationships**

Architecture Development Method (ADM) – Overview (3/9)

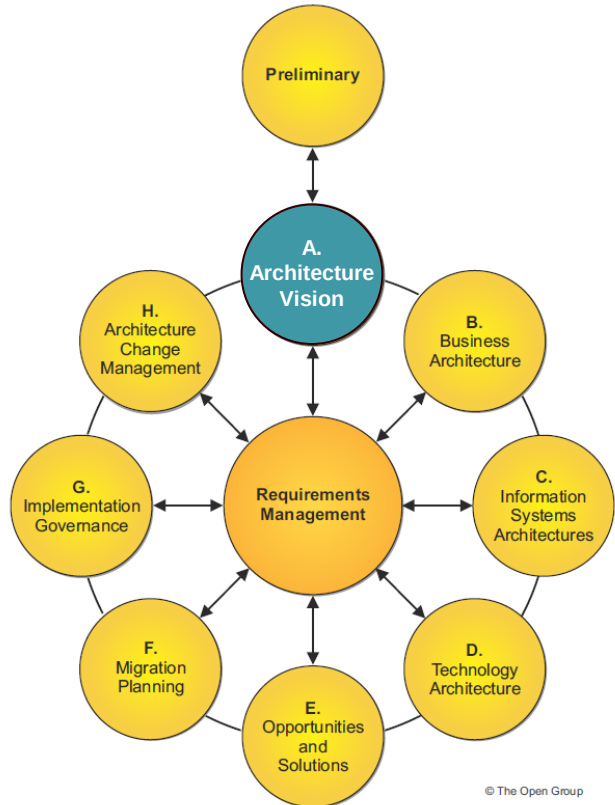
Preliminary Phase creates the architecture capability of an organization



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Preliminary Phase

- Preparation and initiation activities that are required to create an Architecture Capability
- Customize the TOGAF framework
- Define Architecture Principles



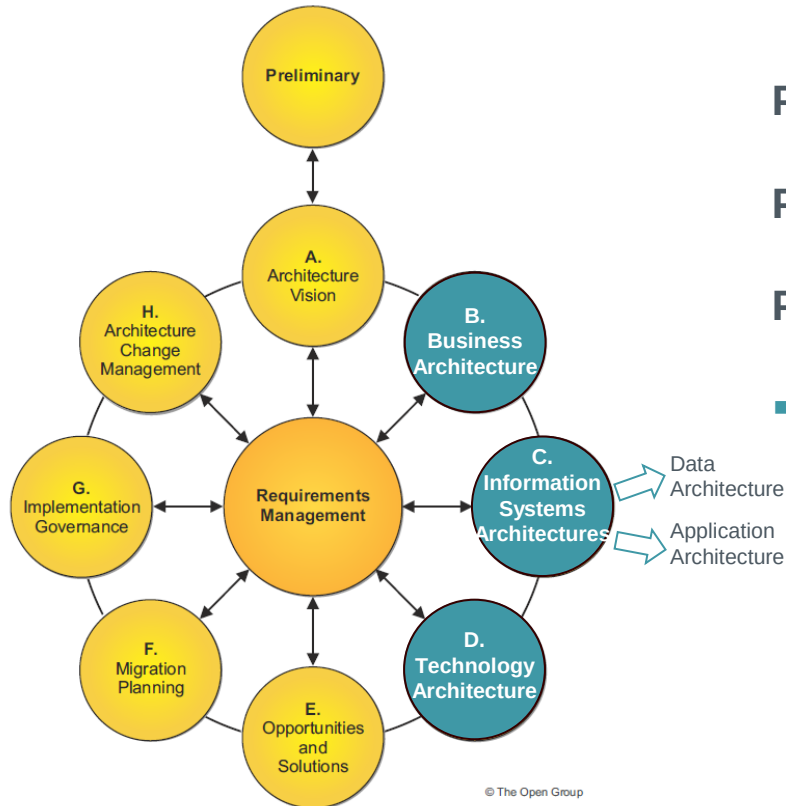
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Phase A: Architecture Vision

- Initial phase of an architecture development cycle
- Define the scope of the architecture development initiative
- Identify the stakeholders
- Create the Architecture Vision
- Obtain approval to proceed with the architecture development

Architecture Development Method (ADM) – Overview (5/9)

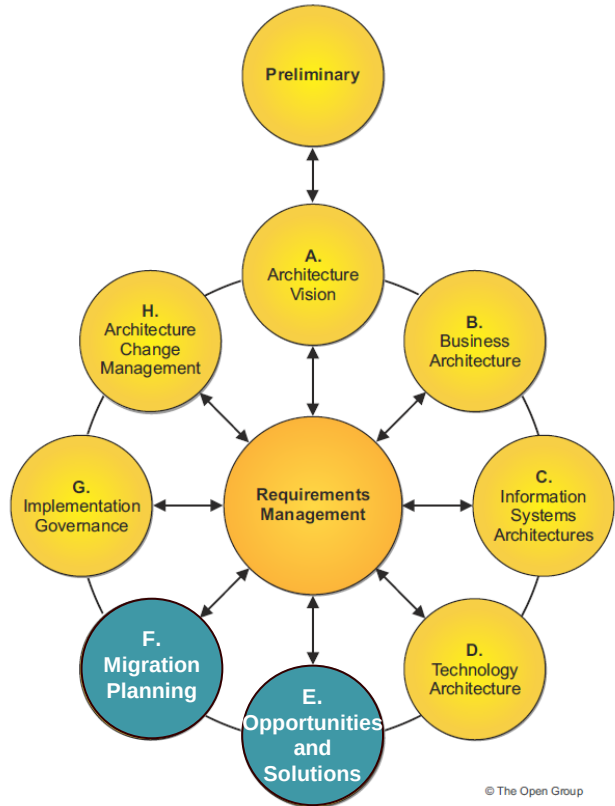
Phases B,C & D develop architectures to support the agreed Architecture Vision



Phase B: Business Architecture & Phase C: Information Systems Architectures & Phase D: Technology Architecture

- Develop Baseline & Target...
- Business Architecture (Phase B),
- Data & Application Architectures (Phase C) &
- Technology Architecture (Phase D)

to support the agreed Architecture Vision



Phase E: Opportunities and Solutions

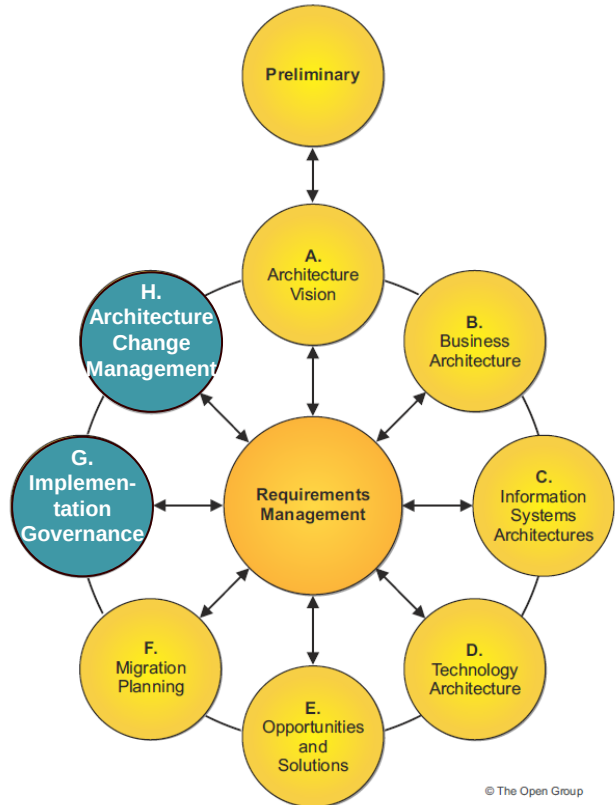
- Conduct the initial implementation planning
- Identify delivery vehicles for the architecture defined in the previous phases

Phase F: Migration Planning

- Describe how to move from the Baseline to the Target Architectures
- Finalize a detailed Implementation and Migration Plan

Architecture Development Method (ADM) – Overview (7/9)

Phase G governs the implementation & Phase H supports the architecture change



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Phase G: Implementation Governance

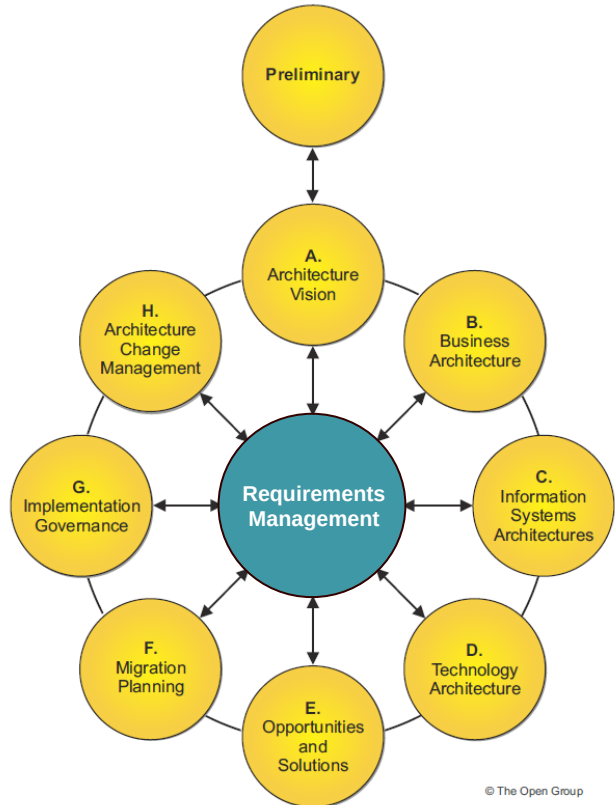
- Provide an architectural oversight of the implementation

Phase H: Architecture Change Management

- Establish procedures for managing change to the new architecture

Architecture Development Method (ADM) – Overview (8/9)

Requirements Management manages the requirements during an ADM cycle

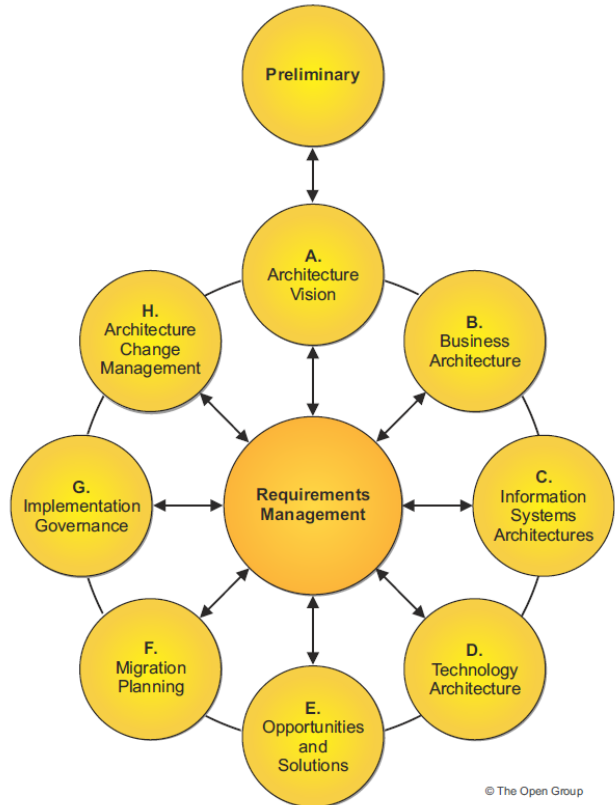


Requirements Management

- Operates the process of managing architecture requirements throughout the ADM
- Continuous phase for requirements management
- Ensures that any changes to requirements are handled through appropriate governance processes and are reflected in all other phases

Architecture Development Method (ADM) – Overview (9/9)

The method develops and manages the lifecycle of an Enterprise Architecture



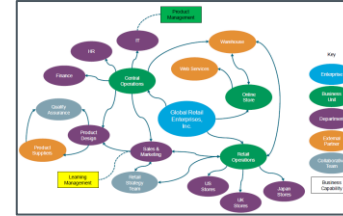
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- **Each phase** generates an **output**
- **Status of outputs** at each stage is **defined**
- **Output** from an early phase may be **modified** in a **later phase**

The architectural work product describes aspects of the Enterprise Architecture

➤ **Artifacts** can be classified as **catalogs, matrices or diagrams**

ID	Business Capability	Category
BC_01	Customer Relationship Management	Sales and Services
BC_02	Customer Portfolio Management	Sales and Services
BC_03	Business Partner Management	Sales and Services
BC_04	Human Resources Management	Support
BC_05	Facility Management	Support
BC_06	Risk Management	Support
BC_07	Accounting	Support



Catalogs: List of things
(e.g., Business Capability Catalog)

Diagrams: Picture of things
(e.g., Organization Map)

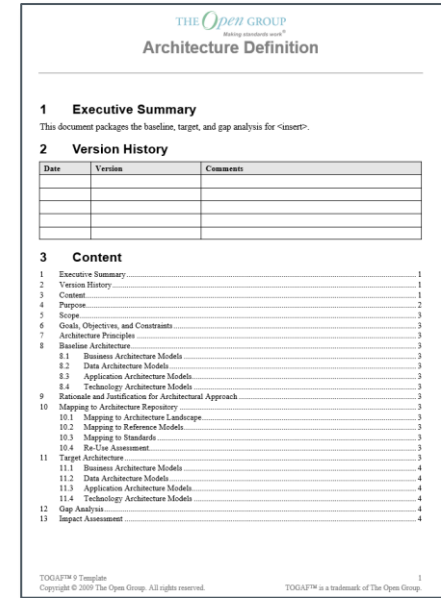
	Organizational Units			
Business Capabilities	Human Resources	Real Estate	Information Technology	Sales and Marketing
Project Management				
Learning Management				

Matrices: Relationships between things (e.g., Capability / Organization Matrix)

ADM – Deliverable

The architectural work products is contractually specified and approved

- Is formally reviewed, approved and signed off by the **stakeholders**
- Is usually a **document** (e.g., Architecture Definition Document)
- Represents the **output of projects** and is archived at completion
- May be **transitioned into an Architecture Repository** (e.g. Reference Model, Standard)



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Making standards work™
Architecture Definition

1 Executive Summary
This document packages the baseline, target, and gap analysis for <insert>.

2 Version History

Date	Version	Comments

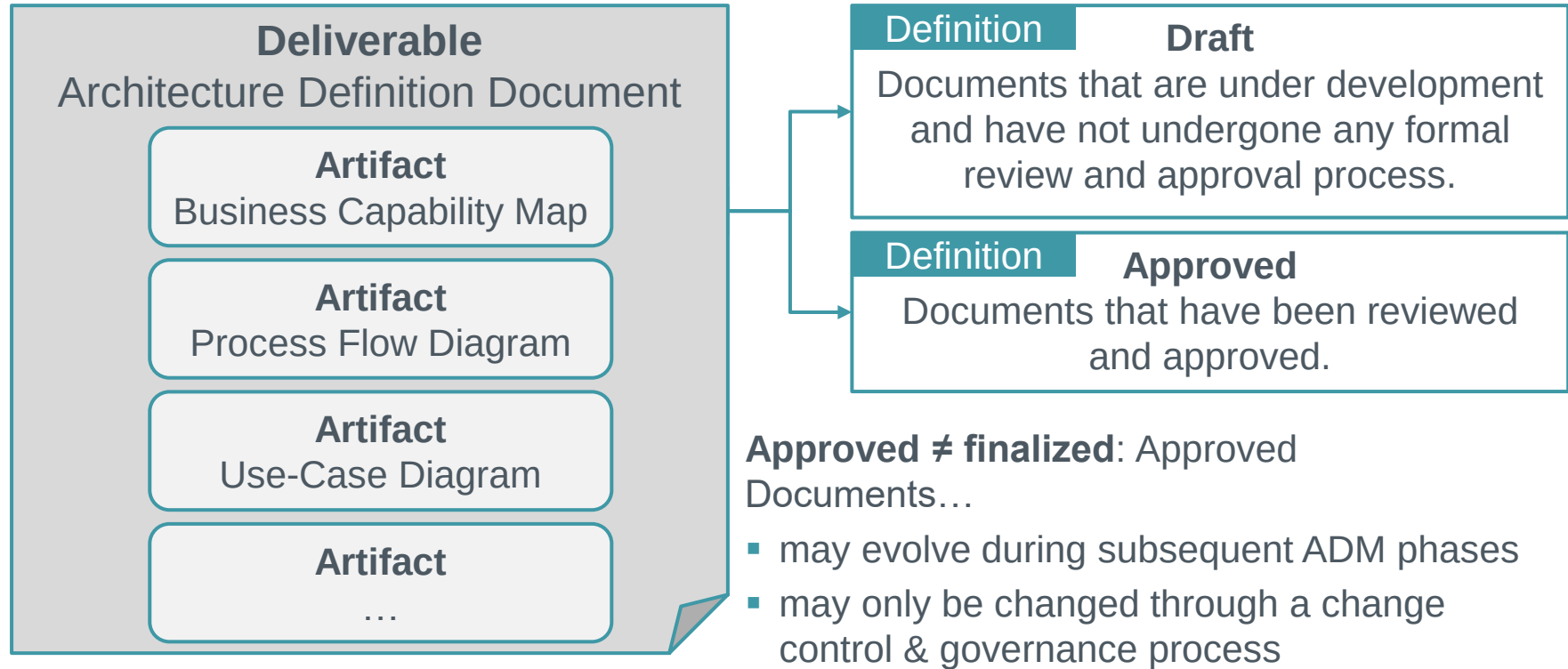
3 Content

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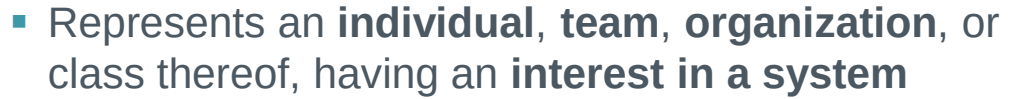
TOGAF® 9 Template
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ADM – Relationship between Artifacts & Deliverables

Deliverables contain Artifacts & are managed through a version numbering policy

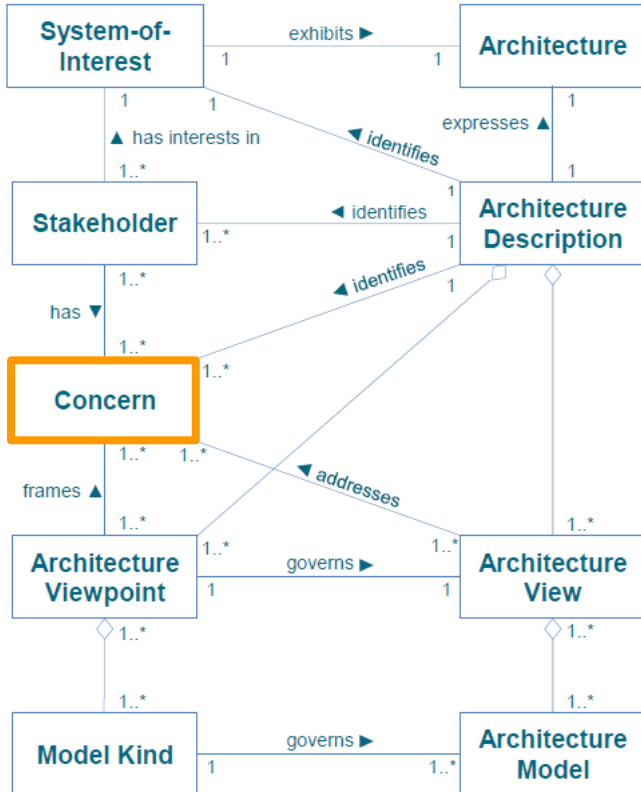


Stakeholder have interest in a system



Key concepts – Concern

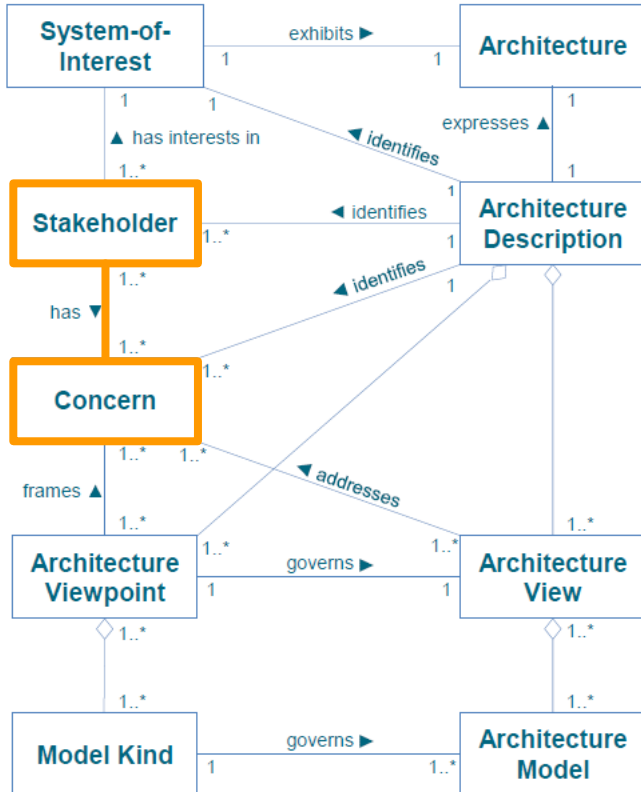
Concern is an interest in a system to stakeholders



- Represents an **interest** in a **system** relevant to one or more of its **stakeholders** and their goals
- May pertain to **any aspect of the system's** functioning, development or operation, including considerations such as performance, reliability, security, distribution and evolvability
- May determine the **acceptability of the system**
- Can be a general **requirement type** (e.g., availability)
- May lead to the **definition of several requirements**

Key concepts – Relationship between stakeholder and concern

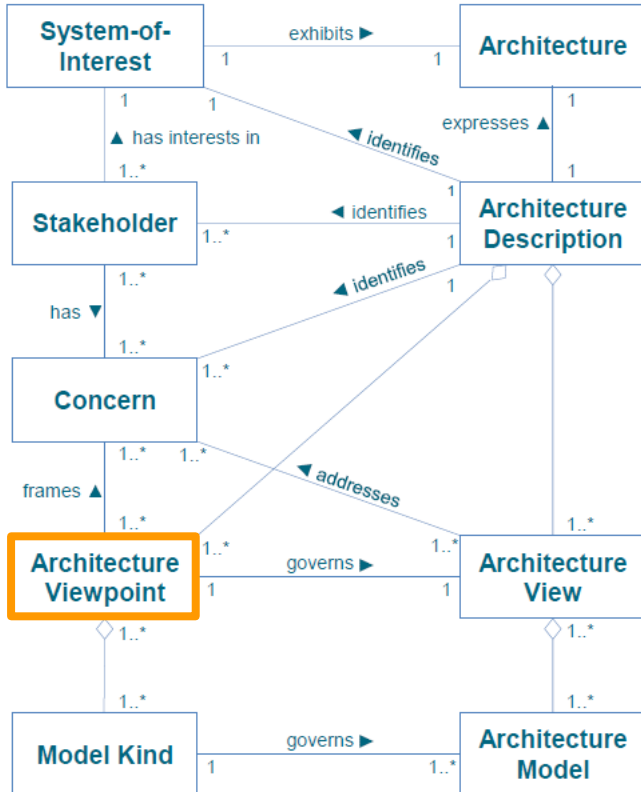
Stakeholders' concerns lead to requirements



- Identifying **concerns** helps ensure **stakeholders' interests** are addressed and requirements are identified

Key concepts – Architecture Viewpoint

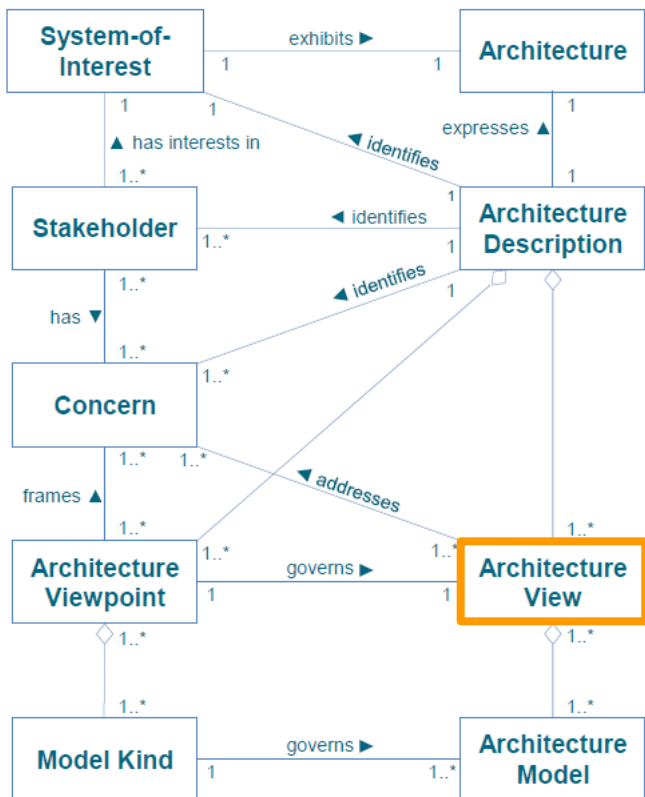
(Architecture) Viewpoint represents where you are looking from on a system



- Represents the **vantage point or perspective** that determines what can be seen of a system
- Defines **conventions for constructing, interpreting and using an architecture view** to address specific concern(s) about a system-of interest
- Can also be called the **definition or schema** for that kind of architecture view
- Is **generic** and can be stored in **viewpoint libraries** for re-use

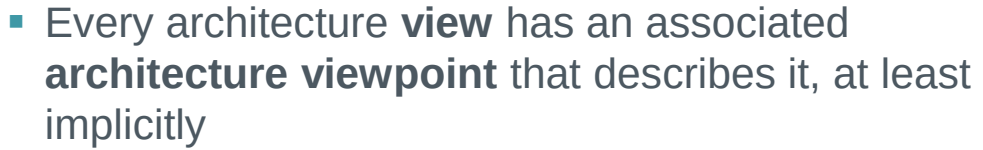
Key concepts – Architecture View

(Architecture) View represents what you see of a system



- Is a **representation of a system** from the **perspective of a related set of concerns**
- **Consists of one or more architecture models** of the system
- Is always **specific to the architecture** for which it is created and **meaningful** to stakeholders
- Enables the **architecture to be communicated to and understood by the stakeholders**

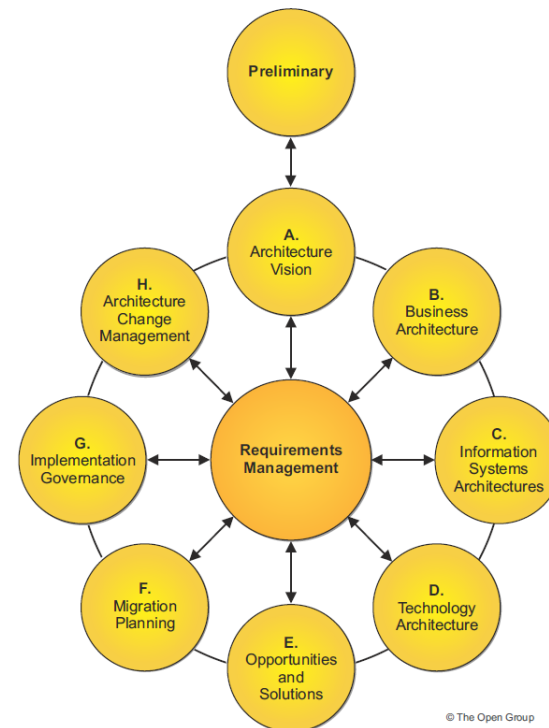
An (Architecture) View is rooted in an (Architecture) Viewpoint



Document outline

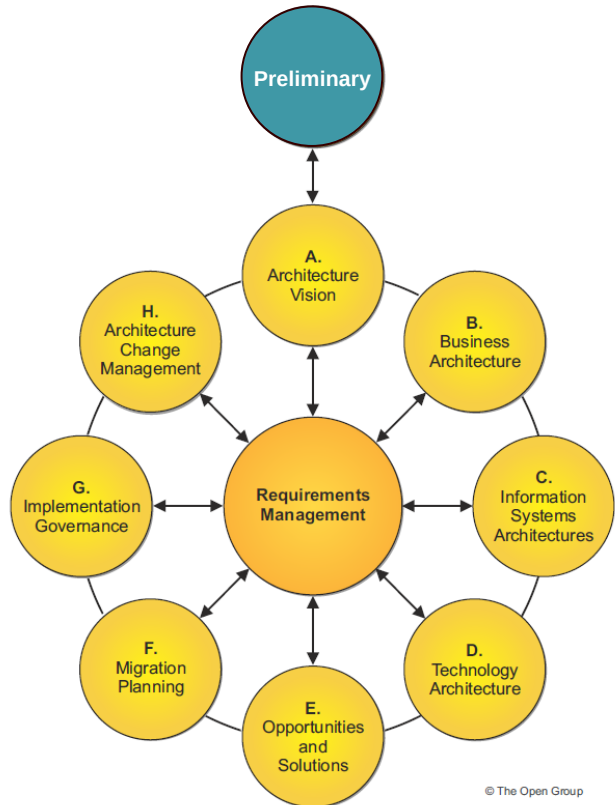
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ADM – Preliminary Phase – Purpose

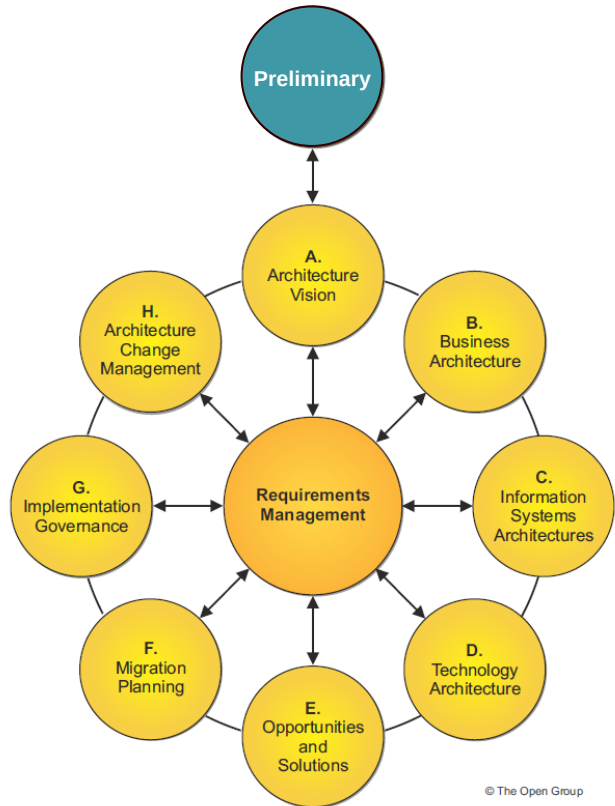
The phase creates the Architecture Capability of an enterprise



- Preparation and initiation activities to **create an Architecture Capability**
- **Customizing** the TOGAF framework
- Definition of **Architecture Principles**

ADM – Preliminary Phase – Objectives

The phase determines the target maturity & establishes the Architecture Capability



1. Determine the Architecture Capability desired by the organization

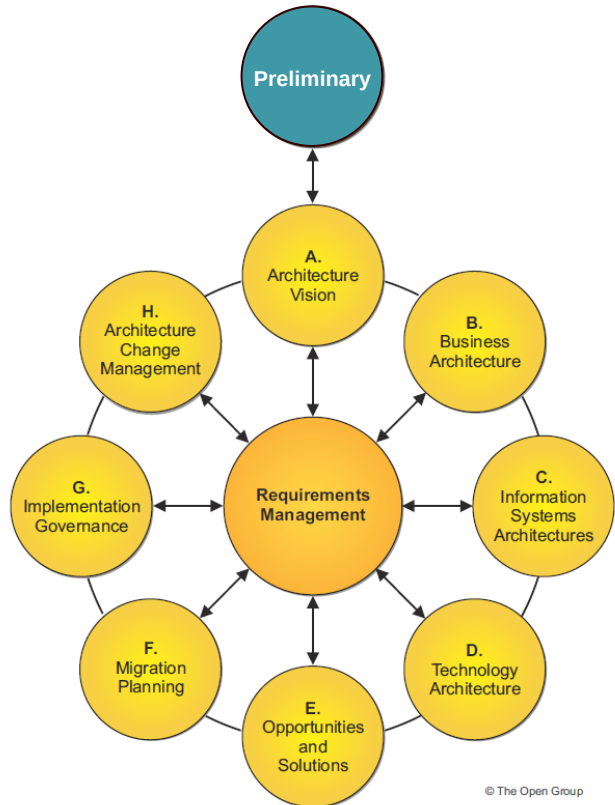
- Review the organizational context for conducting EA
- Identify and scope the elements of the enterprise organizations affected by the Architecture Capability
- Identify the established frameworks, methods and processes that intersect with the Architecture Capability
- Establish a Capability Maturity target

2. Establish the Architecture Capability

- Define and establish the Organizational Model for Enterprise Architecture
- Define and establish the detailed process and resources for Architecture Governance
- Select and implement tools that support the Architecture Capability
- Define the Architecture Principles

ADM – Preliminary Phase – Steps

The phase defines "where, what, why, who and how you do architecture"



1

Scope the enterprise organizations impacted

2

Confirm governance and support frameworks

3

Define and **establish** Enterprise Architecture **team and organization**

4

Identify and **establish** **Architecture Principles**

5

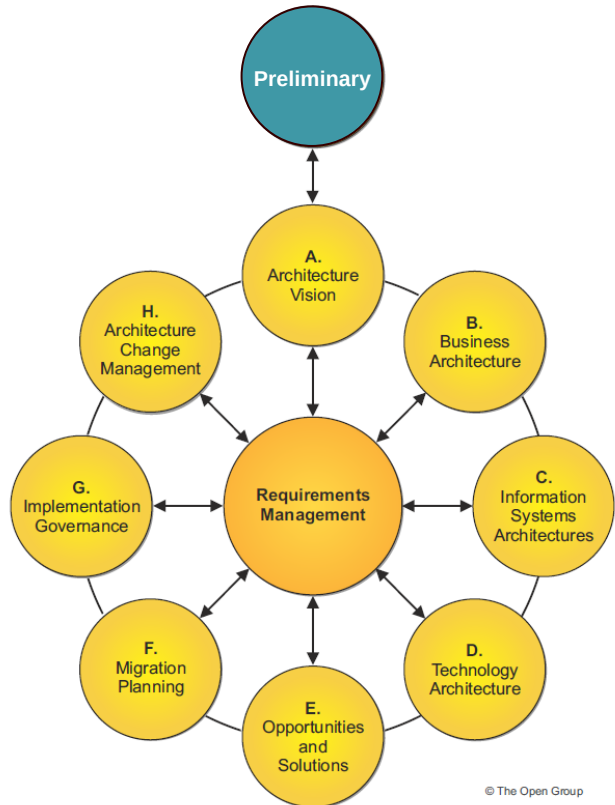
Tailor the **TOGAF framework** and, if any, **other** selected architecture **frameworks**

6

Develop a **strategy and implementation plan** for **tools and techniques**

ADM – Preliminary Phase – Architecture Principles

Architecture Principles are developed in the Preliminary Phase



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1

Scope the enterprise organizations impacted

2

Confirm governance and support frameworks

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Define and **establish** Enterprise Architecture **team and organization**

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Identify and **establish Architecture Principles**

5

Tailor the **TOGAF framework** and, if any, **other** selected architecture **frameworks**

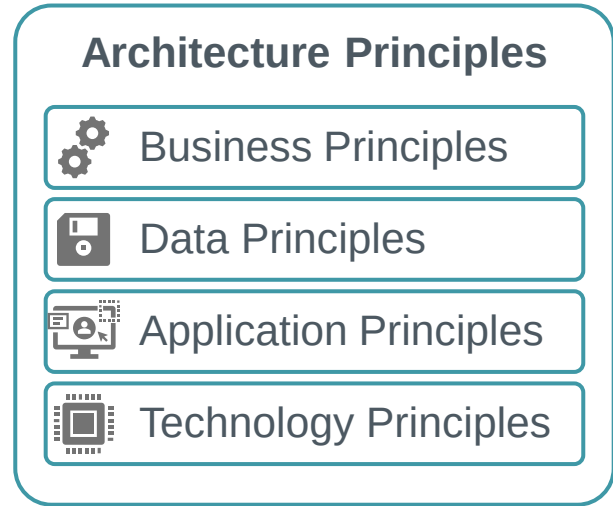
6

Develop a **strategy and implementation plan** for **tools and techniques**

Architecture Principles – Definition (1/2)

Architecture Principles guide the development & maintenance of the EA

- Guide **architecture development** as part of the **Principles Catalog**
- Define the underlying general **rules & guidelines** for the use of **resources and assets**
- Form the **basis** for making **architecture decisions**
- Are intended to be **enduring** and **seldom amended**
- Reflect a **level of consensus** among the various elements of the enterprise
- Relate back to the **business objectives** and **key architecture drivers**



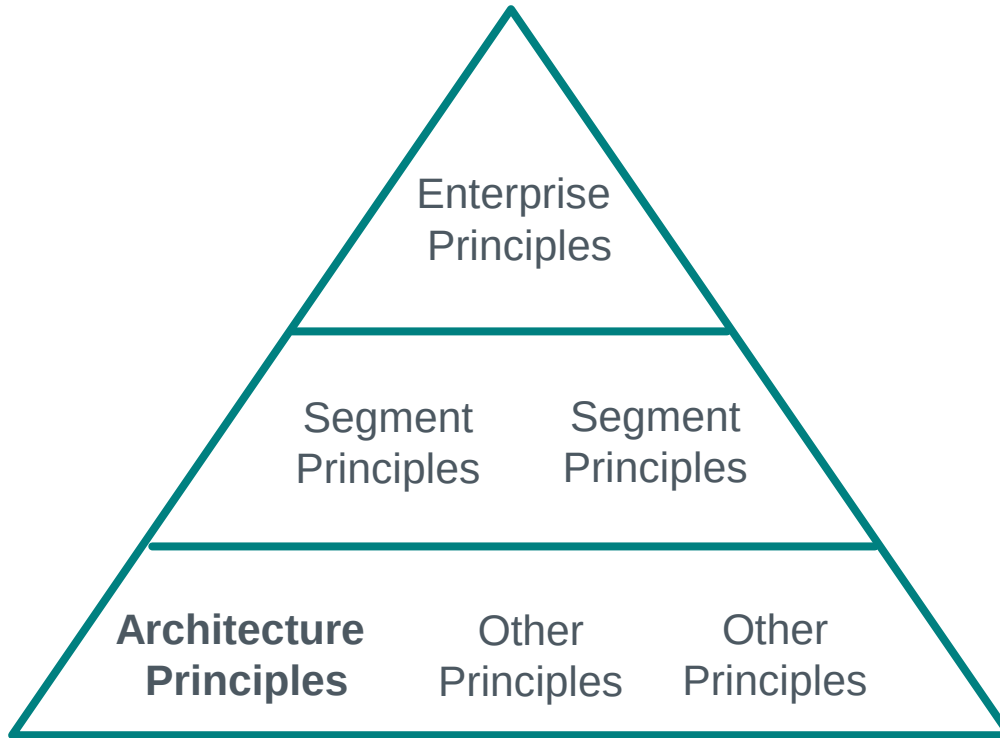
Architecture Principles – Definition (2/2)

The ADM technique aligns target architecture & implementation with the business

- Influenced by **emerging industry trends, enterprise mission & plans, enterprise strategic initiatives, current systems & technology** and **external constraints**
- Provide a foundation for making **architecture & planning decisions**, framing **policies, procedures & standards** and supporting resolution of **conflicting situations**
- Typically developed by the **Enterprise Architects** in conjunction with **key stakeholders**, being approved by the **Architecture Board** afterwards
- Altered and removed through an **amendment process** after initial ratification
- **Few in number, future-oriented** and endorsed & championed by senior management

Architecture Principles – Principle hierarchy

Enterprise Principles set bounds for principles on lower levels



- The hierarchy of principles starts with the **overarching Enterprise Principles**
- At each hierarchical level, the principles will be **elaborate on the principles inherited from the level above**
- A lower-level principle cannot overstep the **boundaries of a higher-level principle**

Architecture Principles – Principle key domains

Principles may be established within different domains and at different levels

Enterprise Principles

- Provide a **basis for decision-making** throughout an enterprise
- Inform & support how an organization sets about **fulfilling its mission**
- Are a key element in a successful **Architecture Governance** strategy
- Can have **subsidiary principles** within a business or organizational unit

Architecture Principles

- Relate to **Architecture Work**
- Govern the **architecture process** (i.e., development, maintenance and use of the Enterprise Architecture)
- Embody the **spirit and thinking** of existing Enterprise Principles

Architecture Principles – Structure

A template helps to define and use principles in a standard way

Name	▪ Represents the essence of the principle ▪ Is easy to remember 
State-ment	▪ Communicates succinctly and unambiguously the fundamental rule 
Rationale	Implications
▪ Highlights the business benefits of adhering to the principle in business terms ▪ Describes relationship to other principles, esp. where tradeoffs will be required ▪ Gives guidance toward a balanced and consistent interpretation 	▪ Highlight the requirements for carrying out the principle in terms of resources, costs and activities ▪ State the impact to business and consequences of adopting the principle ▪ Answer the “<i>How does this affect me?</i>” question of the reader 

Architecture Principles – Example

Business Principle: Compliance

Name	Compliance with Law 
State-ment	Enterprise information management processes comply with all relevant laws, policies and regulations. 
Rationale	Implications
▪ Enterprise policy is to abide by laws, policies and regulations. ▪ This will not preclude business process improvements that lead to changes in policies and regulations. 	▪ Comply with laws, regulations and external policies regarding the collection, retention and management of data ▪ Education and access to the rules ▪ Changes in the law and changes in regulations may drive changes in the business processes or applications 

Architecture Principles – Characteristics

Five criteria distinguish a good set of principles



Complete

Each principle is defined and every situation perceived is covered



Consistent

Expressed in a way that allows a balance of flexible interpretations without contradictions



Stable

Principles should be enduring, yet able to accommodate changes



Understandable

Intentions of the principles are clear, understandable & unambiguous so that (unintended) violations are minimized



Robust

Enable good quality and consistent decisions about architectures, plans enforceable policies as well as standards to be created in complex situations

Architecture Principles – Purpose

The enduring guidelines address different purposes



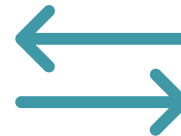
Enable decision-making

Set precedence during trade-off discussions and authority of tie-breaking



Support governance

Ensure that the right decisions are made at the right time with the right decision-makers and that these decisions are monitored



Align the enterprise

Take subjectivity & bias out and drive critical conversations aligned with the enterprise's value

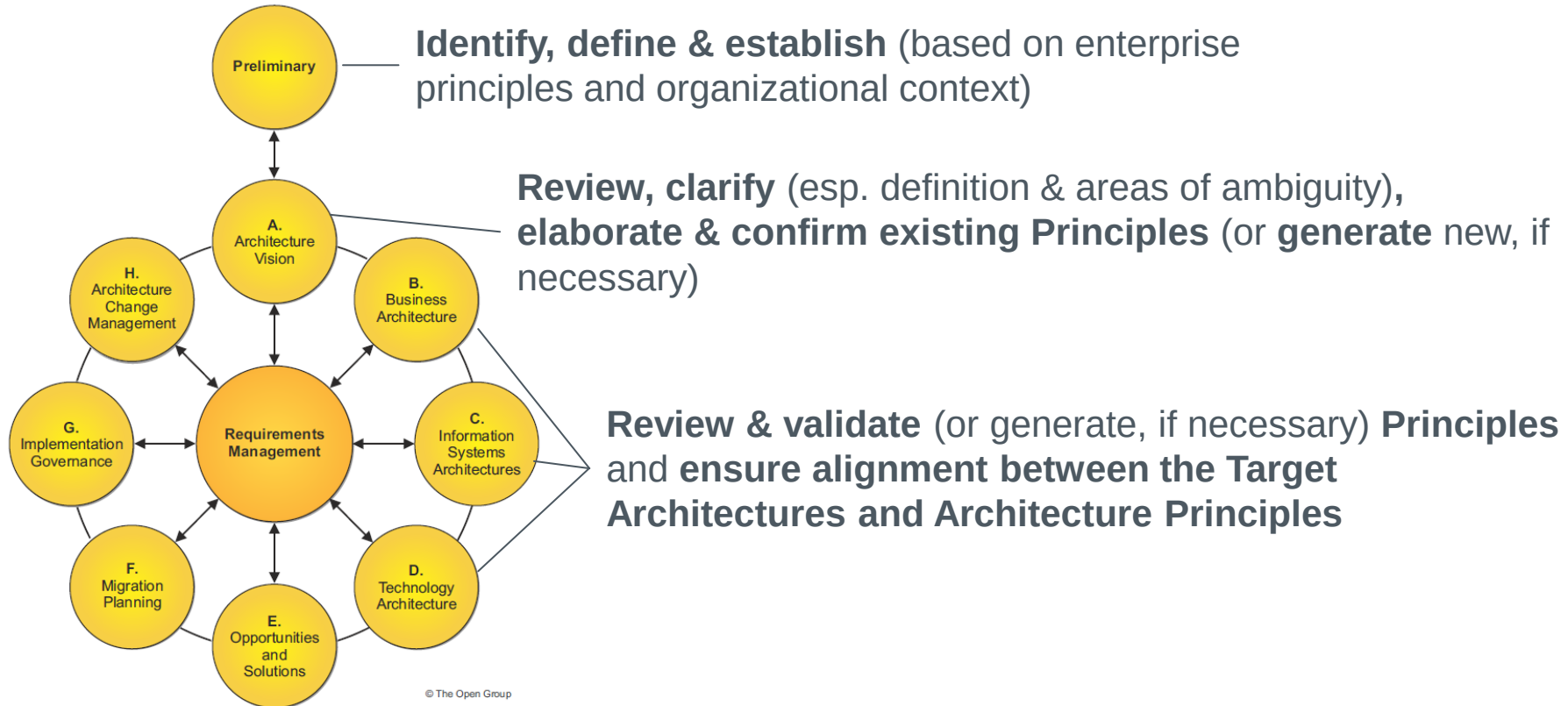


Reflect value & culture

Provide understanding about the enterprise's culture and values as well as an insight into how well the enterprise reacts to change

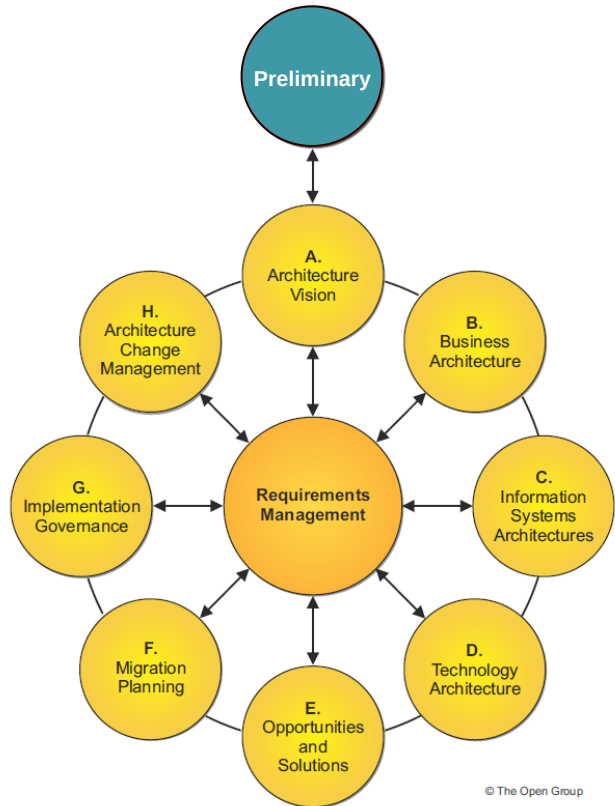
Architecture Principles – Usage within the TOGAF ADM

Principles are created in the Preliminary Phase and used in Phases A, B, C & D



ADM – Preliminary Phase – Approach

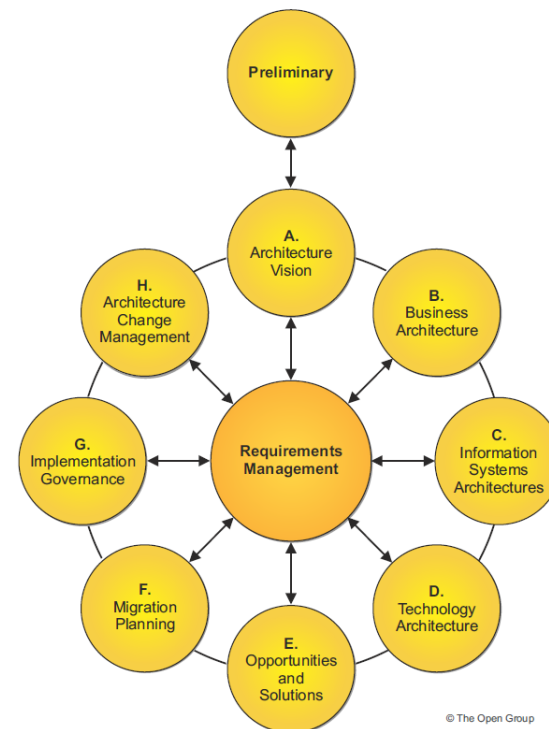
The phase defines "where, what, why, who and how we do architecture"



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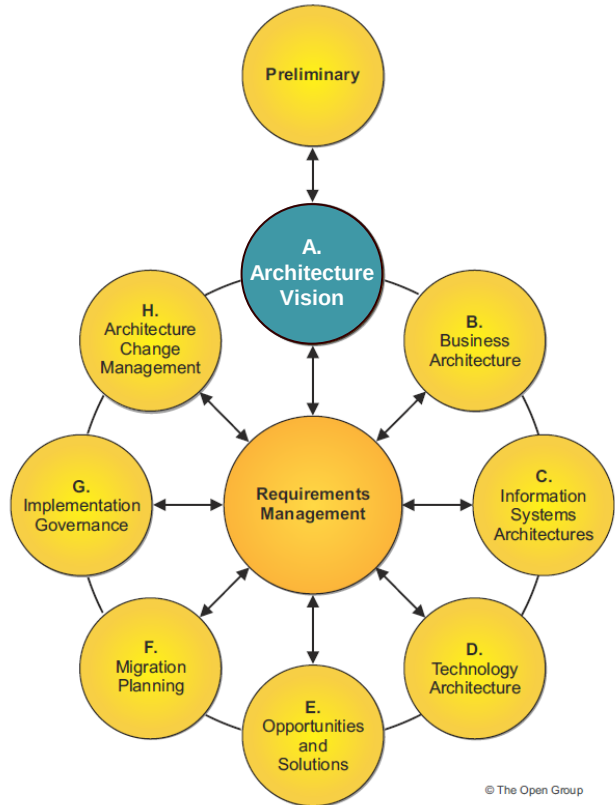
- **Define the enterprise**
- Identify **key drivers** and the **organizational context**
- Define the **requirements for architecture work**
- Define the **Architecture Principles**
- Define the **framework to be used**
- Define the **relationships between management frameworks**
- **Evaluate the Enterprise Architecture maturity**
- **Determine the Architecture Capability desired by the organization**

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ADM – Phase A: Architecture Vision – Purpose

The phase creates the Architecture Vision & establishes the Architecture Project

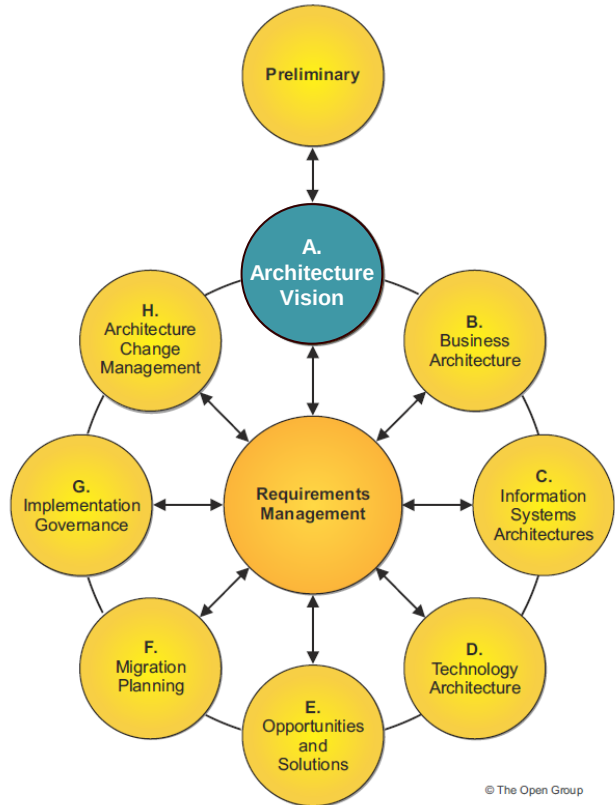


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- **Initial phase** of the Architecture Development Cycle
- Defining the **scope** (problem & project)
- Identify **stakeholders & concerns**
- Create an **Architecture Vision** (summary answer to the problem & value)
- **Obtain approval** to proceed

ADM – Phase A: Architecture Vision – Objectives

The phase develops the Architecture Vision & Statement of Architecture Work

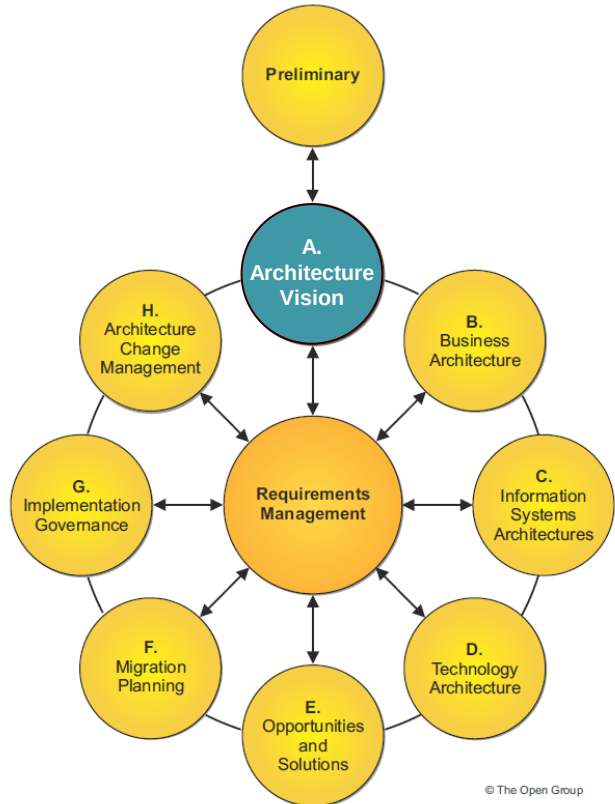


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- 1. Develop a high-level aspirational vision of the capabilities and business value to be delivered as a result of the proposed Enterprise Architecture**
- 2. Obtain approval for a Statement of Architecture Work that defines a program of works to develop and deploy the architecture outlined in the Architecture Vision**

ADM – Phase A: Architecture Vision – Steps (1/2)

The phase creates the Vision, Statement of Architecture Work & secures approval



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1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

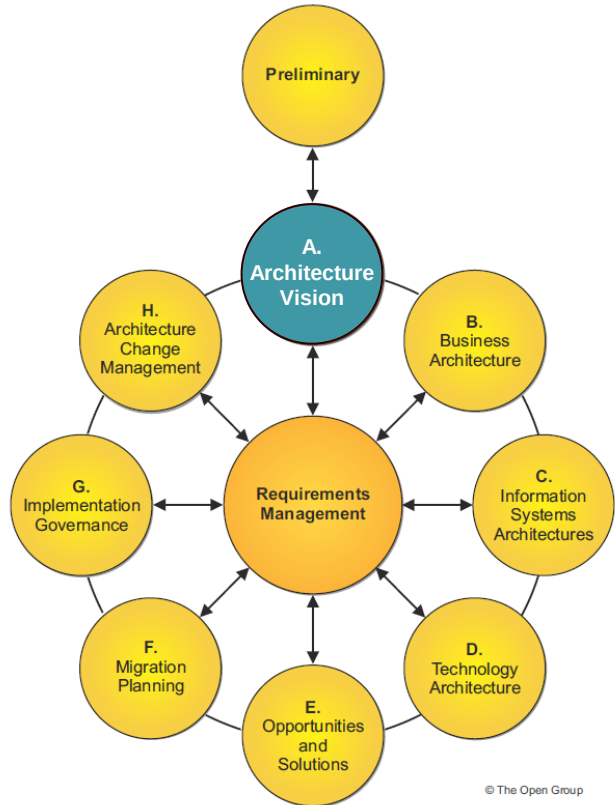
Assess readiness for business transformation

6

Define scope

ADM – Phase A: Architecture Vision – Steps (2/2)

The phase creates the Vision, Statement of Architecture Work & secures approval



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7

Confirm and elaborate **Architecture Principles**, including business principles

8

Develop Architecture Vision

9

Define the Target Architecture **value propositions** and KPIs

10

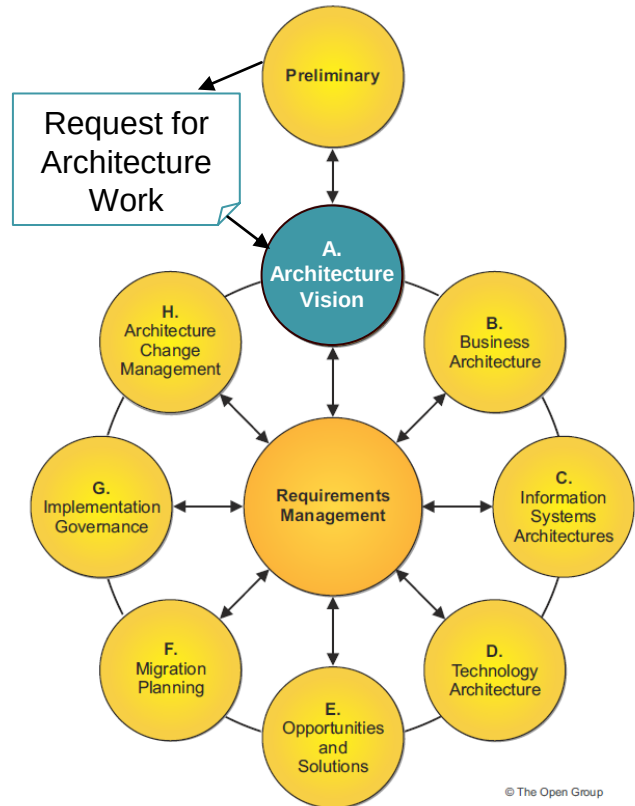
Identify the business transformation risks and mitigation activities

11

Develop Statement of Architecture Work; secure approval

ADM – Phase A: Architecture Vision – Request for Architecture Work

The phase begins with the receipt of a Request for Architecture Work



1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Request for Architecture Work – Definition

The official document triggers the start of an architecture development cycle

Purpose

- High-level document that is sent from the **sponsoring** to the **architecture organization** triggering the **start of an architecture development cycle**
- Created as **output of the Preliminary Phase**, results of an **approved architecture Change Requests** or is generated as terms of reference for architecture work originating from **migration planning**

Content

- Organization **sponsors**, **mission statements**, **budget information**, **constraints**
- Business **goals**, strategic **plans**, **changes** in environment, **time limits**, **constraints**
- **Business system**, **IT system** and **architecture** description
- **Developing organization** including resources

Request for Architecture Work – Structure

All the information in the document should be at a high level



Request for Architecture Work

Project XXXX
Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

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4	Key Constraints	6
5	Additional Information	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
Prepared By:		Document Version Date:	
Title:	Request for Architecture Work	Review Date:	
Reviewed By:			

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3 Business Imperative

<<Provide a brief overview of the business context, with the focus on describing the key business opportunity or issue to be addressed.>>

<<Topics to consider include:

- Organization's mission statement
- Business goals (and changes)
- Strategic plans of the business
- Changes in the operating environment
- Outline of the specific opportunity or issue underlying the need for architecture work>>

3.1 Organization Mission Statement

3.2 Business Goals (and Changes)

3.3 Strategic Plans of the Business

3.4 Changes in the Business Environment

3.5 Purpose of Architecture Work

<<Outline the purpose of the architecture work, highlighting its anticipated contribution to the business issue described above.>>

3.6 Success Criteria

<<Indicate what a "good" outcome of the architecture work will look like.>>

<<This should be considered on two levels:

1. Short-term – the desired content and usage of the architecture work products
2. Long-term – the desired eventual business improvements resulting from this architecture work

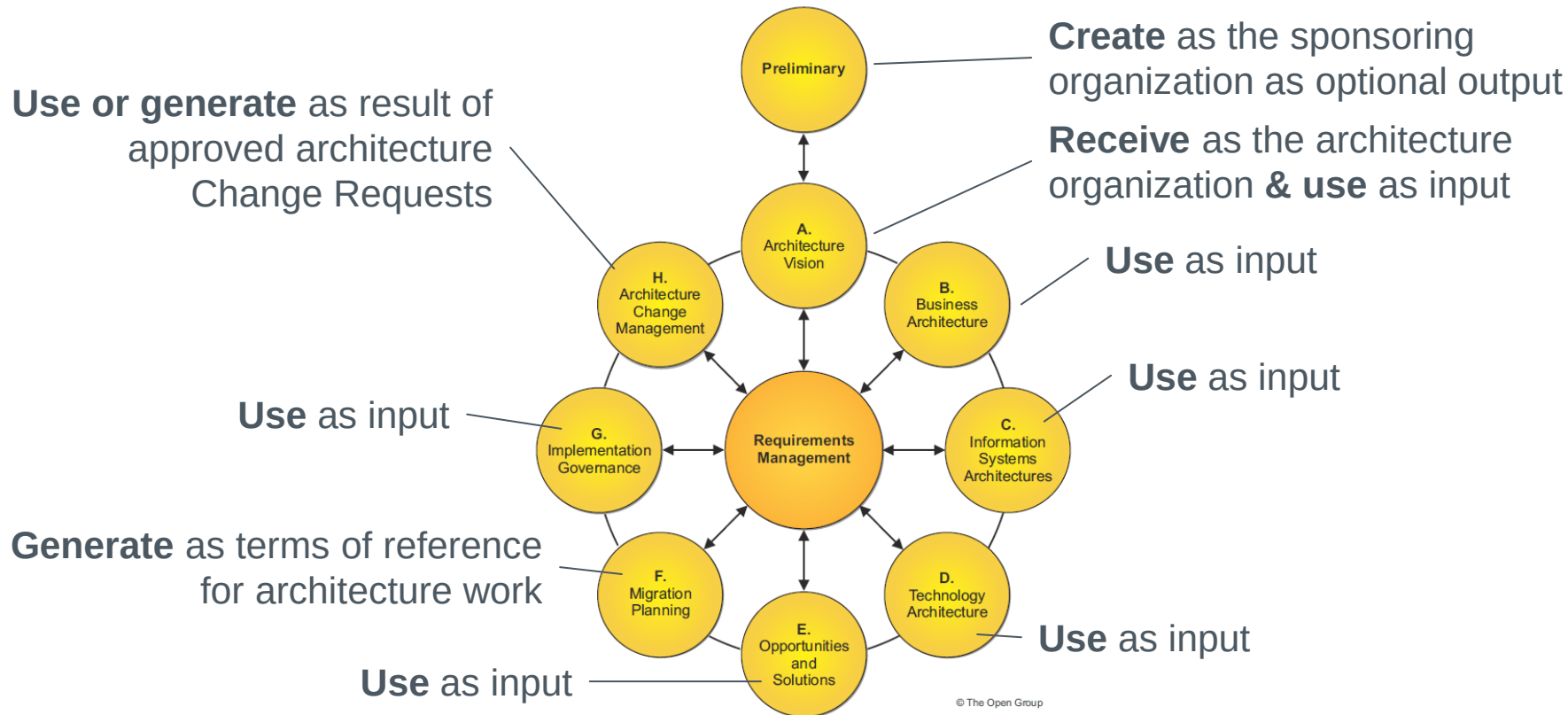
Both qualitative and, ideally, quantitative success metrics should be noted.>>

3.7 Timescale

<<When are the results of the architecture work needed? If a deadline is stated, please explain its significance.>>

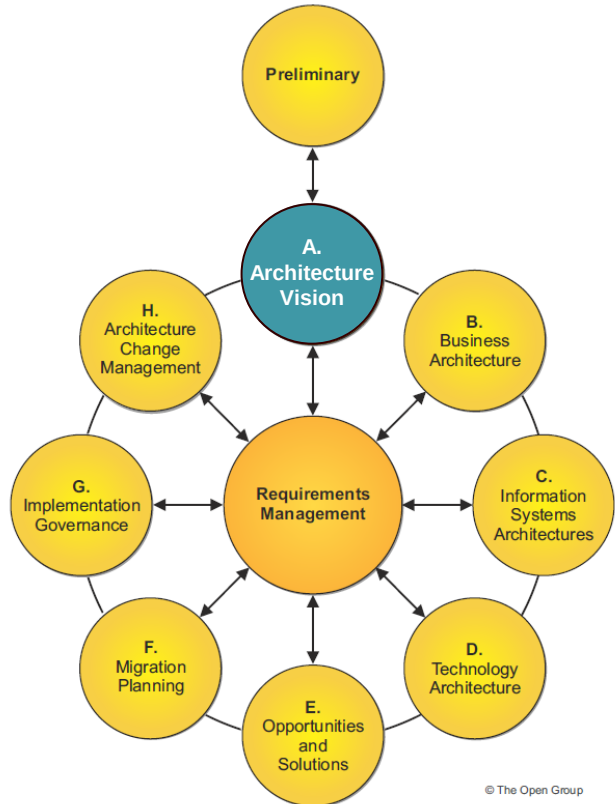
Request for Architecture Work – Usage within the TOGAF ADM

The deliverable is created in the Preliminary Phase, Phase F and H



ADM – Phase A: Architecture Vision – Communications Plan

The communications plan is derived from a stakeholder map



1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Communications Plan – Definition

The deliverable defines how architecture information reaches the stakeholders

Purpose

- Allows **communication** for to be carried out within a **planned and managed process**
- Ensures **effective communication** of **targeted information** to the **right stakeholders** at the **right time** as critical success factor for Enterprise Architecture

Content

- **Stakeholders** (e.g., grouping by communication requirements)
- **Information** (e.g., needs, key messages for the vision, risks, success factors)
- **Mechanisms** (e.g., meetings, newsletters, repositories)
- **Timetable** (e.g., stakeholder <> time <> location)

Communications Plan – Structure

The document defines the “who, what, how and when” of communication



Communications Plan

Project XXXX
Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Stakeholders	4
3 Communication Requirements	5
4 Communication Mechanism	6
5 Communication Timetable	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
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Reviewed By:			

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2 Stakeholders

2.1 Stakeholders

Item Name	Stakeholders
Stakeholder	Sponsor
Concerns	Who will be impacted by the architecture? Who must be consulted about what aspects of the architecture?
Description	There are two types of stakeholders: - Those who can influence a business activity or project - Those who are impacted by a business activity or project The two types of stakeholder should be addressed separately, with particular care and attention paid to any stakeholder who falls into both groups. An understanding of the benefits case for all stakeholders is crucial to ensuring the success of a project. If a group of stakeholders exists that will not benefit from, or worse will be disadvantaged by, a project then the scope, mission, strategy, and objectives of the business unit/project may need to be revisited.
Guidance	For a stakeholder analysis the next steps are necessary: - Define all relevant stakeholders - Assess the nature of each stakeholder's interest/concern - Assess the nature of each stakeholder's influence - Map stakeholder relationships - Map stakeholder conditions - Construct a matrix of stakeholder priorities When defining the stakeholders, a stakeholder mapping (as a visual representation of stakeholders and their relationship to the mission and strategy) can be a vital tool. During a project it is important to monitor the stakeholder interests, relationships, and conditions.

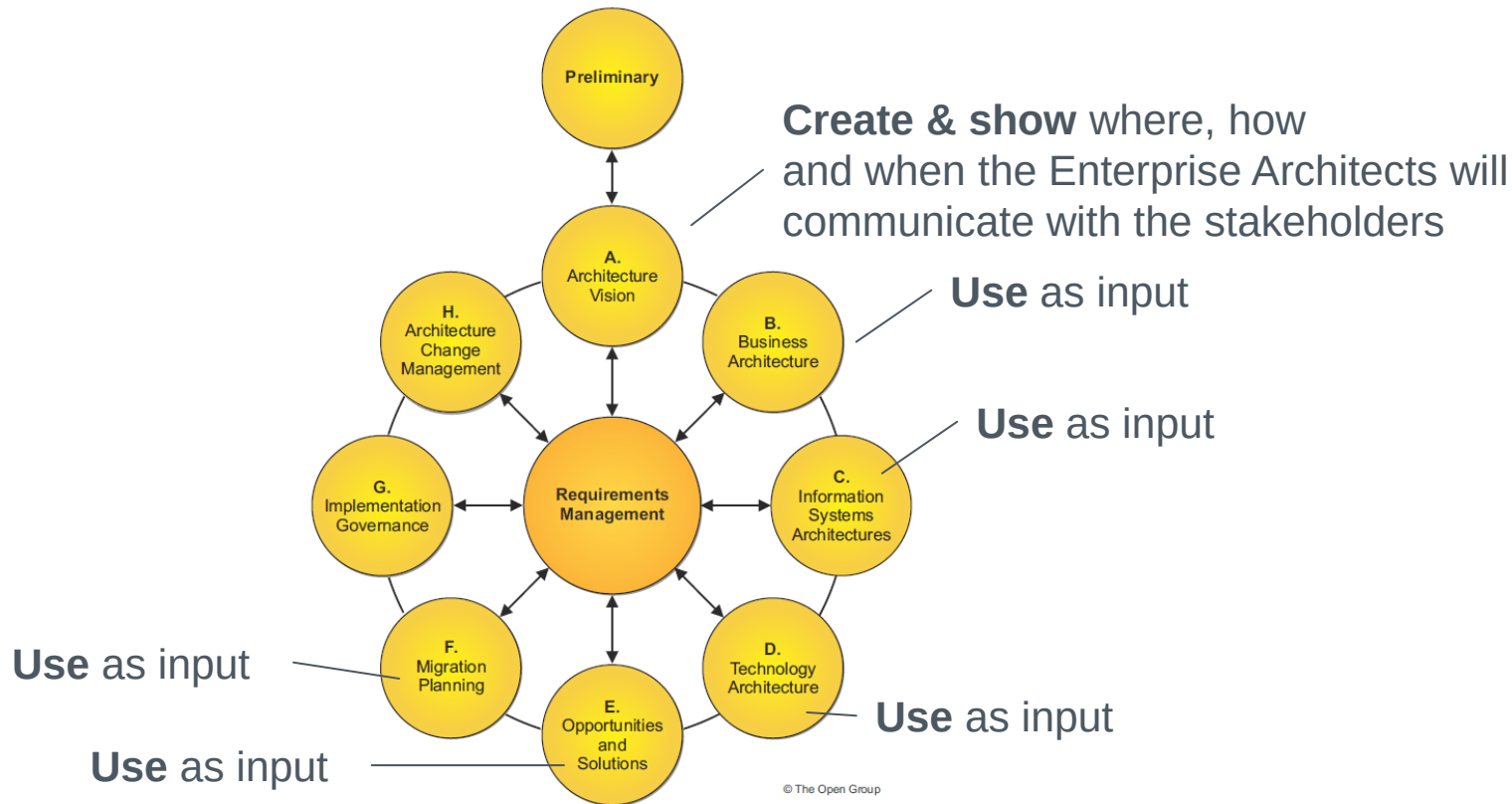
Reference-ID*	Title*	Stakeholder Description	Concerns

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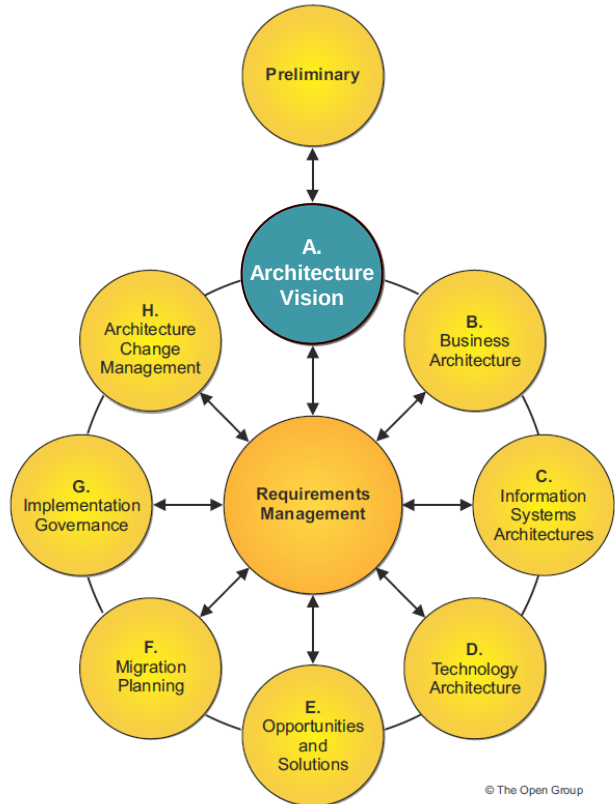
Communications Plan – Usage within the TOGAF ADM

The deliverable is created in Phase A and used throughout Phase B till F



ADM – Phase A: Architecture Vision – Business Scenarios

Business Scenarios are a source for business requirements



1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

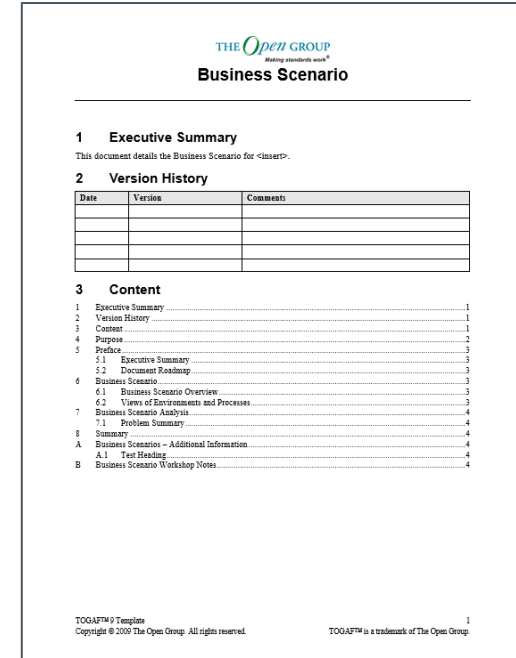
6

Define scope

Business Scenarios – Definition

The technique identifies, documents & understands the business needs

- Derive the **characteristics of the architecture** from the high-level **requirements of the business** to achieve **desired business outcomes**
- Consider **business value, outcomes & Value Streams** and the required **resources** to understand the **requirements** in any industry (e.g., healthcare, transportation, oil)
- Can be **applied** to very **large general problems areas** (e.g., standards for national lotteries) and very **small focused problem areas** (e.g., retail point of sale upgrades)



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Making standards work™
Business Scenario

1 Executive Summary
This document details the Business Scenario for <insert>.

2 Version History

Date	Version	Comments

3 Content

1	Executive Summary	1
2	Version History	1
3	Context	1
4	Purpose	2
5	Preface	3
6	Business Scenario	3
6.1	Executive Summary	3
6.2	Document Roadmap	3
6.3	Business Scenario Overview	3
6.4	Views of Environments and Processes	3
7	Business Scenario Analysis	4
7.1	Problem Summary	4
8	Summary	4
A	Business Scenario – Additional Information	4
A.1	Test Heading	4
B	Business Scenario Workshop Notes	4

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1

Business Scenarios – Key elements (1/2)

The technique uses a uniform description for business requirements



Real **business problem**



Business and technology environment in which the problem occurs



Desired outcome(s) of proper execution



Human actor(s)
who provide the capabilities



Computer actor(s)
who support the capabilities

- **Not examining all elements of a Business Scenario** carries a risk of producing an incomplete solution

Business Scenarios – Key elements (2/2)

A good Business Scenario contains SMART outcomes

S



Specific

by defining what needs to be done in the business

M



Measurable

through clear metrics for success

A



Actionable

by segmenting the problem and providing the basis for determining elements & plans for the solution

R



Realistic

if the problem can be solved within the bounds of physical reality, time and cost constraints

T



Time-bound

with a clear statement of when the solution opportunity expires

Business Scenarios – Development phases & steps

Each key element is iteratively refined through phases, steps & activities

Steps:

- Plan
- Gather Information
- Analyze/Process
- Document
- Review

1- Problem

2- Environment

3- Outcomes

4- Human Actors

5- Computer Actors

- Three successive **phases** capture, examine and improve the **key elements** of the **Business Scenario**
- Each **phase** requires the **steps** planning, information gathering, analysis, documentation and review consisting of several **activities**
- All phases are repeated through iterations until the **understanding is fit-for-purpose** for deciding to move forward
- The resulting Business Scenario is valid **at a specific point in time** and should be updated to **reflect changes**

- **Use-cases (e.g., OMG, Rational Software):** Detailed descriptions of human to computer interaction being typically used in software development
- **Business Models/cases/scenario plans:** A Business Scenario can be informed by a company's Business Model and a SMART Business Scenario can inform a Business Model/case/scenario plan
- **SPIN (Situation, Problem, Implication, Need-Payoff):** Sales technique that can be used to gather information for a Business Scenario and the Business Scenario can provide relevant “context” for a SPIN engagement
- **Engineering specifications (e.g., IEEE):** Detailed, material-specific and tied to science

Business Scenarios – Benefits

The technique creates value at an early stage in the Architecture Project



Provide a common **language** with which the commercial (software) **vendor community** can link customer problems to their solutions while drawing on open standards & skills



Enable **all requirements** to be viewed in relation to one another **in the context of the overall problem**, its **solution** and the **business value**



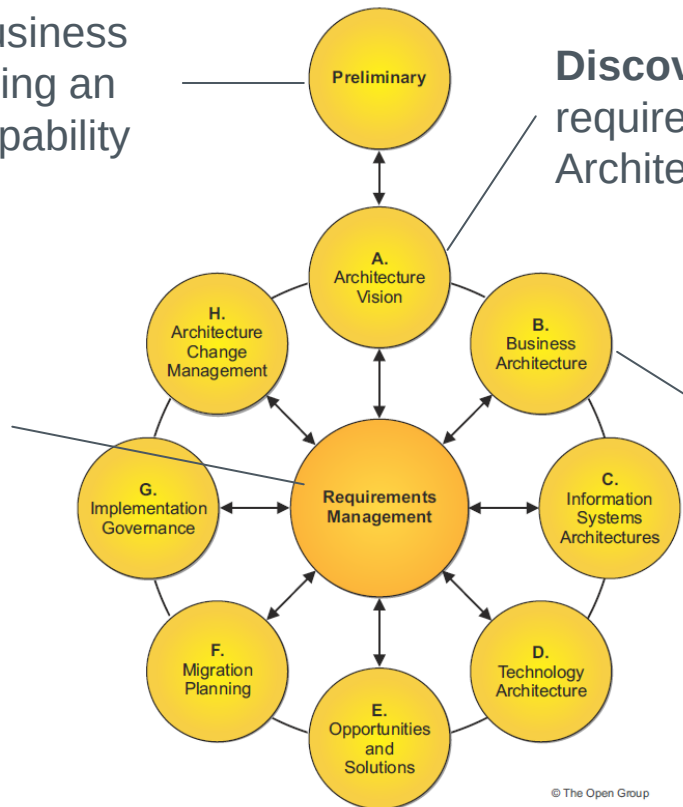
Gain the **buy-in of stakeholders** (esp.: business line management) to the Architecture Project and the resulting Enterprise Architecture

Business Scenarios – Usage within the TOGAF ADM

The technique is used in ADM Phases A & B and Requirements Management

Discover & document business requirements for establishing an Enterprise Architecture capability

Discover & document business requirements and provide them for all phases of an ADM lifecycle



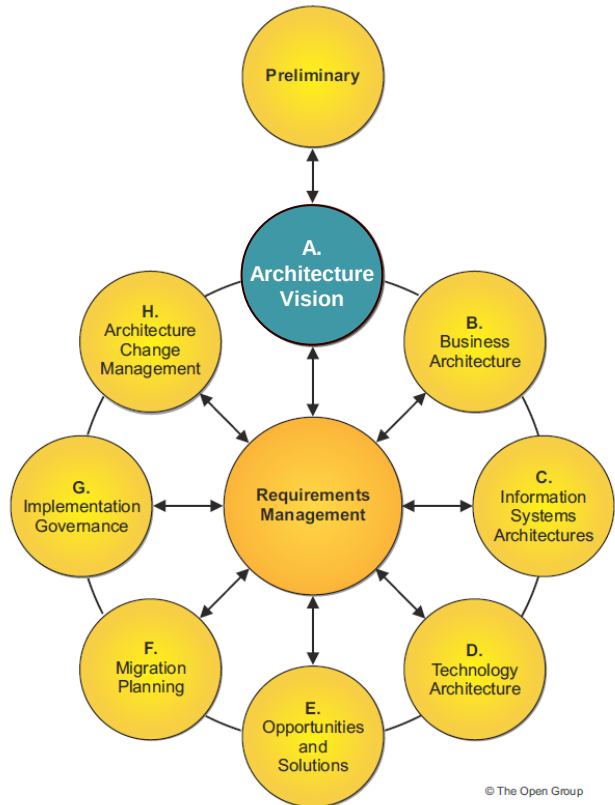
Discover & document business requirements and deriving an Architecture Vision

Discover & document business requirements and iteratively decomposing the Business Architecture at different levels of detail

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ADM – Phase A – Architecture Requirements Specification

The phase documents the requirements that are in scope of the project



1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Architecture Requirements Specification – Definition

The quantitative view of the solution states the criteria that must be met

Purpose

- Defines a set of **quantitative statements** that outline what an **implementation project** must do to comply with the **architecture**
- Is a major part of an **implementation contract** or contract for **Architecture Definition**
- Acts as **companion** to the **Architecture Definition Document** (= qualitative view)

Content

- Architecture, Interoperability & IT Service Management **requirements**
- Business & Application **service contracts**
- **Implementation** guidelines, specifications & standards
- **Success measures, Constraints, Assumptions**

Architecture Requirements Specification – Structure

The deliverable complements the Architecture Definition Document



Architecture Requirements Specification

Project XXXX
Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Architecture Requirements	4
3 Service Contracts	5
4 Implementation	6

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
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Reviewed By:			

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TOGAF™ 9 Template: Architecture Requirements Specification
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2 Architecture Requirements

2.1 Architecture Requirements

2.2 Interoperability Requirements

2.3 Constraints

2.4 Assumptions

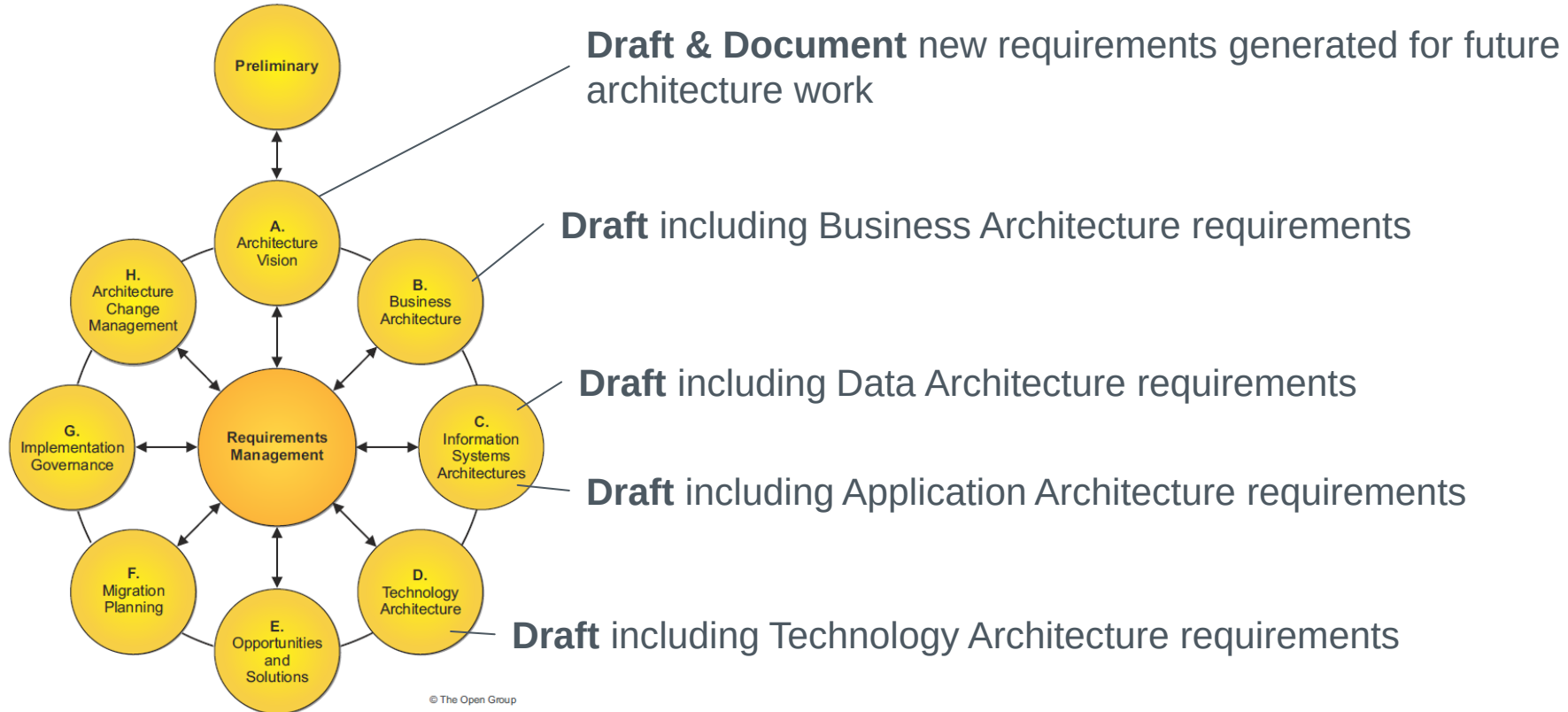
2.5 Success Measures

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Architecture Requirements Spec. – Usage within the TOGAF ADM (1/2)

The deliverable is created in Phase A and complemented during Phase B till D



Architecture Requirements Spec. – Usage within the TOGAF ADM (2/2)

The deliverable is used in Phase G & H and populated in Requirements Mgmt.

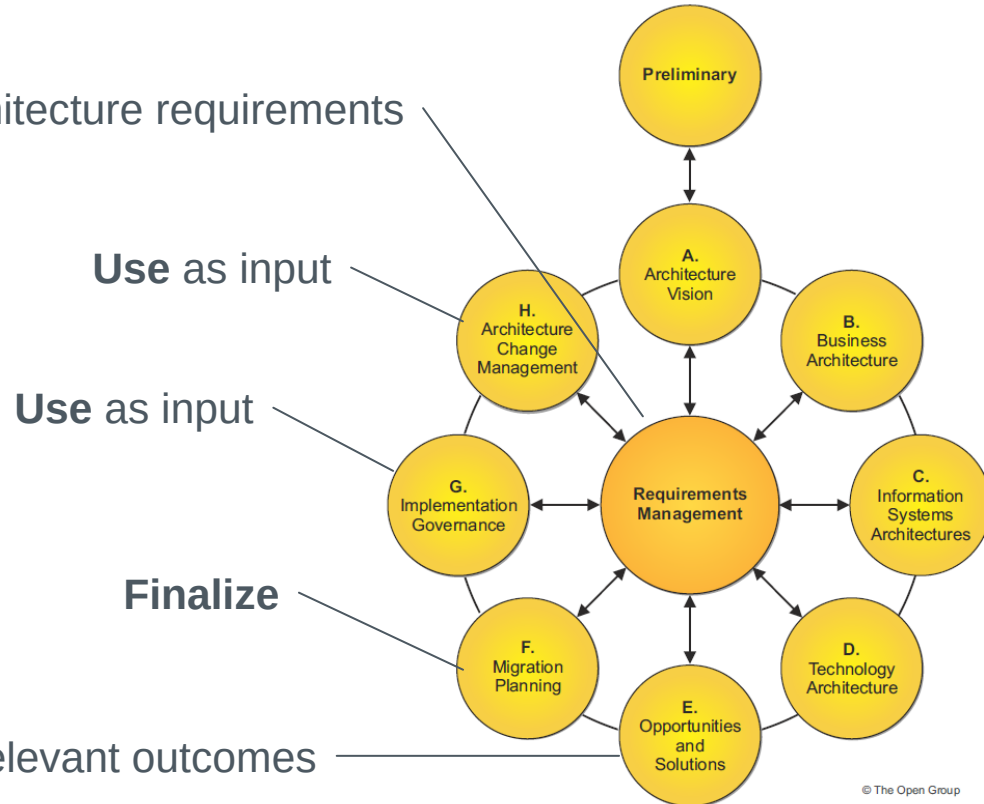
Populate & update with architecture requirements

Use as input

Use as input

Finalize

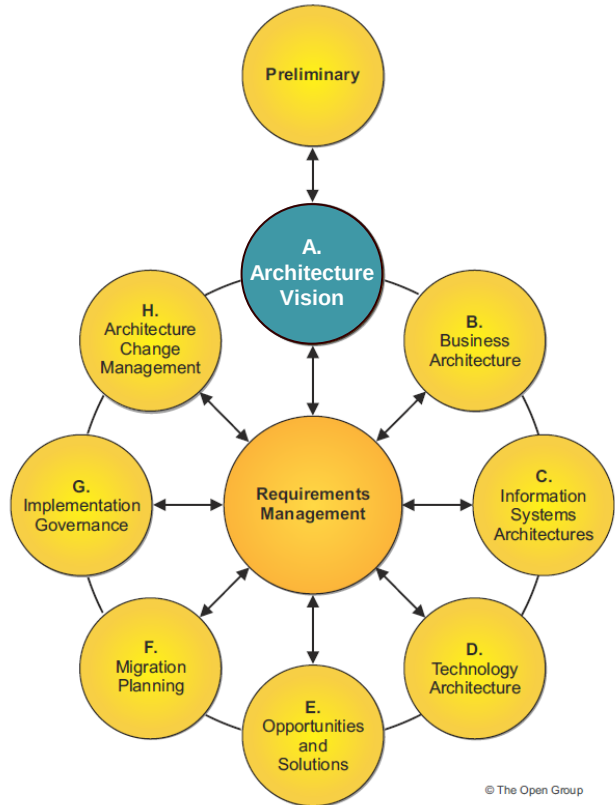
Update with any additional relevant outcomes



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ADM – Phase A: Architecture Vision – Principles, Goals & Drivers

The phase explores the contextual surroundings of the Architecture Project



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1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Business Principles, Goals & Drivers – Definition

The concepts provide the business context for the architecture work

Purpose

- Describe the **needs and ways of working** within the enterprise
- May have **significant implications** for the way that architecture is developed, although being outside the consideration of the architecture discipline

Content

- **Business principles, business goals and business drivers**
- **Vary considerably** from one organization to the next

Business Principles, Goals & Drivers – Structure

The content & structure of business context for architecture is likely to vary



Business Principles, Goals, Drivers

Project XXXX Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Business Principles	4
3 Business Goals	5
4 Business Drivers	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
Prepared By:		Document Version Date:	
Title:	Business Principles, Goals, Drivers	Review Date:	
Reviewed By:			

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TOGAF™ 9 Template: Business Principles, Goals, Drivers
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4 Business Drivers

View Name	Business Drivers
Stakeholder	Sponsor
Concern	Do the Business Drivers to which the architecture is aligned, match my expectations?
Description	The impacts of environmental trends on an organization are described as the business drivers. Another way to view business drivers is that they represent the business' understanding of the way the business must itself change in response to changes and trends in the environment. The business drivers drive the creation of business strategies, and shape the architecture principles. The business vision is the business' estimate of where a set of expected changes and trends in the environment and the responses to them will position the business at some future time. The business vision can crystallize the business drivers, and thus the organization is in control of and changes its own environment. An example of an environment change creating business drivers: "A new low-cost airline enters the market and the existing, high-cost carriers have to change their business strategies to the new operating environment." An example of changing the environment creating business drivers: "A company realizes that the airline market is dominated by high-cost carriers. It decides to enter the market, defining a new niche for low-cost no-frills travel."
Guidance	To be able to find all relevant business drivers for the project, the following list can help: - Competitive environment – e.g., emergence of powerful competitors with different business models. - Regulatory environment – e.g., impact of US reporting requirements on foreign owned businesses or trading partners. - Ownership environment – e.g., emergence of private equity firms, leading to different expectations of returns. - Demand environment – e.g., high growth in acceptability and use of e-channels amongst many customer segments. - Supply Environment – e.g., M&A in the supply base, leading to a shift in bargaining power. - Resources environment – e.g., increased availability and capability of call centres in Eastern Europe and Asia. The attributes of the drivers themselves are described within the appropriate artifact template.

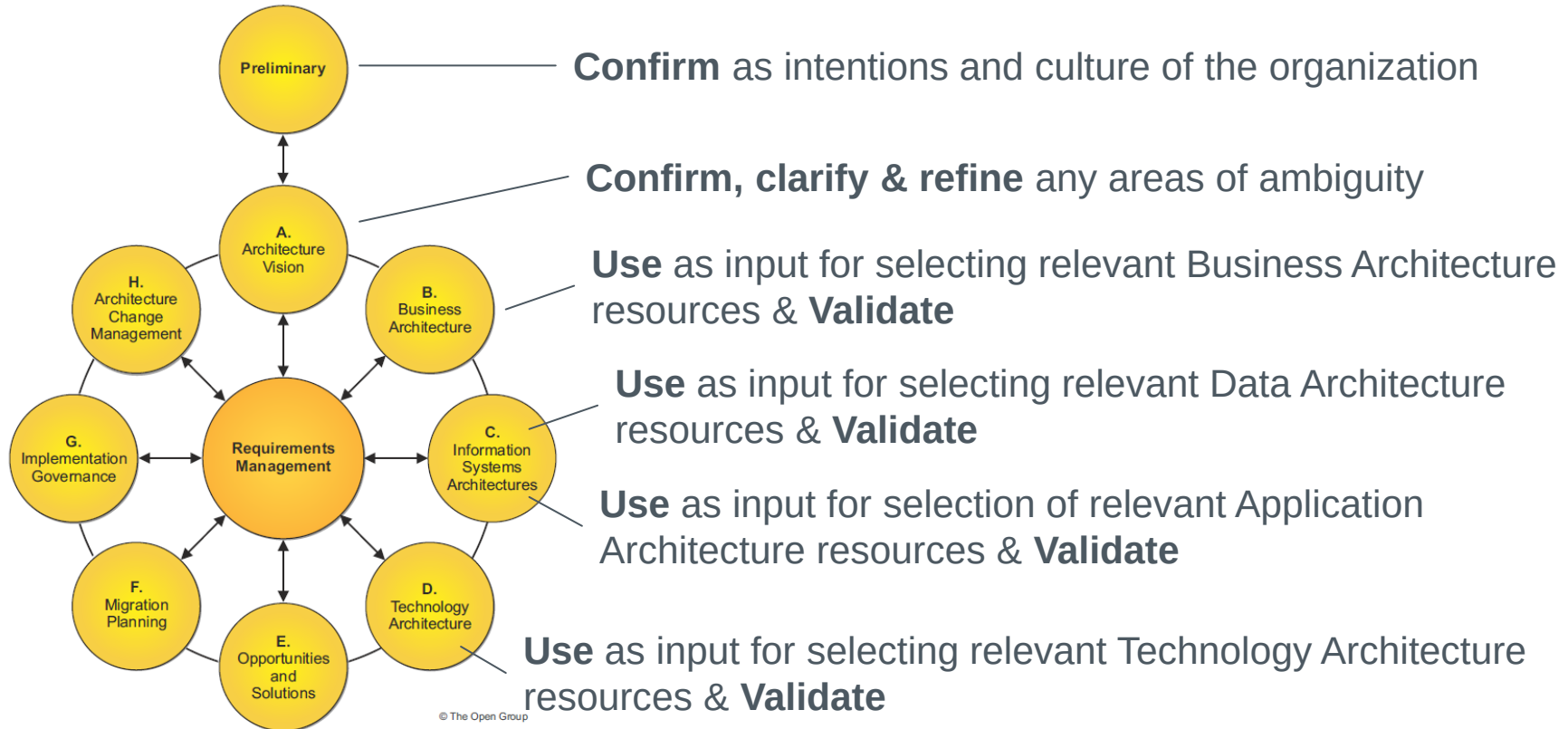
Reference-ID	Title	Business Driver

TOGAF™ 9 Template: Business Principles, Goals, Drivers
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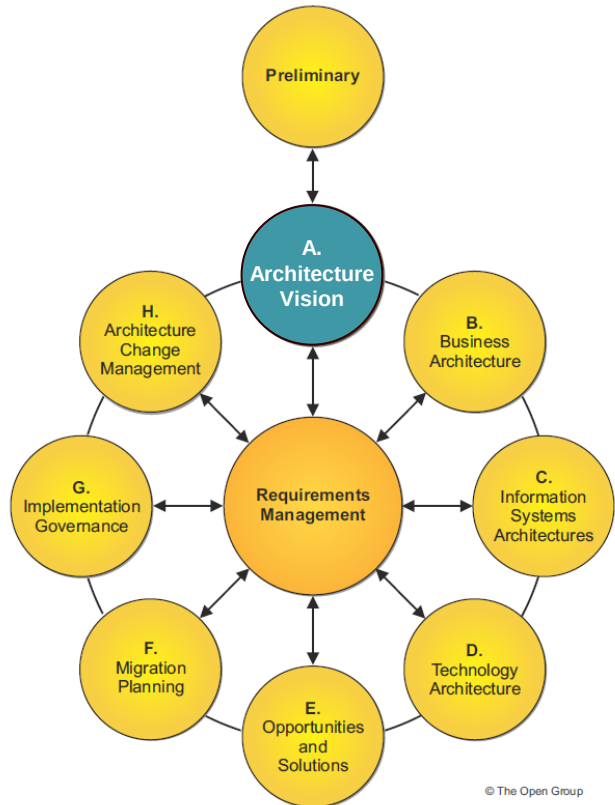
Business Principles, Goals & Drivers – Usage within the TOGAF ADM

The concepts are clarified in Phase A and used throughout Phase B till D



ADM – Phase A: Architecture Vision – Capability Assessment

The phase performs a Capability Assessment to evaluate capabilities



1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Capability Assessment – Definition (1/2)

The deliverable captures the baseline & target capability level of the enterprise

Purpose

Can be examined on several **capability levels**:

- **Enterprise**, e.g., architectural focus areas
- **IT function**, e.g., design governance, operational governance, skills and organization structure or style, formality and detail for the architecture project
- **Architecture function**, e.g., architectural assets, standards and reference models, re-usable assets
- **Capability gaps**, e.g., business readiness, risks to transformation, cultural barriers

Capability Assessment – Definition (2/2)

The deliverable is a prerequisite for a detailed architecture definition

- **Business Capability Assessment**, including baseline state assessment & future state aspiration regarding performance and realization; impacts to the organization
- **IT Capability Assessment**, including baseline & target maturity level of change and operational processes; baseline capacity & capability; impacts to the organization
- **Architecture maturity assessment**, including Architecture Governance; re-use potential; Architecture skills; landscape, standards and reference model definition
- **Business Transformation Readiness Assessment**, including readiness factors and particular vision; current and target readiness ratings; readiness risks

Capability Assessment – Structure

A template supports the capturing & documentation of the capabilities



Capability Assessment

Project XXXX
Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents

1	Purpose of this Document	3
2	Business Capability Assessment	4
3	IT Capability Assessment	5
4	Architecture Maturity Assessment	6
5	Business Transformation Readiness Assessment	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
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Document Version History

Version Number	Version Date	Revised By	Description	Filename

2 Business Capability Assessment

2.1 Capabilities of the Business

2.2 Baseline Performance Assessment

<<Baseline state assessment of the performance level of each capability.>>

2.3 Future Performance Aspiration

<<Future state aspiration for the performance level of each capability.>>

2.4 Baseline Capability Assessment

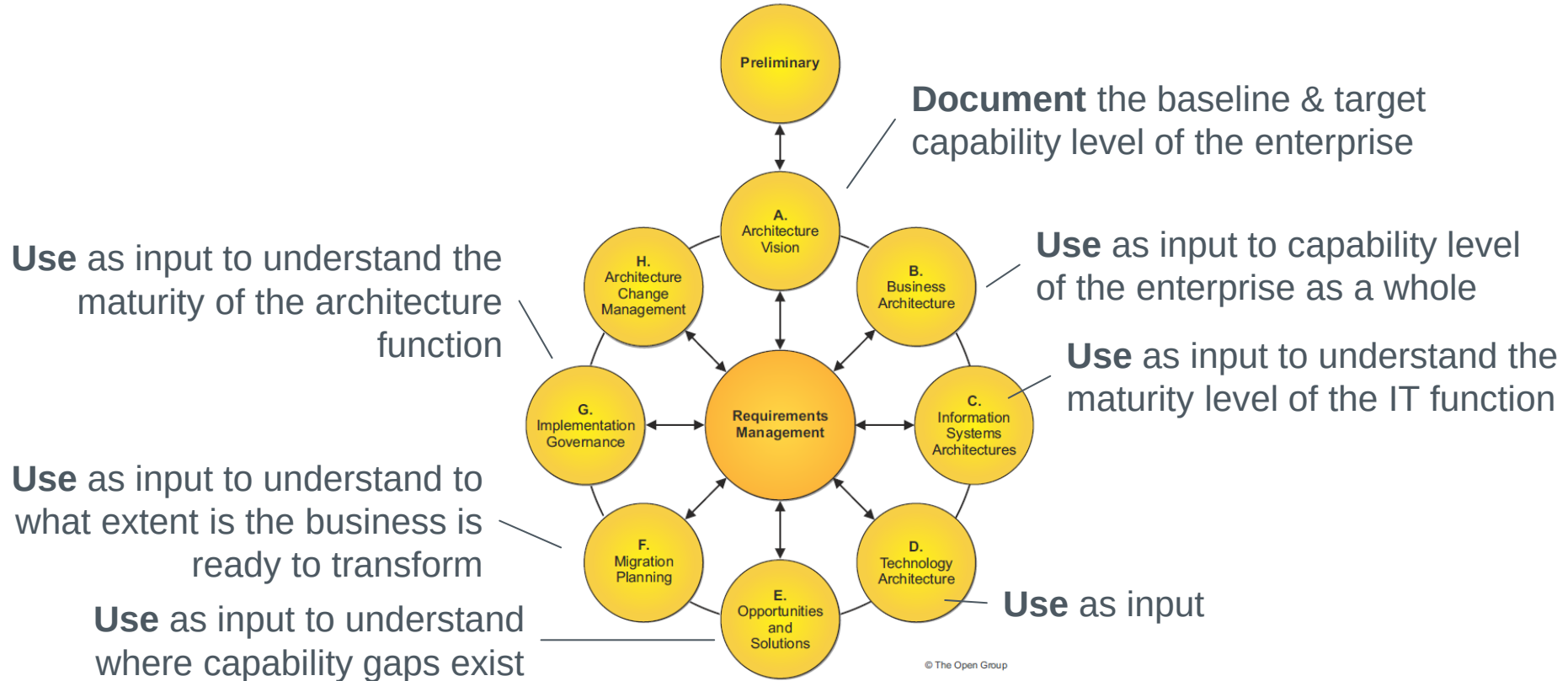
<<Baseline state assessment of how each capability is realized.>>

2.5 Future Capability Aspiration

<<Future state aspiration for how each capability should be realized.>>

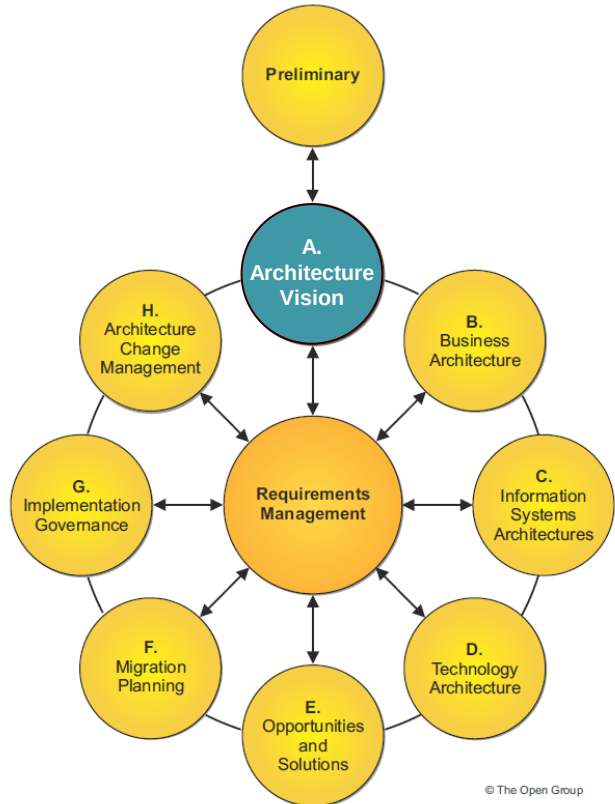
Capability Assessment – Usage within the TOGAF ADM

The deliverable is created in Phase A and used throughout Phase B till G



ADM – Phase A – Business Transformation Readiness Assessment

The technique evaluates the organization's readiness to undergo change



1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Business Transformation Readiness Assessment – Motivation

The technique assesses an organization's readiness for business transformation

- **Enterprise Architecture** will entail **considerable change**
- Innovative and sound architecture plans could lead to nowhere, if there is a **change-averse culture** and a **narrowly skilled workforce**
- Need for understanding the **readiness of the organization to accept change**, identifying the issues and then dealing with them
- **Business Transformation Readiness Assessment** as a **joint effort** between **corporate staff** (especially human resources), **lines of business** and **IT planners**



Image source: [Pexels.com](https://www.pexels.com/photo/sailboat-on-water-at-sunset/) / Tom Fisk

Business Transformation Readiness Assessment – Approach

The ADM technique assesses an organization's readiness through five activities

1. Determine the **readiness factors** that will impact the organization
2. Present the readiness factors using **maturity models**
3. Assess the readiness factors, including determination of **readiness factor ratings**
4. Assess the **risks** for each readiness factor and identify **mitigation actions**
5. Work these actions into **Phase E and F Implementation and Migration Plan**

> There are **many dimensions to change**,
but by far the **most important** is the **human element**

Business Transformation Readiness Assessment – Example

The TOGAF Standard provides templates & tools for the ADM technique



Business Transformation Readiness Assessment - Maturity Model					
Factor 2: Need for Enterprise Information Architecture			Class		Organizational Context
			BTEP Readiness Factor		YES
Definition		There is recognition by the organization that information is a strategic corporate asset requiring stewardship. There is also recognition that the data is not universally understandable, of requisite quality, and accessible.			
Maturity Model Levels					
0 Not defined	1 Ad Hoc	2 Repeatable	3 Defined	4 Managed	5 Optimized
Information is not recognized as an asset. There is no clear stewardship of data.	Data Management (DM) concepts are intuitively understood and practiced on an <i>ad hoc</i> basis. Stewardship of the data is informal. Data is recognized by certain internal experts and senior management as being of strategic importance to the organization. Focus is primarily on technically managing redundant data at the applications level.	Many parts of the organization value information/data as a strategic asset. Internal DM experts maintain clear lines of responsibility and stewardship of the data, organized along lines of business and at all senior levels. Staff put into practice DM principles and standards in their daily activities.	Data is recognized as a strategic asset in most parts of the organization, and throughout most levels from operations to senior management. Resources are committed to ensuring strong stewardship of data at the lower management and information expert levels.	Data is recognized as a strategic asset in all parts of the organization, and throughout most levels from operations to senior management. Resources are committed to ensuring strong stewardship of data at the senior management and information expert levels.	Data is treated in all levels throughout the organization as a strategic asset to be exploited and re-used. Data products and services are strongly integrated with the management practice of the organization. All staff are empowered and equipped to take stewardship of information, and are seen as "knowledge workers".
				Recommended Target State	

Maturity Model for the Data Architecture

Business Factor Assessment Summary				
Ser	Readiness Factor	Urgency	Readiness Status	Degree of Difficulty to Fix
1	Vision			
2	Desire/willingness/resolve			
3	Need			
4	Business case			
5	Funding			
6	Sponsorship and leadership			
7	Governance			
8	Accountability			
9	Workable approach and execution model			
10	IT capacity to execute			
11	Departmental capacity to execute			
12	Ability to implement and operate			

Summary table as management overview

Business Transformation Readiness Assessment – BTEP Method

The complementary method helps identifying business transformation issues

- Canadian Government **Business Transformation Enablement Program** (BTEP) method provides guidance on how to identify **business transformation-related issues**
- All projects conduct a **transformation readiness assessment** based on **defined readiness factors** to **uncover the business transformation issues**
- Outcome is a deep **understanding of the challenges and opportunities** with many of them translating into risks

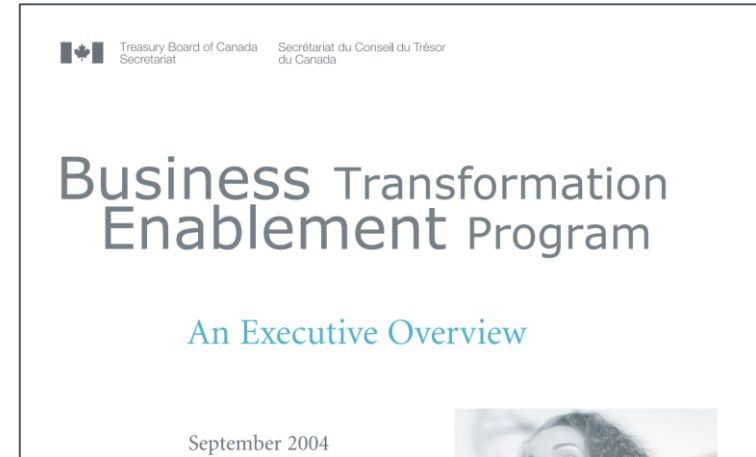
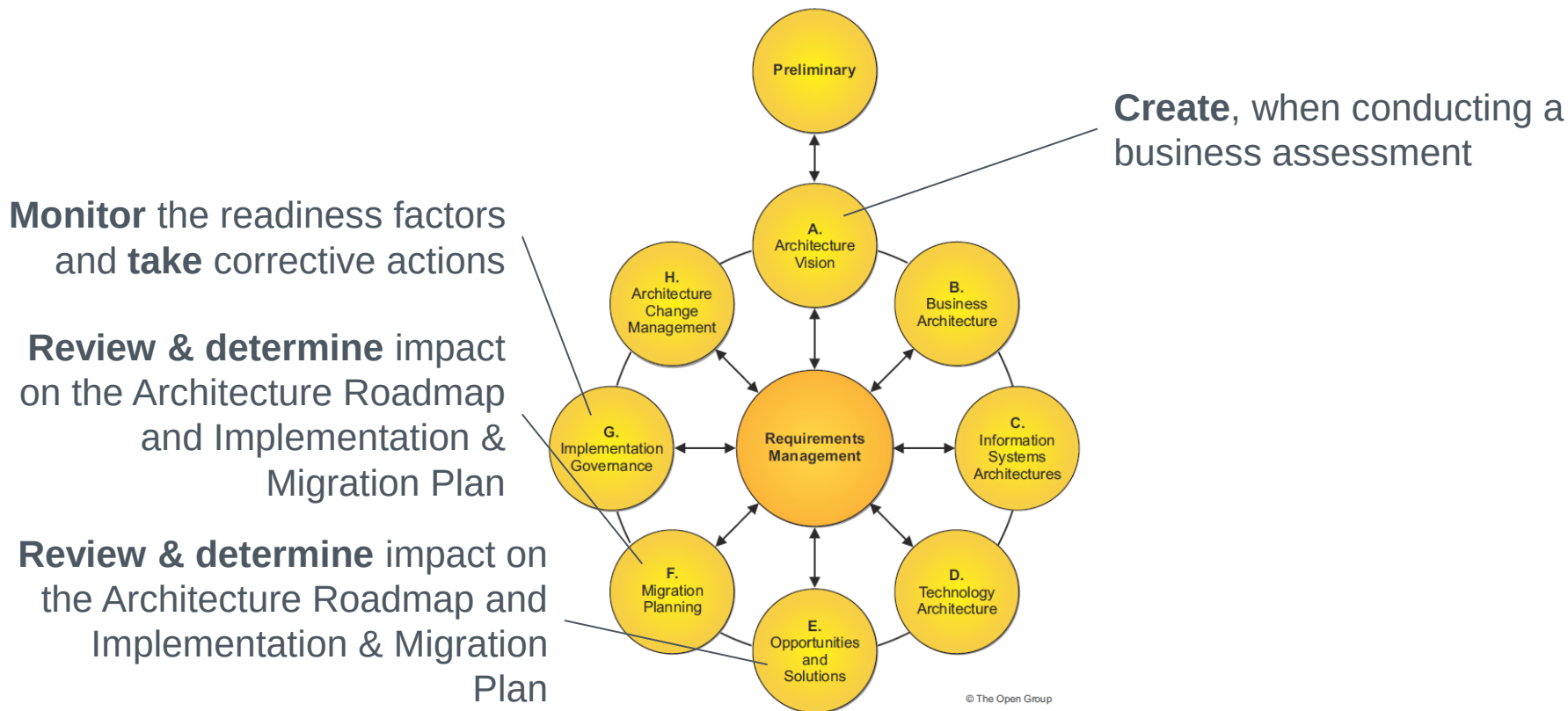


Image source: [Government of Canada Publications](#)

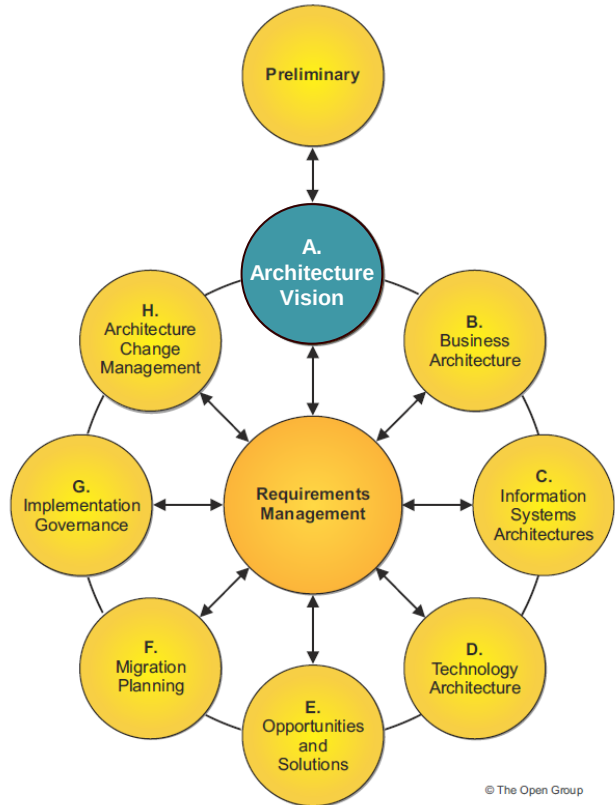
Business Tr. Readiness Assessment – Usage within the TOGAF ADM

The deliverable is created in Phase A and used in Phase E, F and G



ADM – Phase A: Architecture Vision – Architecture Vision

The deliverable contains a high-level aspirational vision of the capabilities & value



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7

Confirm and elaborate **Architecture Principles**, including business principles

8

Develop Architecture Vision

9

Define the Target Architecture **value propositions** and KPIs

10

Identify the business transformation risks and mitigation activities

11

Develop Statement of Architecture Work; secure approval

ADM Phase A: Architecture Vision – Definition

The early summary version of the full Architecture Definition eases communication

Purpose

- Contains a **summary of the changes** to the enterprise after the successful deployment of the Target Architecture
- Provides **stakeholders** with a formally **agreed outcome** and **communication basis** early in the ADM cycle
- Serves as a basis to develop the **Domain Architectures (BDAT)**

Content

- **Problem description** (stakeholder and concerns, list of issues/scenarios)
- **Objective of the Statement of Architecture Work**
- **Summary views** (e.g., value chain diagram, solution concept diagram)
- Mapped **requirements** and **reference** to draft **Architecture Definition Document**

Architecture Vision – Structure

The document provides key stakeholders with a formally agreed outcome



Architecture Vision

Project XXXX Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

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1 Purpose of this Document	3
2 Problem Description	4
3 Detailed Objectives	6
4 Environment and Process Models	7
5 Actors and their Roles and Responsibilities	9
6 Resulting Architecture Model	10
7 End Vision Statement	11

Document Information

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* Action Types: Approve, Review, Inform, File, Action Required, Attend Meeting, Other (please specify)

Document Version History

Version Number	Version Date	Revised By	Description	Filename

TOGAF™ 9 Template: Architecture Vision
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2 Problem Description

2.1 Stakeholders and their Concerns

<<The purpose of this section is to describe the stakeholders for the vision together with any concerns they may have.

Mandatory/optional: This section is mandatory.

In terms of quality criteria, this section should make clear:

- Any stakeholder concerns about the business
- Any stakeholder concerns about the exercise being undertaken>>

2.2 List of Issues/Scenarios to be Addressed

<<The purpose of this section is to define the business context and problem description for which the vision of the target architecture has been produced.

Mandatory/optional: This section is mandatory.

In terms of quality criteria, this section should make clear:

- Business vision statement for the target architecture
- Pictorial representation of business context
- Change drivers and opportunities behind this vision for the target architecture>>

2.3 Business Vision Statement

<<The purpose of this section is to define the business vision statement of the target architecture.

Mandatory/optional: This section is mandatory.

In terms of quality criteria, this section should make clear:

- Business vision statement(s) for the target architecture>>

2.4 Business Vision Diagram

<<The purpose of this section is to provide one or more suitable pictorial representation(s) that explain the business context and business problem. They are required in order to obtain senior executive engagement with this activity.

Mandatory/optional: This section is optional but recommended.

In terms of quality criteria, this section should make clear:

- Diagram(s) explaining the problem statement for the target architecture
- Description of business vision diagram>>

2.5 Change Drivers & Opportunities

<<The purpose of section is to identify the change drivers and opportunities behind this vision for the target architecture.

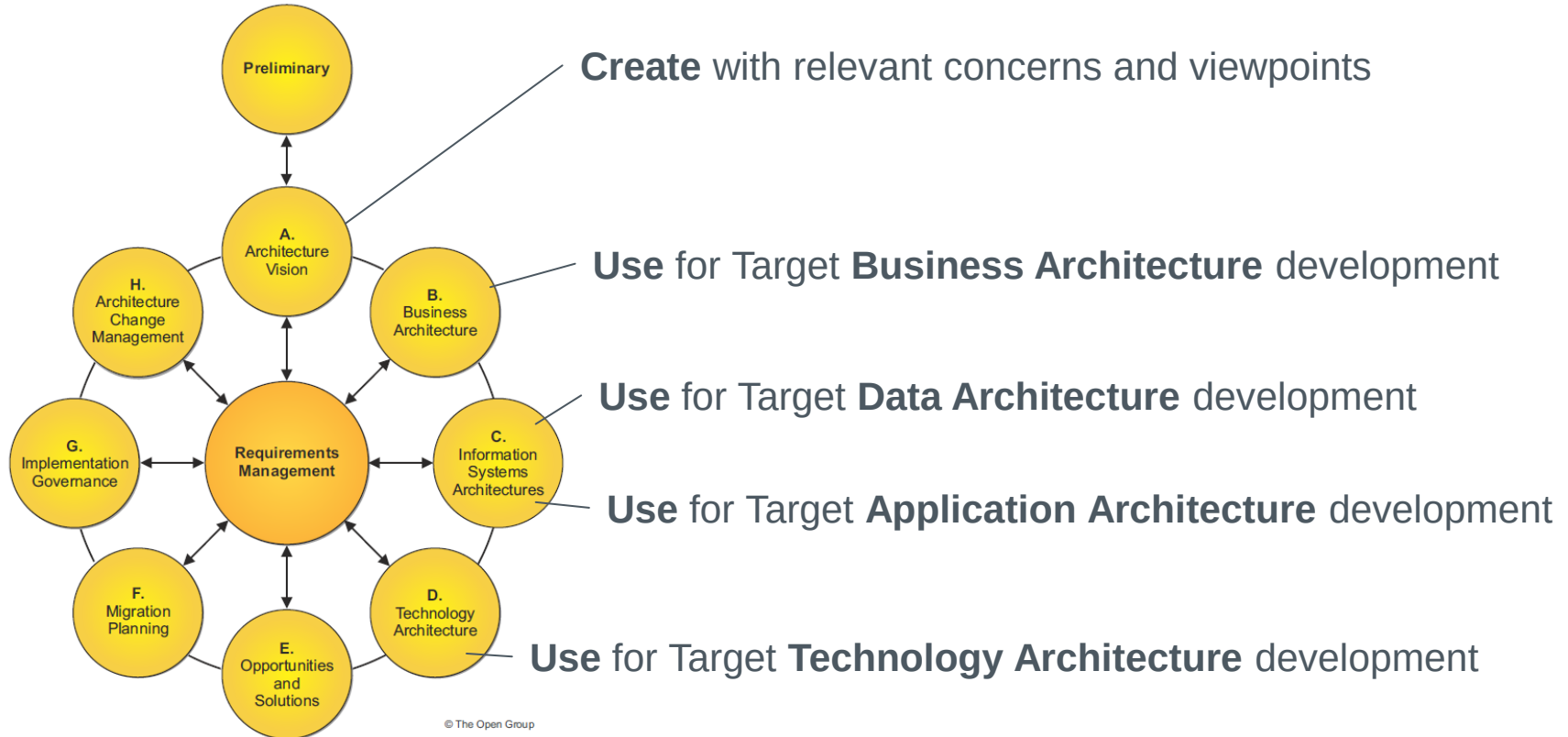
Mandatory/optional: This section is mandatory.

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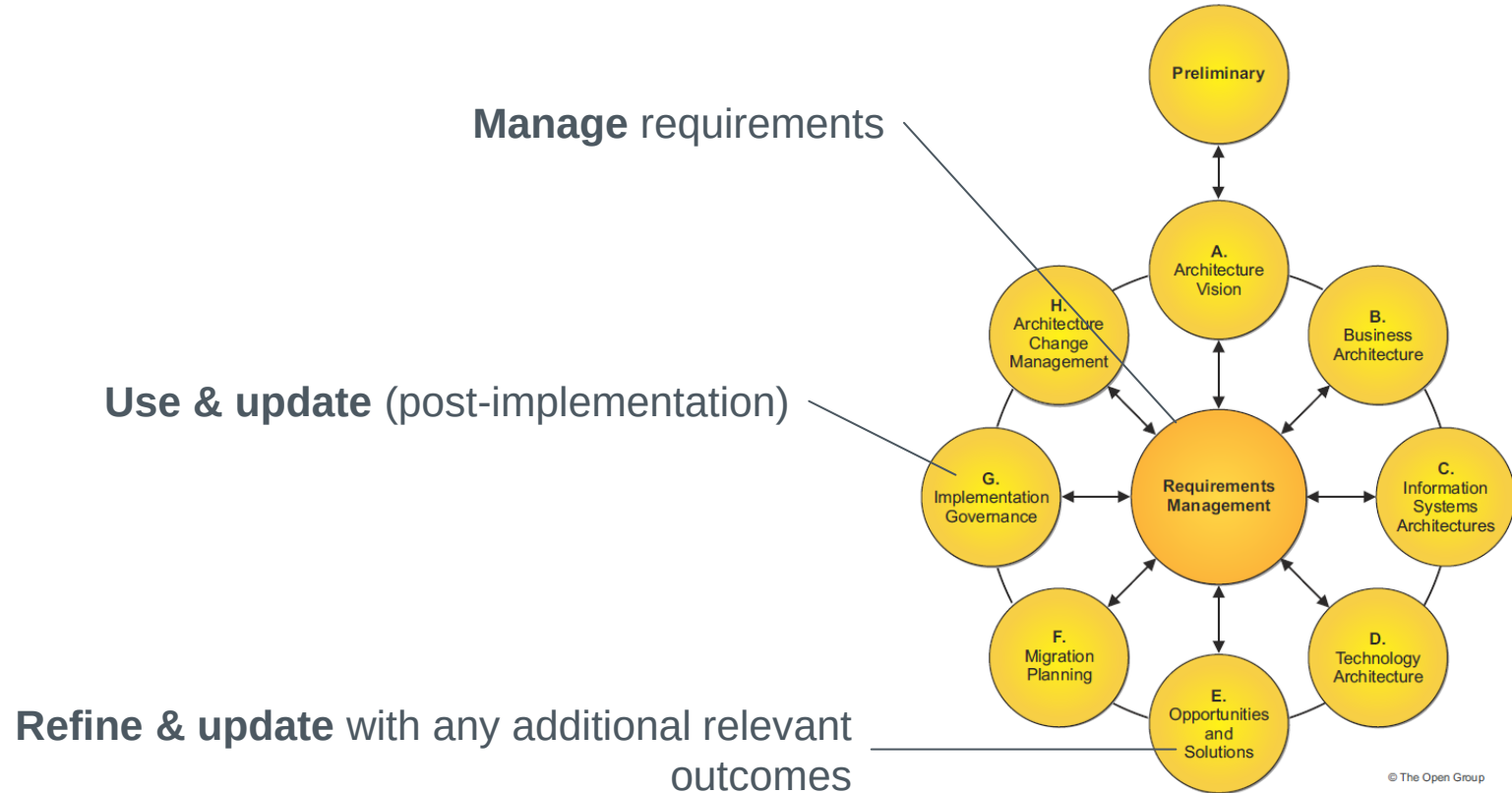
Architecture Vision – Usage within the TOGAF ADM (1/2)

The deliverable is created in Phase A and used throughout Phase B till D



Architecture Vision – Usage within the TOGAF ADM (2/2)

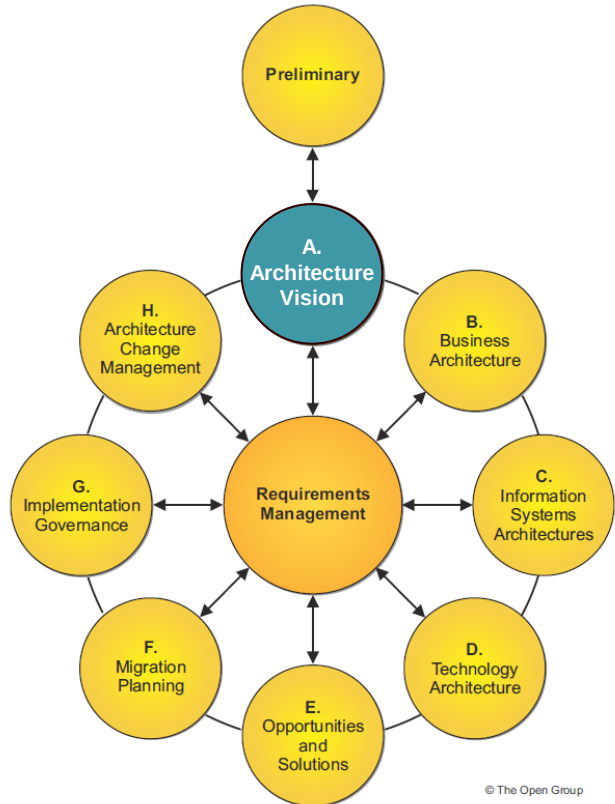
The deliverable is refined in Phase E and used in Phase G



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ADM – Phase A: Architecture Vision – Architecture Definition Document

The deliverable acts as container for the core architectural artifacts



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1

Establish the architecture project

2

Identify stakeholders, concerns and business requirements

3

Confirm and elaborate business goals, business drivers and constraints

4

Evaluate capabilities

5

Assess readiness for business transformation

6

Define scope

Architecture Definition Document – Definition (1/2)

The deliverable is a qualitative view of the solution communicating the intent

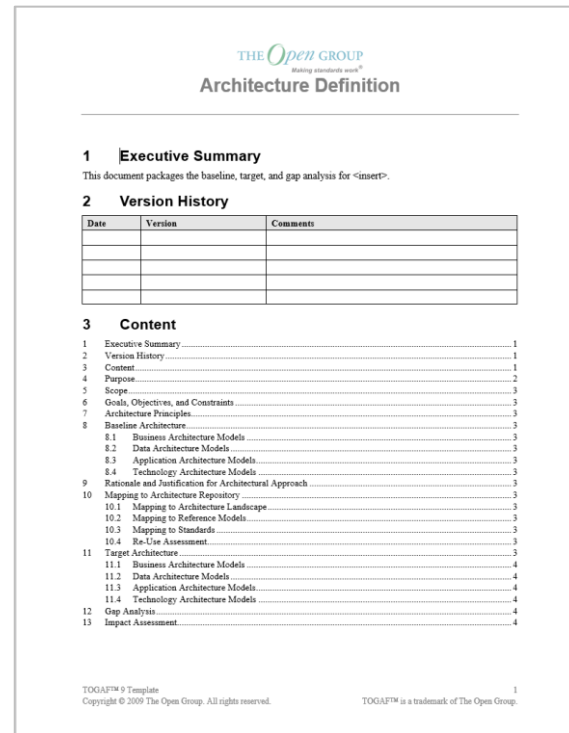
Purpose

- Works as **deliverable container** for the core **architectural artifacts** created during a project and **important related information**
- **Qualitative view of the solution**
- Spans **all architecture domains** (Business, Data, Application and Technology)
- Examines **all relevant states of the architecture** (Baseline, Transition and Target)
- Aims to communicate the **intent of the architects**
- Acts a **companion** to the **Architecture Requirements Specification** (= quantitative view)

Architecture Definition Document – Definition (2/2)

The deliverable is a qualitative view of the solution communicating the intent

- **Scope, Goals, Objectives & Constraints, Architecture Principles, Baseline & Target Architecture, Gap analysis, Impact assessment**
- **Architecture Models** for all domains
- **Rationale and justification** for architectural approach
- **Mapping to Architecture Repository** (Architecture Landscape, reference models, standards, re-use assessment)
- **Transition architectures** (definition, business, data, application, technology)



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Making standards work™
Architecture Definition

1 Executive Summary
This document packages the baseline, target, and gap analysis for <insert>.

2 Version History

Date	Version	Comments

3 Content

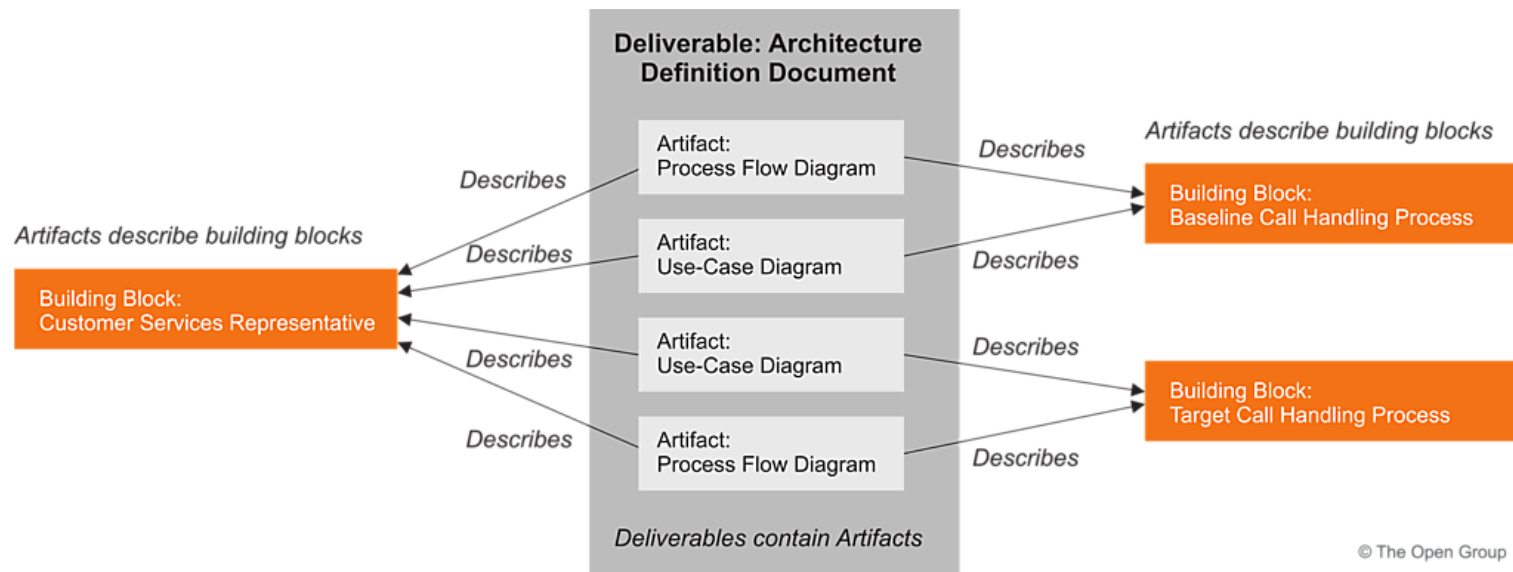
1	Executive Summary.....	1
2	Version History.....	1
3	Content.....	1
4	Purpose.....	2
5	Scope.....	3
6	Goals, Objectives, and Constraints.....	3
7	Architecture Principles.....	3
8	Baseline Architecture.....	3
8.1	Business Architecture Models.....	3
8.2	Data Architecture Models.....	3
8.3	Application Architecture Models.....	3
8.4	Technology Architecture Models.....	3
9	Rationale and Justification for Architectural Approach.....	3
10	Mapping to Architecture Repository.....	3
10.1	Mapping to Architecture Landscape.....	3
10.2	Mapping to Reference Models.....	3
10.3	Mapping to Standards.....	3
10.4	Re-Use Assessment.....	3
11	Target Architecture.....	3
11.1	Business Architecture Models.....	4
11.2	Data Architecture Models.....	4
11.3	Application Architecture Models.....	4
11.4	Technology Architecture Models.....	4
12	Gap Analysis.....	4
13	Impact Assessment.....	4

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Architecture Definition Document – Structure

The container keeps architectural artifacts created during the project

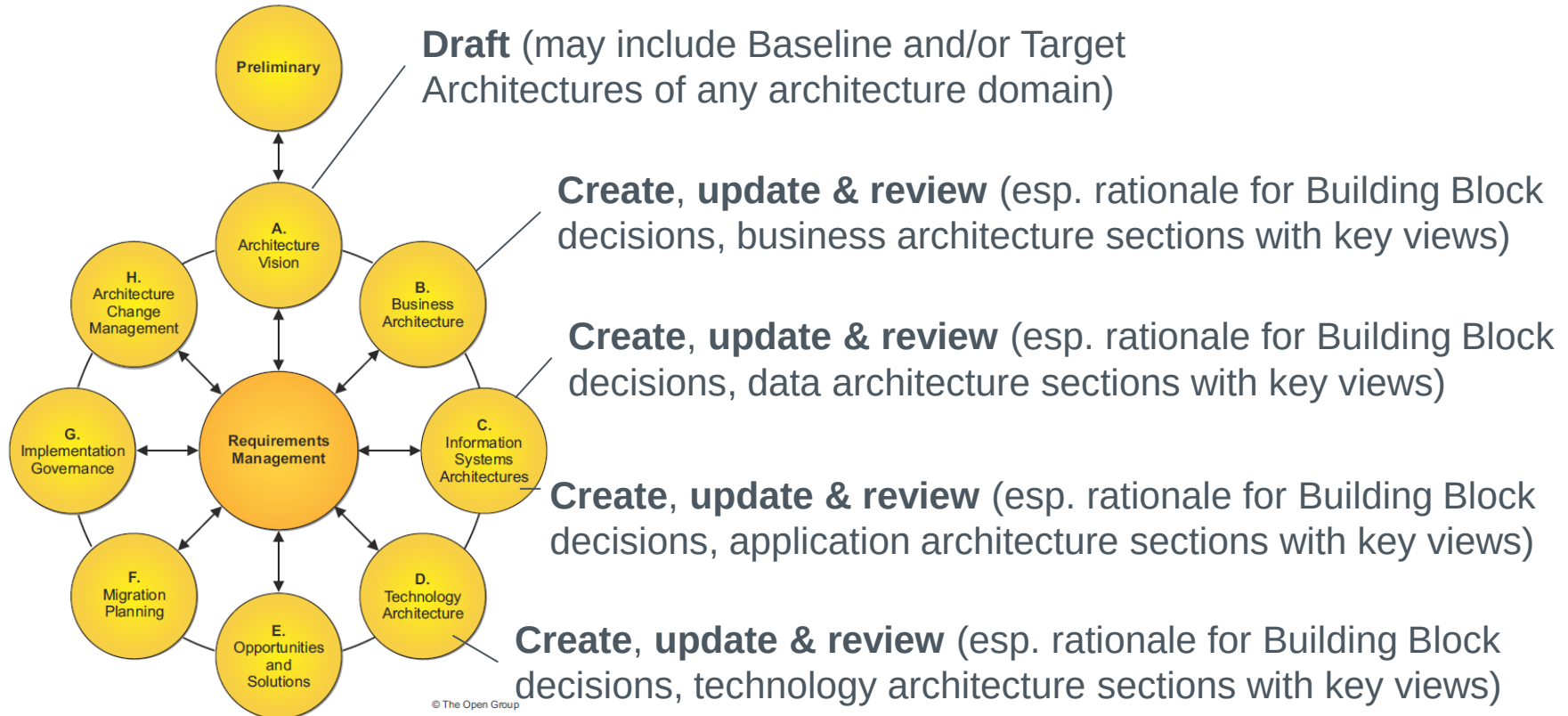


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- The document contains **complementary artifacts** representing **Architecture Views** on the Building Blocks

Architecture Definition Document – Usage within the TOGAF ADM (1/2)

The deliverable is created in Phase A and enriched through Phase B, C and D



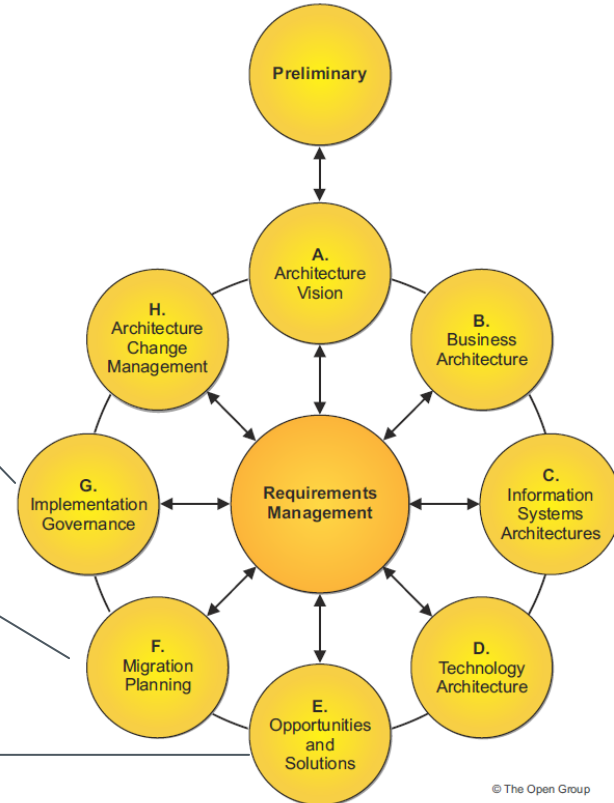
Architecture Definition Document – Usage within the TOGAF ADM (2/2)

The deliverable is finalized in Phase F and used throughout Phase G

Use & update (post-implementation)

Finalize (may include assigning project objectives and aligning projects and their deliverables with the Transition Architectures)

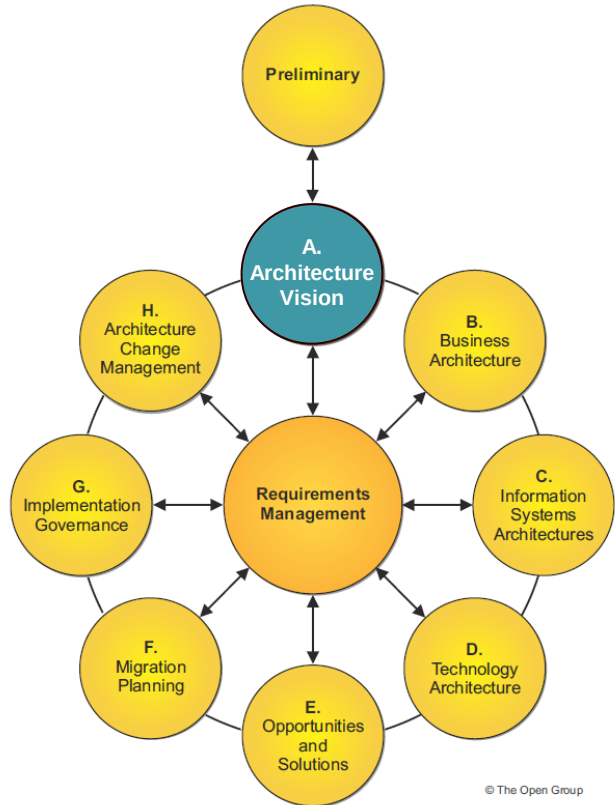
Update with any additional relevant outcomes



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ADM – Phase A: Architecture Vision – Statement of Architecture Work

The deliverable documents all findings & results and serves as a contract



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7

Confirm and elaborate **Architecture Principles**, including business principles

8

Develop Architecture Vision

9

Define the Target Architecture **value propositions** and KPIs

10

Identify the business transformation risks and mitigation activities

11

Develop Statement of Architecture Work; secure approval

Statement of Architecture Work – Definition

The deliverable measures the execution success of the Architecture Project

Purpose

- Defines **scope & approach** being used for an **architecture development cycle**
- May form the **basis for a contractual agreement** between the **supplier and consumer** of architecture services

Content

- **Title, approvals, change of scope procedures**
- Architecture project **request, background, description, scope, plan & schedule**
- **Acceptance criteria and procedures, roles, responsibilities and deliverables**
- **Overview of Architecture Vision**

Statement of Architecture Work – Structure

The deliverable answers the Request for Architecture Work



Statement of Architecture Work

Project XXXX Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Statement of Architecture Work	4
3 Objectives and Scope	5
4 Roles and Responsibilities	6
5 Architecture Approach	7
6 Work Plan	10
7 Risks and Mitigations	12
8 Acceptance Criteria and Procedures	13
9 Signature Approvals	14

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
Prepared By:		Document Version Date:	
Title:	Statement of Architecture Work	Review Date:	
Reviewed By:			

Distribution List

From	Date	Phone/Fax/Email

To	Action*	Due Date	Phone/Fax/Email

* Action Types: Approve, Review, Inform, File, Action Required, Attend Meeting, Other (please specify)

Document Version History

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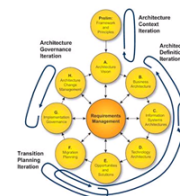
TOGAF™ 9 Template: Statement of Architecture Work
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5 Architecture Approach

5.1 Architecture Process

The TOGAF Architecture Development Method (ADM) defines a best-practice methodology for architecture development. However, not all phases are necessarily equally relevant to every project. The table below describes the usage of the ADM for this particular project.



Phase	In/Out	Notes
Preliminary		
A – Architecture Vision		
B – Business Architecture		<<Baseline and/or Target>>
C – Information Systems Architectures		<<Baseline and/or Target>>
D – Technology Architecture		<<Baseline and/or Target>>
E – Opportunities and Solutions		
F – Migration Planning		
G – Implementation Governance		
H – Architecture Change Management		
Requirements Management		

<<Provide any further notes on phasing, iterations, etc.>>

5.2 Architecture Content

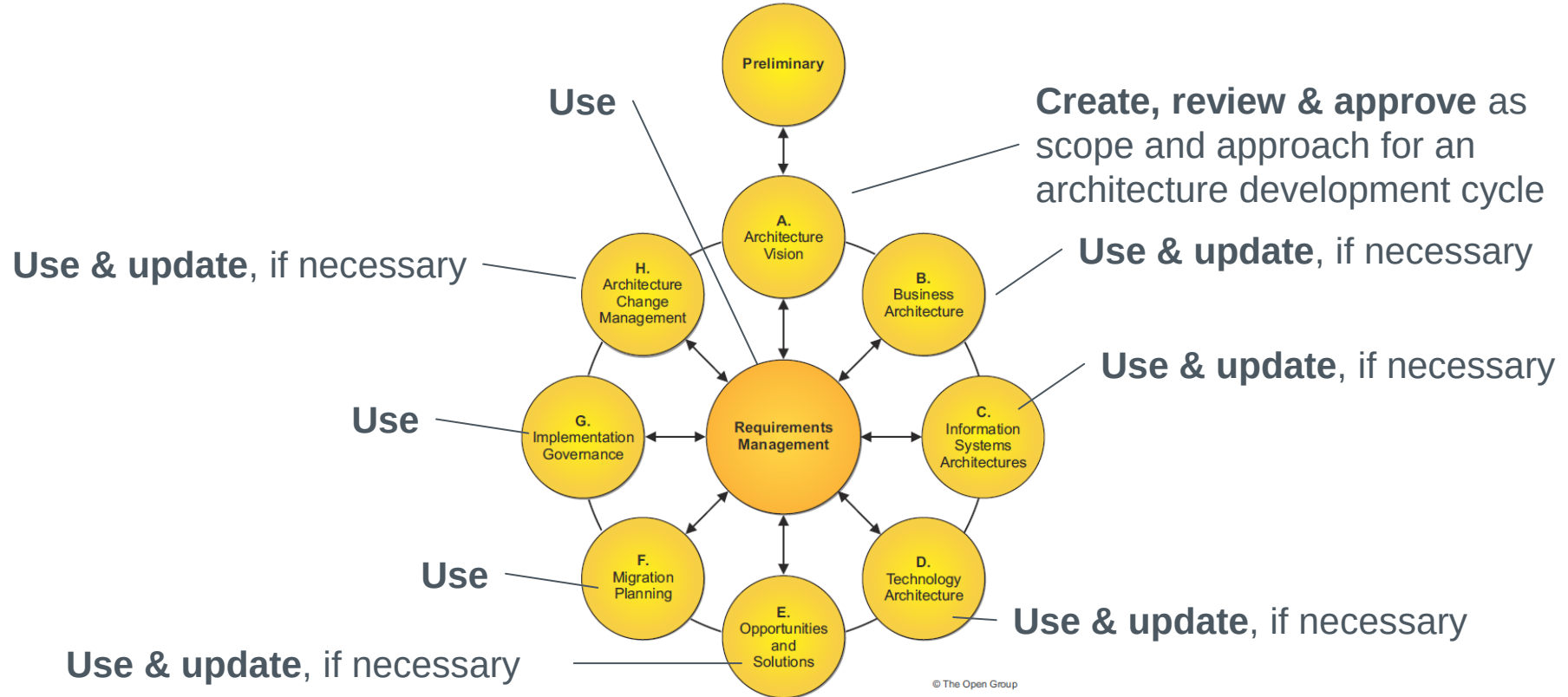
The TOGAF Architecture Content Framework (ACF) provides a best-practice categorization of architecture content. However, not all items are necessarily equally relevant to every project. The table below describes the content areas relevant to this particular project.

TOGAF™ 9 Template: Statement of Architecture Work
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Statement of Architecture Work – Usage within the TOGAF ADM

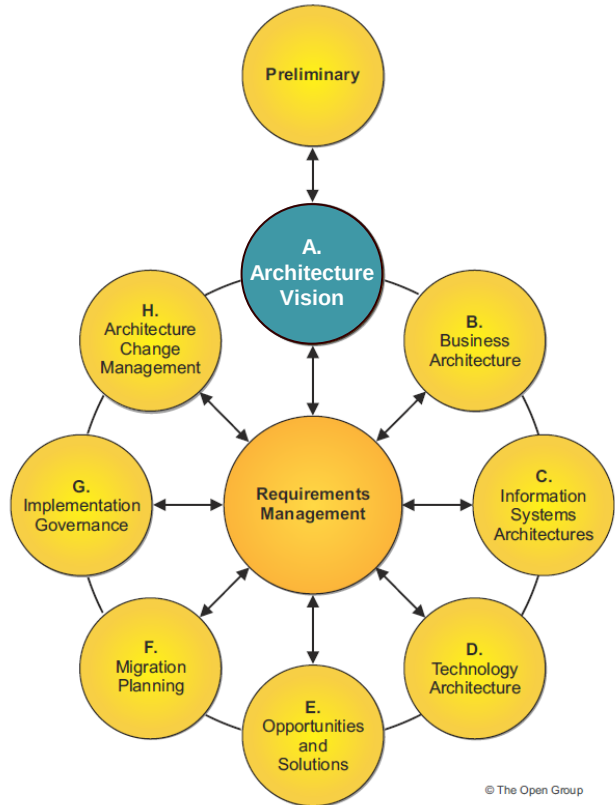
The deliverable is created in Phase A and used throughout Phase B till H



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ADM – Phase A: Architecture Vision – Approach (1/2)

The phase builds consensus about the Architecture Vision with the sponsors

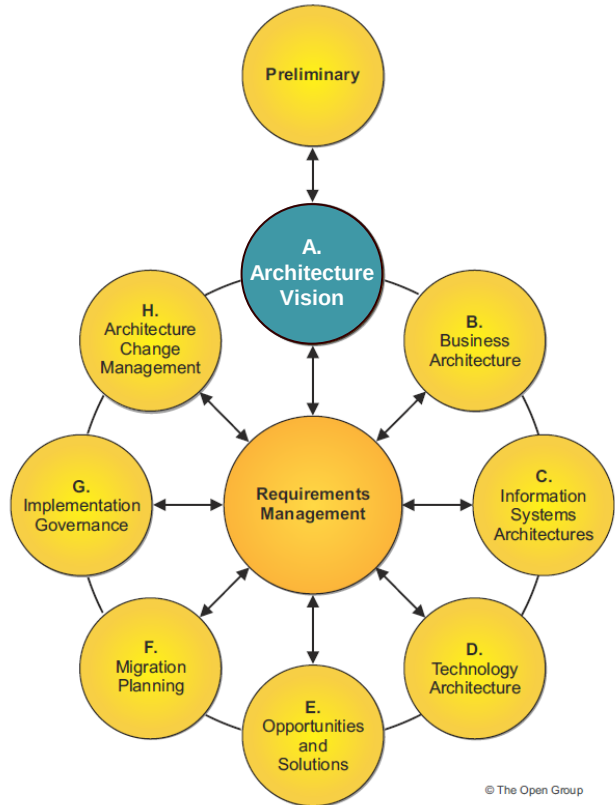


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- Receive a Request for Architecture Work
- Define what is in and what is outside the scope of the architecture effort and constraints
- Obtain, verify and understand the documented business principles, business goals, strategic drivers and architecture principles (preliminary phase) and ensure that existing definitions are current
- Describe Baseline and Target Architectures for all domains (Business, Data, Application, Technology) on a High-level

ADM – Phase A: Architecture Vision – Approach (2/2)

The phase builds consensus about the Architecture Vision with the sponsors



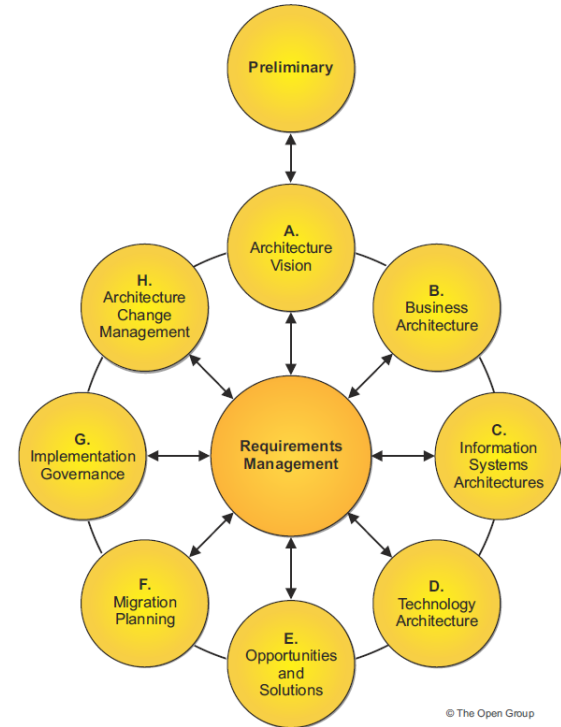
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- Define and document an overview of the Architecture Vision in the Statement of Architecture Work (incl. how new business capabilities will meet business goals & strategic objectives and address stakeholder's concerns)
- Use Architecture Vision to communicate the benefits of the proposed business capability to stakeholders and decision-makers
- Get the Statement of Architecture Work signed by the sponsoring organization

Document outline

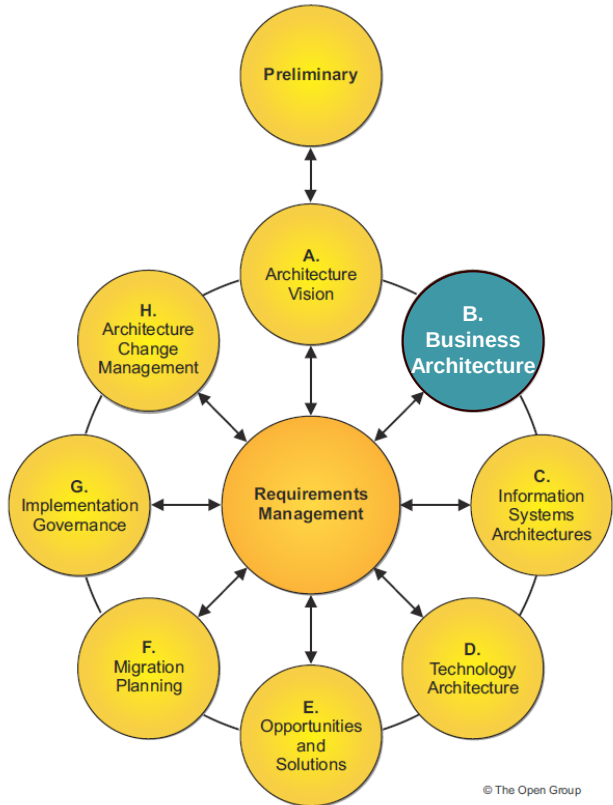
Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition

1	Welcome
2	Introduction to Enterprise Architecture
3	Architecture Development Method (ADM)
4	Preliminary Phase
5	Phase A: Architecture Vision
6	Phase B: Business Architecture
7	Phase C: Information Systems Architecture
8	Phase D: Technology Architecture



ADM – Phase B: Business Architecture – Purpose (1/2)

The phase is a prerequisite for architecture work in any other domain

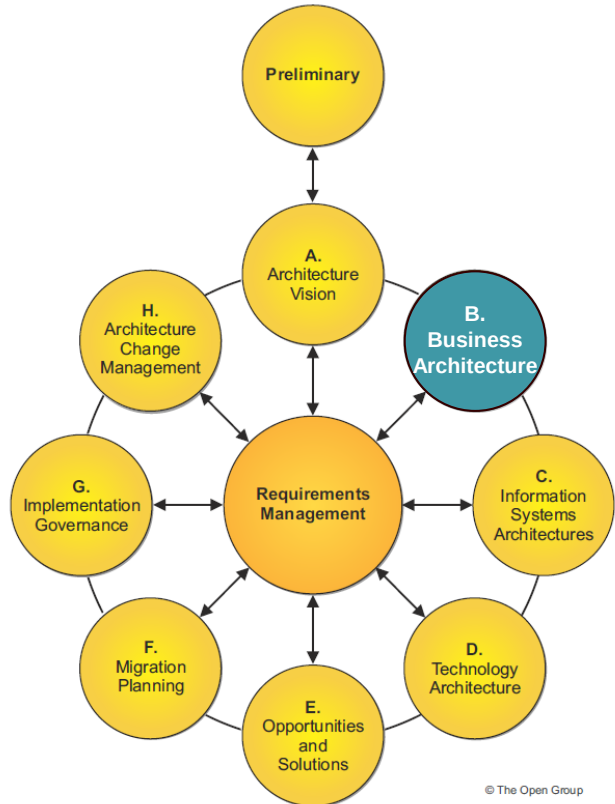


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- **Develop a Business Architecture to support an agreed Architecture Vision**
- **Show** how the current Enterprise **fails to meet** the **Stakeholders preferences**
- Identify necessary changes to meet Stakeholder preferences (**Gaps**)
- Definition of the required work to realize the changes and the added value of the changes (**Work Packages**)

ADM – Phase B: Business Architecture – Purpose (2/2)

The phase is a prerequisite for architecture work in any other domain



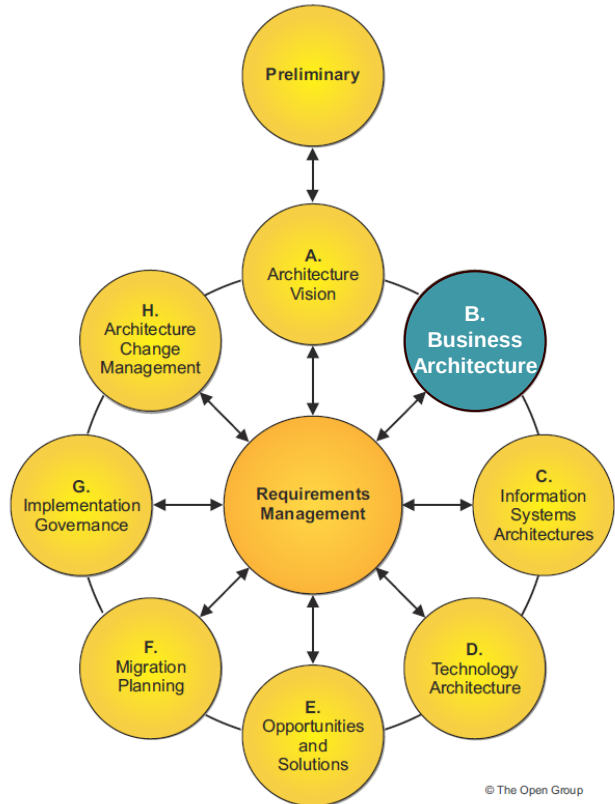
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- Manage stakeholder priority and preference adjusted to value, effort and risk (**Stakeholder Requirements**)
- Make sure, that Gaps, Work, Added Value and Risks are **understood by the stakeholders**
- **Output & Outcome:** Business Architecture approved by the stakeholders for the problem being addressed, with a set of gaps and work to clear the gaps understood by the stakeholders

Phase B, C and D have the “same” purpose

ADM – Phase B: Business Architecture – Objectives

The phase develops the Target Architecture & identifies Roadmap components

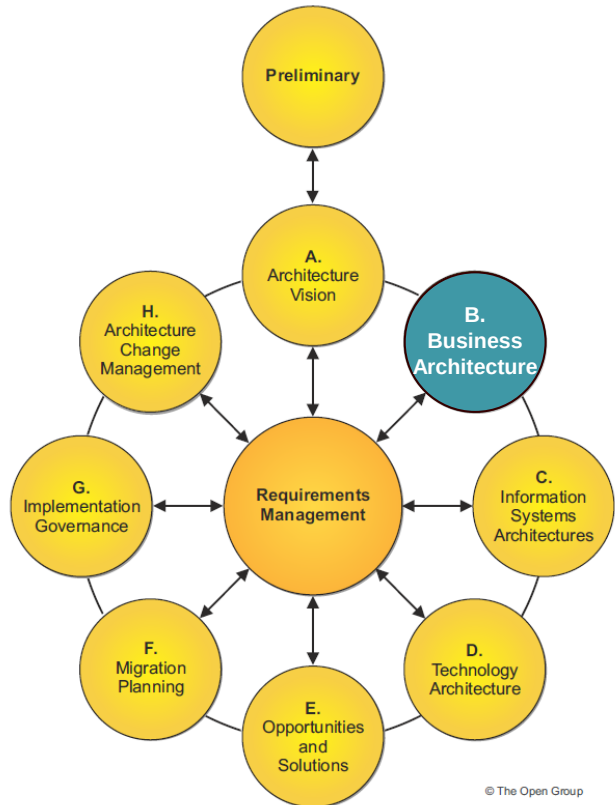


1. Develop the **Target Business Architecture** that describes **how the enterprise needs to operate** to **achieve the business goals** and **respond to the strategic drivers** set out in the **Architecture Vision**, in a way that **addresses the Statement of Architecture Work** and stakeholder concerns

2. Identify candidate **Architecture Roadmap components based upon gaps** between the Baseline and Target Business Architectures

ADM – Phase B: Business Architecture – Steps (1/2)

The phase performs a gap analysis between the Baseline & Target Architecture



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1

Select reference models, viewpoints and tools

2

Develop Baseline Business Architecture Description

3

Develop Target Business Architecture Description

4

Perform gap analysis

5

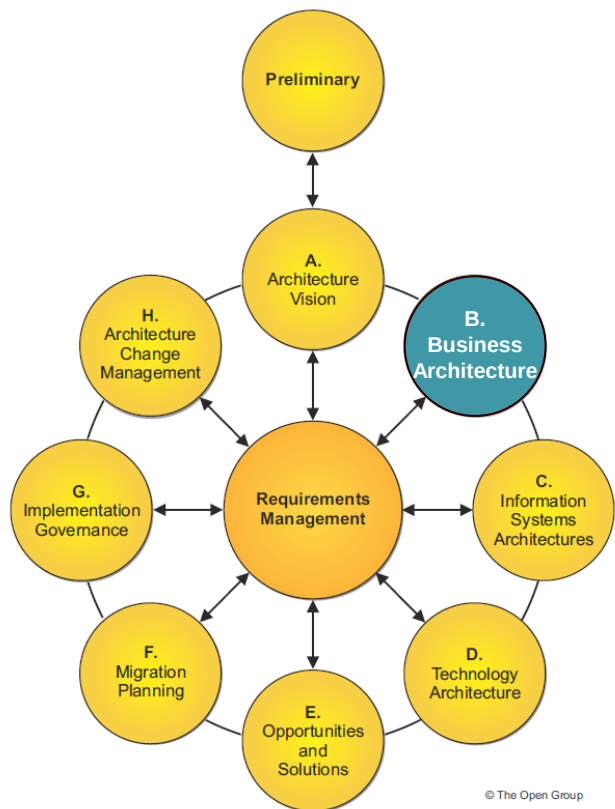
Define candidate roadmap components

6

Resolve impacts across the Architecture Landscape

ADM – Phase B: Business Architecture – Steps (2/2)

The phase performs a gap analysis between the Baseline & Target Architecture



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7

Conduct formal stakeholder review

8

Finalize the Business Architecture

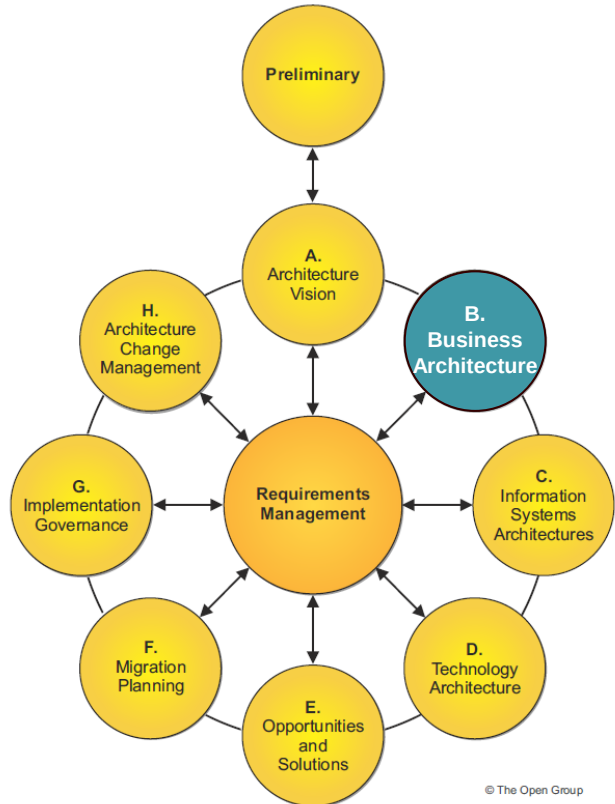
9

Create/Update the Architecture Definition Document

In phase B, C and D, follow the same steps

ADM – Phase B: Business Architecture – Gap Analysis

The technique identifies gaps between the Baseline & Target Business Architecture



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1

Select reference models, viewpoints and tools

2

Develop Baseline Business Architecture Description

3

Develop Target Business Architecture Description

4

Perform gap analysis

5

Define candidate roadmap components

6

Resolve impacts across the Architecture Landscape

Gap Analysis – Technique

A Gap defines what must be changed within the enterprise



Baseline Architecture
Current state acting as
reference for all change



Gap



Target Architecture
Future state that
stakeholders have approved

Shortfall between **Baseline Architecture** and **Target Architecture**
with **items** that have been deliberately
omitted, accidentally **left out** or **not yet defined**

Gap Analysis – Gap sources

Architecture gaps can be found in every Architecture Domain

Business

People, process, tools, information, measurement, financial, facility

Data

Currency, location, availability, creation, consumption, relations, need

Application

Creation, elimination, impact

Technology

Creation, elimination, impact

➤ **Ignored stakeholder concerns** are the **most critical source of gaps**

Gap Analysis – Example 1 (1/2)

The technique visually compares the Baseline with the Target Architecture

Target Architecture Baseline Architecture ↓	Video Conferencing Services	Enhanced Telephony Services	Mailing List Services	Eliminated Services ↓
Broadcast Services				Intentionally eliminated
Video Conferencing Services	Included			
Enhanced Telephony Services		Potential match		
Shared Screen Services				Unintentionally excluded - a gap in Target Architecture
New →		Gap: Enhanced services to be developed or produced	Gap: To be developed or produced	© The Open Group

1. Create **Matrix** (ABBs of Target Architecture on horizontal axis, ABBs Baseline Architecture on vertical axis)
2. Add row “**New**” & column “**Eliminated**”
3. For each **Architecture Building Block (ABB)** of the **Baseline Architecture**, check whether it is included in the **Target Architecture**
 - **Yes?** Record “**Included**” at the intersection
 - **No?** Record “**Intentionally eliminated**” or “**Unintentionally eliminated**” at the column “**Eliminated**”

Gap Analysis – Example 1 (2/2)

The technique visually compares the Baseline with the Target Architecture

Target Architecture Baseline Architecture ↓	Video Conferencing Services	Enhanced Telephony Services	Mailing List Services	Eliminated Services ↓
Broadcast Services				Intentionally eliminated
Video Conferencing Services	Included			
Enhanced Telephony Services		Potential match		
Shared Screen Services				Unintentionally excluded - a gap in Target Architecture
New →		Gap: Enhanced services to be developed or produced	Gap: To be developed or produced	© The Open Group

4. For each **Architecture Building Block (ABB)** of the **Target Architecture**, check whether it is included in the **Baseline Architecture**

4. **Yes?** You should have already recorded **"Included"** in the intersection

5. **No?** Record the gap at the row **"New"**

➤ Anything in the row **"New"** or column **"Eliminated"** is a **gap** between Baseline and Target Architecture

Gap Analysis – Example 2

The technique supports long-term compliance reporting



Reporting against constraints, expected value and known gaps

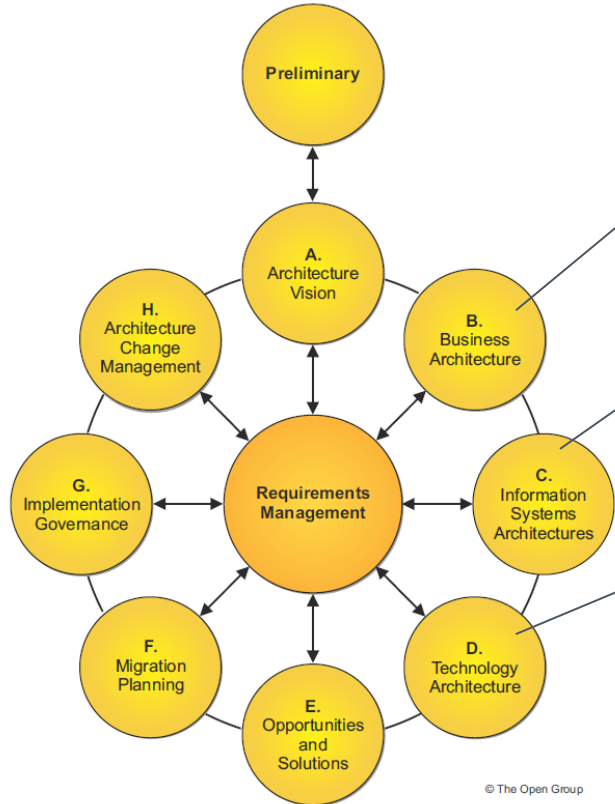
Considering
of Baseline,
Project and
Planning

	Constraint (Architecture Principle, Architecture Requirements Specification, or Control)	Value (Best done in terms of the Enterprise's mandatory concerns)	Gap
Current state: assess what the Enterprise has	Conforms	Fails to Deliver	Not Applicable
Implementation Project: assess project, design, and implementation	Violates	Not Applicable	Filling
Roadmap, portfolio, or program: assess plans and directions	Not Applicable	Delivers	Leaving Open

Assessment
results with
color coding

Gap Analysis – Usage within the TOGAF ADM (1/2)

The technique is widely used throughout all cyclic ADM phases



Perform Gap Analysis between Baseline and Target Business Architecture

Perform Gap Analysis between Baseline and Target Information Systems Architectures

Perform Gap Analysis between Baseline and Target Technology Architecture

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Gap Analysis – Usage within the TOGAF ADM (2/2)

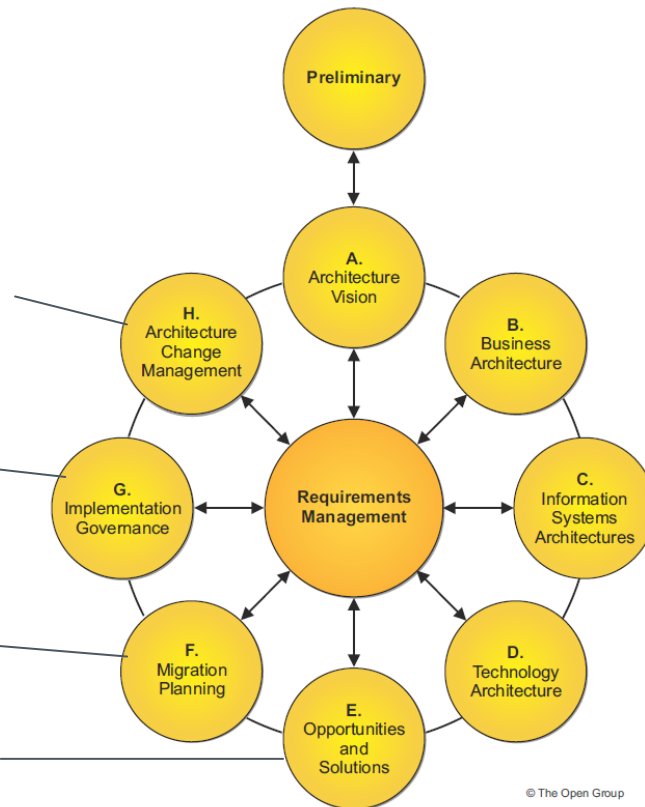
The technique is widely used throughout all cyclic ADM phases

Use as input & **Perform** Gap Analysis of the performance of the Enterprise Architecture

Use as input & **Perform** Gap Analysis on Enterprise Architecture and solutions framework as well as Solution Architecture and operations

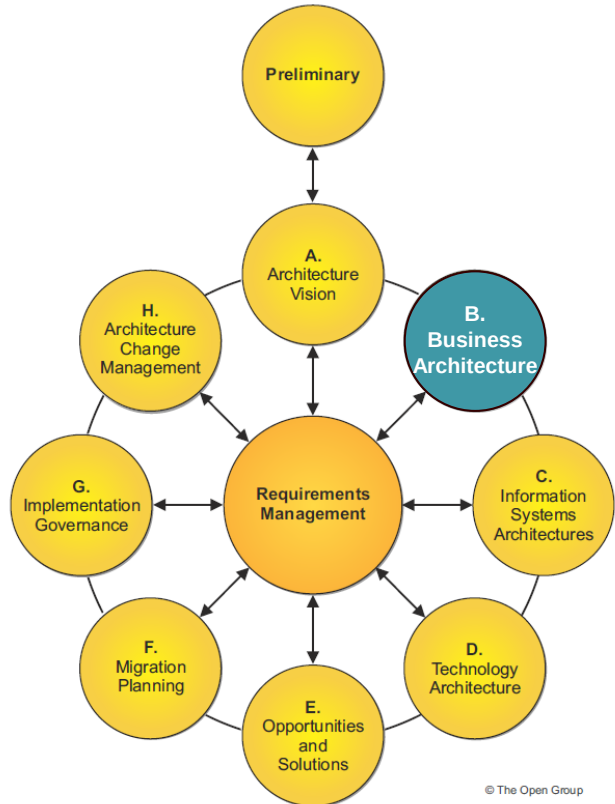
Use as input

Review & Consolidate Gap Analysis results



ADM – Phase B: Business Architecture – Approach (1/2)

Business Architecture is the first architecture activity that needs to be done

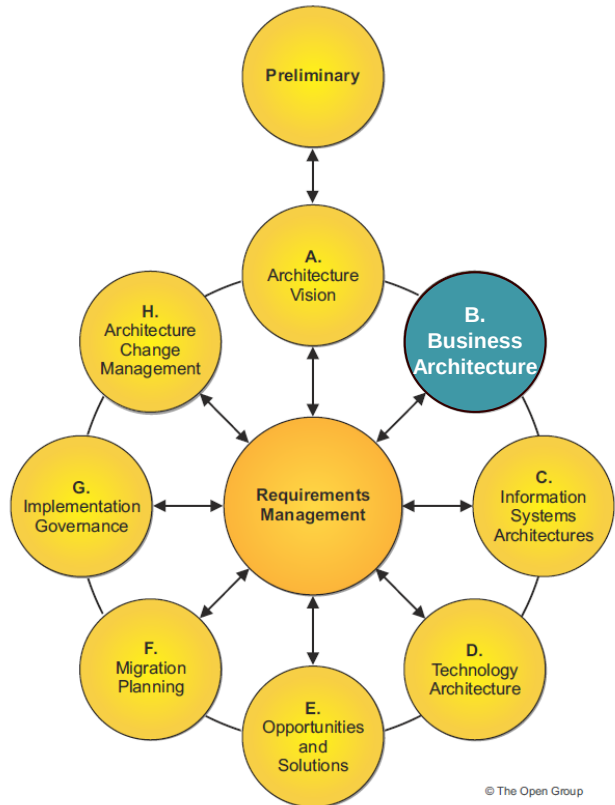


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- Prerequisite for architecture work of BDAT-Architecture
- Means of demonstrating the business value and return on investment of subsequent architecture work to key stakeholders
- Case 1: Parts of the Business Architecture are already done >> Verify and update the documented material and bridge between high-level and specific requirements
- Case 2: Little or no Business Architecture is done >> Research, verify and gain buy-in to key business objectives and processes

ADM – Phase B: Business Architecture – Approach (2/2)

The re-use of internal and external material saves time

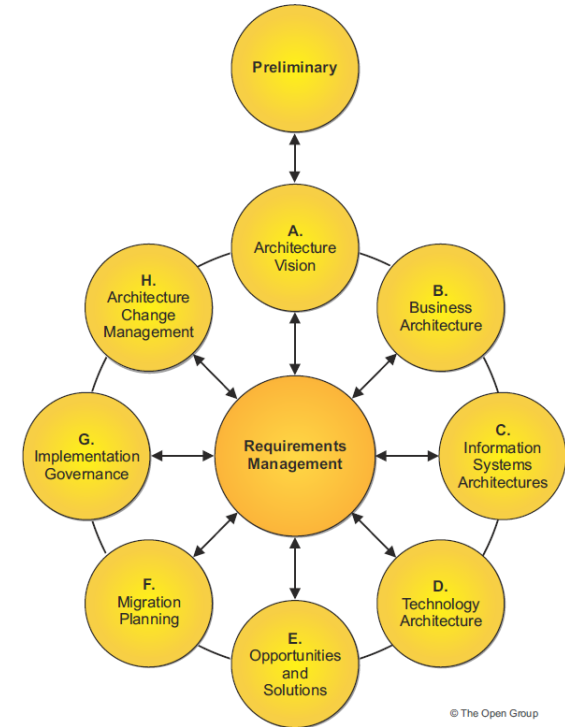


- Re-use existing material and descriptions (e.g., Architecture Definition Document)
- Use and update existing architecture descriptions (Phase A) and/or gather information to develop the Business Architecture models
- Determine if fundamental view of the business is changing
- Develop Business Architecture models (e.g. Business Capability Map, Value Stream Model, Organization Map, Business Process Models) and applying (industry specific) Modeling Techniques (e.g., UML®)
- Use resources from the Architecture Repository (industry reference models, applicable standards, enterprise-specific business architecture views and Building Blocks)

Document outline

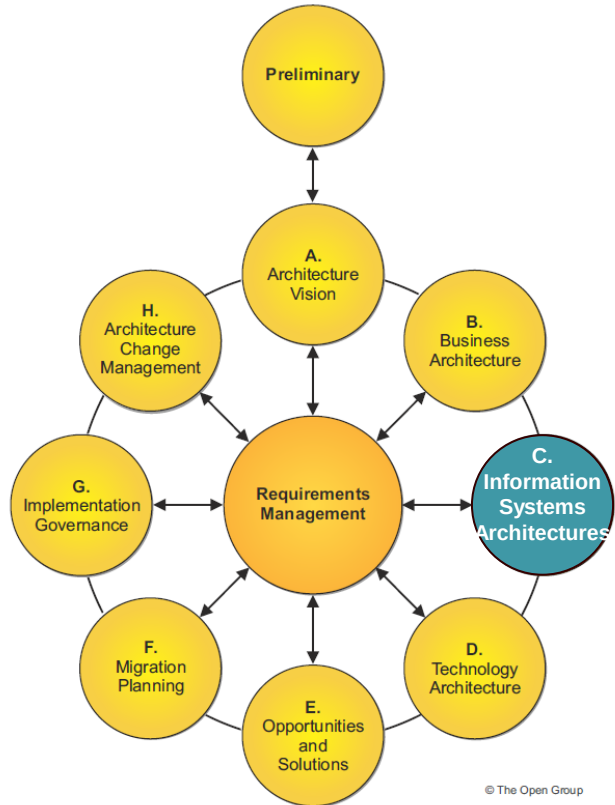
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8	Phase D: Technology Architecture



ADM – Phase C: Information Systems Architectures – Purpose

The Data & Application Architectures enable the Business Architecture & Vision



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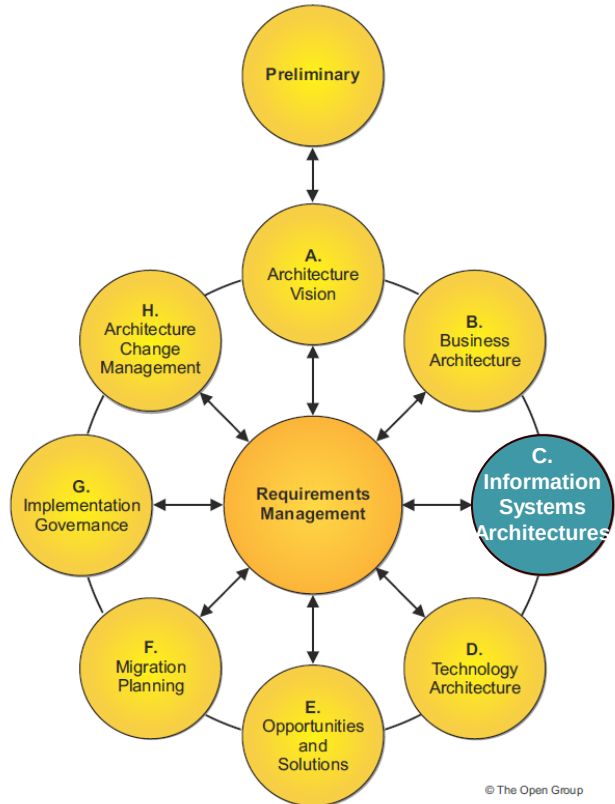
Phase C consists of **two parts** (in either order):

1. Development of the **Application Architecture**
2. Development of the **Data Architecture**

- **Develop the Target Information Systems Architectures** (Application & Data Architectures) to support an agreed **Architecture Vision**
- ...
- **See Purpose of Phase B: Business Architecture** (they are the same)

ADM – Phase C: Information Systems Architectures – Objectives

The phase develops the Target Architectures & identifies Roadmap components



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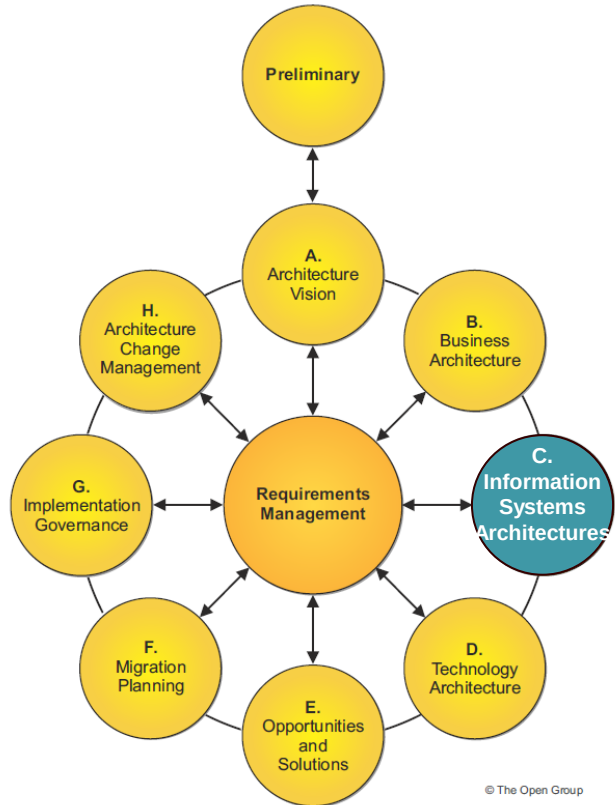
1. Develop the Target Information Systems Architectures

(Data & Application Architecture), describing **how** the enterprise's Information Systems Architectures will **enable the Business Architecture and the Architecture Vision**, in a way that **addresses the Statement of Architecture Work and stakeholder concerns**

2. Identify candidate Architecture Roadmap components

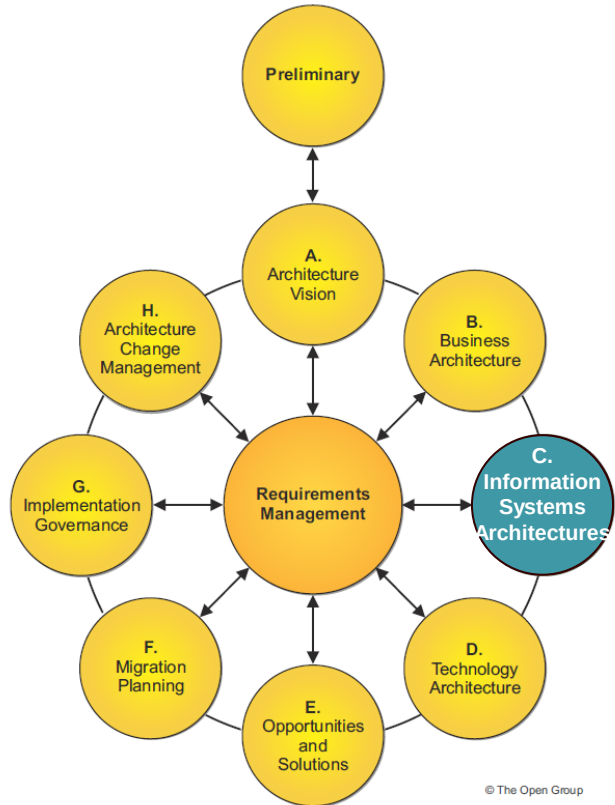
based upon gaps between the Baseline and Target Information Systems Architectures (Data & Application Architecture).

The phase identifies gaps between Baseline & Target Data & Application Arch.



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The phase describes the Baseline and Target Application Architecture



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- **Application Portfolio Catalog**
- Interface Catalog
- **Application/Organization Matrix**
- Role/Application Matrix
- Application/Function Matrix
- Application Interaction Matrix
- **Application Communication Diagram**
- Application and User Location Diagram
- Application Use-Case Diagram
- Enterprise Manageability Diagram
- Process/Application Realization Diagram
- Software Engineering Diagram
- Application Migration Diagram
- Software Distribution Diagram

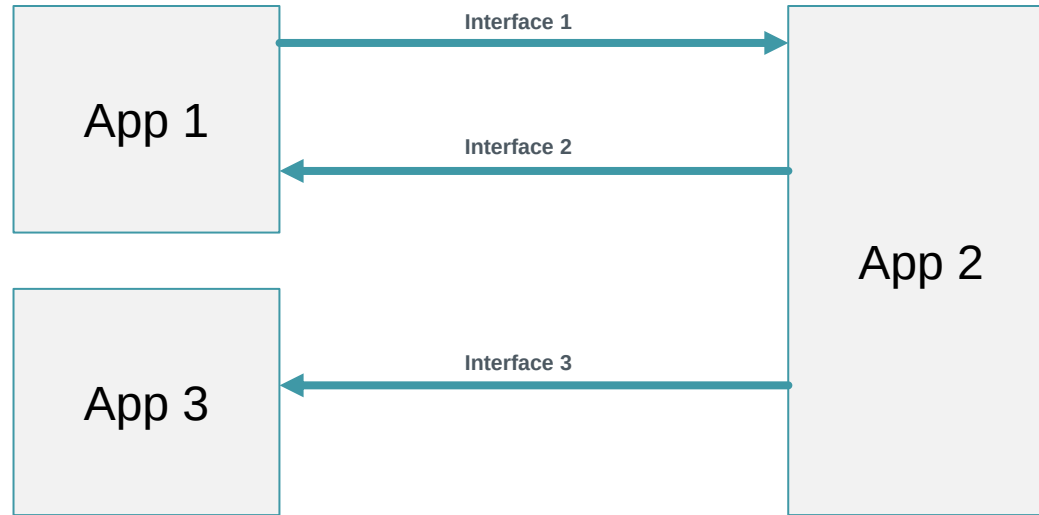
ADM – Phase C: Application Architecture – Artifacts (2/4)

The Application Portfolio Catalog lists all logical application components

ID	Name	Description	Category	Source
ISA_LAC_01	Enterprise Resource Planning Platform	An Enterprise Resource Planning (ERP) system is a comprehensive software solution used by organizations to manage and streamline various aspects of their business operations. It integrates core functions such as finance, human resources, inventory management, and supply chain management into a single unified platform, enabling data sharing and real-time collaboration across different departments.	Platform	Interview
ISA_LAC_02	Customer Relationship Management Platform	A Customer Relationship Management (CRM) platform is a software tool that helps businesses effectively manage their interactions and relationships with customers. It centralizes customer data and communication history, allowing organizations to track and analyze customer interactions, preferences, and feedback.	Platform	Interview

- Provides a **foundation for other matrices and diagrams**
- Contains **logical application components** and **physical application components**
- Often the **starting point** of the Application Architecture phase

The Application Communication Diagram shows App components & interfaces



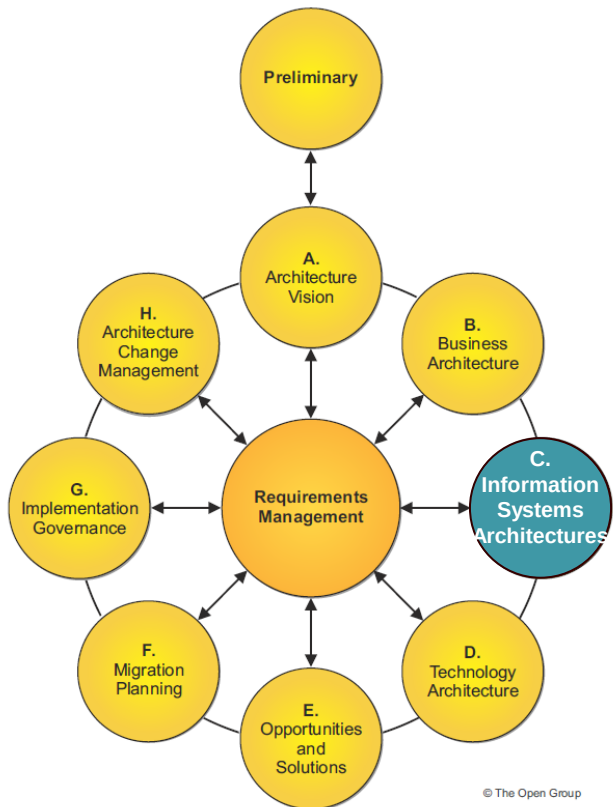
- **Interfaces** may be associated with **data entities**
- **Applications** may be associated with **business services**
- **Communication** should be logical and only show **technology** where it is relevant

The Application / Organization Matrix shows usage of applications

Logical Application Components Map to Organization's Units				
	Sales	HR	Production	Development
Enterprise Ressource Planning Platform	x	x	x	x
Customer Relationship Management Platform	x			
Human Resource Platform		x		
Product Lifecycle Management System			x	x

- Defines the **application set** used by a particular organization unit
- Shows which **organizational units** are supported by which **applications**
- Helps to understand the **application support requirements**
- Supports the **gap analysis** by determining **missing or inappropriate applications**

The phase describes the Baseline and Target Data Architecture



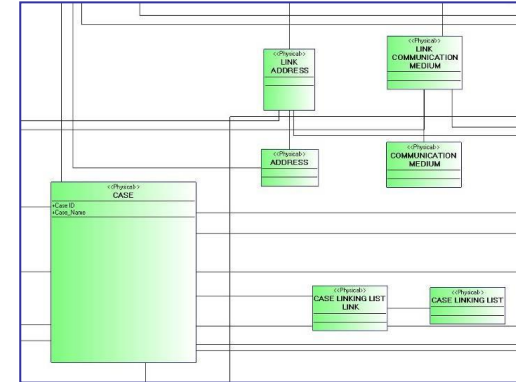
- | | |
|---|--|
| 1 | Select reference models, viewpoints and tools |
| 2 | Develop Baseline Application & Data Architecture Description |
| 3 | Develop Target Application & Data Architecture Description |
| 4 | Perform gap analysis |
| 5 | Define candidate roadmap components |
| 6 | Resolve impacts across the Architecture Landscape |

ADM – Phase C: Data Architecture – Artifacts

Catalogs, matrices and diagrams visualize the Data Architecture

Logical Data Components Map to Business Functions		
	Sales	HR
Product Margin Data	X	
Customer Data	X	
Employee Data		X
Payroll Data		X
Insurance Data		X
Revenue Data	X	

Data Entity/Business Function Matrix

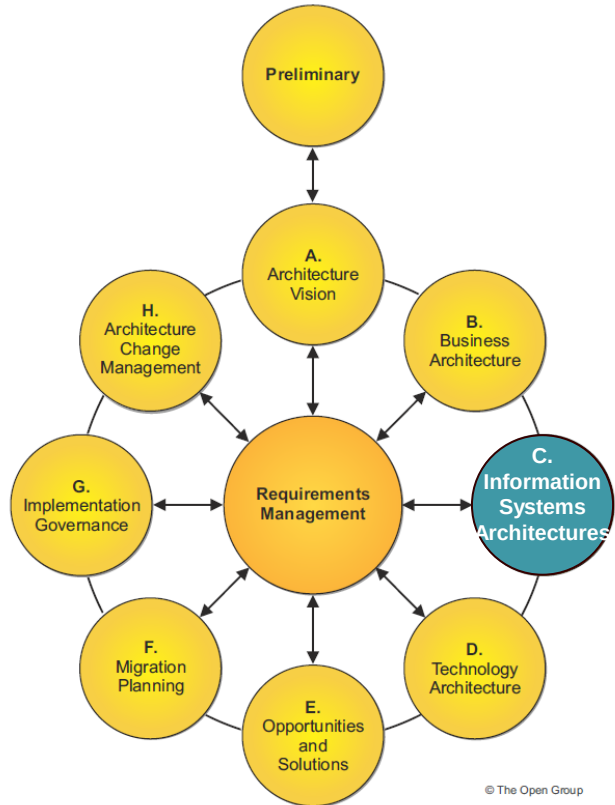


Logical Data Diagram

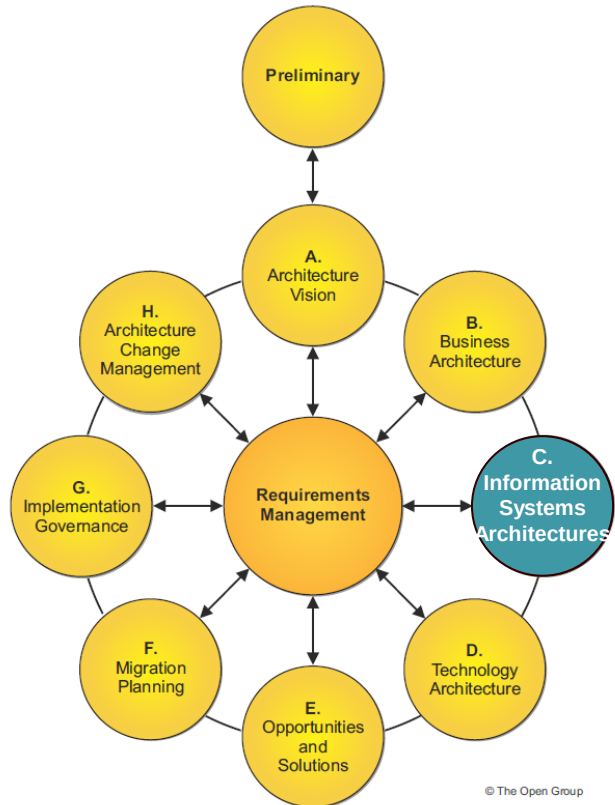
- Data Entity/Data Component Catalog
- Application/Data Matrix
- Conceptual Data Diagram
- Data Dissemination Diagram
- Data Security Diagram
- Data Migration Diagram
- Data Lifecycle Diagram

ADM – Phase C: Application Architecture – Approach

The phase considers the relevant & available Application Architecture resources



- **Use Application models** relevant to common high-level business functions (e.g., e-commerce)
- **Use generic business models and related application models** relevant to the organization's industry sector provided by
 - The Open Group Forum
 - Object Management Group (<https://www.omg.org/>)
 - TM Forum (<https://www.tmforum.org/>)
- The Open Group has a Reference Model for Integrated Information Infrastructure (III-RM) focuses on the application-level components and services necessary to provide an integrated information infrastructure



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Three key considerations for developing a Data Architecture:

1. Data Management:

- Define which application components serve as enterprise master data
- Understand how data entities are utilized by business capabilities, business functions, processes and business and application services
- Understand how and where enterprise data entities are created, stored, transported and reported

2. Data Migration:

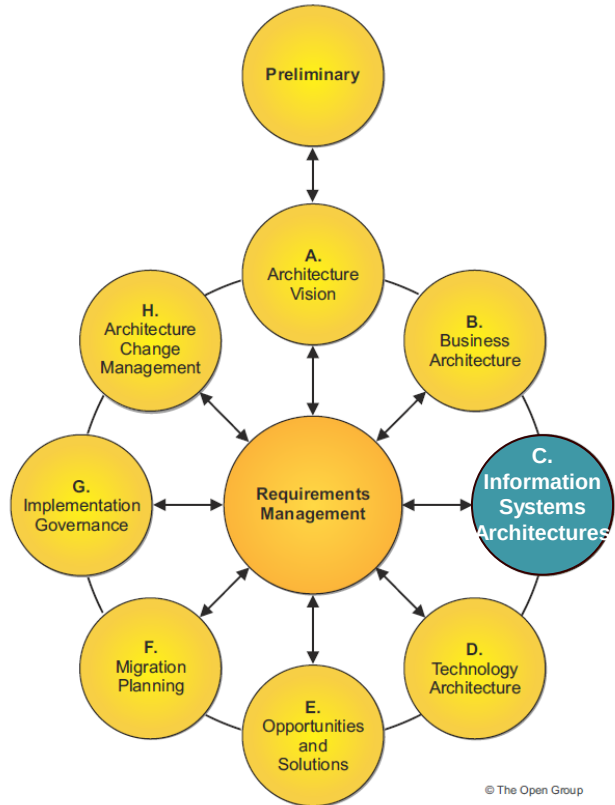
- Identify data migration requirements
- Provide indicators for data transformation, weeding and cleansing
- Ensure an enterprise-wide common data definition to support the transformation & migration

3. Data Governance:

- Ensure that the enterprise has the necessary structures, management systems and people in place to enable the transformation

ADM – Phase C: Data Architecture – Approach (2/2)

Conceptual, logical and physical data models provide a structural view



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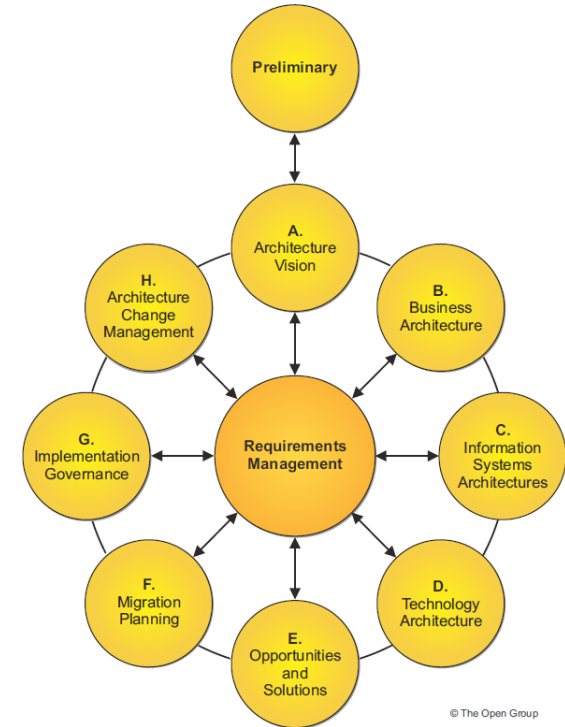
Data Architecture is created by using three metamodel entities:

- **Data entity:** used to create conceptual data models to help the IT developers understand the concepts they will be dealing with
- **Logical data entity:** used to create logical data models as a requirement on the data stored in applications (at rest), data moved between applications (in motion), or data at the user interface of applications (data in use)
- **Physical data entity:** used to create data models that have been implemented

Data Architecture should be able to handle:

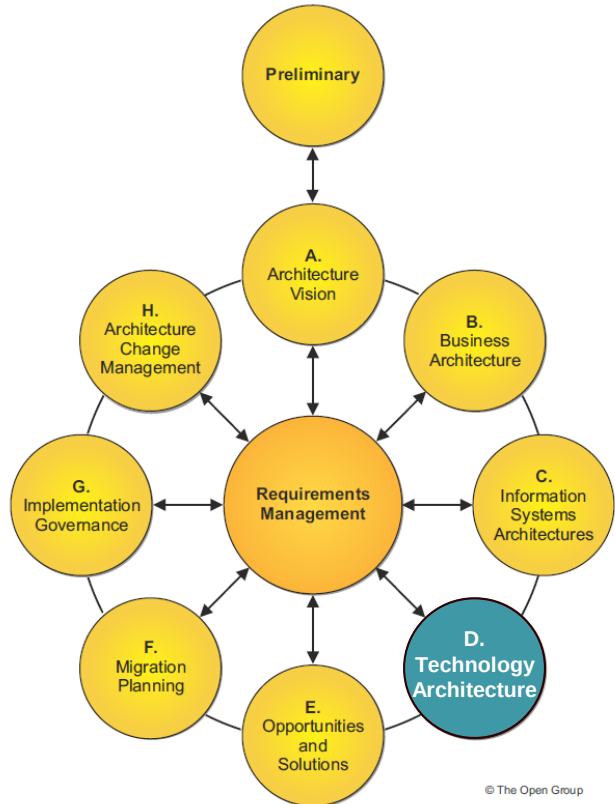
- Data at rest - data in stores
- Data in motion - data in transactions or services/APIs
- Data in use - data at the border of the application (e.g. GUI)
- Open data - data that the organization provides for public usage

1	Welcome
2	Introduction to Enterprise Architecture
3	Architecture Development Method (ADM)
4	Preliminary Phase
5	Phase A: Architecture Vision
6	Phase B: Business Architecture
7	Phase C: Information Systems Architectures
8	Phase D: Technology Architecture



ADM – Phase D: Technology Architecture – Purpose

The Technology Architecture supports Business, Data & Application Architecture

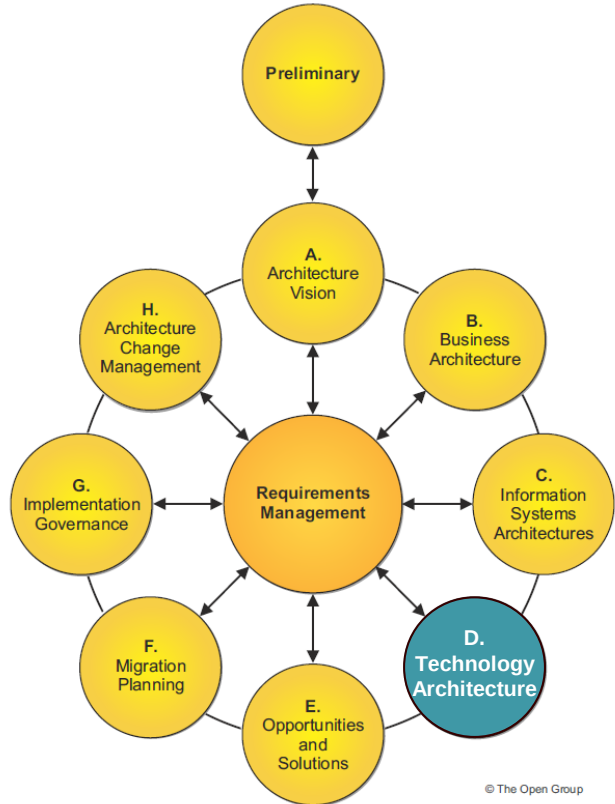


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- **Develop the Target Technology Architecture to support an agreed Architecture Vision**
- ...
- **See Purpose of Phase B: Business Architecture (they are the same)**

ADM – Phase D: Technology Architecture – Objectives

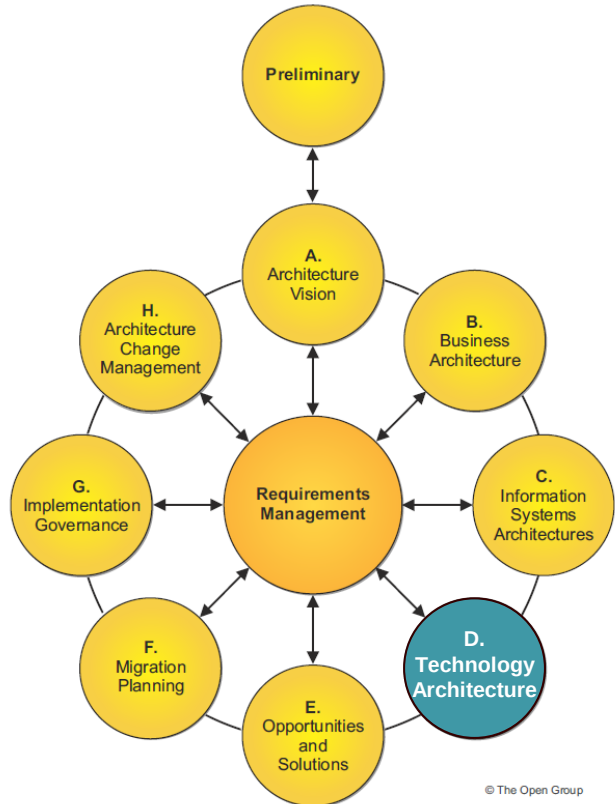
The phase develops the Target Architecture & identifies Roadmap components



- **Develop the Target Technology Architecture** that enables the Architecture Vision, target business, data and application Building Blocks to be delivered through technology components and technology services, in a way that addresses the Statement of Architecture Work and stakeholder concerns
- **Identify candidate Architecture Roadmap components** based upon gaps between the Baseline and Target Technology Architectures

ADM – Phase D: Technology Architecture – Steps

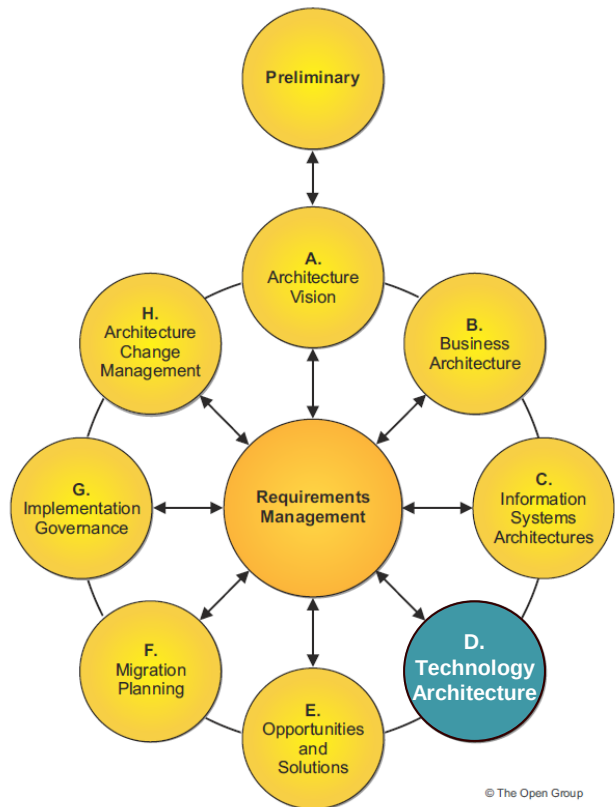
The phase identifies gaps between Baseline & Target Technology Architecture



1	Select reference models, viewpoints and tools
2	Develop Baseline Technology Architecture Description
3	Develop Target Technology Architecture Description
4	Perform gap analysis
5	Define candidate roadmap components
6	...

ADM – Phase D: Technology Architecture – Artifacts

The phase identifies gaps between Baseline & Target Technology Architecture



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1

Select reference models, viewpoints and tools

2

Develop Baseline Technology Architecture Description

3

Develop Target Technology Architecture Description

4

Perform gap analysis

5

Define candidate roadmap components

6

Resolve impacts across the Architecture Landscape

ADM – Phase D: Technology Architecture – Artifacts (1/3)

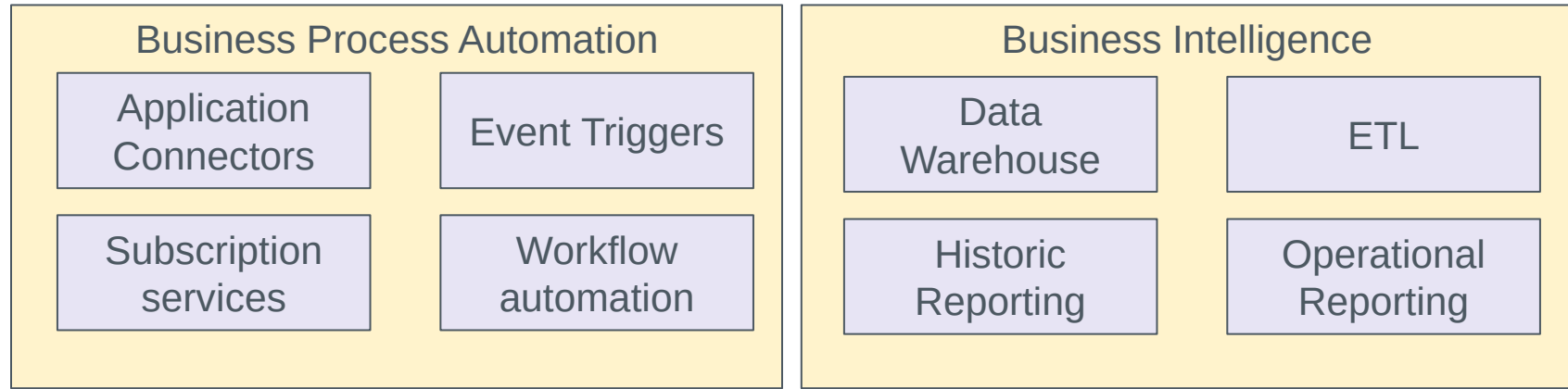


The Technology Portfolio Catalog lists all the logical technology in use

ID	Name	Description	Date	Category
TA_LTC_01	Workflow automation	Automate repetitive tasks and processes, improving efficiency and reducing human intervention.	10/15/2023	Business Process Automation
TA_LTC_02	Event trigger	Mechanism that initiates an action or process in response to a specific occurrence or predefined condition within a system or software.	10/15/2023	Business Process Automation
TA_LTC_03	Application Connector	Software component or interface that facilitates seamless communication and integration between different applications or systems, enabling data exchange and interoperability.	10/15/2023	Business Process Automation
TA_LTC_04	Data Warehouse	Centralized repository that stores large volumes of structured, unstructured, and semi-structured data from various sources, designed for analysis, reporting, and business intelligence purposes	10/16/2023	Business Intelligence
TA_LTC_05	ETL	Extract, Transform, Load, is a process in data integration where data is gathered from various sources, transformed into a consistent format, and loaded into a target data repository for analysis and reporting	10/16/2023	Business Intelligence

- Includes **hardware**, infrastructure software and application software
- Is a typical **starting point** for ADM Phase D Technology Architecture
- Provides a **basis** for other matrices and diagrams

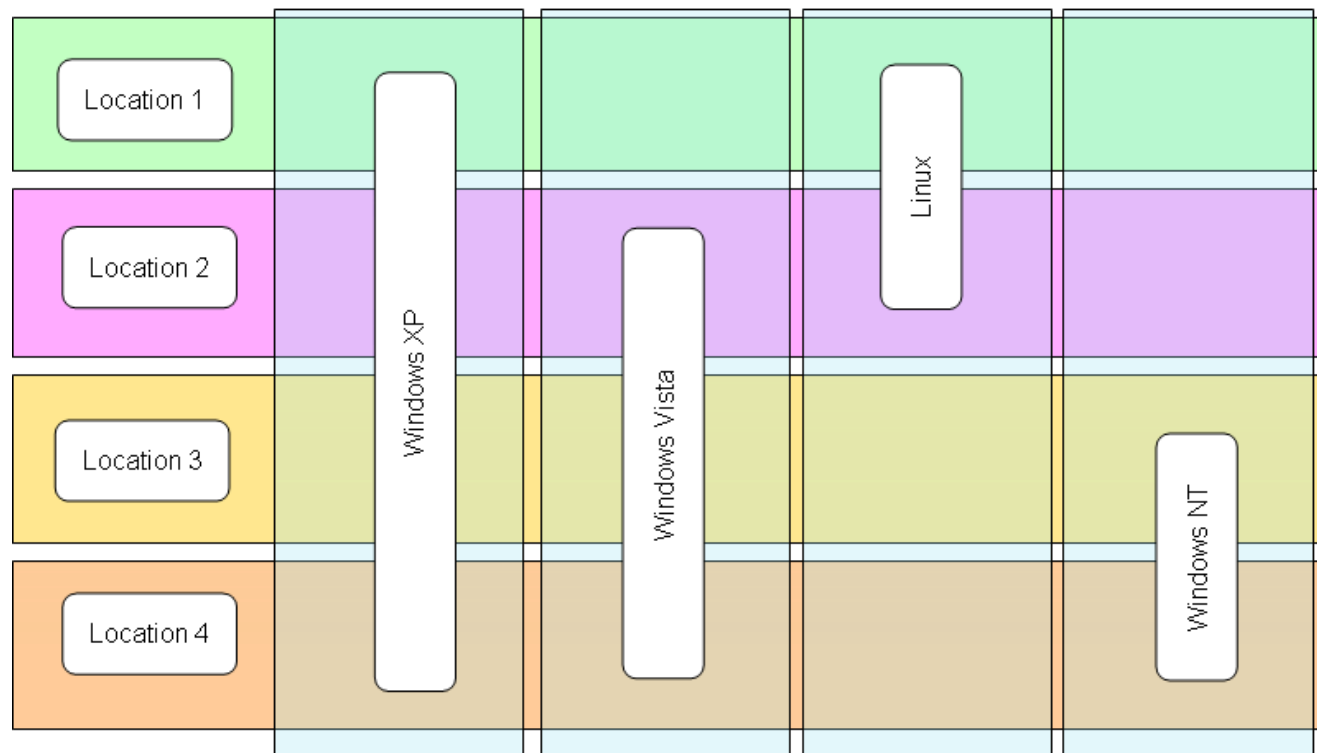
The Platform Decomposition Diagram depicts the used technology platforms



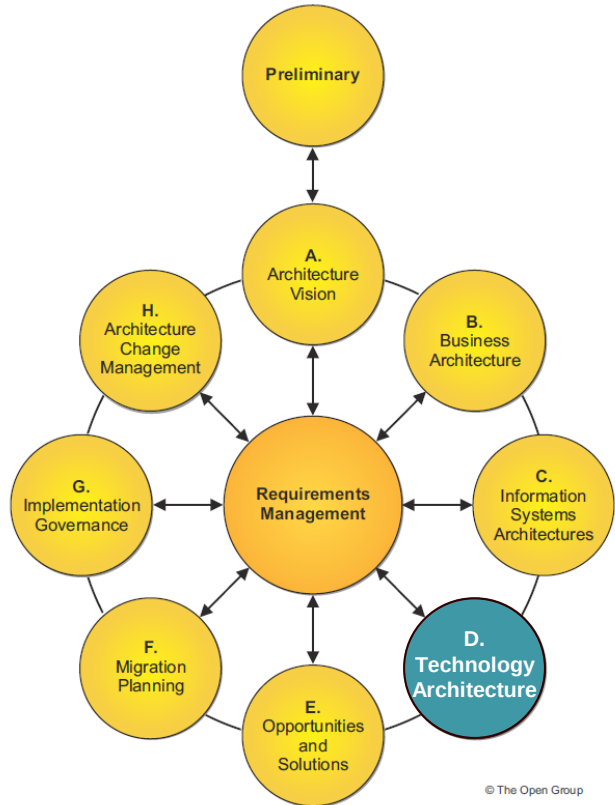
- Shows the **enterprise applications** and the **decomposed technology components**
- Provides an **overview of the enterprise's technology platforms**
- May map technology platforms to application components within a specific functional or process area
- May show **further specification details** (e.g., product versions, number of CPUs)

ADM – Phase D: Technology Architecture – Artifacts (3/3)

The Environments & Location Diagram shows technology of the different sites



- Depicts what **technologies** are used at which **locations**

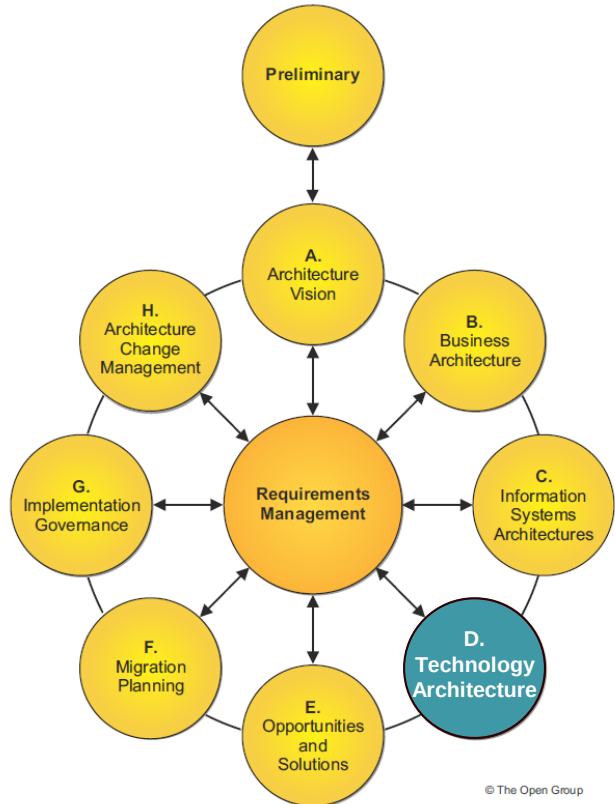


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Emerging Technologies:

- **Anticipate** and be open to **technology-driven change**
- Look for new **innovative ways of operating and improve** the business
- Technology Architecture needs to **capture the transformation opportunities** available through adoption of **new technology**
- The TOGAF ADM enables technology change to **become a driver and strategic resource rather than a recipient of Change Requests**
- **Technology Architecture** may both **drive business capabilities and respond to information system requirements** at the same time
- **Rapid adoption** of changing technologies **can cause discontinuities** across the enterprise.

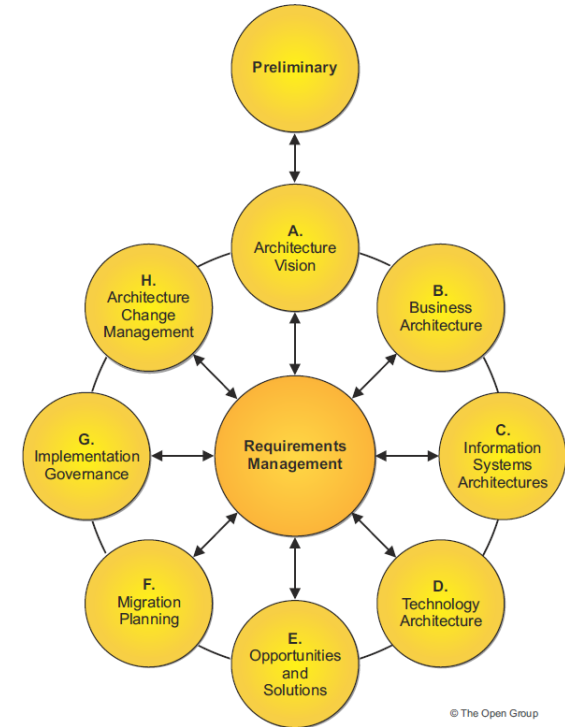
The re-use of internal and external material saves time



Use resources from the Architecture Repository:

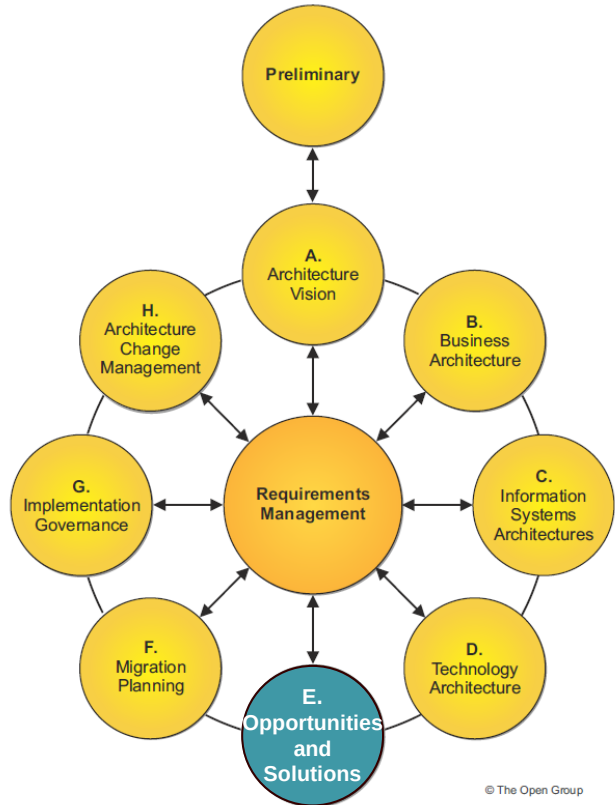
- **Existing IT services** documented in the **IT repository** or **IT service catalog**
- Generic technology models relevant to the organization's industry sector
- **Reference Models** from The Open Group Technology (e.g. III-RM, Technical Reference Model TRM)

9	Phase E: Opportunities and Solutions
10	Phase F: Migration Planning
11	Phase G: Implementation Governance
12	Phase H: Architecture Change Management
13	Requirements Management Phase
14	Applying the ADM
15	Architecture Content & Core Concepts
16	Architecture Governance & Architecture Capability



ADM – Phase E: Opportunities and Solutions – Purpose (1/2)

The phase is directly concerned with solutions & implementation

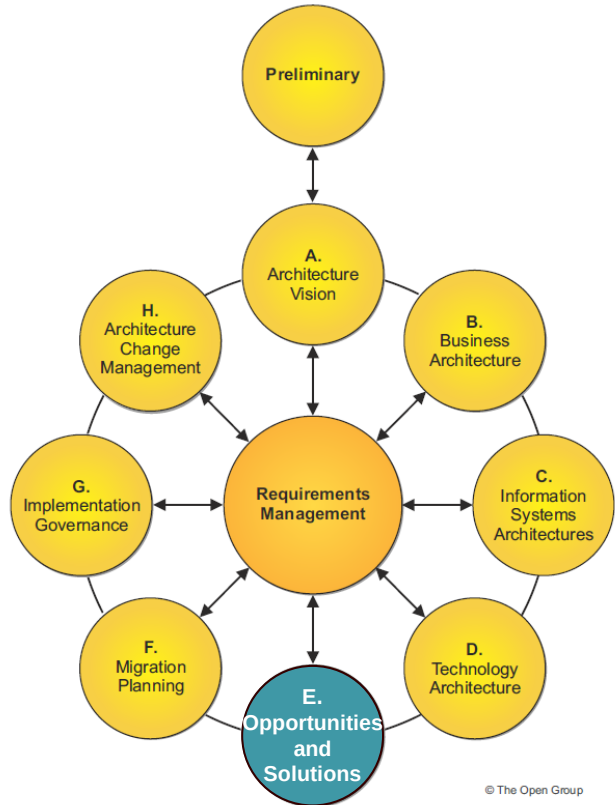


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- Find **Solution Building Blocks (SBBs)**
- Conduct **initial implementation planning**
- Validate **dependencies** (Work Package & Gap dependency)
- Identify **delivery vehicles** (projects, programs, or portfolios) that effectively deliver the Target Architecture

ADM – Phase E: Opportunities and Solutions – Purpose (2/2)

The phase is directly concerned with solutions & implementation

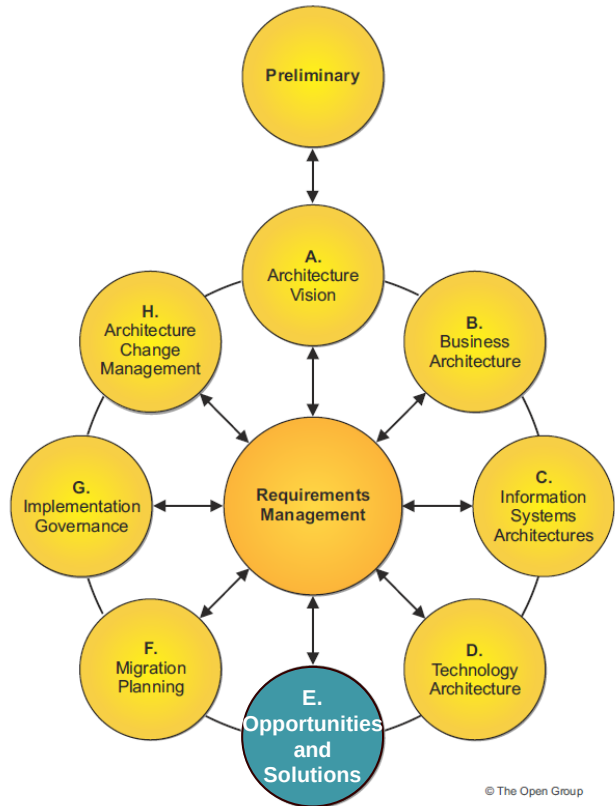


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- Assess value, effort and risk associated with each change and work package
- Manage stakeholder priority and preference adjusted to value, effort and risk (Stakeholder Requirements)
- Output & Outcome:
 - **Work packages**
 - **Addressed gaps**
 - **Added value**
 - **Required effort**
 - **Dependencies**

ADM – Phase E: Opportunities and Solutions – Objectives

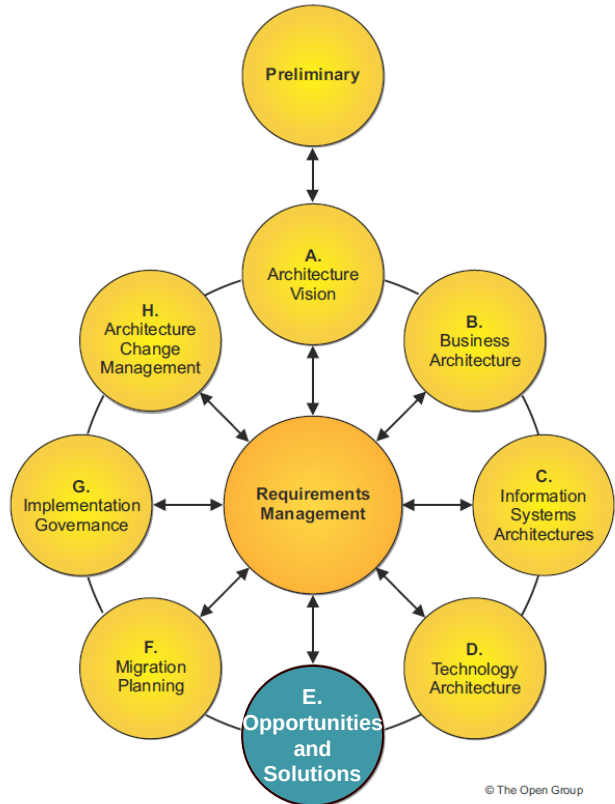
The phase concentrates on how to deliver the architecture



1. Generate the initial complete version of the Architecture Roadmap, based upon the gap analysis and candidate Architecture Roadmap components from Phases B, C and D
2. Determine whether an incremental approach is required, and if so, identify Transition Architectures that will deliver continuous business value
3. Define the overall Solution Building Blocks (SBBs) to finalize the Target Architecture based on the Architecture Building Blocks (ABBs)

ADM – Phase E: Opportunities and Solutions – Steps (1/2)

The phase consolidates the gaps and creates an initial implementation plan



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1

Determine/confirm the key corporate change attributes

2

Determine the business constraints for implementation

3

Review and consolidate the gap analysis results from Phases B to D

4

Review the consolidated requirements across related business functions

5

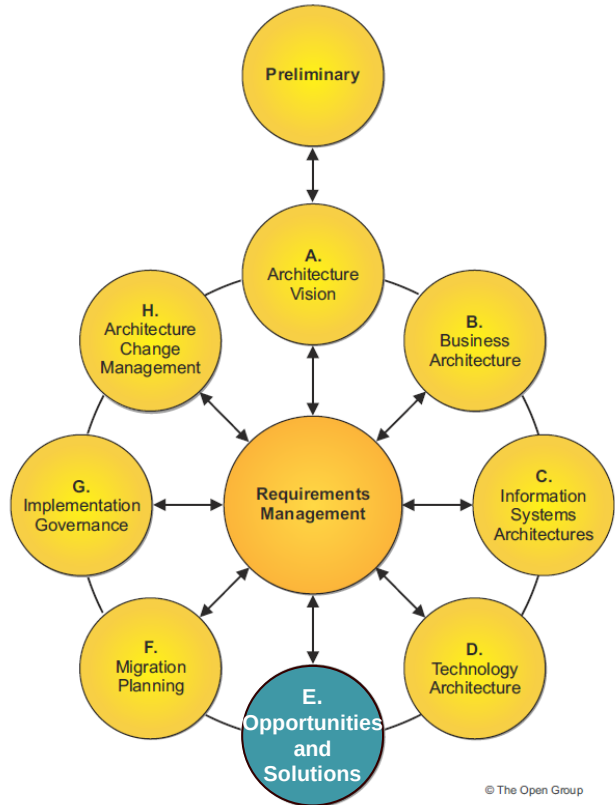
Consolidate and reconcile interoperability requirements

6

Refine and validate dependencies

ADM – Phase E: Opportunities and Solutions – Steps (2/2)

The phase consolidates the gaps and creates an initial implementation plan



7

Confirm readiness and risk for business transformation

8

Formulate Implementation and Migration Strategy

9

Identify and group major work packages

10

Identify Transition Architectures

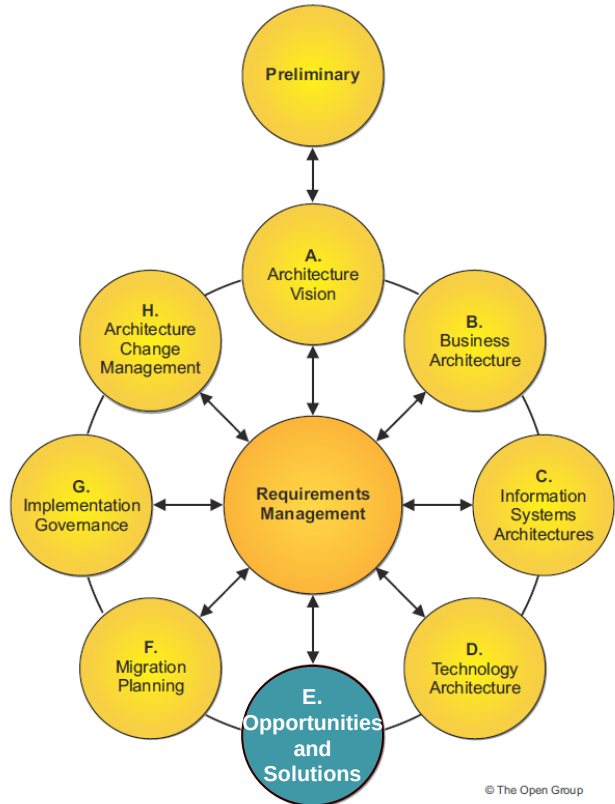
11

Create the Architecture Roadmap & Implementation and Migration Plan

Phase E is the bridge between the Target Architecture and the Solution. It concentrates on how to deliver the architecture.

ADM – Phase E: Opportunities and Solutions – Interoperability

The phase minimizes interoperability conflicts between solutions



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1

Determine/confirm the key corporate change attributes

2

Determine the business constraints for implementation

3

Review and consolidate the gap analysis results from Phases B to D

4

Review the consolidated requirements across related business functions

5

Consolidate and reconcile interoperability requirements

6

Refine and validate dependencies

Interoperability – Definition

Interoperability is the ability to share information and services



Operational or Business Interoperability

How different parts of the enterprise work together at the business level?



Information Interoperability

How information is to be shared?



Technical Interoperability

How technical resources are to be shared or at least connect to one another?



Presentation: Common look-and-feel approach through a common portal-like solution guides the user through the system functions



Information: Information is seamlessly shared between applications based on a corporate ontology and shared services (e.g., security, access, structure, quality)



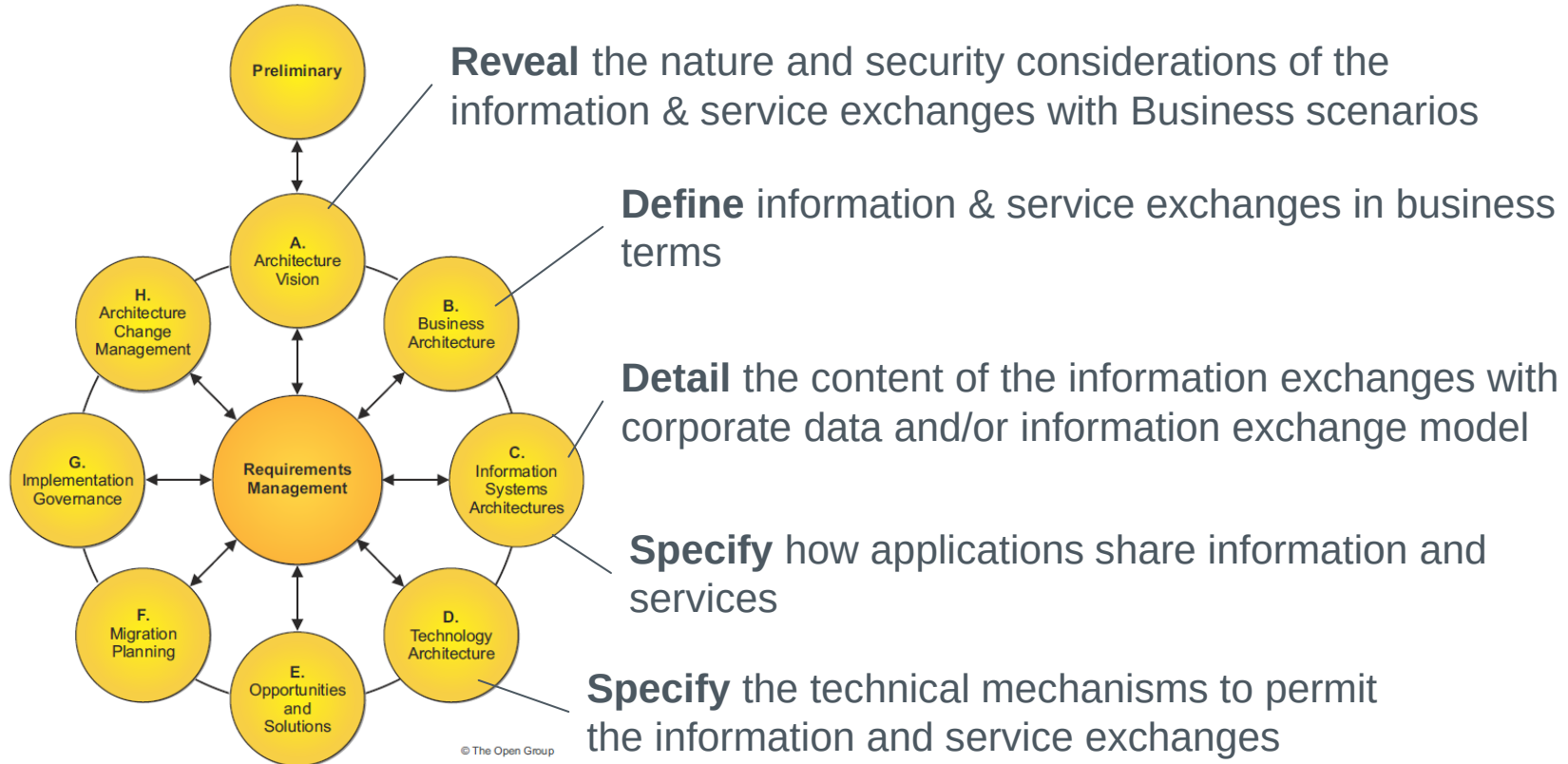
Application: Corporate functionality is integrated & shareable and business & infrastructure applications are seamlessly linked together



Technical: Common methods and shared services for the communication, storage, processing and access to data in platforms & infrastructure

Interoperability – Usage within the TOGAF ADM (1/2)

The technique is applied in Phase A till Phase F

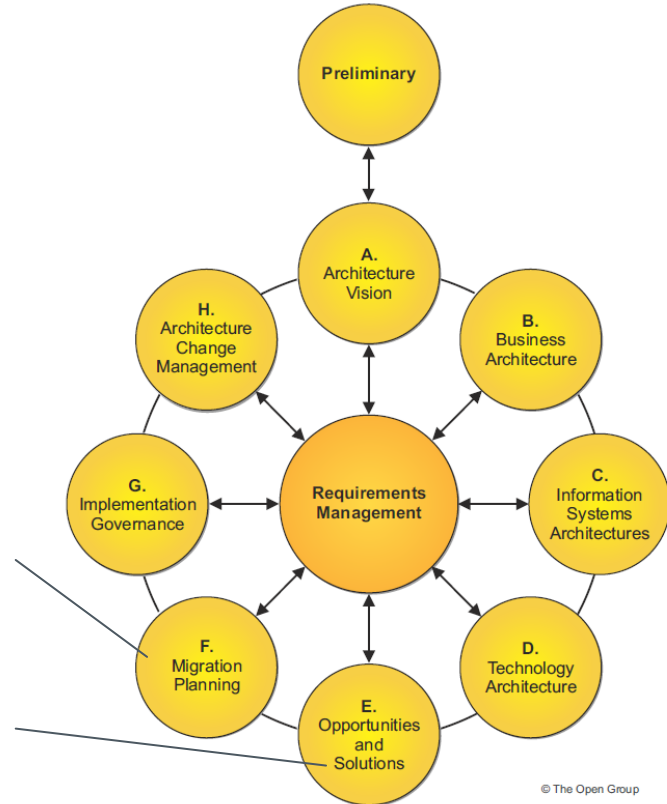


Interoperability – Usage within the TOGAF ADM (2/2)

The technique is applied in Phase A till Phase F

Logically implement the interoperability

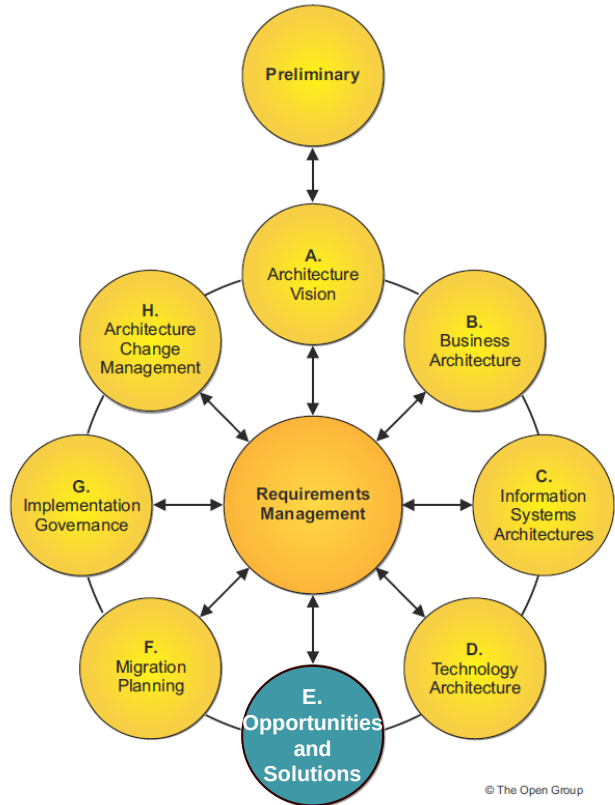
Select actual solutions (e.g., Commercial Off-The-Shelf packages)



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ADM – Phase E: Opportunities and Solutions – Risk Management

The technique identifies, classifies & mitigates risks of the transformation effort



7

Confirm readiness and risk for business transformation

8

Formulate Implementation and Migration Strategy

9

Identify and group major work packages

10

Identify Transition Architectures

11

Create the Architecture Roadmap & Implementation and Migration Plan

Risk Management – Definitions

There is always risk with any architecture/business transformation effort

- **Risk:** “*Effect of uncertainty on objectives*” (ISO 31000:2009) with uncertainty as any deviation from what is expected (positive and negative)
- **Risk triggers:** May be inside or outside the scope of the transformation
- **Risk Management:** Identifying and assessing the likelihood and magnitude of potential positive or negative events on strategic, tactical & operational level
- **Initial Level of Risk:** Risk categorization prior to determining & implementing mitigation
- **Residual Level of Risk:** Risk categorization after implementation of mitigation

➤ Main goal: **Maximizing business benefit** and **minimizing business loss**

Risk Management – Classification

The TOGAF Standard uses risk concepts from The SABSA Institute

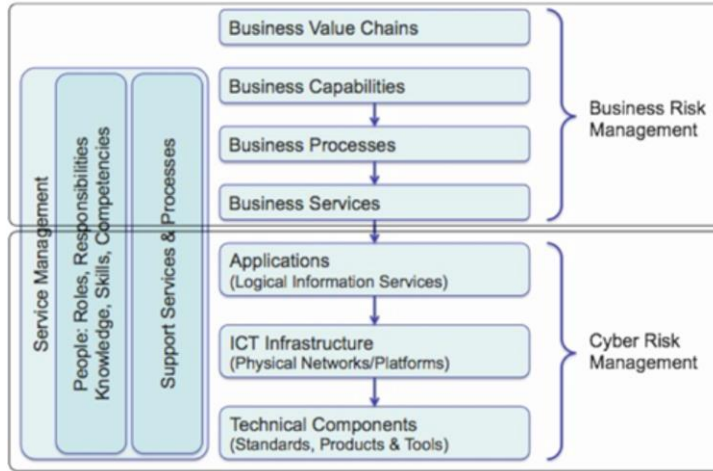


Image source: © The SABSA Institute

Differentiate
Business risk vs. Cyber risk

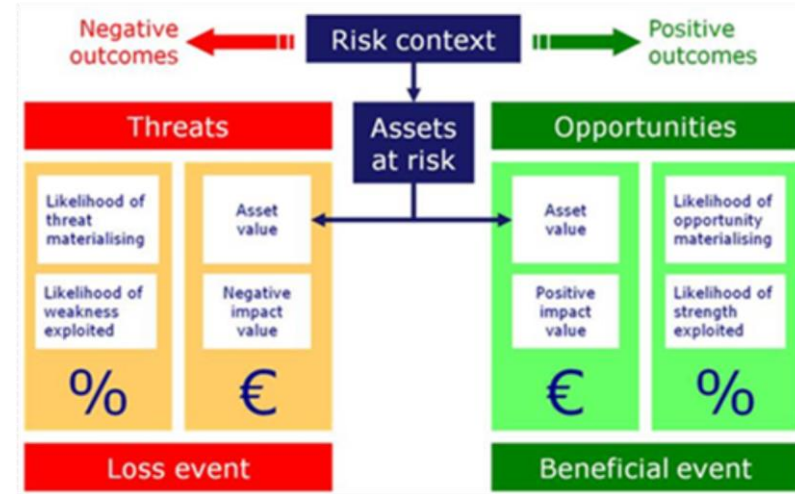


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Balanced view
Control **effects of threats** vs. Leveraging
benefits of opportunities

Risk Management – Assessment

Top-down assessment starts with business value and its optimization

1. Risk identification

2. Risk classification

3. Initial Risk Assessment

4. Risk mitigation

5. Residual risk assessment

6. Risk monitoring & governance

Initial Risk Assessment – Effect & Frequency

The method classifies risk with respect to effect & frequency by using scales



Effect on (parts of) the organization

- **Catastrophic:** critical financial loss that could result in bankruptcy
- **Critical:** serious financial loss in more than one line of business
- **Marginal:** minor financial loss in one line of business
- **Negligible:** minimal impact on one line of business



Frequency during a transformation cycle

- **Frequent:** likely to occur very often and/or continuously
- **Likely:** occurs several times
- **Occasional:** occurs sporadically
- **Seldom:** occur not more than once
- **Unlikely:** probably not occur



Initial Risk Assessment – Risk impact

Effect & frequency are combined using a heuristically-based classification



Extremely High Risk

Transformation will most likely fail with severe consequences

High Risk

Failure of parts of the transformation resulting in goals not being achieved

Corporate Risk Impact Assessment					
Effect	Frequency				
	Frequent	Likely	Occasional	Seldom	Unlikely
Catastrophic	E	E	H	H	M
Critical	E	H	H	M	L
Marginal	H	M	M	L	L
Negligible	M	L	L	L	L

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Low Risk

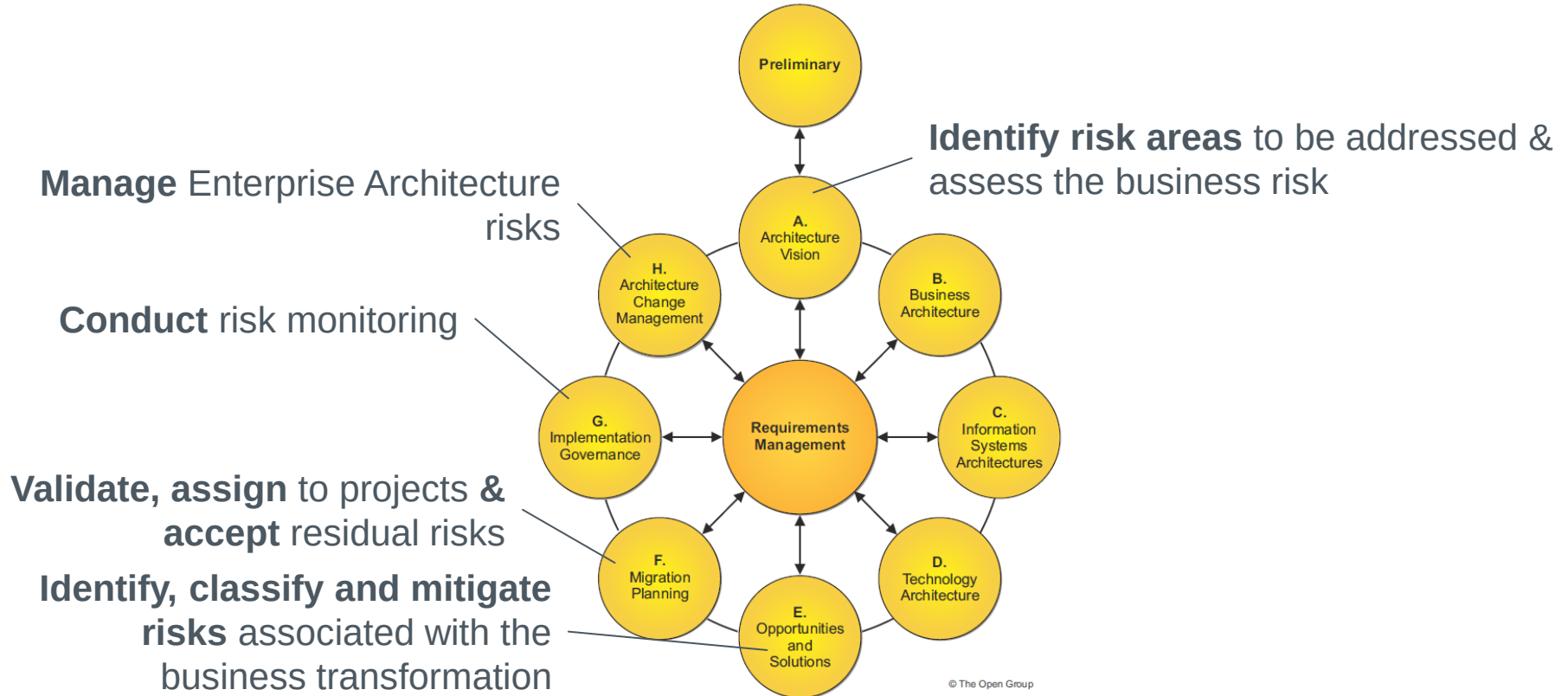
Certain goals of the transformation will not be wholly successful

Moderate Risk

Failure of parts of the transformation threatening the success of goals

Risk Management – Usage within the TOGAF ADM

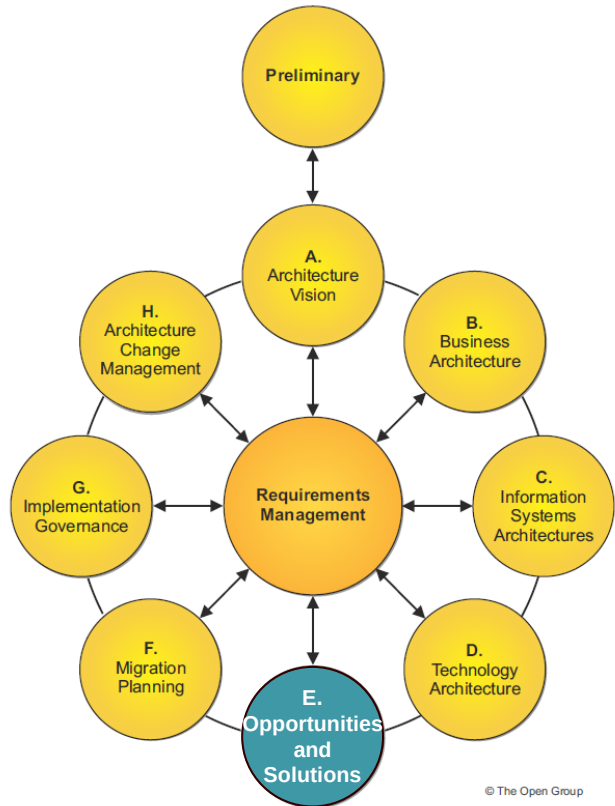
The technique is conducted throughout Phase E till H



ADM – Phase E: Opportunities and Solutions – Architecture Roadmap



The phase consolidates work packages & transitions architectures into a roadmap



7

Confirm readiness and risk for business transformation

8

Formulate Implementation and Migration Strategy

9

Identify and group major work packages

10

Identify Transition Architectures

11

Create the Architecture Roadmap & Implementation and Migration Plan

Architecture Roadmap – Definition (1/2)

The deliverable lists all work packages realizing the Target Architecture

Purpose

- Shows progression from the **Baseline Architecture** to the **Target Architecture**
- Lists **work packages** and their **business value** realizing the **Target Architecture**
- Incrementally developed throughout **ADM phases E and F** and informed by readily identifiable roadmap components from Phase B, C and D

Architecture Roadmap – Definition (2/2)

The deliverable lists all work packages realizing the Target Architecture

- **Work package portfolio** (i.e., description, functional requirements, dependencies, value, relationship to opportunity/architecture definition document & requirements)
- **Implementation factor catalog** (i.e., risks, issues, assumptions, dependencies, actions, inputs)
- **Consolidated Gaps, Solutions and Dependencies matrix** (i.e., architecture domain, gap, potential solutions, dependencies)
- **Transition Architectures**
- **Implementation recommendations** (i.e., criteria measures of effectiveness, risks and issues, Solution Building Blocks)

Architecture Roadmap – Structure

The deliverable focuses on the work packages realizing the Target Architecture



Architecture Roadmap

Project XXXX Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Project List	4
3 Time-Oriented Migration Plan	5
4 Implementation Recommendations	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
Prepared By:		Document Version Date:	
Title:	Architecture Roadmap	Review Date:	
Reviewed By:			

Distribution List

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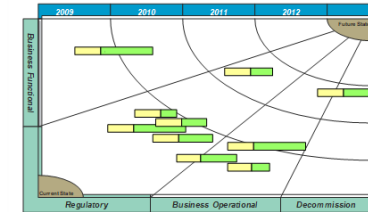
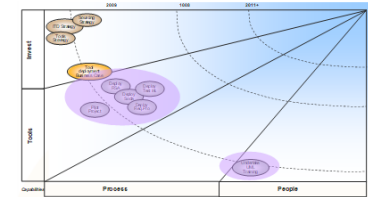
Document Version History

Version Number	Version Date	Revised By	Description	Filename

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3.2 Migration Options



3.3 Benefits of Migration

<<Determined (including mapping to business requirements).>>

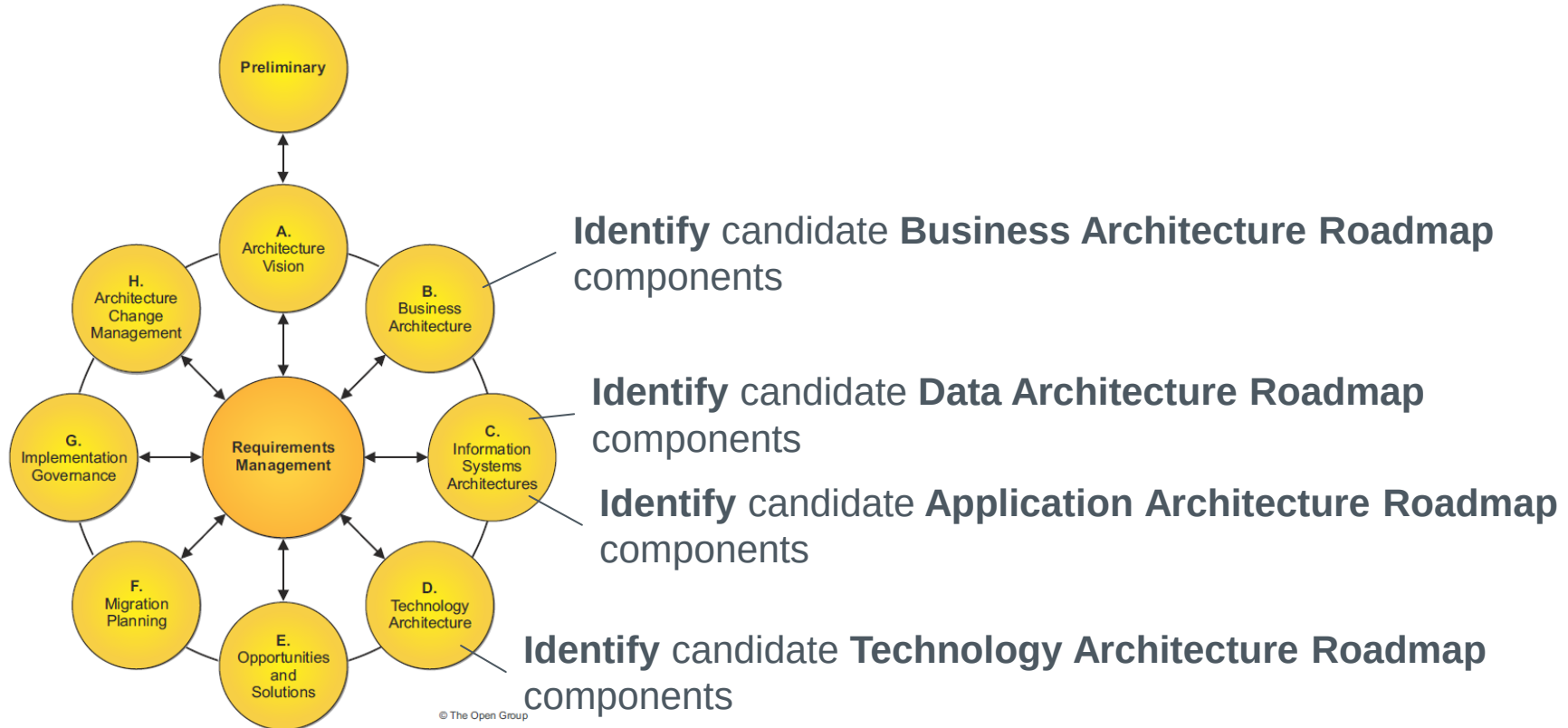
3.4 Estimated Costs of Migration Options

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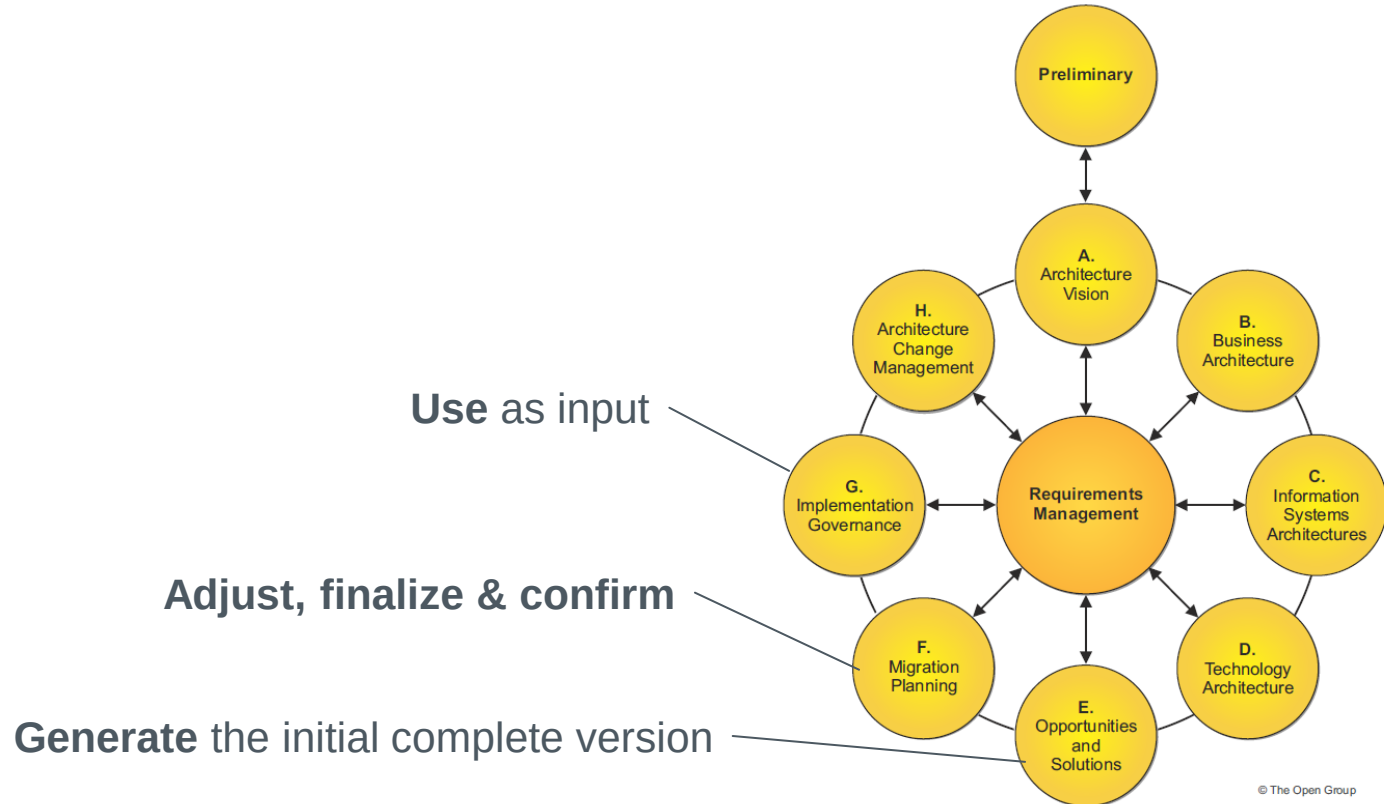
Architecture Roadmap – Usage within the TOGAF ADM (1/2)

The deliverable is based on input from the Phase B throughout D



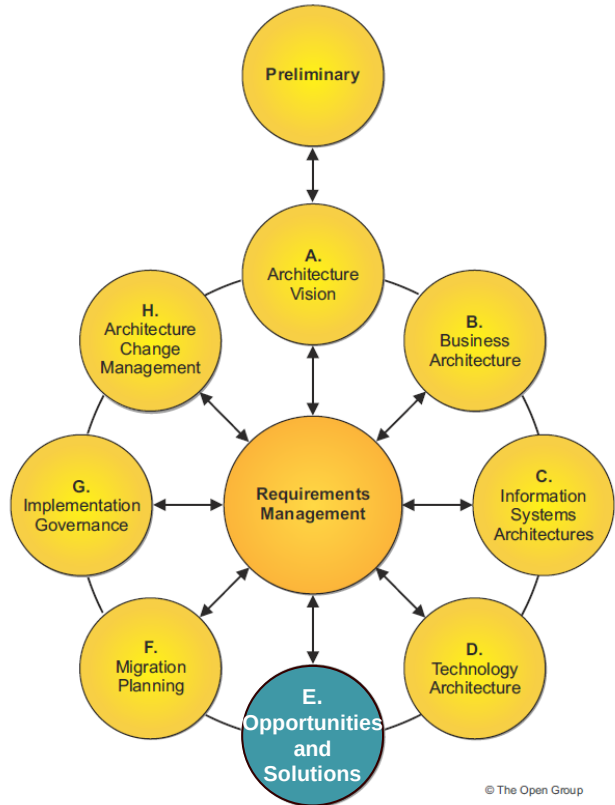
Architecture Roadmap – Usage within the TOGAF ADM (2/2)

The deliverable is finalized in Phase F and used in Phase G



ADM – Phase E – Implementation and Migration Plan

The phase aligns Implementation and Migration Plan to Architecture Roadmap



7

Confirm readiness and risk for business transformation

8

Formulate Implementation and Migration Strategy

9

Identify and group major work packages

10

Identify Transition Architectures

11

Create the Architecture Roadmap & Implementation and Migration Plan

Implementation and Migration Plan – Definition

The deliverable provides a schedule of projects towards the Target Architecture

Purpose

- Contains a **schedule of the projects** realizing the **Target Architecture**
- Includes **executable projects** grouped into managed portfolios and programs
- Shows the **Implementation and Migration Strategy** identifying the change approach

Content

- **Implementation and Migration Strategy** with **direction & sequencing approach**
- **Project and portfolio breakdown** with work packages, delivered capabilities, milestones and timing, work breakdown structure and impact analysis
- **Project charters** with work packages business value, risks, issues, assumptions, dependencies, resources, benefits, costs

Implementation and Migration Plan – Structure

The deliverable focuses on the projects realizing the Target Architecture



Implementation and Migration Plan

Project XXXX
Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents

1	Purpose of this Document	3
2	Implementation and Migration Strategy	4
3	Interaction with Other Management Frameworks	5
4	Project Charters	6
5	Implementation Plan	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
Prepared By:		Document Version Date:	
Reviewed By:	Implementation and Migration Plan	Review Date:	

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Document Version History

Version Number	Version Date	Revised By	Description	Filename

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4 Project Charters

4.1 Capabilities Delivered by Projects

4.2 Included Work Packages

4.3 Business Value

4.4 Risk

4.5 Issues

4.6 Assumptions

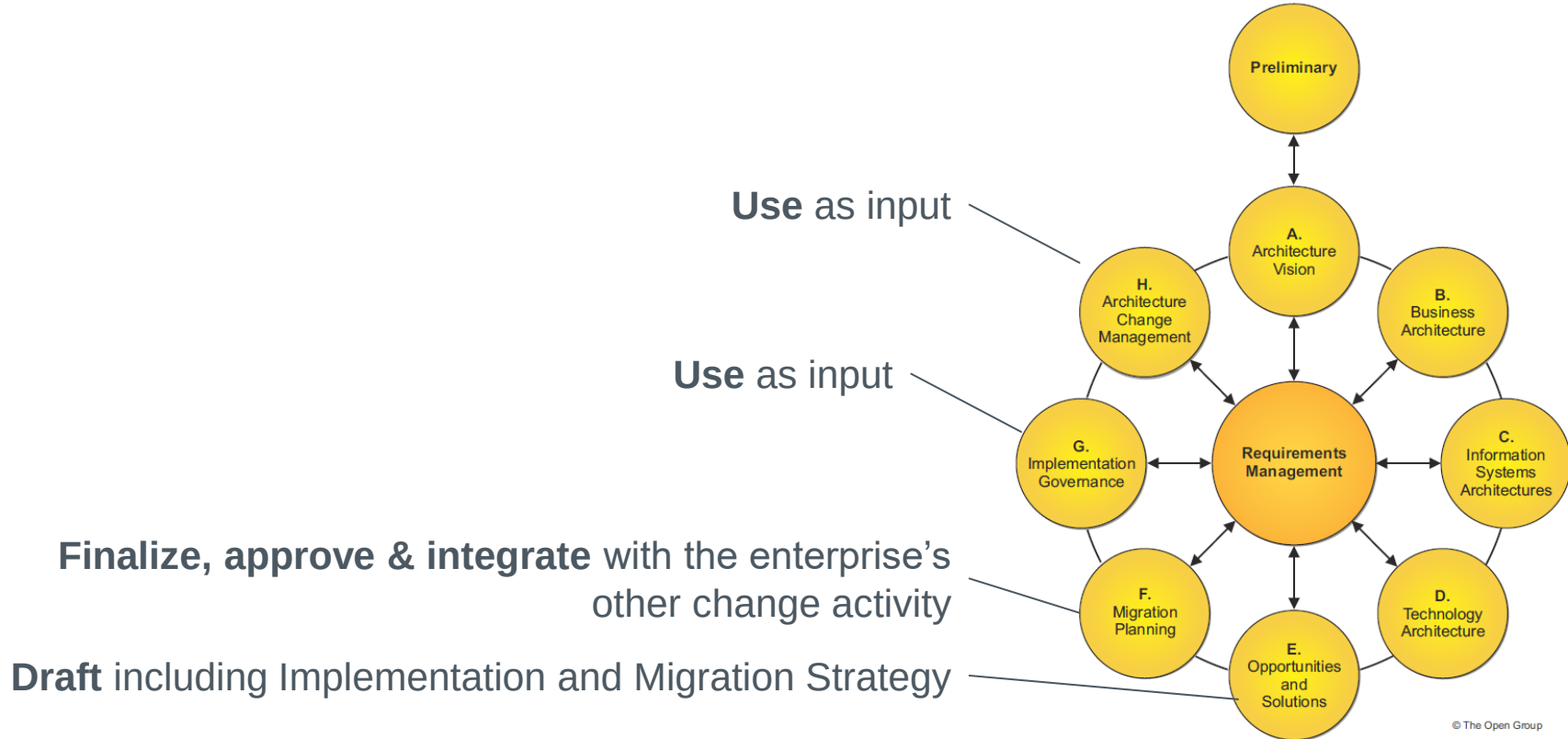
4.7 Dependencies

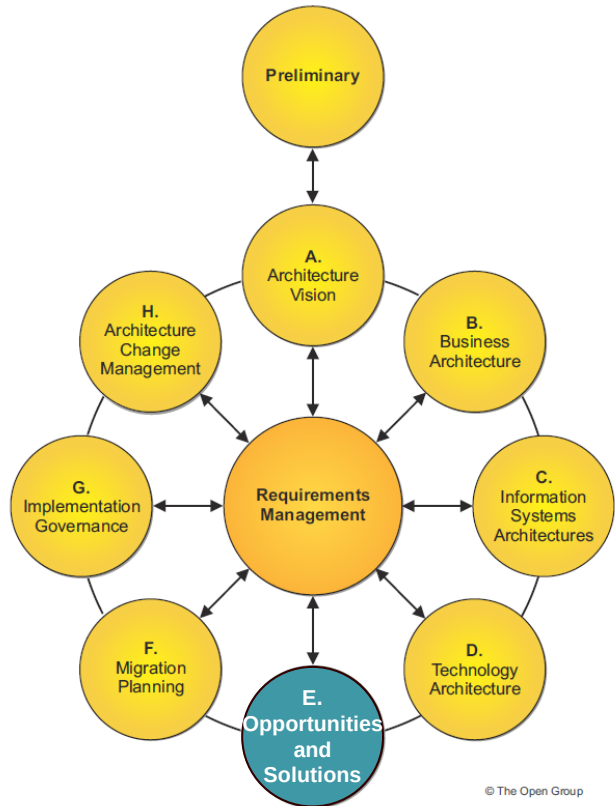
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Implementation and Migration Plan – Usage within the TOGAF ADM

The deliverable is created in Phase E & F and used in Phase G and H

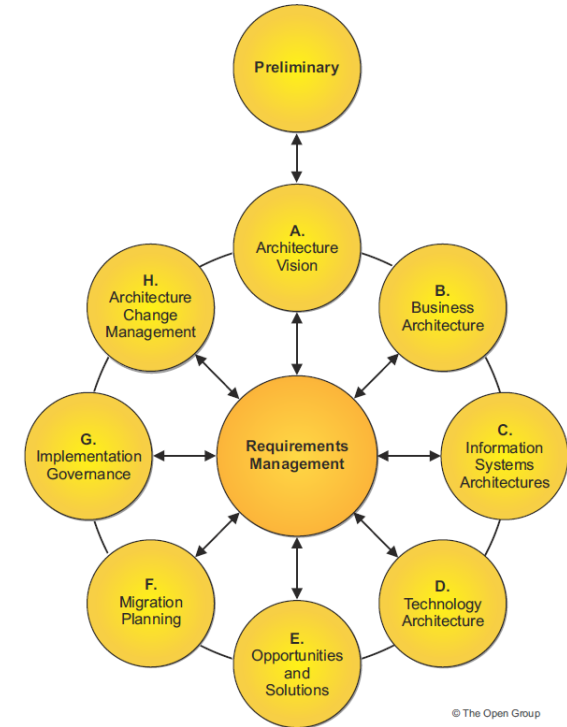




Four concepts for the transition from developing to delivering a Target Architecture:

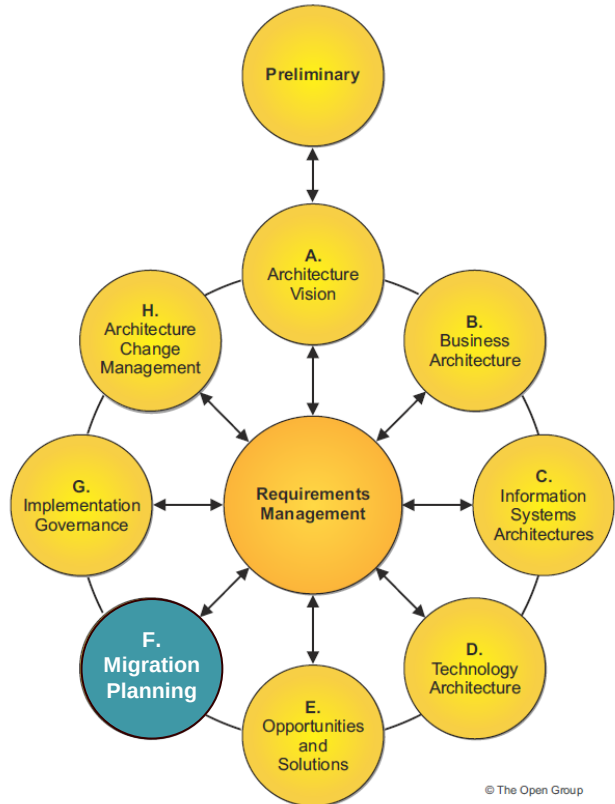
- 1. Architecture Roadmap:** lists work packages in a timeline that will realize the Target Architecture
- 2. Work Packages:** group changes that are necessary to realize the Target Architecture
- 3. Transition Architectures:** significant state between the Baseline and Target Architecture (interim Target Architectures)
- 4. Implementation and Migration Plan:** schedule of projects that will realize the Target Architecture

9	Phase E: Opportunities and Solutions
10	Phase F: Migration Planning
11	Phase G: Implementation Governance
12	Phase H: Architecture Change Management
13	Requirements Management Phase
14	Applying the ADM
15	Architecture Content & Core Concepts
16	Architecture Governance & Architecture Capability



ADM – Phase F: Migration Planning – Purpose

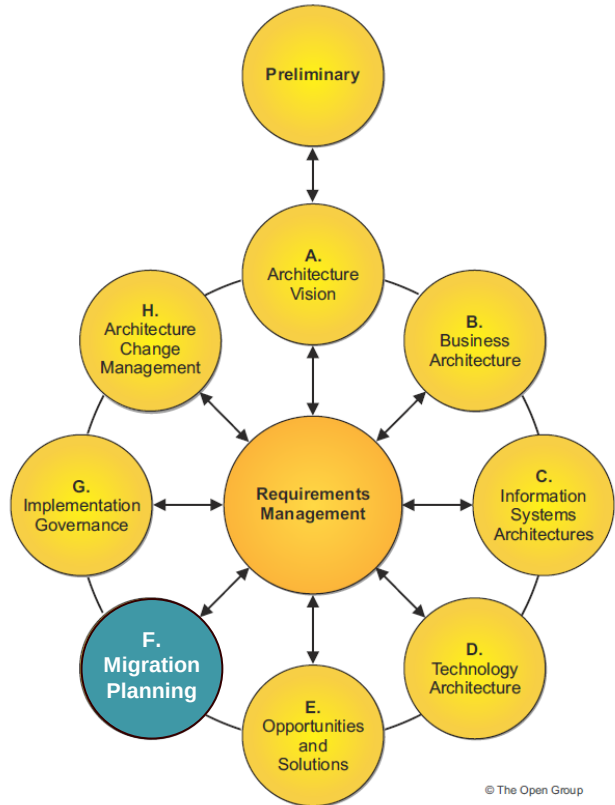
The phase finalizes the Implementation and Migration Plan



- **Finalizing** a detailed **Implementation and Migration Plan** (how to move from the baseline to the target architectures)
- **Estimating** the **resources** required to undertake the change **and ensuring availability**
- **Manage stakeholder priority** and preference adjusted to value, effort and risk
- Output & Outcome: **Approved set of projects incl.**
 - Objectives
 - Constraints
 - Resources required
 - Start & finish date

ADM – Phase F: Migration Planning – Objectives

The phase finalizes the Implementation and Migration Plan

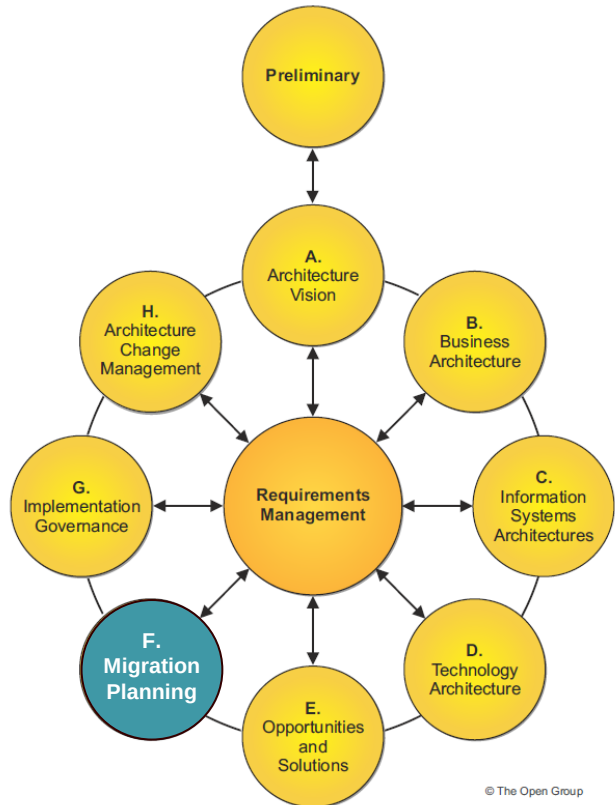


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1. **Finalize** the **Architecture Roadmap** and the supporting **Implementation and Migration Plan**
2. **Ensure** that the **Implementation and Migration Plan** is **coordinated with the enterprise's approach** to managing and implementing change in the enterprise's overall change portfolio
3. **Ensure** that the **business value, cost of work packages and Transition Architectures** is **understood by key stakeholders**

ADM – Phase F: Migration Planning – Steps

The phase completes the architecture development cycle



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1

Confirm management framework interactions for the Implementation and Migration Plan

2

Assign a business value to each work package

3

Estimate resource requirements, project timings and availability/delivery vehicle

4

Prioritize the migration projects through conduct of a cost/benefit assessment and risk validation

5

Confirm Architecture Roadmap and update Architecture Definition Document

6

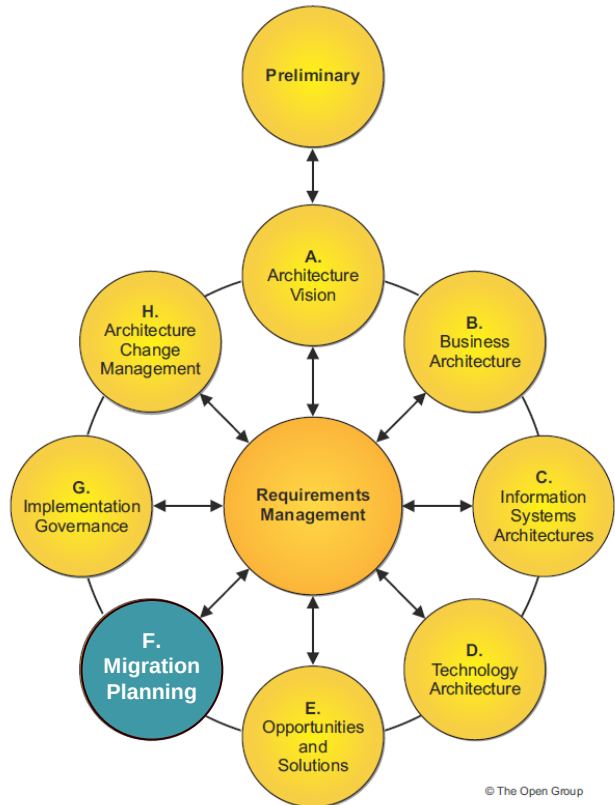
Complete the Implementation and Migration Plan

7

Complete the architecture development cycle and document lessons learned

ADM – Phase F – Implementation Governance Model

The phase transitions governance from the development to realization



1

Confirm management framework interactions for the Implementation and Migration Plan

2

Assign a business value to each work package

3

Estimate resource requirements, project timings and availability/delivery vehicle

4

Prioritize the migration projects through conduct of a cost/benefit assessment and risk validation

5

Confirm Architecture Roadmap and update Architecture Definition Document

6

Complete the Implementation and Migration Plan

7

Complete the architecture development cycle and document lessons learned

Implementation Governance Model – Definition

The deliverable enables Architecture Governance on a project basis

Purpose

- Ensures that a **project** transitioning into **implementation** also smoothly transitions into appropriate **Architecture Governance**
- Supports also organizations having established architecture functions and a **governance framework** on **project-by-project basis**

Content

- Governance **process**
- Governance **organization structure**
- Governance **roles and responsibilities**
- Governance **checkpoints** and **success/failure criteria**

Implementation Governance Model – Structure

The deliverable defines the governance for the implementation projects



Implementation Governance Model

Project XXXX
Client YYYY

<<Note: This document provides a generic template. It may require tailoring to suit a specific client and project situation.>>

Table of Contents

1	Purpose of this Document	3
2	Governance Processes	4
3	Governance Organisation Structure	5
4	Governance Roles and Responsibilities	6
5	Governance Checkpoints and Success/Failure Criteria	7

Document Information

Project Name:	Project XXXX	Document Version No:	0.1
Prepared By:		Document Version Date:	
Title:	Implementation Governance Model	Reviewed By:	
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Document Version History

Version Number	Version Date	Revised By	Description	Filename

TOGAF® 9 Template: Implementation Governance Model
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5 Governance Checkpoints and Success/Failure Criteria

5.1 Governance Checkpoints

5.2 Metrics and KPIs

In addition, the following metrics will be used to determine the success of this architecture work:

Metric	Measurement Technique	Target Value	Rationale/Further Notes

<<Refer back to the Request for Architecture Work and restate/refine.>>

5.3 Success/Failure Criteria

5.4 Acceptance Procedure

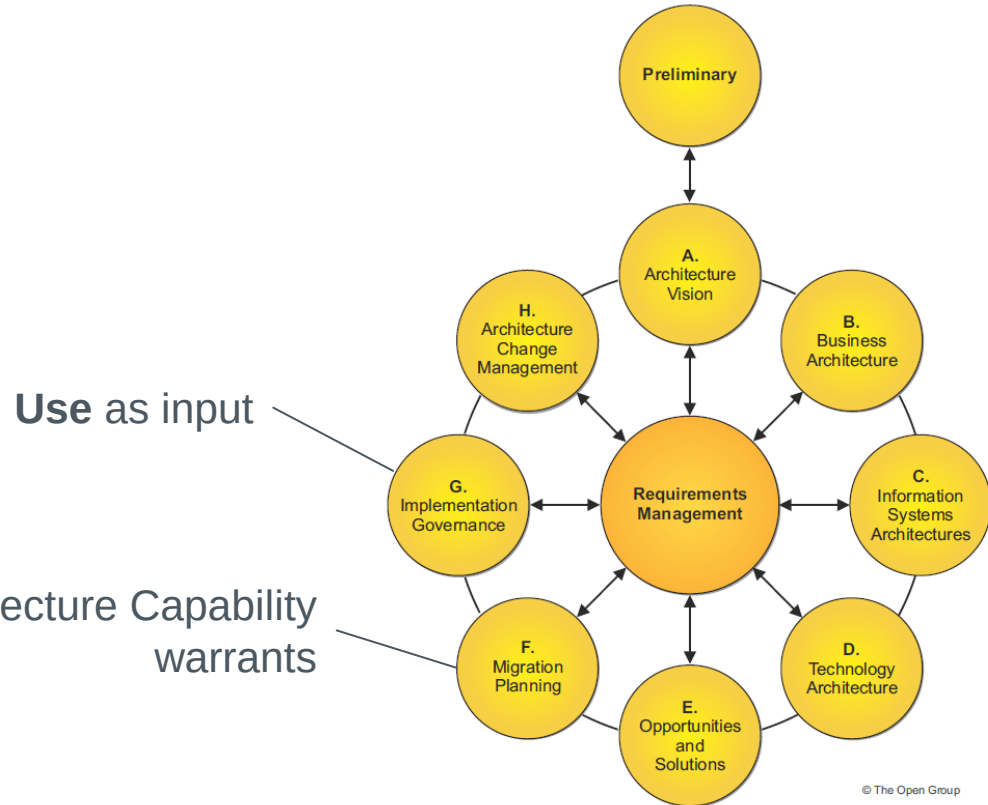
<<Describe the process to be used for acceptance/sign-off.>>

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Implementation Governance Model – Usage within the TOGAF ADM

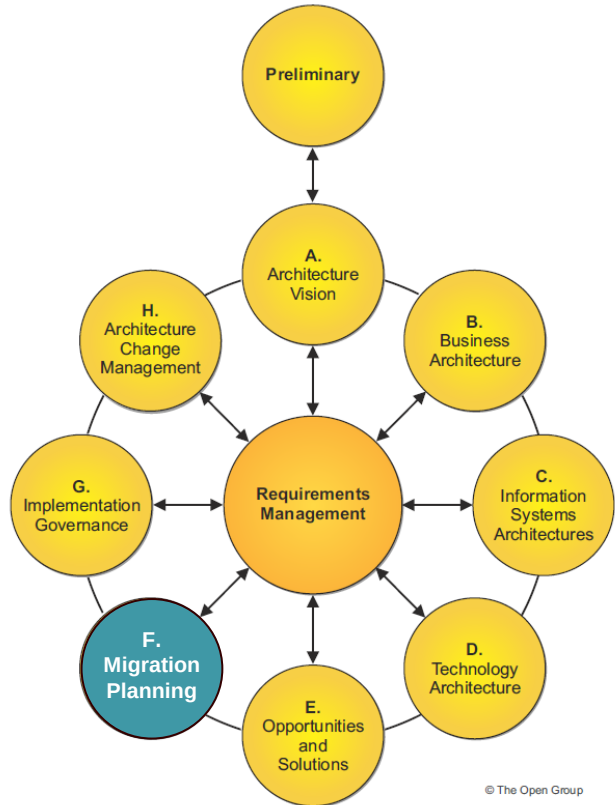
The deliverable is created in Phase F and used in Phase G



Created, if the maturity of the Architecture Capability warrants

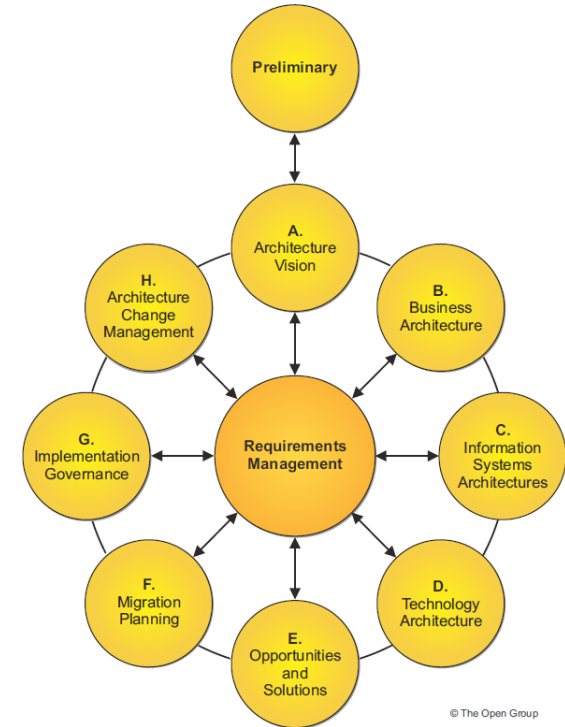
ADM – Phase F: Migration Planning – Approach

The phase closely aligns with the work of project and portfolio managers



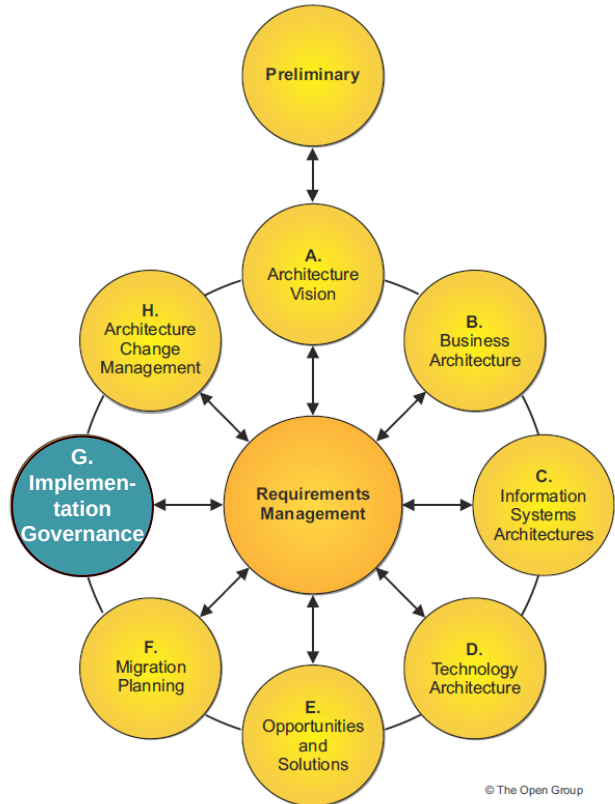
- **Finalize Implementation and Migration Plan**
(based on Draft Architecture Roadmap and Draft Implementation and Migration Plan from Phase E)
- **Include portfolio and project-level details** to the Implementation and Migration Plan
- **Integrate** the Roadmap, the Implementation and Migration Plan **with** the enterprise's **change activities**
- **Assess the dependencies, costs and benefits** of the various migration projects
- **Transition to Implementation Governance & Document the lessons learned** to enable continuous process improvement

9	Phase E: Opportunities and Solutions
10	Phase F: Migration Planning
11	Phase G: Implementation Governance
12	Phase H: Architecture Change Management
13	Requirements Management Phase
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ADM – Phase G: Implementation Governance – Purpose

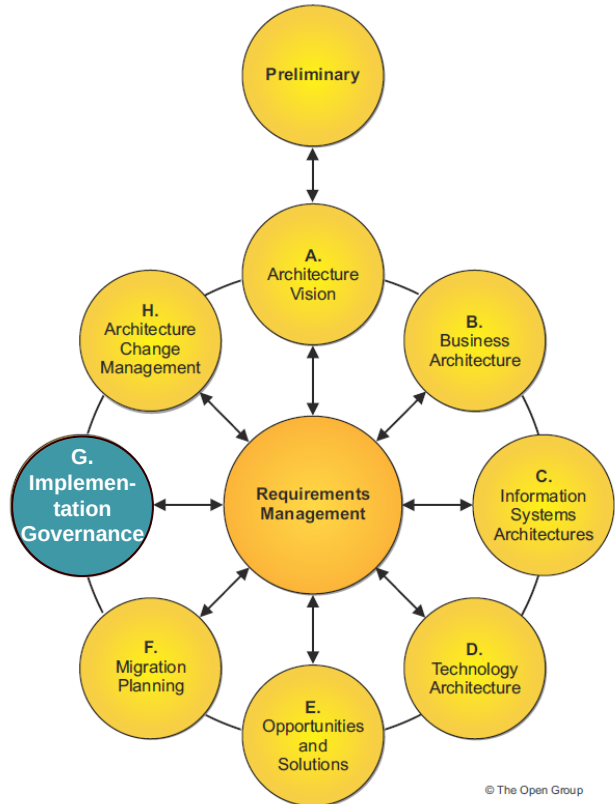
The phase provides the architectural oversight of the implementation



- **Ensure the compliance** of the (implementation) projects with the defined architectures
- **Guide the implementation teams** (Purpose, Constraints, Gap, Architecture Requirement Specification, Control)
- **Manage stakeholder priority and preference adjusted to success**, value, effort and risk
- **Output & Outcome:**
 - **Completed projects**
 - **Implemented changes**
 - **Adjusted target state**

ADM – Phase G: Implementation Governance – Objectives

The phase ensures conformance and performs Architecture Governance



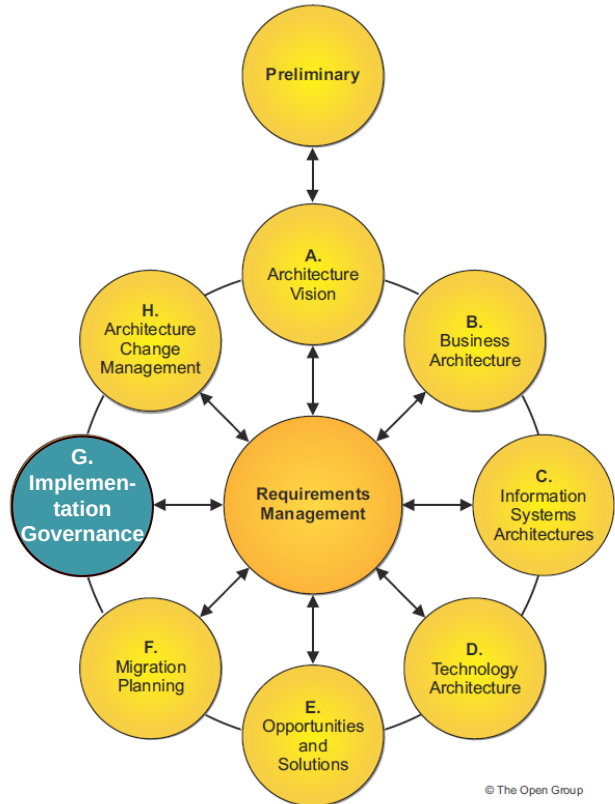
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1. Ensure conformance of the implementation projects with the Target Architecture

2. Perform appropriate Architecture Governance functions for the solution and any implementation-driven architecture Change Requests

ADM – Phase G: Implementation Governance – Steps

The phase runs parallel to the implementation projects



1

Confirm the scope and the priorities for the deployment **with the development management**

2

Identify deployment resources and skills

3

Guide development of solutions deployment

4

Perform Enterprise Architecture Compliance reviews

5

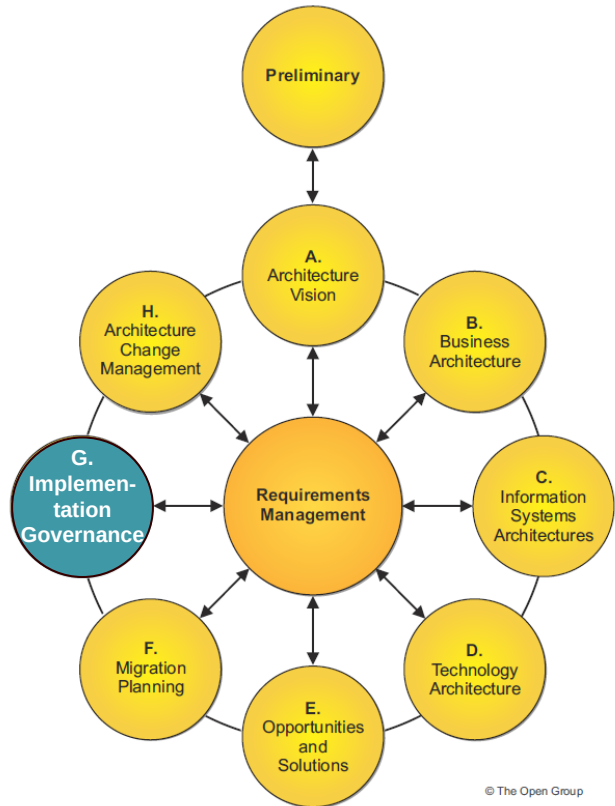
Implement business and IT operations

6

Perform post-implementation review and close the implementation

ADM – Phase G: Implementation Governance – Architecture Contract

Architecture Contracts guide the development of solutions deployment



1

Confirm the scope and the priorities for the deployment **with the development management**

2

Identify deployment resources and skills

3

Guide development of solutions deployment

4

Perform Enterprise Architecture Compliance reviews

5

Implement business and IT operations

6

Perform post-implementation review and close the implementation

Architecture Contract – Development Contract

The contract type governs the collaboration of sponsors and implementers

Purpose

- Reflects a joint agreement between **development partners** and **sponsors** on the **deliverables**, **quality** and **fitness-for-purpose** of an **architecture**
- Delivered through effective **Architecture Governance**
- Enables a **dynamic Enterprise Architecture** allowing for flexible evolution in response to changing technology and business drivers
- **Traditional:** Agreements between the **sponsor** and the **architecture function** or **Information Systems (IS) department**
- **Today:** Agreements between **all parties involved** in the architecture development and delivery (e.g., systems integrators, applications providers & service providers)

Architecture Design & Development Contract

- **Introduction and background, Nature** of the agreement, **Scope** of the architecture, Architecture and strategic **principles** and **requirements**, Conformance requirements, Architecture development and management **process and roles**, Target Architecture **measures**, Defined **phases of deliverables**, Prioritized joint **workplan**, **Time window(s)**, Architecture **delivery and business metrics**

Business Users' Architecture Contract

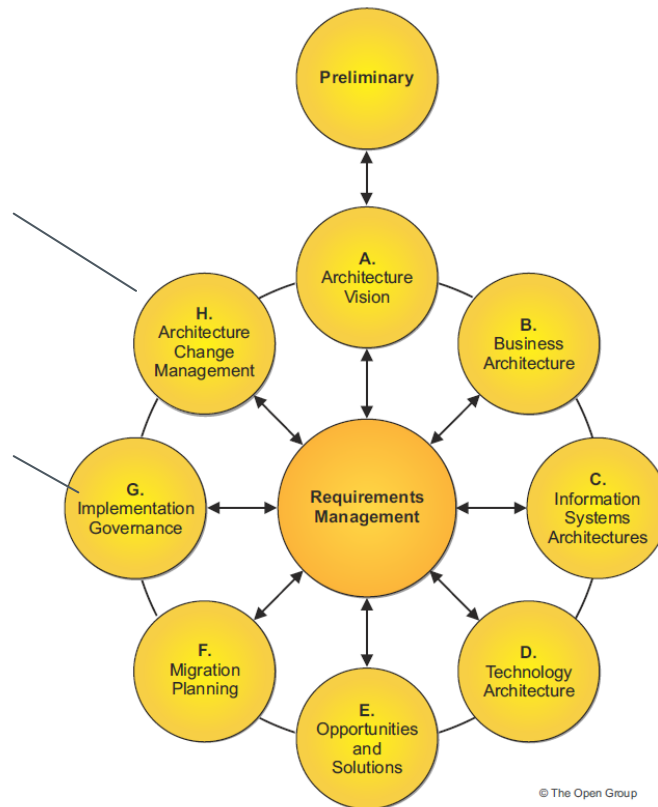
- **Introduction and background, Nature** of the agreement, **Scope**, Strategic **requirements**, Conformance requirements, Architecture **adopters**, **Time window**, Architecture business **metrics**, **Service architecture** (including Service-Level Agreement)

Development Contract – Usage within the TOGAF ADM

The contract is created mainly in Phase G and used throughout Phase H

**Connect architecture and implementation
organization &
& update, if necessary**

**Document &
Obtain signature** from all developing organizations
and sponsoring organization



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Architecture Contract – Statement of Architecture Work



The contract type governs the collaboration between sponsor & architects

Purpose

- Represents an Architecture Contract between the **architecting organization** and the **sponsor of the Enterprise Architecture**, created as a deliverable of **Phase A**
- Defines **scope & approach** being used for an **architecture development cycle**
- May form the **basis for a contractual agreement** between the **supplier and consumer** of architecture services

Content

- **Title, approvals, change of scope procedures**
- Architecture project **request, background, description, scope, plan & schedule**
- **Acceptance criteria and procedures, roles, responsibilities and deliverables**
- **Overview of Architecture Vision**

Architecture Contract – Architecture Design & Development Contract



The contract type governs the architecture and implementation work

Purpose

- Is a **statement of intent** on designing & implementing (parts of) the **Enterprise Architecture** from partners (e.g., system integrators, application providers)
- Defines the **deliverables**, **quality** and **fitness-for-purpose** of the developed architecture and the **collaboration processes** of the partners
- Is defined at the **appropriate stage of the ADM**, depending on the contracted work

Content

- **Introduction, background and nature of the agreement**
- Architecture **scope**, Target Architecture **measures** and **delivery/business metrics**
- **Architecture** & strategic **Principles** and (conformance) **requirements**
- Architecture development & management **process** and **workplan & time window**

Architecture Contract – Business Stakeholder's Architecture Contract



The contract type governs the collaboration between architects & business

Purpose

- Drawn up between the **architecting function** and the **business stakeholders** who will subsequently be building and deploying the architected business solutions
- Created at the **end of Phase F** with the agreed Implementation and Migration Plan
- Also used to **manage changes** to the **Enterprise Architecture** in **Phase H**

Content

- **Introduction, background and nature of the agreement**
- **Scope**, strategic and conformance **requirements** and **Service Level Agreements**
- **Architecture deliverables & business metrics** meeting business requirements
- **Time window** and **architecture adopters**

Architecture Contract – Structure

The content of the document depends on the contract's type

Architecture Contract with Business Users

Project XXXX Client YYYY

<<Note: This document provides a gen
project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Architecture Contract	4
3 Objectives and Scope	5
4 Architectural Deliverables	6
5 Risks and Mitigations	7
6 Communications Plan	8
7 Acceptance Criteria and Procedures	9
8 Signature Approval	10

Document Information

Project Name:	Project XXXX		
Prepared By:		Document Version No:	0.1
Title:	Architecture Contract with Business Users	Document Version Date:	
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Architecture Contract with Development Partners

Project XXXX Client YYYY

<<Note: This document provides
project situation.>>

Table of Contents	
1 Purpose of this Document	3
2 Architecture Contract	4
3 Objectives and Scope	5
4 Architectural Deliverables	6
5 Projected Joint Work Plan	7
6 Communications Plan	8
7 Risks and Mitigations	9
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Architecture Contract – Benefits

A governed approach to the management of contracts has many advantages



Accountability, responsibility and discipline using architectural artifacts



Continuous monitoring of all architecture-related activities (e.g., integrity, changes, decision-making, audit)



Adherence to the principles, standards & requirements of architectures



Identification of risks of the development (e.g., standards, policies, technologies) **and operation** (e.g., resilient environment) of **architectures**



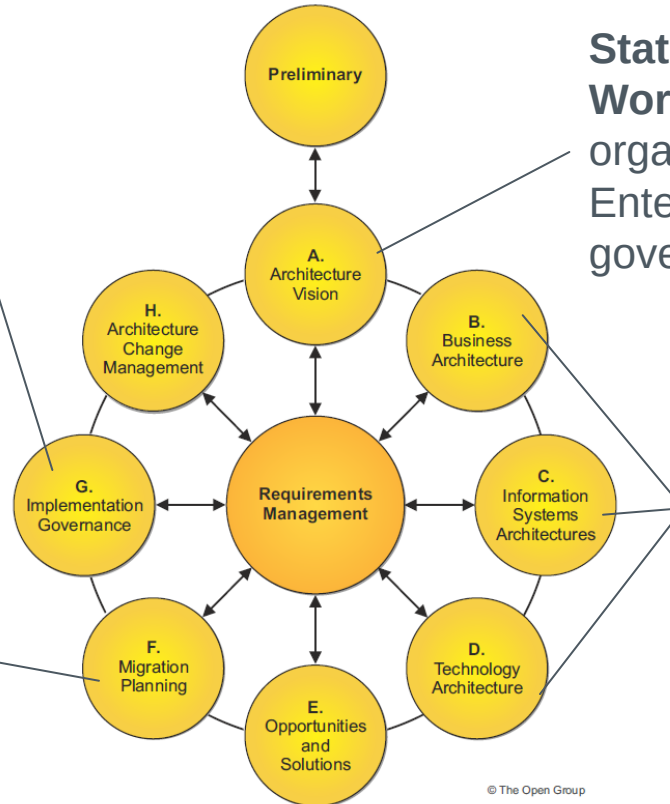
Understanding of the **governance organization** responsible for the contract, their level of authority and scope of the architecture

Architecture Contract – Usage within the TOGAF ADM

Contracts may be used during various ADM phases

Contract of architecture function <> implementation function for implementing the EA (inhouse, contractor), e.g., strategic & enterprise-wide technology infrastructure, applications & data management

Contract of architecture function <> business stakeholders building and deploying application systems in the architected environment



Statement of Architecture Work as contract of architecting organization <> sponsor of Enterprise Architecture (or IT governance function)

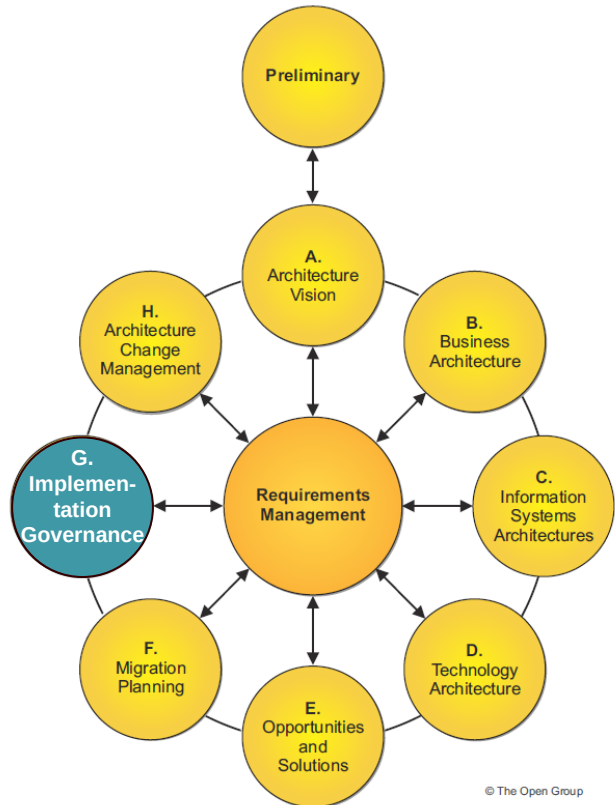
Contract of systems integrators, applications providers and/or service providers regarding the architecture and work

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ADM – Phase G: Implementation Governance – Compliance Assessment



The phase performs compliance reviews to fulfill Architecture Governance



1

Confirm the scope and the priorities for the deployment **with the development management**

2

Identify deployment resources and skills

3

Guide development of solutions deployment

4

Perform Enterprise Architecture Compliance reviews

5

Implement business and IT operations

6

Perform post-implementation review and close the implementation

Architecture Compliance Assessment (Deliverable) – Definition

The deliverable governs the architecture through the implementation

Purpose

- Ensures that the realized **design and implementation** is in line with the **strategic and architectural objectives** and the **Architecture Vision**
- Ensures that **implementation learnings** are fed back into the **architecture process**

Content

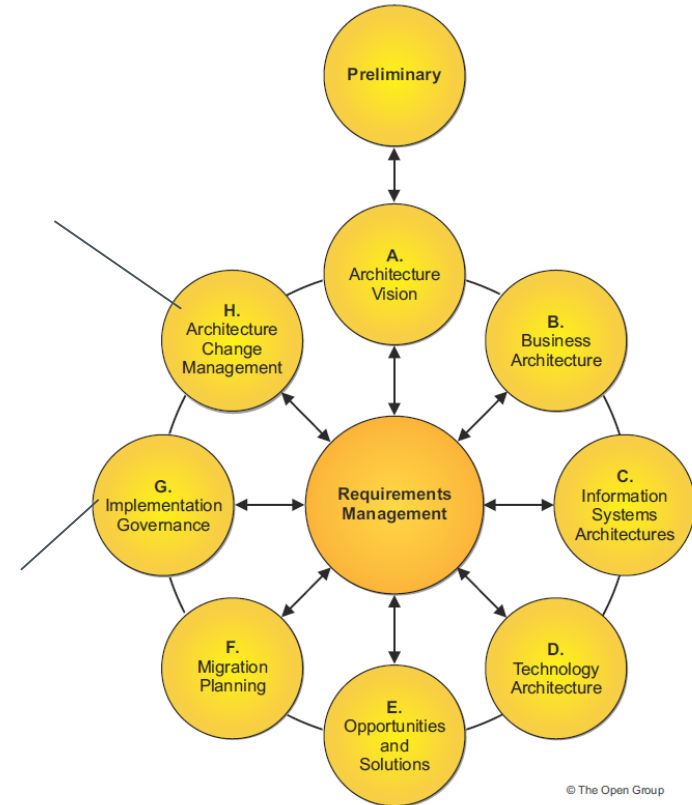
- Overview of **project progress, status** and **architecture/design**
- **Architecture checklists** (e.g., hardware & operating system, software services and middleware, applications, information management, security, system management, system engineering, methods and tools)

Architecture Compl. Assessment – Usage within the TOGAF ADM

The deliverable is used in Phase G and H

Assess realized value using the linkage between stakeholder requirements & specified constraints

Consider the scope of the Implementation Project and the work packages filling the gaps & **Confirm** whether the Architecture Requirements Specifications have been followed



Architecture Compliance Review (Process) – Timing

The review is held at project milestones or checkpoints in the project's lifecycle

Architecture development

- Development of the **architecture** itself (ADM compliance)
- When **business requirements** and the **Enterprise Architecture** are reasonably firm and the project architecture is taking shape



Project implementation

- **Implementation** of the architecture(s) (architecture compliance)
- Specific **checkpoints** within the project (e.g., project initiation, initial design, major design changes, ad hoc)
- Stage when there is still **time to correct any major errors** or shortcomings



Architecture Compliance Review (Process) – Resources (1/2)



The Target Architecture Checklist ensures architecture development compliance

#	Checklist questions (excerpt)
1	Were the correct stakeholders identified?
2	Were constraints and guidance from the superior architecture taken into account?
3	Do appropriate SMEs agree with the facts and interpretation of the facts in the architecture?
4	Do any constraints or guidance produced reflect the views produced for stakeholders and any underpinning architecture models and analysis?
...	

- The **EA Practitioner** assesses the Target Architecture and **stakeholders** approve it

Upon **negative response**...

1. the EA Practitioner **reworks the Target Architecture** or
2. the **Architecture Project** is canceled

Architecture Compliance Review (Process) – Resources (2/2)



The Implementation & other Change Checklist ensures implementation compliance

#	Checklist questions (excerpt)
1	Did the organization embarking on a change reasonably interpret the Target Architecture's guidance and constraints ?
2	Do appropriate SMEs agree with the facts and interpretation of the facts in the impact assessment?
3	Do appropriate SMEs agree with the recommendation to enforce the target, grant time-bound relief, or change the architecture?
...	...

The **EA Practitioner** acts as...

1. **auditor** (compliance assessment) and
2. **architect** (recommendations)

Recommendation is based on impact assessment

1. Enforce **compliance**
2. Provide **temporary relief** or
3. Change **Target Architecture**

ADM – Apply Architecture to Support agile software development

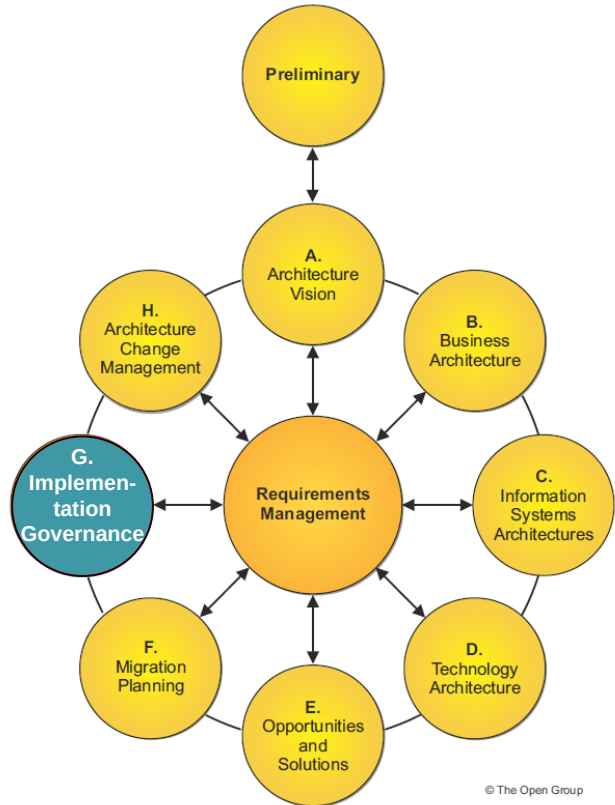
The TOGAF Standard aligns to agile development in Phase G

Portfolio	Project	Solution Delivery
▪ Identify the products needed by the enterprise ▪ Identify the boundaries of the products ▪ Set constraints for the product owner ▪ Define the enterprise's backlog	▪ Identify the products needed by the enterprise ▪ Identify the boundaries of the products ▪ Set constraints for the product owner ▪ Set constraints for the agile teams	▪ Set constraints for the agile teams

➤ The **Enterprise Architect** serves the **stakeholders** guarding the mission, vision, goals and investment roadmap, hence guarding **enterprise value**

ADM – Phase G: Implementation Governance – Approach (1/2)

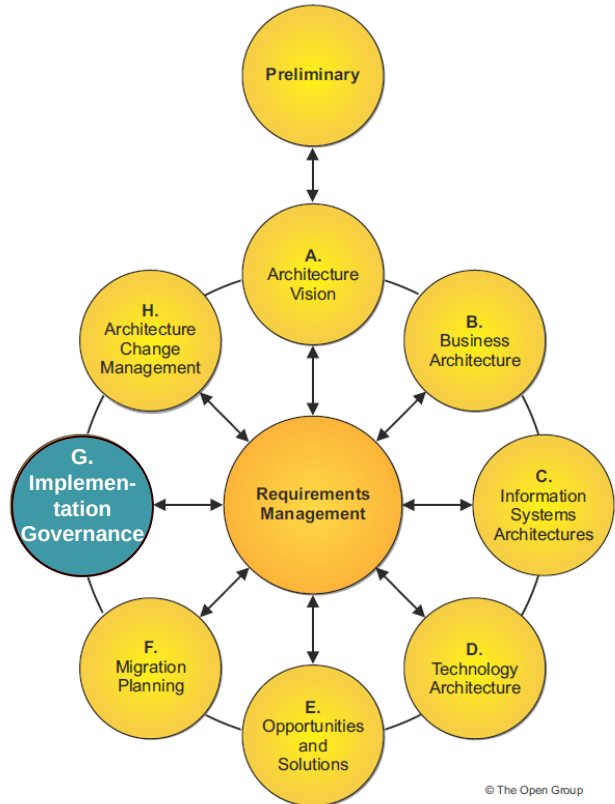
The phase deploys the Target Architecture as a series of transitions



- **Incremental steps** towards the Target Architecture:
 - Each step **delivers business value**
 - **Early realization of business value** and benefits
 - **Minimize risks** in the transformation and migration program
- **Establish an implementation program**
- **Adopt a phased deployment schedule** based on business priorities

ADM – Phase G: Implementation Governance – Approach (2/2)

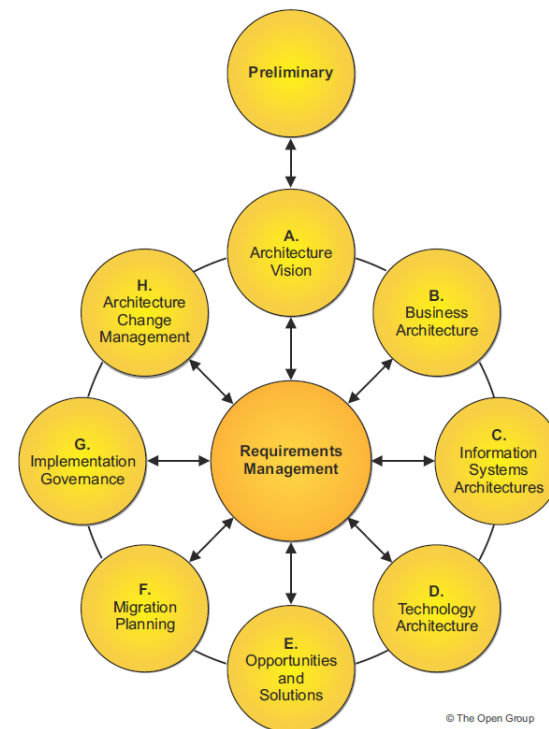
The phase deploys the Target Architecture as a series of transitions



- Follow the organization's standard for corporate, IT and Architecture Governance
- Use the organization's established portfolio/program management approach
- Define an operations framework to ensure the effective long life of the deployed solution
- Develop project details (name, descriptions, acceptance criteria, measures of effectiveness)

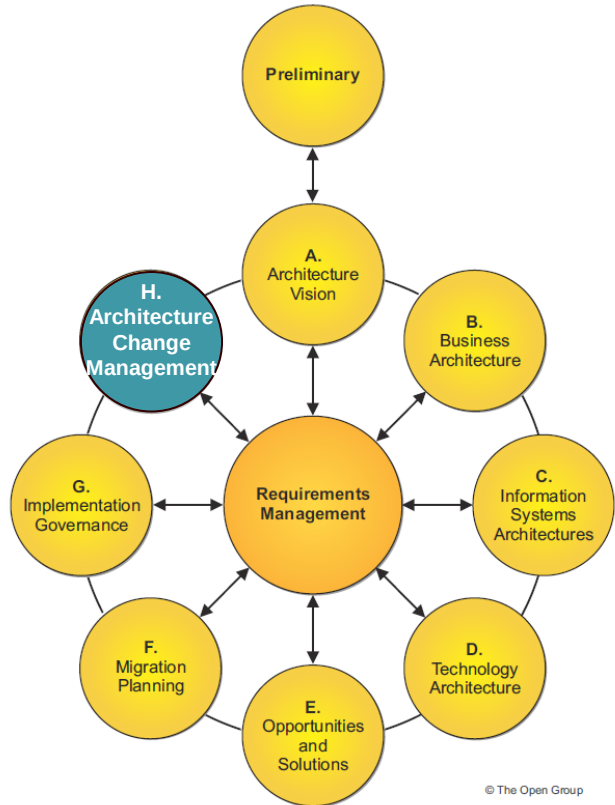
Phase G establishes the connection between architecture and implementation organization, through the Architecture Contract.

9	Phase E: Opportunities and Solutions
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ADM – Phase H: Architecture Change Management – Purpose

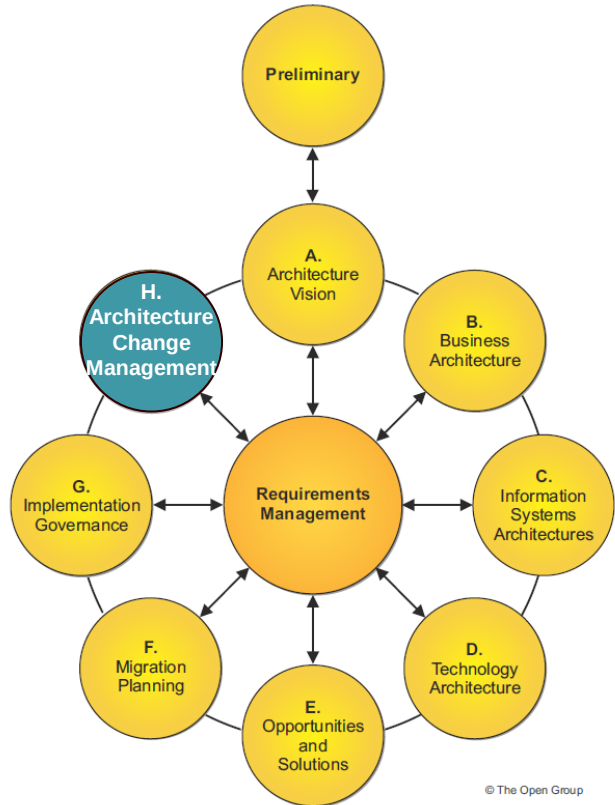
The phase establishes procedures for managing change to the new architecture



- **Ensure** that the architecture achieves its original target business value (**Value Realization**)
- **Identify Gaps between approved target**, or preference **and realization** from prior work
- **Manage Changes** (preference, priority or stakeholder requirements)
- **Output & Outcome:** Direction to proceed and start **developing a Target Architecture that addresses** perceived, real, or anticipated **shortfalls** in the Enterprise relative to stakeholder preferences.

ADM – Phase H: Architecture Change Management – Objectives

The phase maintains the Architecture Development Cycle

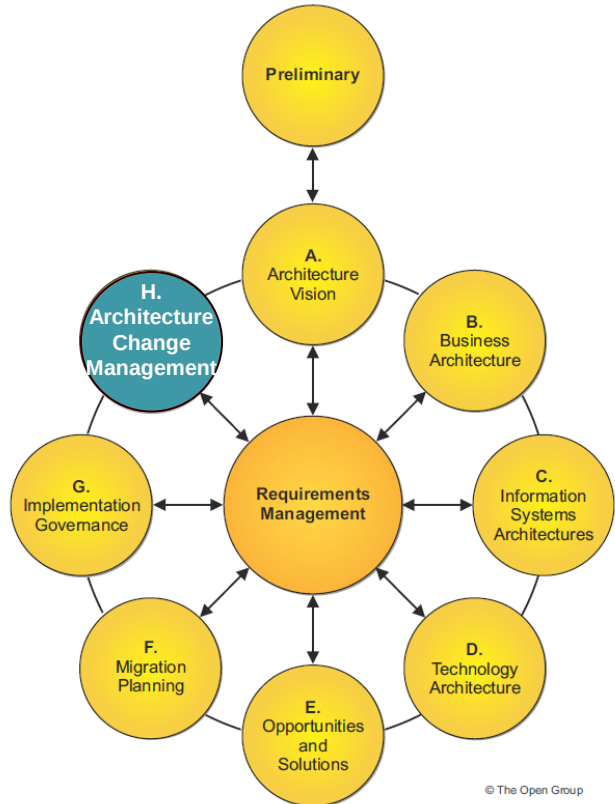


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1. Ensure that the **architecture development cycle is maintained**
2. Ensure that the **Architecture Governance Framework is executed** (regarding changes)
3. Ensure that the Enterprise **Architecture Capability meets current requirements**

ADM – Phase H: Architecture Change Management – Steps

The phase waits for changes and manages them



1

Establish value realization process

2

Deploy monitoring tools

3

Manage risks

4

Provide analysis for architecture change management

5

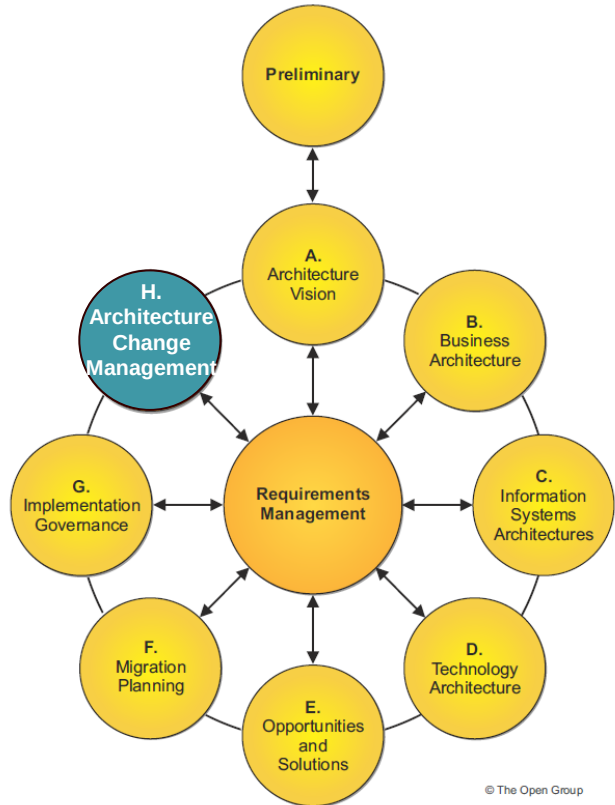
Develop change requirements to meet performance targets

6

Manage governance process

7

Activate the process to implement change



■ Change Requests (technology)

- New technology reports
- Asset management cost reduction initiatives
- Technology withdrawal reports
- Standard initiatives

■ Change Request (business)

- Business developments
- Business exceptions
- Business innovations
- Business technology innovations
- Strategic change developments

Change Request – Definition

The request extends or refines the architecture during its implementation

Purpose

- Describes a **deviation from the suggested architectural approach** (e.g., definition, scope, requirements, solution)
- Created due to **gained knowledge** or **external factors** (e.g., market factors, changes in business strategy, new technology opportunities)
- Kick-starts a further cycle of **architecture work**

Content

- **Description and rational** for the proposed change
- **Impact assessment** (i.e., reference to requirements, stakeholder priority, phases to be revisited, recommendations)
- **Repository reference number**

Change Request – Structure

A change request should be captured in a structured manner



2 Basic Details

Change ID	<<ID to uniquely identify this change request>>
Change Title	<<One-line summary title for the change>>
Change Requestor	<<Name>> <<Position>> <<Organization>> <<Email>> <<Tel>>
Change Sponsor(s)	<<Details of business sponsors for this change>>
Deadline	<<Any deadline for implementing the change, including reason>>

3 Change Description

<<Describe the change that is proposed. (Other documents may be referenced where necessary.)>>

4 Change Rationale and Impacts

4.1 Drivers for Change

The following drivers for change are relevant (provide notes on all that apply).

Business Drivers	
Business-as-usual developments	
Business exceptions	
Business innovations	
Business technology innovations	
Strategic change	
Technology Drivers	
New technology reports	
Asset management cost reductions	
Technology withdrawal	
Standards initiatives	
Operational Drivers	
Improvements based on operational experience	
Capacity planning	

4.2 Change Rationale

<<Any further rationale to justify the change. Include an analysis of the implications of no change.>>

4.3 Anticipated Impacts

<<Provide, as far as possible, a brief initial view on the likely impacts of the change. For example, anticipated architecture rework required, system changes likely to result, affected business areas.>>

<<An initial indication only is required here – a full Requirements Impact Assessment may be done as a follow-up activity if further investigation is deemed necessary.>>

Change Request – ADM Phases

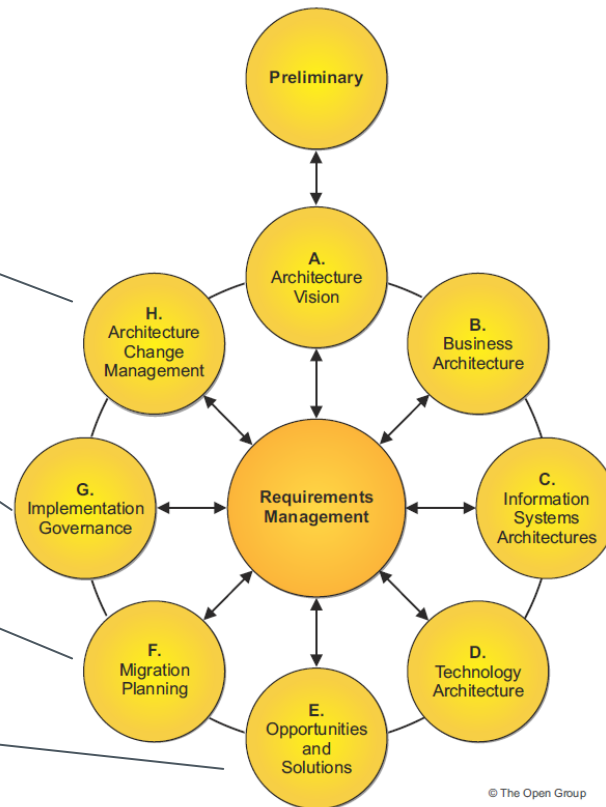
The deliverable is captured & documented in Phase F & G and assessed in H

Assess regarding architecture update vs. new ADM cycle

Create during the implementation and solutions deployment

Create for the Architecture Capability arising from lessons learned

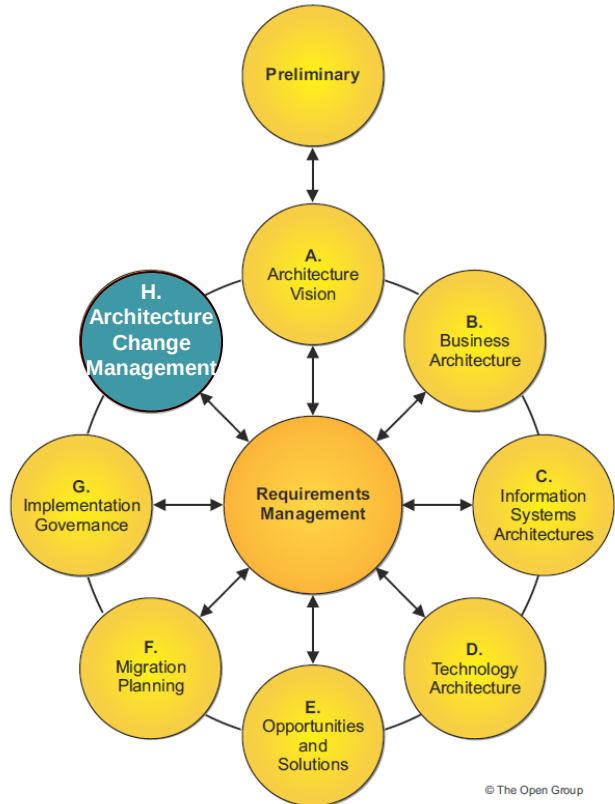
Use as input for existing business programs and projects



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ADM – Phase H: Architecture Change Management – Approach (1/2)

The phase manages changes to the architecture in a cohesive & architected way

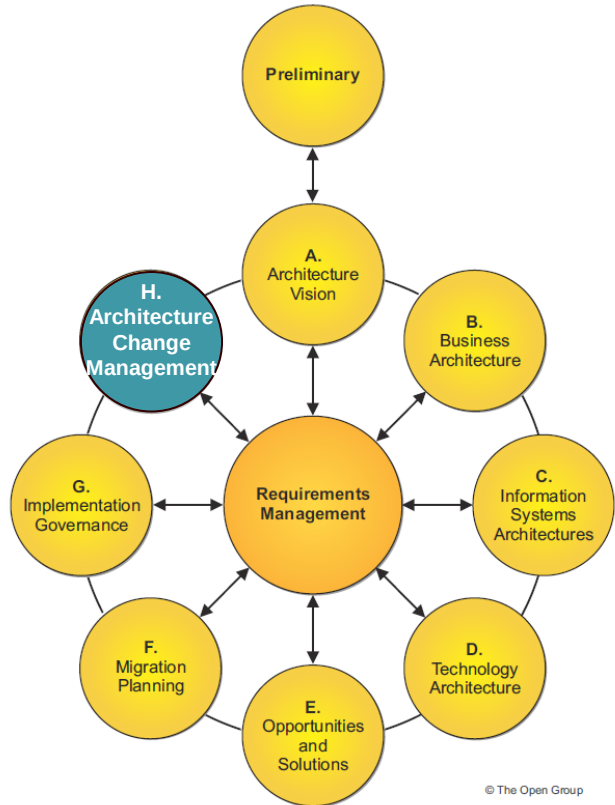


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- The change management process determines circumstances under which...
 - ... the enterprise architecture, or parts of it, will be permitted to **change after implementation** and the **process** by which that will happen
 - ...the enterprise **architecture development cycle** will be **initiated again** to develop a new architecture
- **Establish criteria** to judge (based on **business value**) whether a Change Request...
 - ...warrants an **architecture update**
 - ...warrants **starting a new cycle of the ADM**

ADM – Phase H: Architecture Change Management – Approach (2/2)

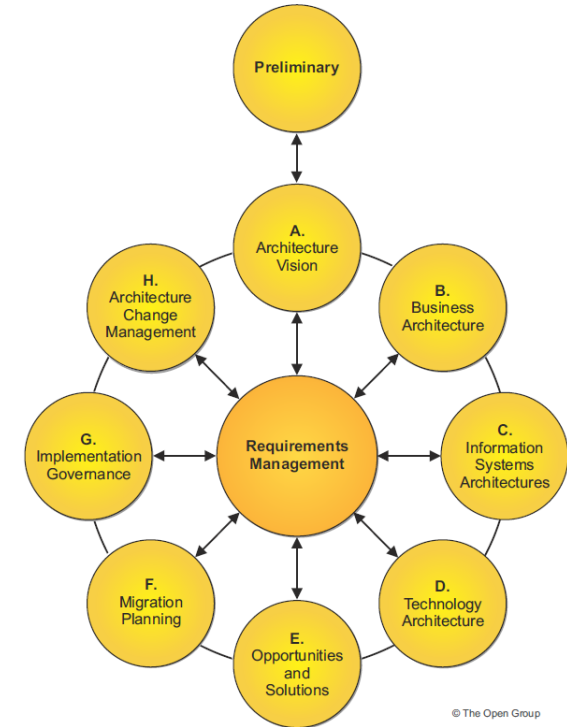
The phase is closely related to Architecture Governance & Architecture Contract



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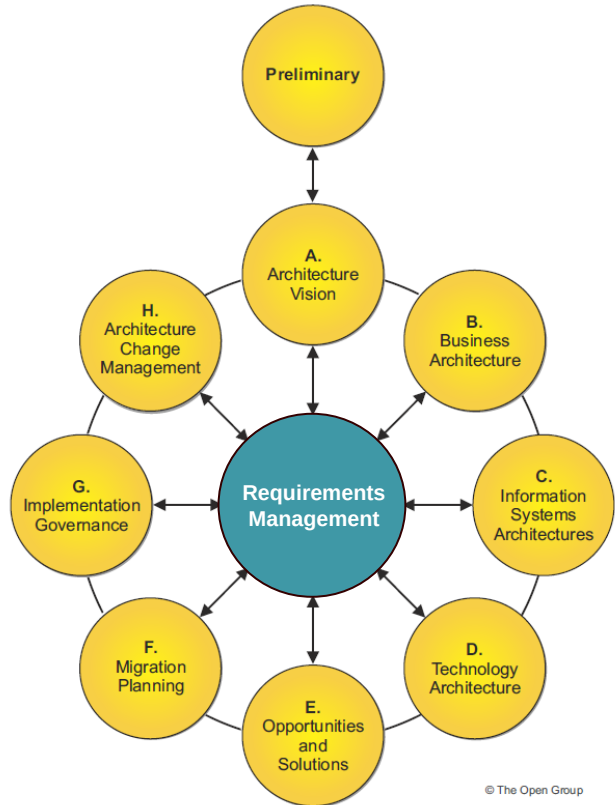
- **Content** of a Change Request:
 - **Description** of the proposed change
 - **Rationale** for the proposed change
 - **Impact assessment** of the proposed change
- **Change classification:**
 - **Simplification change** (handled via change management techniques)
 - **Incremental change** (may require partial re-architecting)
 - **Re-architecting change** (requires a new architecture development cycle)

9	Phase E: Opportunities and Solutions
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ADM – Requirements Management Phase – Purpose

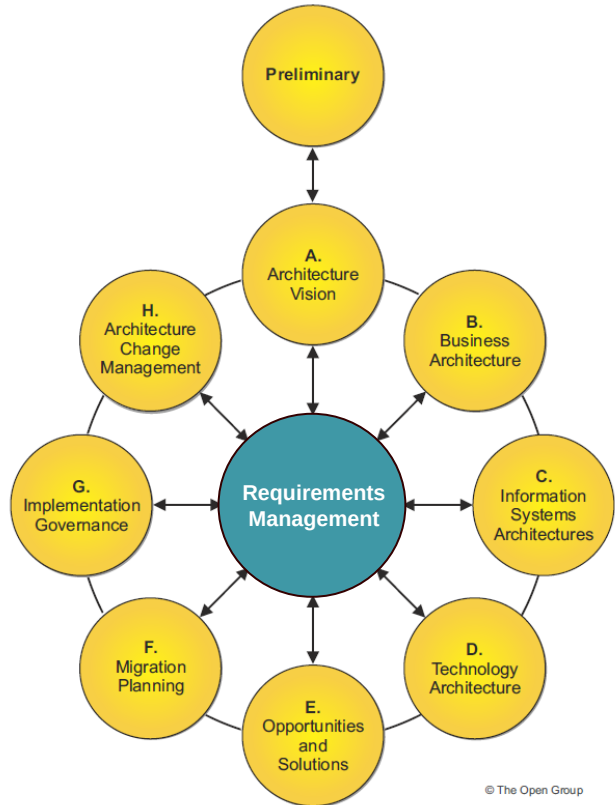
The phase documents, evaluates and distributes architecture requirements



- **Manage architecture requirements** throughout the ADM
- **Operate the requirements management process**
- **Ensure effective requirements management**
- **Provide traceability** from the organization's vision, mission, business model and strategies to the requirements
- **Maintain the complete set of requirements** (incl. stakeholder preferences)

ADM – Requirements Management Phase – Objectives

The phase stores requirements and feeds them in and out of the ADM phases



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- 1. Ensure** that the **Requirements Management process** is **sustained and operates** for all ADM phases
- 2. Manage architecture requirements** identified during any execution of the ADM cycle or a phase
- 3. Ensure** that relevant **architecture requirements are available for use** by each phase as the phase is executed

ADM – Requirements Management Phase – Steps (1/4)

Architecture requirements are available for use in all ADM Phases



#	Requirements Management steps	ADM Phases steps
1		▪ Identify requirements by analyzing business goals & scenarios, management information, value streams & user experience ▪ Document requirements in the Architecture Requirements Specification & Requirements Repository
2	▪ Establish baseline requirements with approved stakeholder priorities	
3	▪ Monitor baseline requirements	

ADM – Requirements Management Phase – Steps (2/4)

Architecture requirements are available for use in all ADM Phases



#	Requirements Management steps	ADM Phases steps
4		Identify new and changed requirements: ▪ Add new requirements and modify or remove existing requirements ▪ Re-assess requirements' priorities
5	▪ Identify changed requirements ▪ Record new requirements' priorities incl. decisions ▪ Solve conflicts among requirements ▪ Generate Requirements Impact Statement (version n)	

ADM – Requirements Management Phase – Steps (3/4)

Architecture requirements are available for use in all ADM Phases



#	Requirements Management steps	ADM Phases steps
6		- Assess impact of changed requirements on previous & current ADM Phase(s) - Decide to implement changed requirements in current/later ADM cycle - Generate Requirements Impact Statement (version n+1)
7		Implement requirements arising from Phase H: Architecture Change Mgmt.
8	Update the Architecture Requirements Repository (incl. changes, affected stakeholder views)	

ADM – Requirements Management Phase – Steps (4/4)

Architecture requirements are available for use in all ADM Phases



#	Requirements Management steps	ADM Phases steps
9		▪ Implement changes
10		▪ Conduct Gap Analysis for past phases ▪ Identify gap requirements (i.e., block being eliminated by accident, causing a change to the Target Architecture) ▪ Record gap requirements in Architecture Requirement Repository ▪ Revise Target Architecture

ADM – Requirements Management – Requirements Impact Assessment

The impact of changed requirements is assessed in the ADM Phases



#	Requirements Management steps	ADM Phases steps
6		- Assess impact of changed requirements on previous & current ADM Phase(s) - Decide to implement changed requirements in current/later ADM cycle - Generate Requirements Impact Statement (version n+1)
7		Implement requirements arising from Phase H: Architecture Change Mgmt.
8	Update the Architecture Requirements Repository (incl. changes, affected stakeholder views)	

Requirements Impact Assessment – Definition

The approach considers new facts that may invalidate parts of the architecture

Purpose

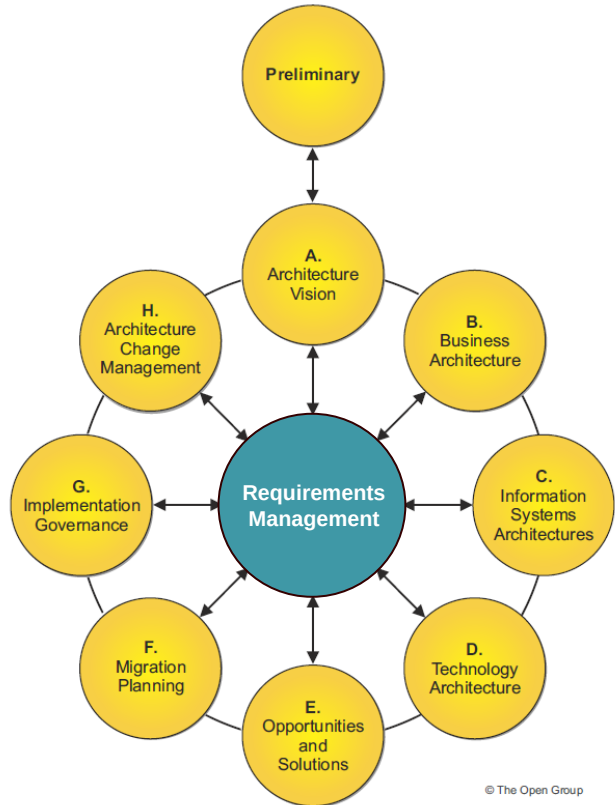
- Assesses the **current architecture requirements and specification** to identify **changes** that should be made as well as their **implications** to the architecture

Content

- Phases** to be **revisited** and to lead on requirements prioritization including **results**
- Investigation results** and **recommendations on management of requirements**
- References** to specific **requirements** and **stakeholder priority**
- Repository reference numbers**

ADM – Requirements Management – Approach (1/2)

The ADM is continuously driven by the Requirements Management process

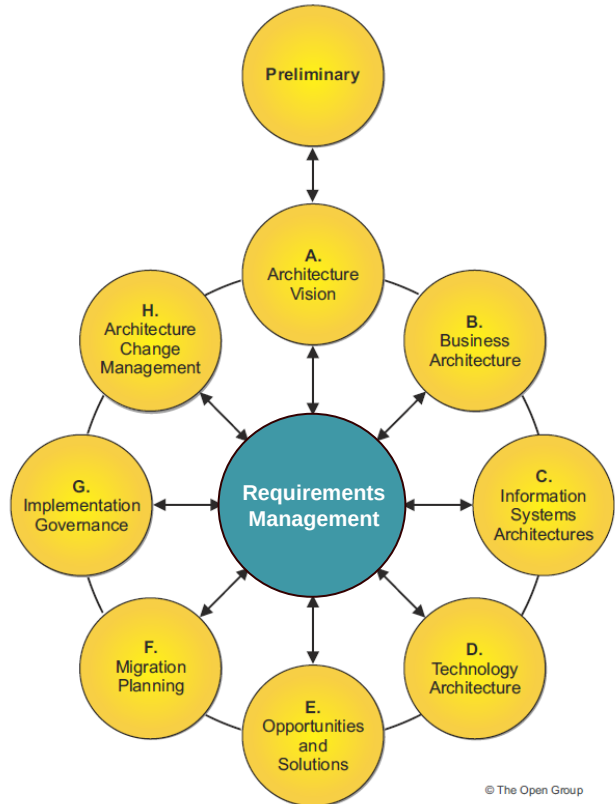


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- Dynamic process to **identify and store requirements** for Enterprise Architecture (**and their changes**)
- The Requirements Management Phase **feeds requirements into and out of the relevant ADM phases and between ADM cycles**
- The relevant **ADM-phases disposes of, addresses, or prioritizes requirements**, not the Requirements Management process

ADM – Requirements Management – Approach (2/2)

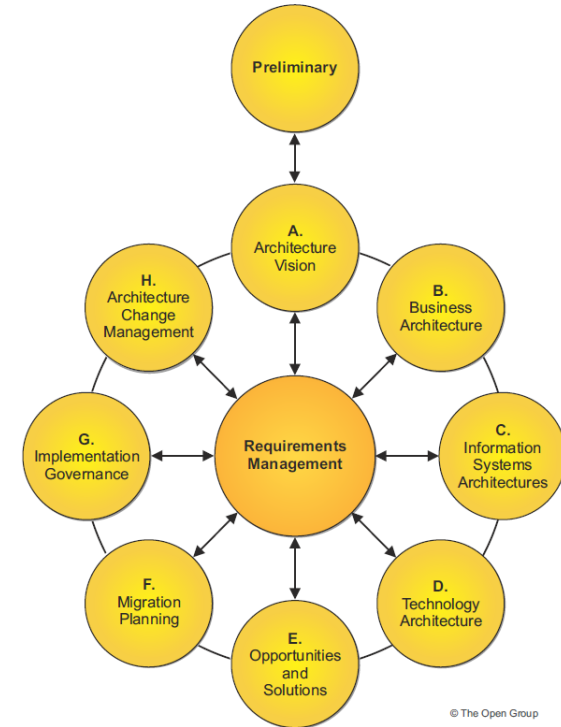
The ADM is continuously driven by the Requirements Management process



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- **Use an Architecture Requirements Repository** to record and manage all architecture requirements (recommendation)
- The **Architecture Requirements Repository can hold information from multiple ADM cycles** (unlike the Architecture Requirements Specification and the Requirements Impact Assessment)

9	Phase E: Opportunities and Solutions
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ADM Iterations – Iterative approach

The Architecture Development Method can be executed in an iterative way

The ADM is iterative...

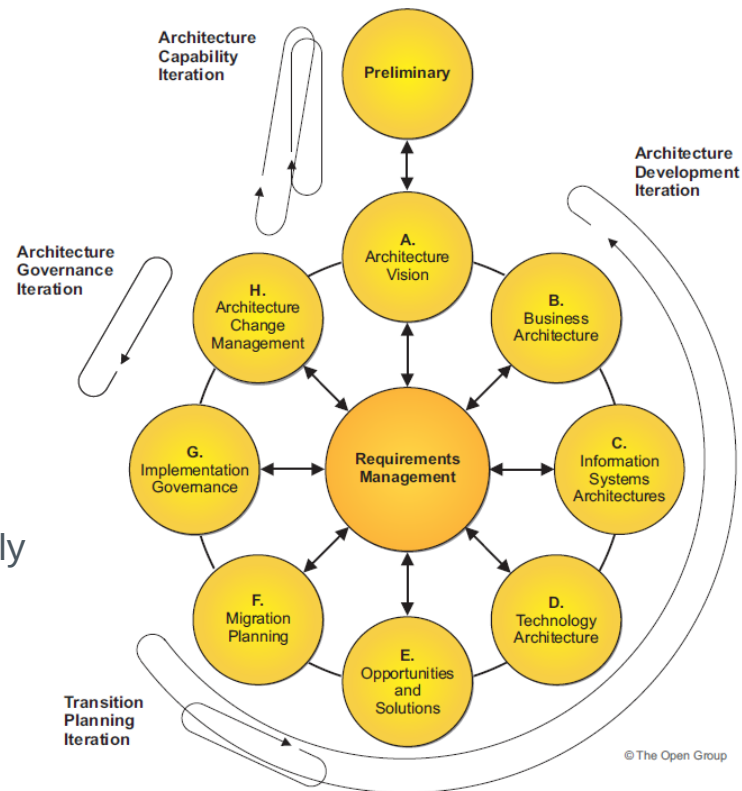
- ...over the whole process,
- ...between phases and
- ...within phases.

Each iteration = new decision:

- Breadth of coverage of the enterprise
- Level of detail
- Aimed time period
- Used architectural assets created previously or externally

Decision based on:

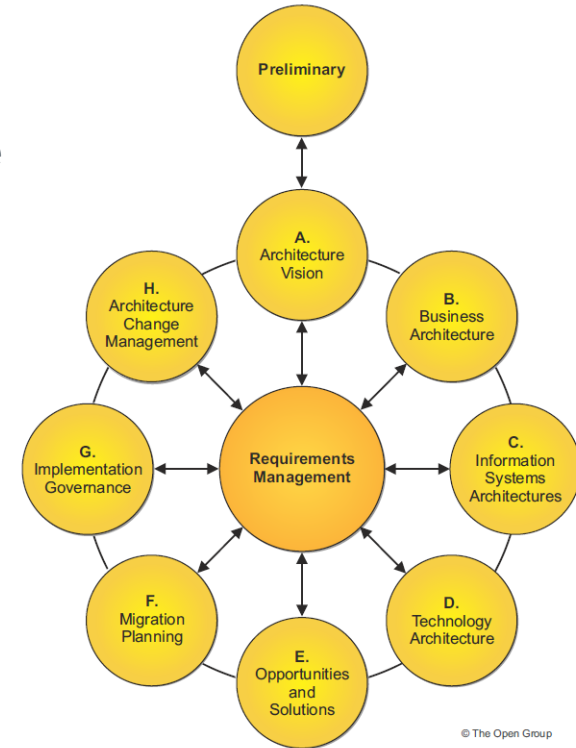
- Availability of resources and competence
- Value to the enterprise from the chosen scope



ADM Iterations – Practice (1/2)

Each Architecture Project starts with Phase A and uses an individual ADM path

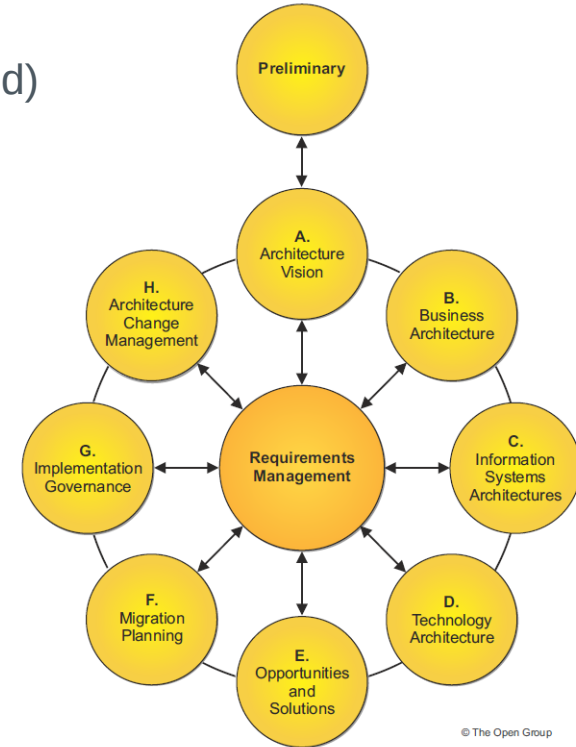
- Need for iterations in the development of an Architecture:
 - High complexity (**manage complexity**)
 - Inter-dependent nature (consider the entire architecture existing gaps and resulting work **simultaneously**)
- Each **Architecture Project**
 - has a **different purpose**
 - traverses a **distinct path**
 - uses a different **configuration of the TOGAF ADM**
 - starts with **Phase A: Architecture Vision** (to initiate a project)



ADM Iterations – Practice (2/2)

Iterations are driven by the information needs of the Architecture Project

- Architecture Projects do not need to enter all phases (if information is not needed or already has been produced)
- **ADM Phases and their steps** can be **executed simultaneously** and repeated at the necessary level of detail
- **Multiple ADM phases may stay open** to produce the information in the required breadth, depth, time and recency



ADM Iterations – Iteration Cycles

The TOGAF Standard suggests iteration cycles to achieve specific purposes

1. Iteration through the entire ADM

Develops a comprehensive Architecture Landscape

2. Architecture Development Iteration

Creates architecture content and ensures the architecture is considered as a whole

3. Architecture Capability Iteration

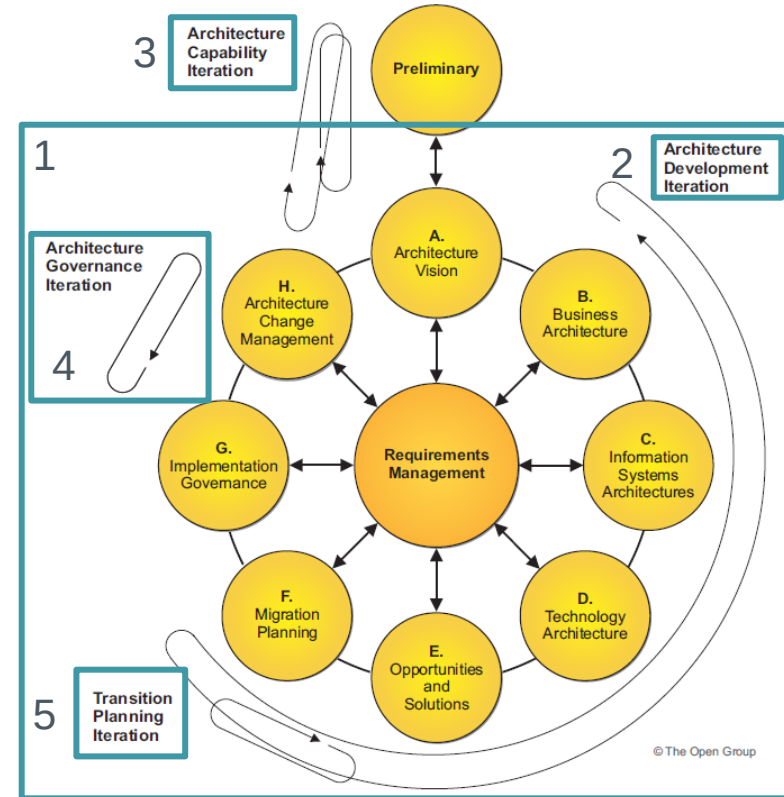
Creates & evolves the Architecture Capability (approach, principle, scope, vision, governance)

4. Architecture Governance Iteration

Supports the governance of change activities

5. Transition Planning Iteration

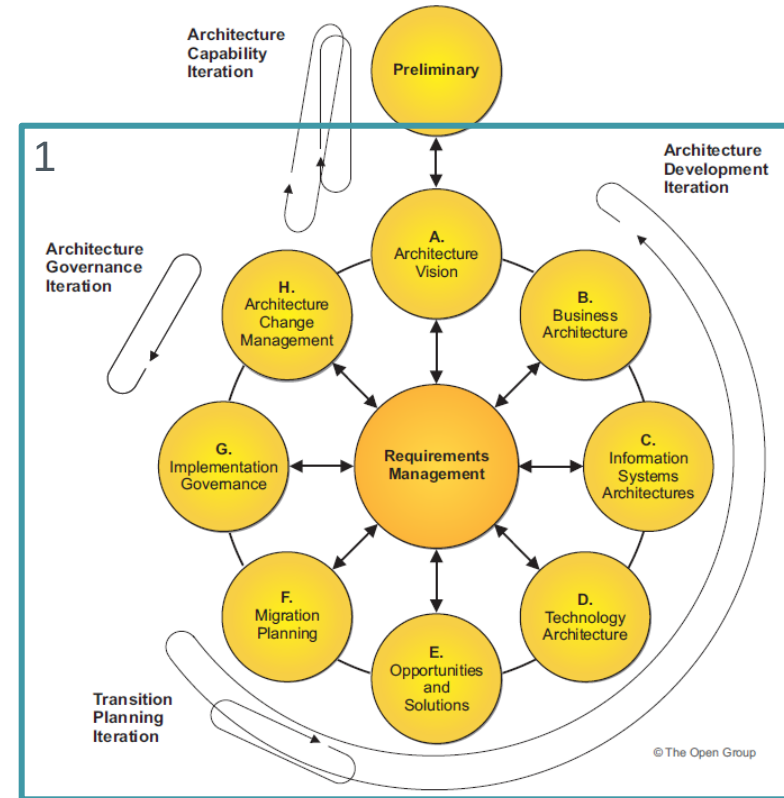
Creates change roadmaps for a defined architecture



ADM Iterations – Iteration through the entire ADM (1/2)

The iteration type develops a comprehensive Enterprise Architecture Landscape

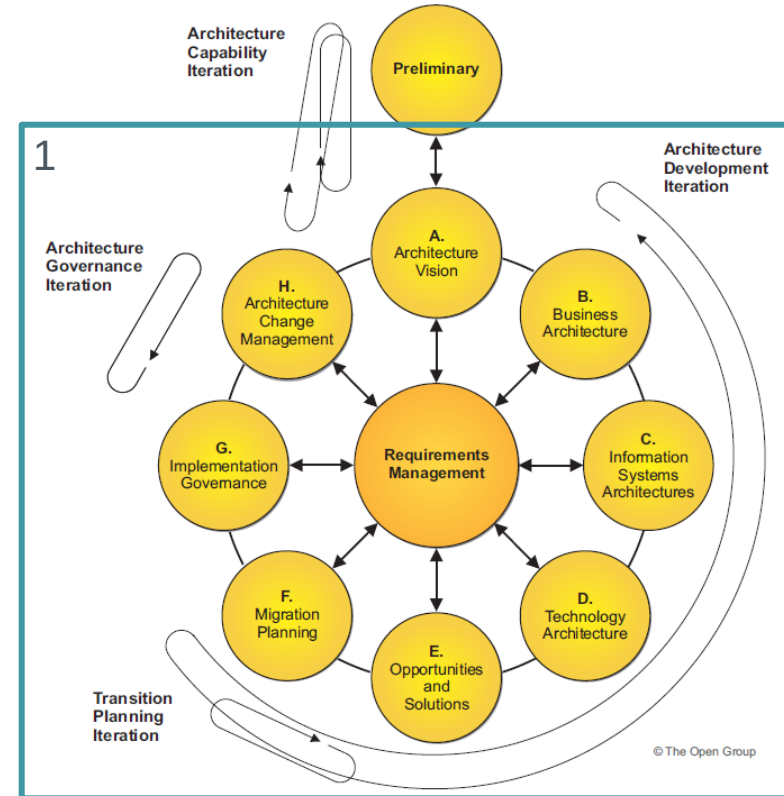
- **Projects** go through the **entire ADM cycle** (starting with Phase A and a Request for Architecture Work)
- Each **project / cycle** is **bound** by a **Request for Architecture Work**
- **Output** of each project / cycle **changes or extends** the **Architecture Landscape**



ADM Iterations – Iteration through the entire ADM (2/2)

The iteration type develops a comprehensive Enterprise Architecture Landscape

- **Separate projects** may **operate** their **own ADM cycles concurrently** with relationships to each other (parallel iteration)
- **A project** may **trigger** another **project** (sequential iteration), if...
 - ...the project identifies opportunities or solutions **requiring a detailed architecture**
 - or
 - ...the project identifies EA landscape **impacts outside the Request for Architecture Work**

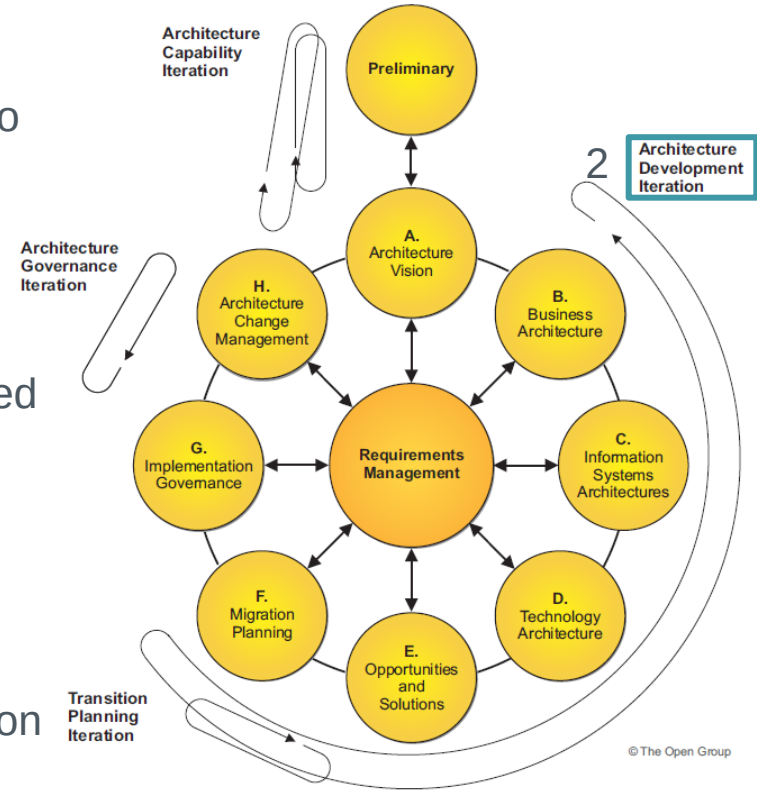


ADM Iterations – Architecture Development Iteration (1/2)

The iteration type is an iteration within an ADM cycle

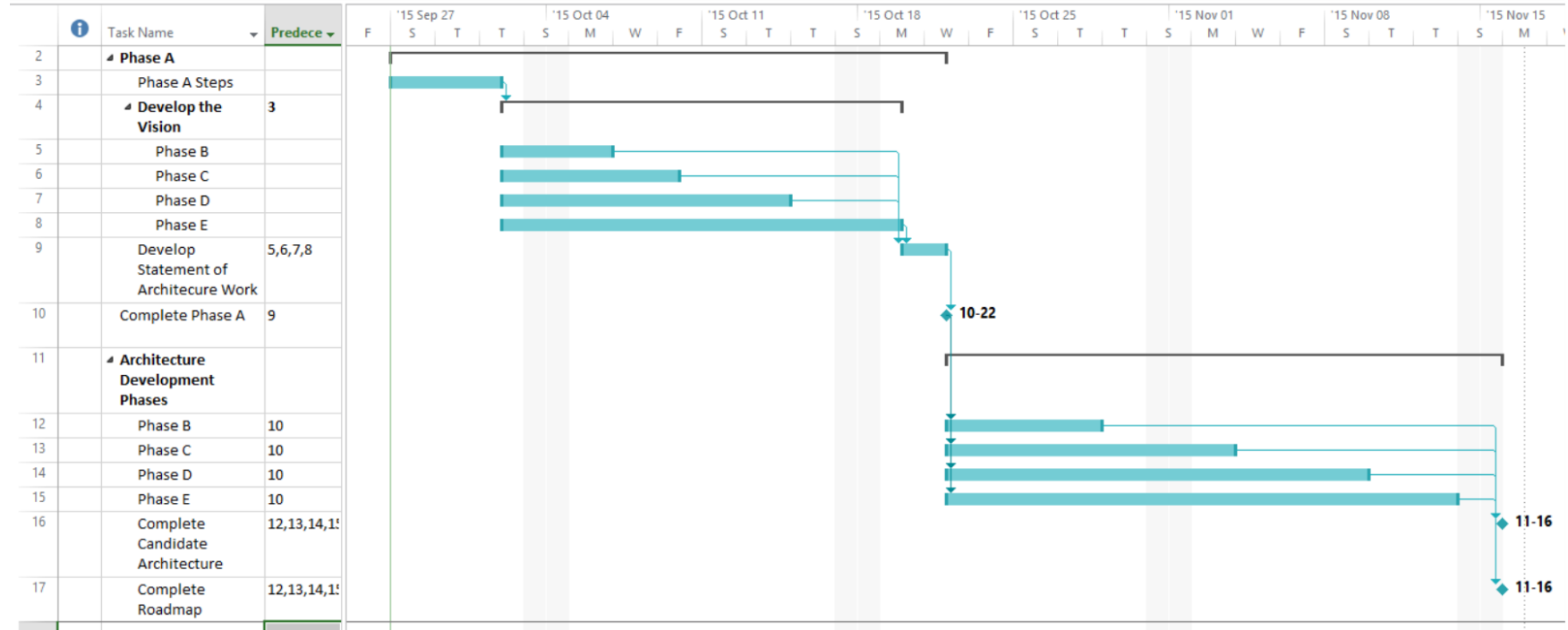
Projects may...

- **...operate multiple ADM phases concurrently** (to manage the inter-relationship between BDAT-Architectures)
- **...cycle between ADM phases** (to cover multiple phases and converge on a detailed Target Architecture, when higher-level architecture does not exist)
- **...return to previous phases** (to update work products, to converge on an Architecture Roadmap or Implementation & Migration Plan, when requirements change)



ADM Iterations – Architecture Development Iteration (2/2)

The ADM can be visualized with a process-oriented view

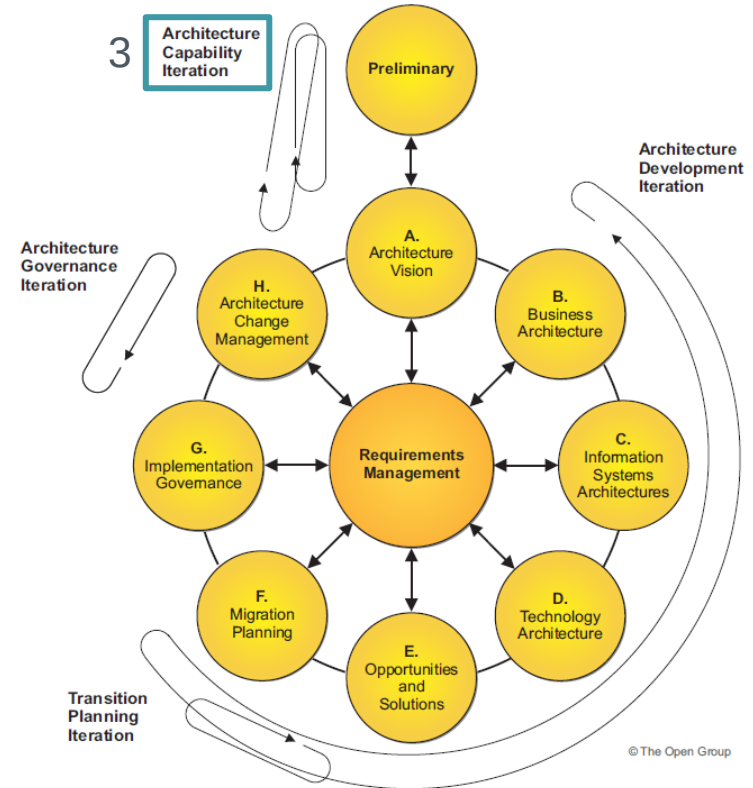


ADM Iterations – Architecture Capability Iteration

The iteration type creates & evolves the Architecture Capability

Projects may require a **new iteration of the Preliminary Phase** to...

- ...(re-)establish aspects of the **Architecture Capability** identified in **Phase A** addressing a **Request for Architecture Work**
- ...adjust the **Architecture Capability** based on new or changed requirements because of a **Change Request in Phase H**.

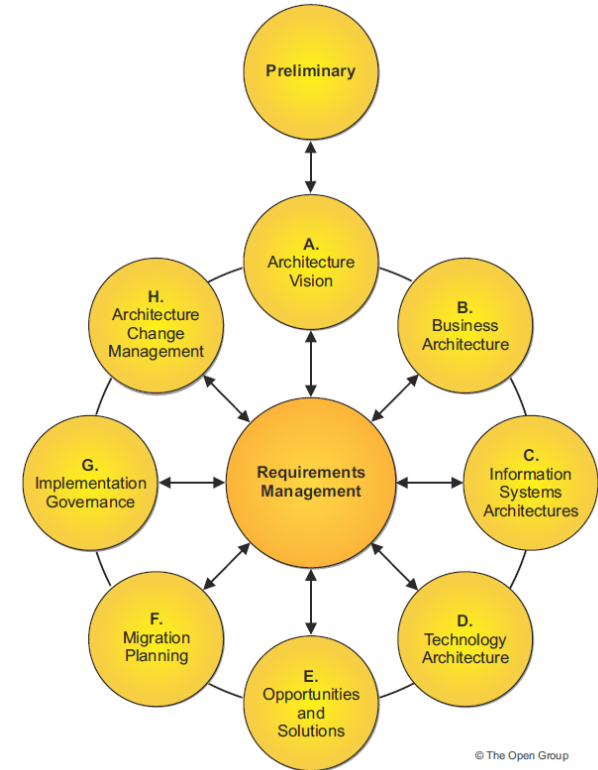


The TOGAF Architecture Development Method (ADM)...

- ...is a **logical method for developing knowledge**
- ...places key activity steps together for **understanding relationships of activities and information flows**

The TOGAF crop-circle diagram...

- is a stylized path that **demonstrates the essential information flow**
- should **not** be understood as a linear processes model
- is **not** a representation of activity sequence



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ADM – Information flow between the ADM phases (2/2)

The ADM is not a linear waterfall process

The TOGAF Standard...

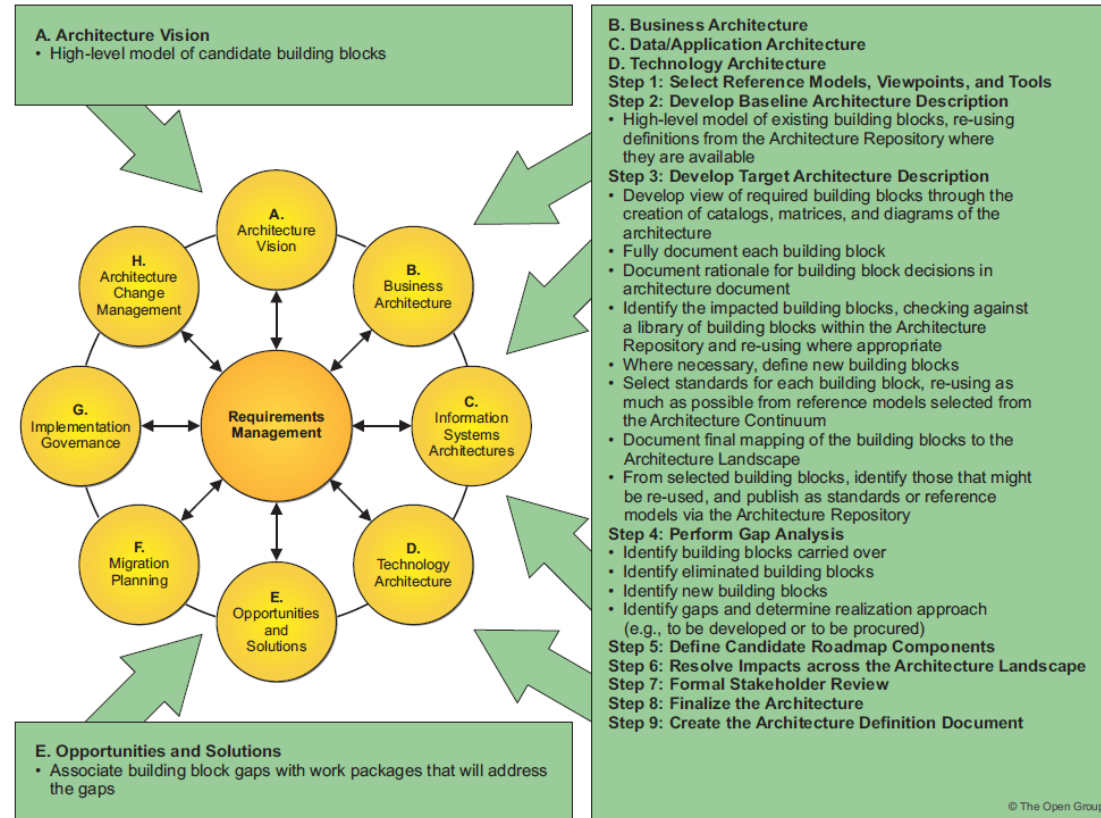
- ...provides an **ADM phase for each essential output** that is necessary to develop an architecture
- ...suggests **what input is necessary at each phase** to develop the required output

Information flow between different ADM phases:

- In every phase of the ADM you **consume inputs to generate outputs**
- The **output** of one phase **can be the input** of a later phase
- The **output** of one phase **can be updated or detailed** in a later phases
- The **work plan** of an Enterprise Architecture team **varies** depending on the problem and development task, that is defined in the Request for Architecture Work
-> **Each request requires a distinct path of the TOGAF ADM and different outputs**

Building Block Specification Process – ADM Phases

ABBs are defined in the phases A, B, C & D and shaped as SBBs in Phase E



Evolutionary and iterative process based on required functions, imposed constraints and available products

1. Identify **Architecture Building Blocks (ABBs)** required to meet business goals and objectives
2. Refine selected ABBs to **Solution Building Blocks (SBBs)** bought from the shelf or custom developed

Architecture Alternatives & Trade-offs – Definition

Multiple Target Architectures could suit the vision, principles and requirements

Sometimes **not a single architecture exists** that will meet all stakeholders' concerns

- Identify **alternative Target Architectures**
- Build an **understanding** of different possibilities
- Identify **trade-offs** between the alternatives (competing forces)
- Extract **hidden agendas, principles and requirements** by presenting the alternatives to the stakeholder

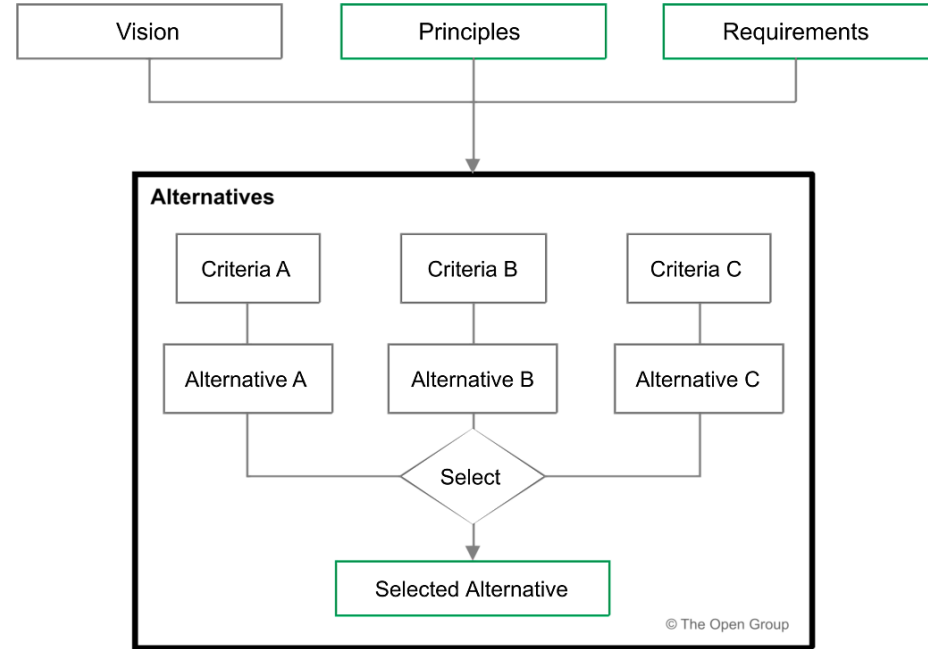


Image source: [Pexels.com](https://www.pexels.com/) / Pavel Danilyuk

Architecture Alternatives & Trade-offs – Technique

The technique investigates Architecture Alternatives

1. Use the vision, principles, requirements and other information to **select sets of criteria** fitting for different alternatives
2. **Define alternatives** based on the criteria and build an understanding of each
3. **Select** one of the **alternatives or combine features** from multiple alternatives to create the proposed target architecture



Architecture Trade-Off Method

Digital Enterprise – Definition

The enterprise type creates, delivers and obtains products and services digital

Definition

Digital Enterprise*

An enterprise characterized by:

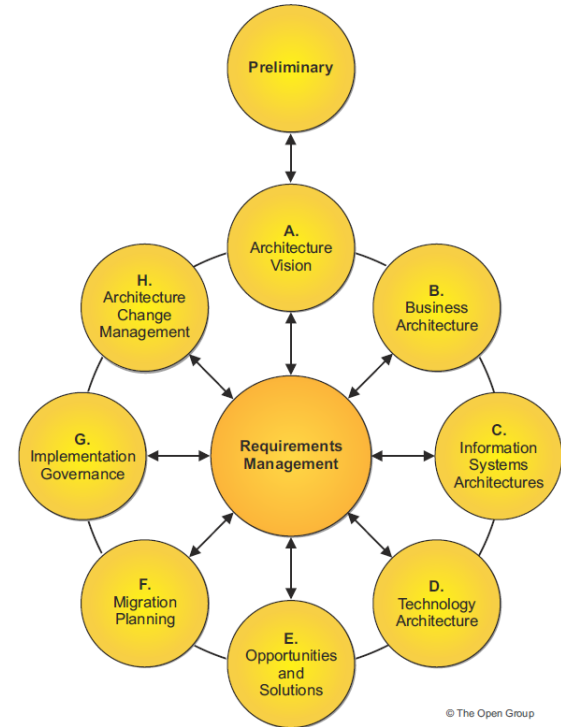
1. The **creation of digitalized products or services** that are **delivered fully digitally** (e.g., digital media or online banking)
2. Where **physical products and services** are **obtained** by the customer by **digital** means (e.g., online car-sharing services)

*based on Digital Practitioner Body of Knowledge (DPBoK)

Enterprise Architecture supports agile environments in **delivering and enhancing digital products and services quicker and easier** by...

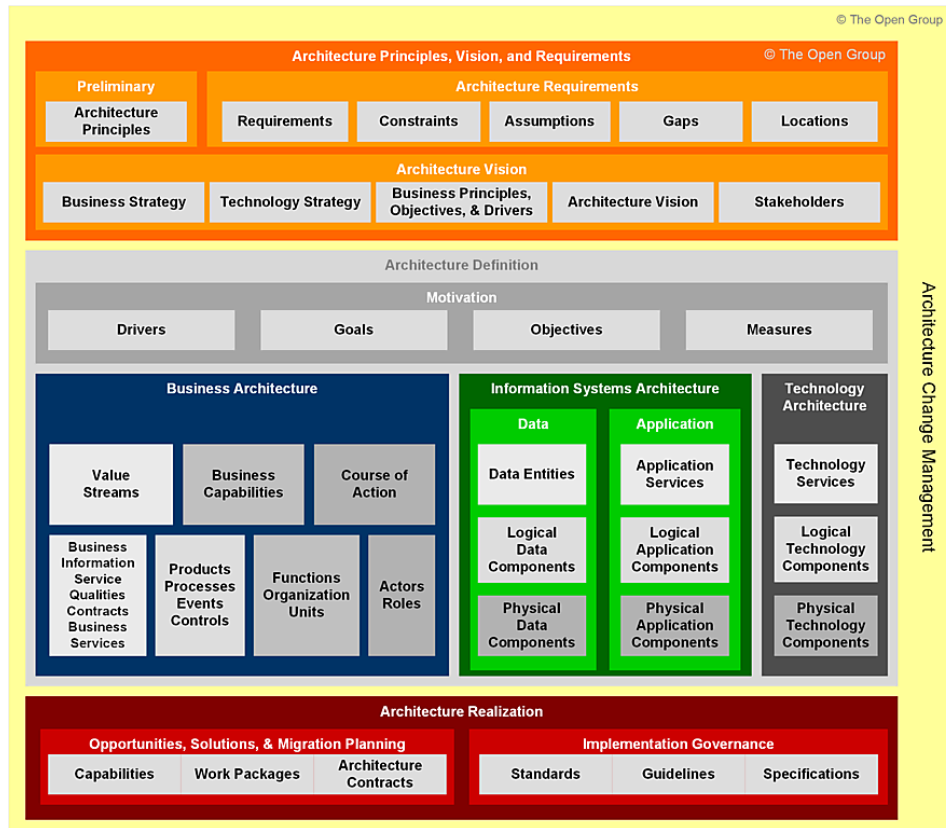
1. ... **managing technical debt** (reactively and proactively)
2. ...**identifying standards** and reusable standard **components**
3. ...providing governance to **oversee the reuse of components**
4. ...**simplifying complexity**
5. ...establishing an Enterprise Capability that drives **operational excellence**
6. ...**institutionalizing agile development methods**

9	Phase E: Opportunities and Solutions
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(Architecture) Content Framework – Introduction (1/2)

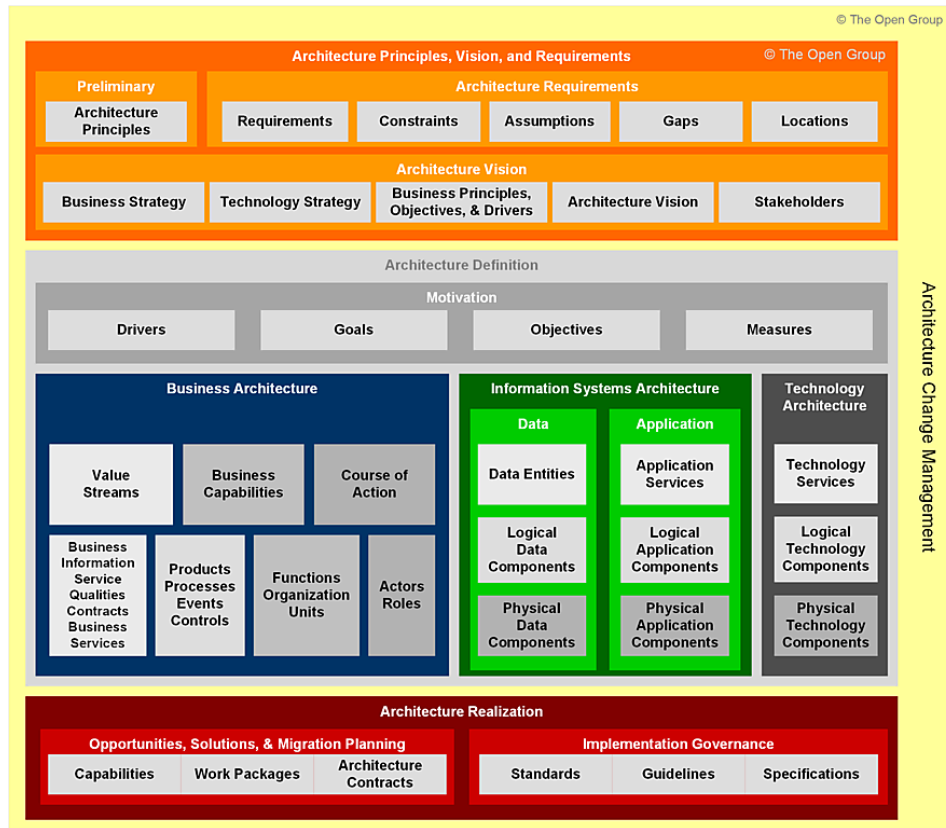
The concept is a categorization framework for Architecture Content



- Structures the **Architecture Description**, work products, artifacts & models
- Defines artifacts and work products to be developed and to describe the Architecture
- Top-level structure mirrors the **ADM phases**
- Provides an underlying structure and **companion** for the **ADM**

(Architecture) Content Framework – Introduction (2/2)

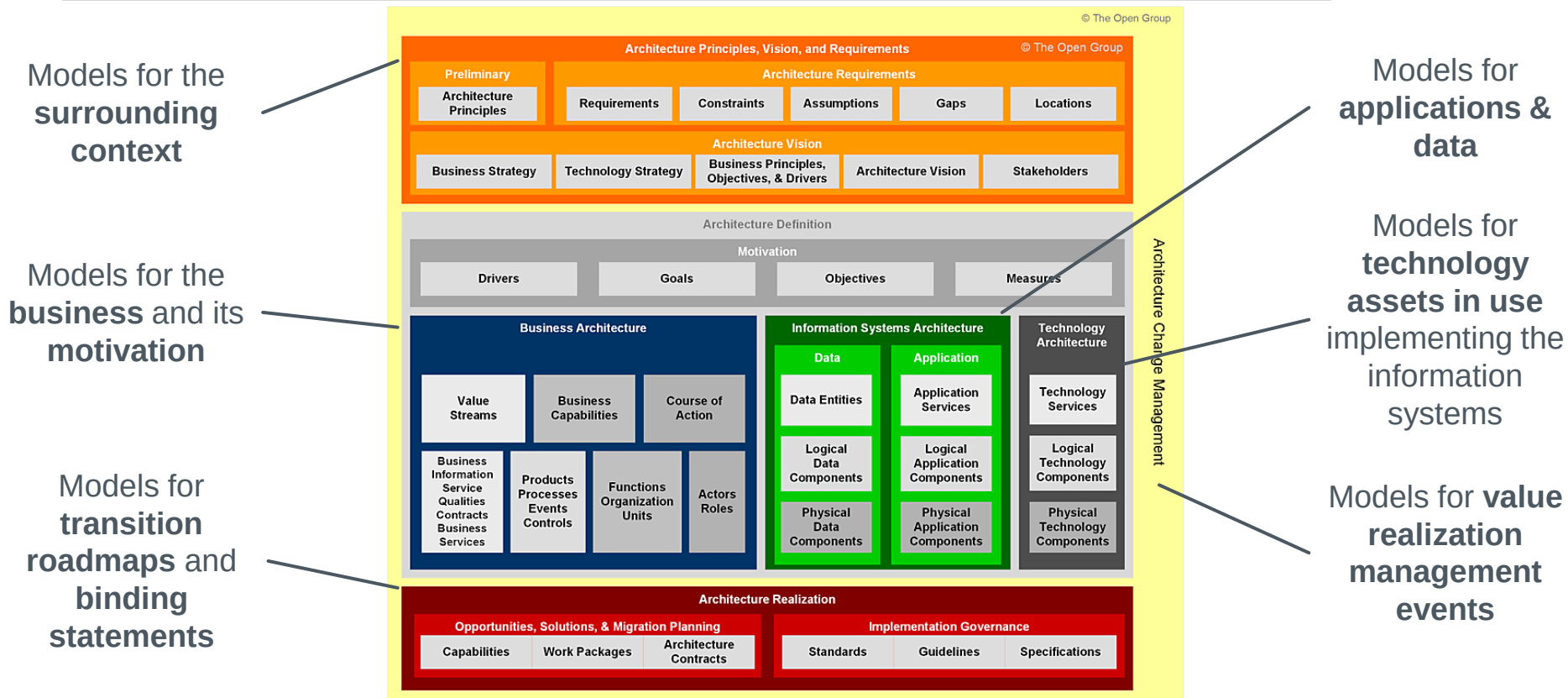
The concept is a categorization framework for Architecture Content



- Puts **each deliverable** into the **context** of the holistic architecture view of the enterprise
- **Multiple Frameworks** are **available** (e.g., TOGAF Content Framework, Zachman Framework, DoDAF, NAF)
- **Framework decision** is **influenced** by the **Architecture Framework** and used **EA software tool**
- Usually **adapted** to fit **organization needs**

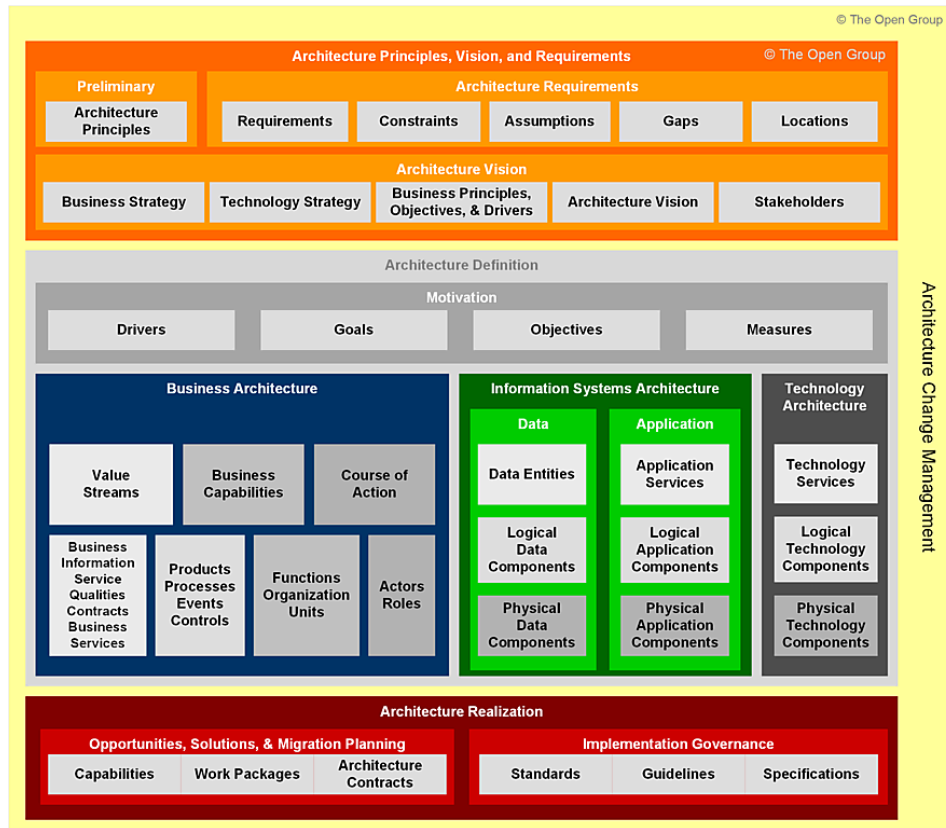
(Architecture) Content Framework – Explanation

The framework is part of the Enterprise-Specific Architecture Framework



(Architecture) Content Framework – Benefits

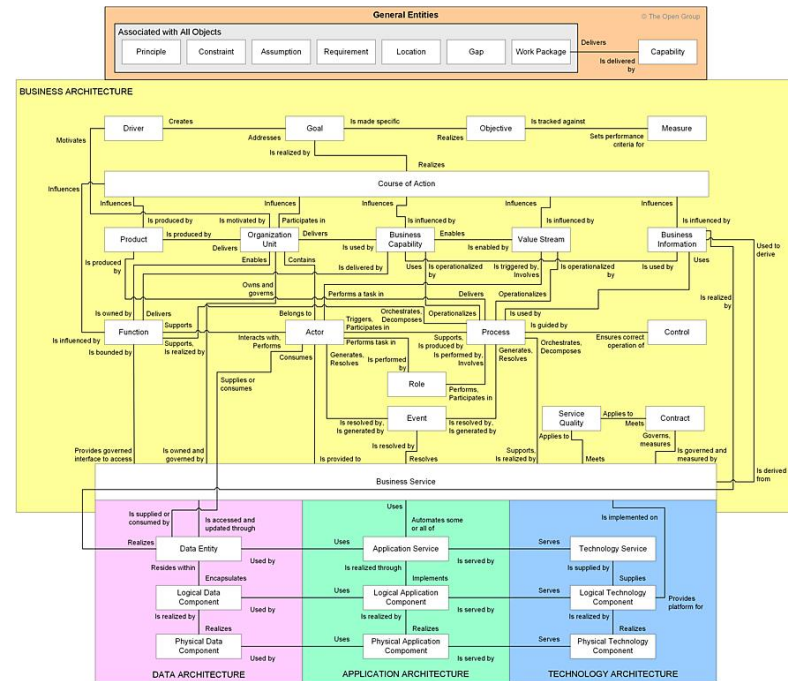
Describes what the architecture should look like once it is done



- **Detailed model** of architectural work products
- **Enhances consistency** in the outputs
- **Comprehensive checklist** of architecture output
- **Reduces the risk of gaps**
- **Defines standards** (i.e., architecture concepts, terms and deliverables)

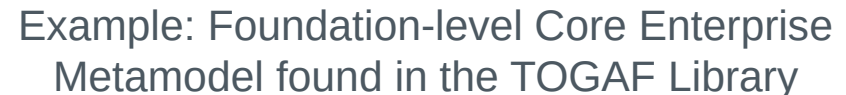
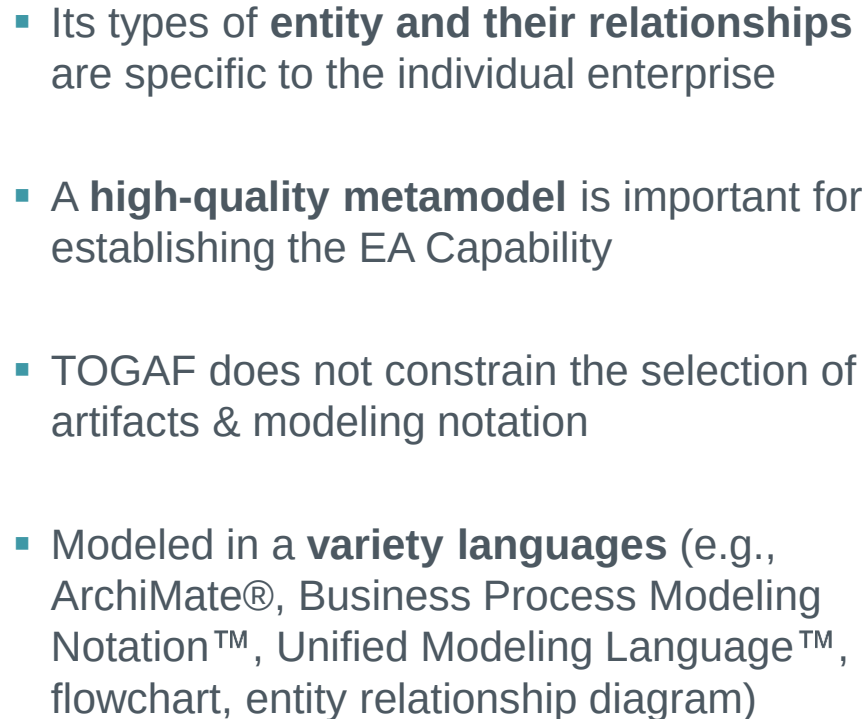
The model defines entities within the enterprise & the relationships between them

- **Starter pack for architects with types of things** to investigate & cover in models
- **Provides context for the artifacts** suggested in the ADM
- **Completeness-check** for architecture modeling languages and metamodels
- Improves **consistency, completeness & traceability**
- **Structures architectural information** to meet stakeholder needs



Example: Foundation-level Core Enterprise Metamodel found in the TOGAF Library

The Enterprise Metamodel is specific to an individual Enterprise

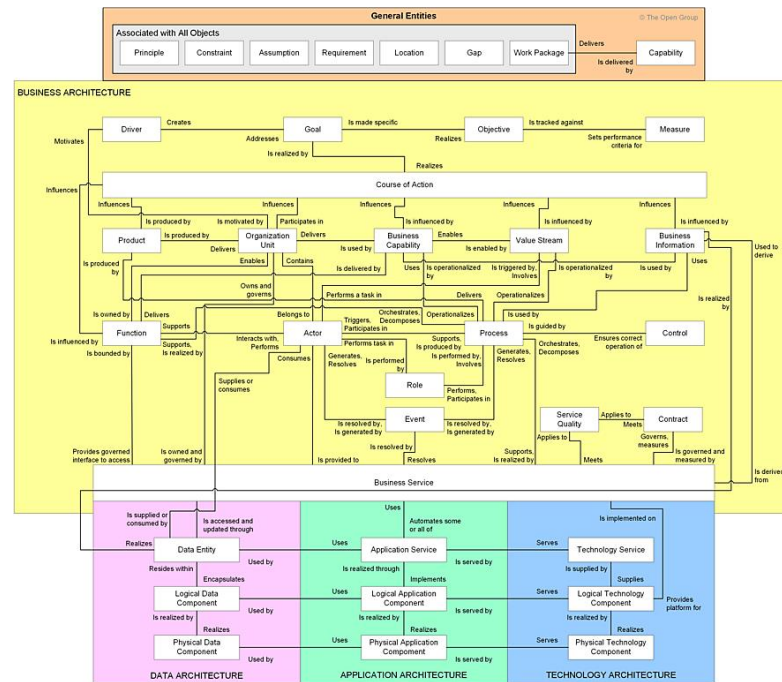


Two approaches to develop an Enterprise Metamodel



- In both cases:

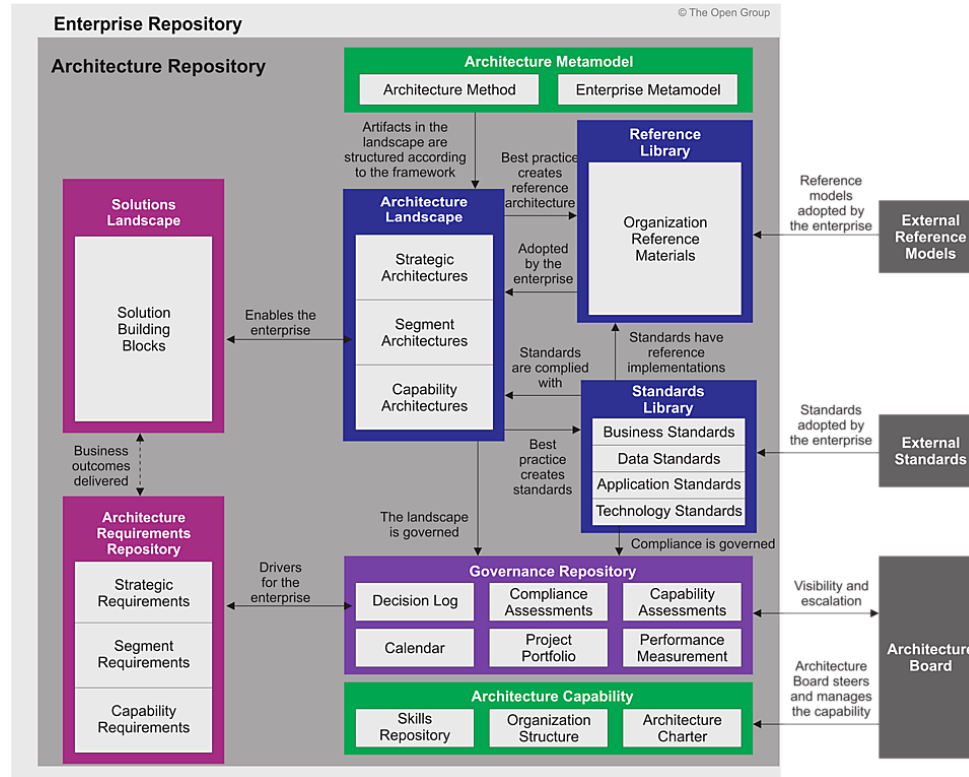
- Minimize the information in the Metamodel according to the purpose of the EA Capability
- Each entity, relationship and their attributes must be maintained and tracked



Example: Foundation-level Core Enterprise Metamodel found in the TOGAF Library

(Enterprise) Architecture Repository – Introduction

The concept stores different classes of architectural output at different levels



- Part of the **Enterprise Repository**
- Often implemented by **modeling** software, **analytic** tools, **file repository** or **document management system**

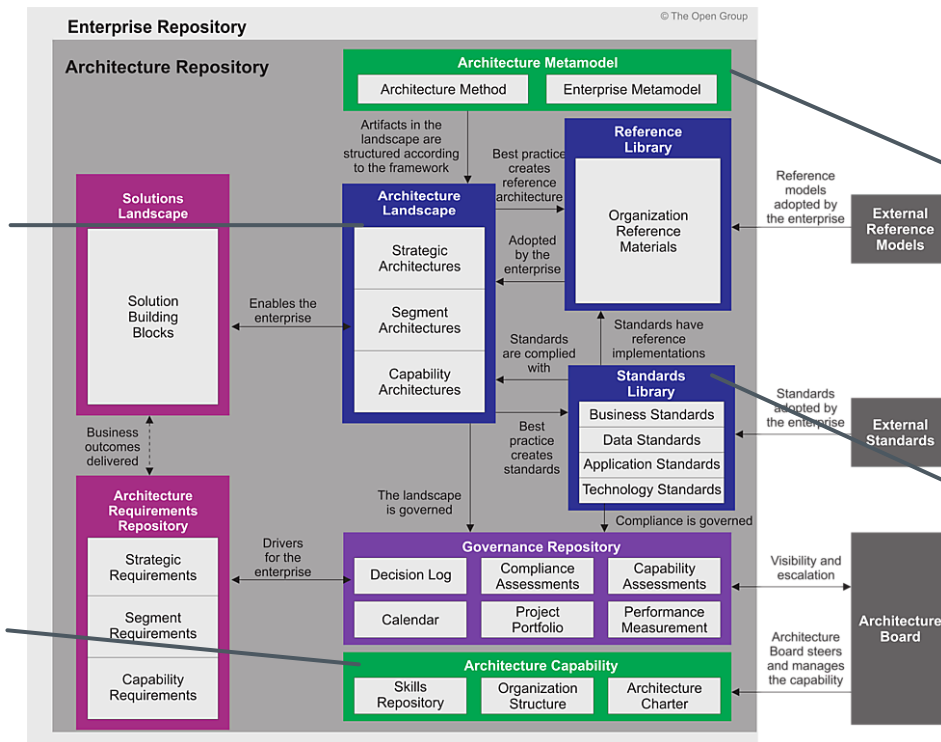
(Enterprise) Architecture Repository – Elements (1/2)

The content facilitates understanding & co-operation of stakeholders



Architectural
**representation of
assets** deployed within
the enterprise at a
particular point in time
(current, target,
transition, candidate)
often at multiple level of
abstractions

**Parameters,
structures and
processes** that support
**governance of the
Architecture
Repository**



Organizationally
tailored **application of
an architecture
framework** including
the Content
Metamodel

Standards with which
new architectures and
implementations must
comply (e.g., industry
standards, selected
products & services
from suppliers, shared
services)

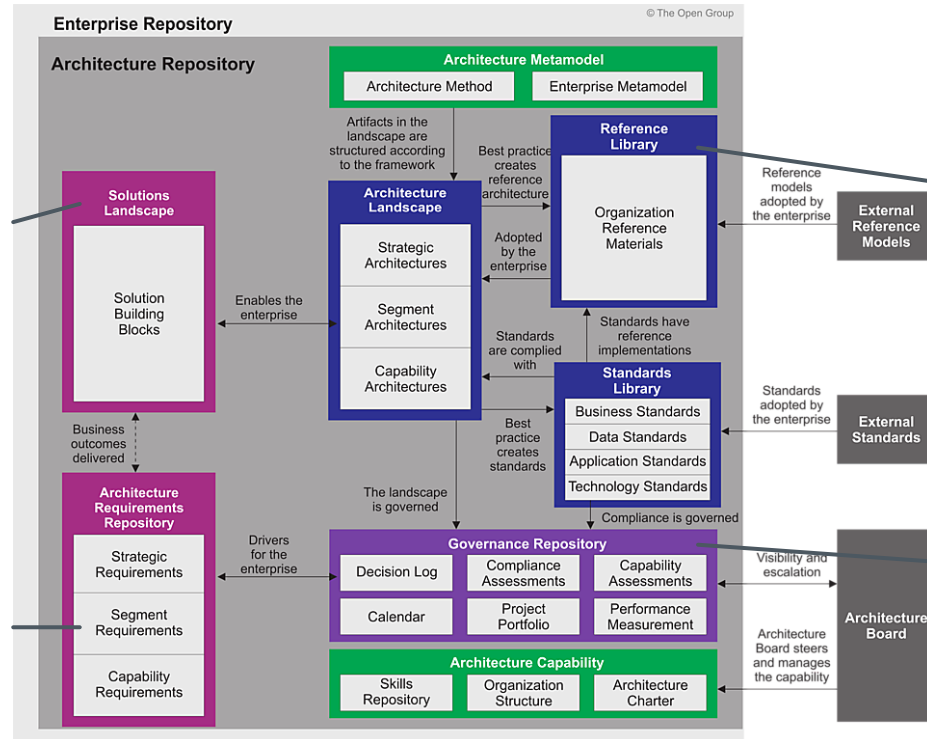
(Enterprise) Architecture Repository – Elements (2/2)

The repository is enriched by the different ADM outputs



Architectural representation of the **Solution Building Blocks (SBBs)** supporting the Architecture Landscape which have been planned or deployed

View of all authorized **architecture requirements** which have been agreed with the Architecture Board

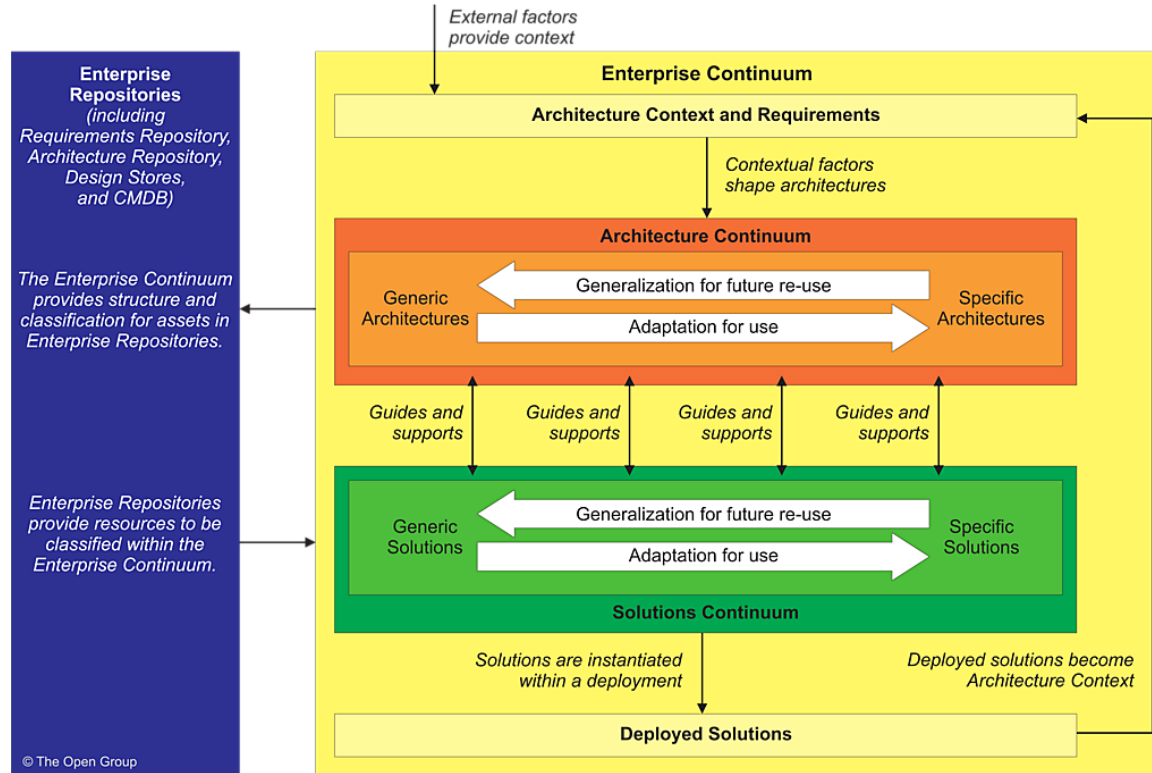


Guidelines, templates, patterns and other forms of reference material to accelerate the creation of new architectures

Record of governance activity across the enterprise

Enterprise Continuum – Introduction & Explanation (1/2)

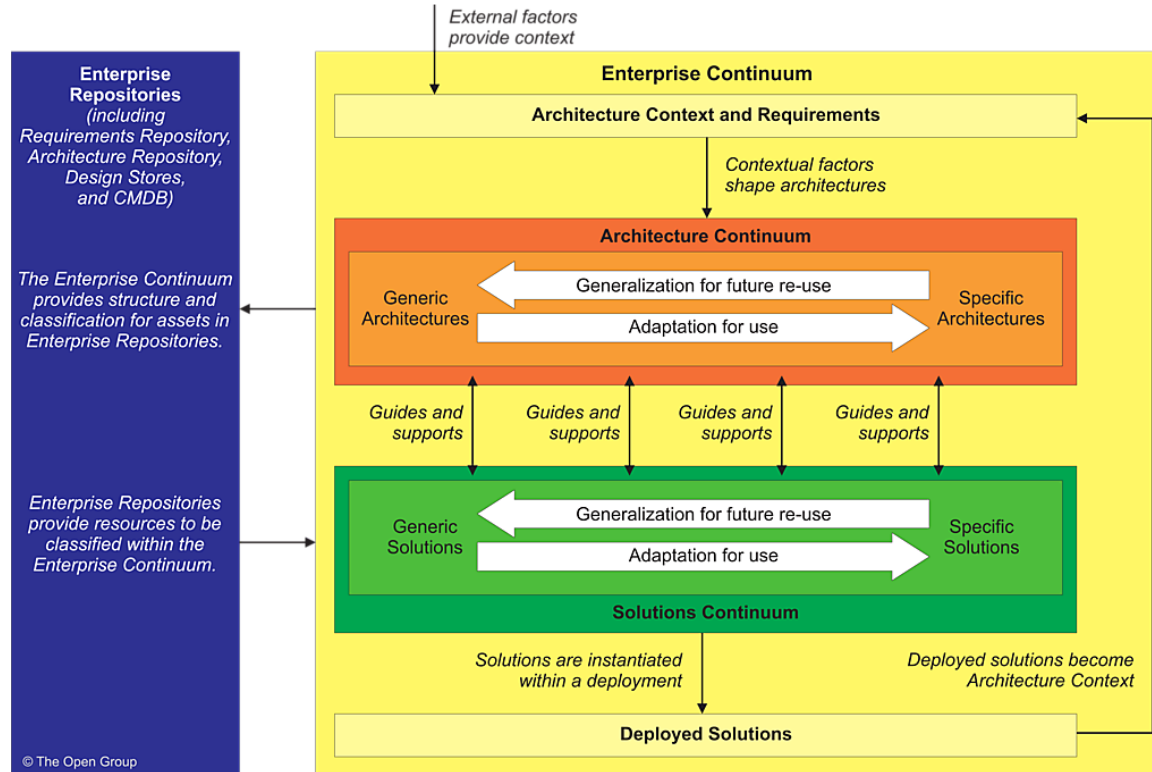
The concept structures & categorizes assets held in Enterprise Repositories



- Sets a **broader context** for **re-using artifacts**
- Explains how **generic solutions** can be leveraged & specialized to support **architecture requirements** of an individual organization
- Comprises the **Architecture & Solution Continuum**

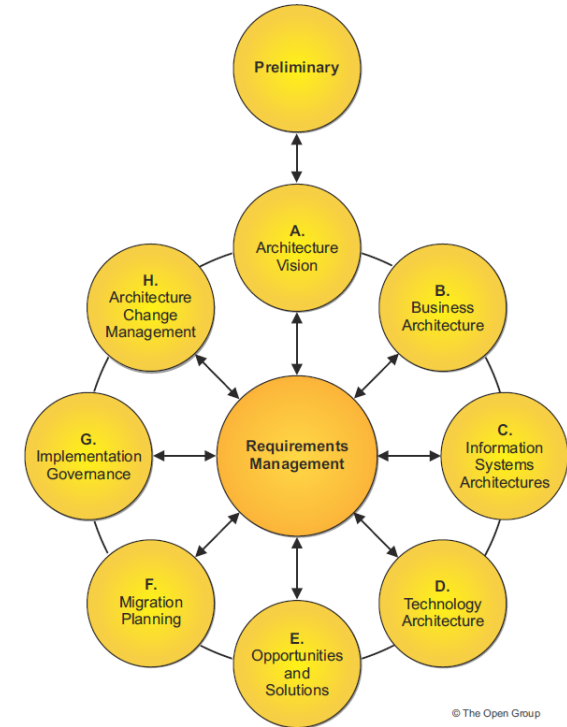
Enterprise Continuum – Introduction & Explanation (2/2)

The concept is a view of the repository of all internal & external architecture assets



- Provides **view** on models, patterns, viewpoints, building blocks and artifacts
- **Internal architecture and solution artifacts:**
Deliverables of previous architecture work, which are available for re-use
- **External architecture and solution artifacts:**
Industry reference models and architecture patterns

9	Phase E: Opportunities and Solutions
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Governance – Definition

A system by which the current and future state are directed and controlled

- Is a committed & hierarchical **decision-making process** with a defined structure of relationships to **direct and control the enterprise** to achieve **stated goals**
 - Accounts for the **equality of concerns and transparency**, protecting the **rights and interests of the business** (and not the ones of narrow-minded interests)
 - Can be used to **drive a set of behaviors** controlled via **measurements and metrics**
-
- A **customized governance approach** should define the Why, What, When and How

Corporate Governance – Definition

A system by which business corporations are directed and controlled



- Ensures that **business is conducted properly** through guidance & usage of resources
- Focuses on the **rights, roles and equitable treatment** of shareholders
- Enacts a **board** to review and guide **corporate strategy** and to set and monitor achievement of **performance objectives**
- Ensures **disclosure and transparency** of the **responsibilities of the board**
- Ensures **strategic guidance, monitoring of management** and **board accountability**

(Enterprise) Architecture Governance – Definition

A system by which (Enterprise) Architectures are directed and controlled



- Is a **practice and orientation** by which **Enterprise Architectures** and other architectures are **managed and controlled** at an **enterprise-wide level**
- Operates within a **hierarchy of governance structures** (= tiers) with their own disciplines and processes (e.g., corporate governance, technology governance, IT governance)
- May exist at **multiple geographic levels** within the enterprise (e.g., global, regional, local)

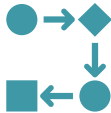
Architecture Governance – Need

The development & maintenance of an Enterprise Architecture must be governed

Why	What	How
- Without control, there is no need to develop a Target Architecture - Direct & control change	- Development of the Target Architecture - All changes within the scope of the Target Architecture	- Architects & implementer are directed & controlled by the stakeholders Architecture Project: Statement of Architecture Work, Target Architecture Checklist Implementation Project: Architecture Contracts, Architecture Requirements Specification

Architecture Governance – Governance Repository

The governance artifacts & processes are part of the Architecture Repository

Reference Data	Process Status	Audit information
▪ Description of the governance procedures ▪ Guidance & instruction in project implementation 	▪ State of the governance processes ▪ Examples: outstanding compliance requests, dispensation requests, compliance assessments investigations 	▪ Completed governance process actions ▪ Key decisions & responsables for any Architecture Project ▪ Architecture & supporting process developments, guidance & precedence 

(Enterprise) Architecture Governance – Benefits

Architecture Governance adopts its benefits from Corporate Governance



Discipline

Commitment to adhere to procedures, processes and authority structures



Responsibility

Each contracted party is required to act responsible



Accountability

Organization groups are authorized & accountable for their actions



Fairness

Decisions, processes and implementations will not be allowed to create unfair advantage to any party



Transparency

Decisions and supporting actions will be available for inspection



Independency

Processes, decision-making and mechanisms will be established as to minimize potential conflicts of interest

Architecture Board – Role

The cross-organizational board is represented by key architecture stakeholders



- Comprises a **group of executives** responsible for the **review and maintenance** of the **architecture** and for **oversee the implementation** of the strategy within Phase H of the Architecture Development Method (ADM)



- Possesses identifiable and articulated **responsibilities, decision-making capabilities, remit** and **authority limits**



- Distinguishes between **local** (domain experts, line responsibility) and **global** (organization-wide responsibility) **representatives** in **larger enterprises**



- May have **global, regional, or business line scope**

Architecture Board – General responsibilities

The board is made responsible and accountable for achieving different goals

- Improves the **maturity level of architecture discipline** while ensuring that the discipline of **architecture-based development** is adopted
- **Ensures consistency of sub-architectures** and **enforces Architecture Compliance**
- Establishes targets for **re-use of components**
- **Ensures flexibility** of the **Enterprise Architecture** regarding business needs & technologies
- Supports a visible **escalation capability for out-of-bounds decisions**

➤ **Basis for all decision-making** regarding the architectures

Architecture Board – Operational responsibilities

The board has many different operational responsibilities



- Monitors, controls and publishes of **Architecture Contracts**



- Ensures the **management & implementation of the architectures**



- Resolves **ambiguities, issues, conflicts** & provides **advice, guidance, infos**



- Ensures **compliance with the architectures** and grants dispensations



- Considers **policy changes** (e.g., schedule, SLAs, new service requirement)



- Validates of **reported service levels, cost savings**, etc.

Architecture Board – Governance responsibilities

The board has many different governance responsibilities



- Provides a **mechanism** for the formal **acceptance and approval of the enterprise architecture**



- Provides a **control mechanism** for ensuring the **implementation of the enterprise architecture**



- Identifies **divergence from the architecture** and planning activities for **realignment** (e.g., dispensations, policy updates)



- Links **strategic objectives of the business, the architectural strategy and objectives** and the **implementation of the architecture**



- Produces usable **governance material** and activities

Architecture Board – Decision Rights

The decision rights about the Target Architecture lies with the stakeholders

1. Did the organization embarking on a change reasonably interpret the target architecture's guidance and constraints? Y/N
 - If yes, their interpretation should be accepted as compliance and any issues addressed through a change to the architecture
 - If no, proceed
2. Do appropriate subject matter experts agree with the facts and interpretation of the facts in the impact assessment? Y/N
 - If yes, proceed
 - If no, either direct the architect to engage with the subject matter experts or develop a recommendation for the stakeholders that they should have limitations in confidence
3. Do appropriate subject matter experts agree with the recommendation to enforce the target, grant time-bound relief, or change the architecture? Y/N
 - If yes, proceed
 - If no, either direct the architect to engage with the subject matter experts or develop a recommendation for the stakeholders that they should have limitations in confidence
4. Do the views produced for the stakeholders reflect the impact assessment and reflect any underpinning architecture models and analysis? Y/N
 - If yes, proceed to the stakeholders for approval
 - If no, direct the architect to develop appropriate views
5. Do the stakeholders understand any limitations in confidence they should have in the impact assessment? Y/N
 - If yes, proceed

- The Architecture Board owns the **process & recommendations** regarding completeness and confidence in the **architecture work** that led to the **Target Architecture**
- **Decision rights about Target Architecture**, relief and enforcement are **always vested in the architecture's stakeholders**, independent of the architecture domain
- TOGAF® provides decision-tree **checklists** to require an **architect** to answer when assessing a **Target Architecture** as well as a **non-compliance report** (cf. extract on left)

(Enterprise) Architecture Compliance – Types

Architecture compliance of projects is essential to Architecture Governance

Architecture function

- Prepares **project architectures** (i.e., project-specific views) illustrating how the **Enterprise Architecture impacts** on **major projects**
- May stipulate the participation in the **technology selection process**, in **external service provision** and **product purchases** in addition to architecture definition and standards selection



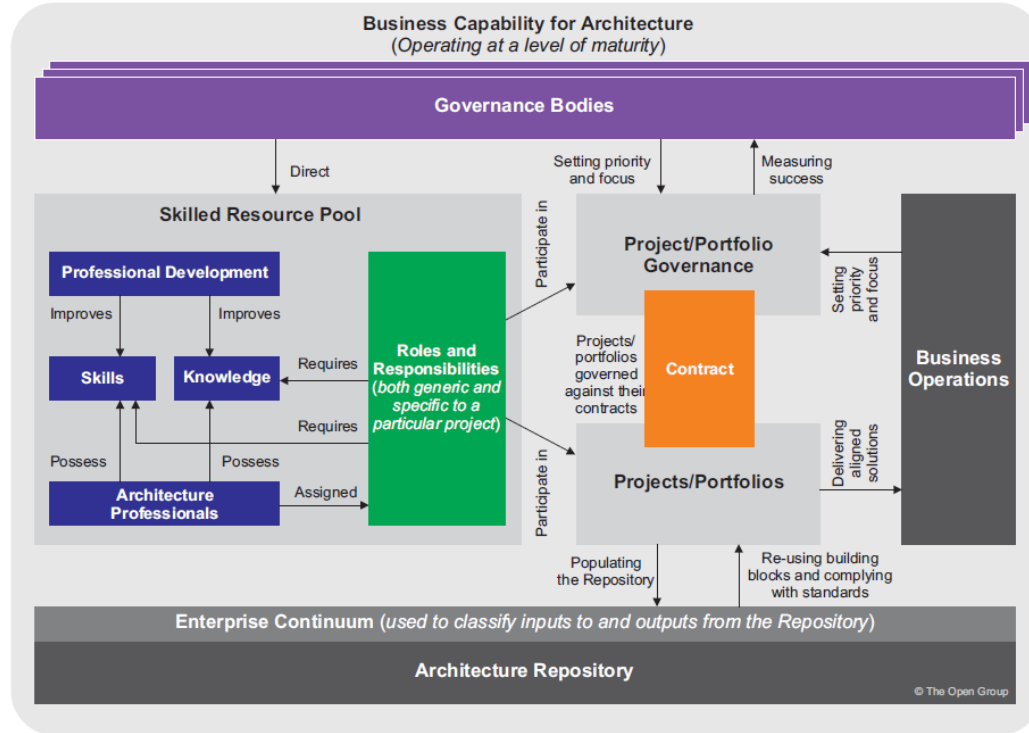
Enterprise & IT Governance function

- Defines a formal **Architecture Compliance Review** process for reviewing the **compliance of projects** to the Enterprise Architecture



Architecture Capability – Overview

The EA capability is a business capability supporting four different purposes

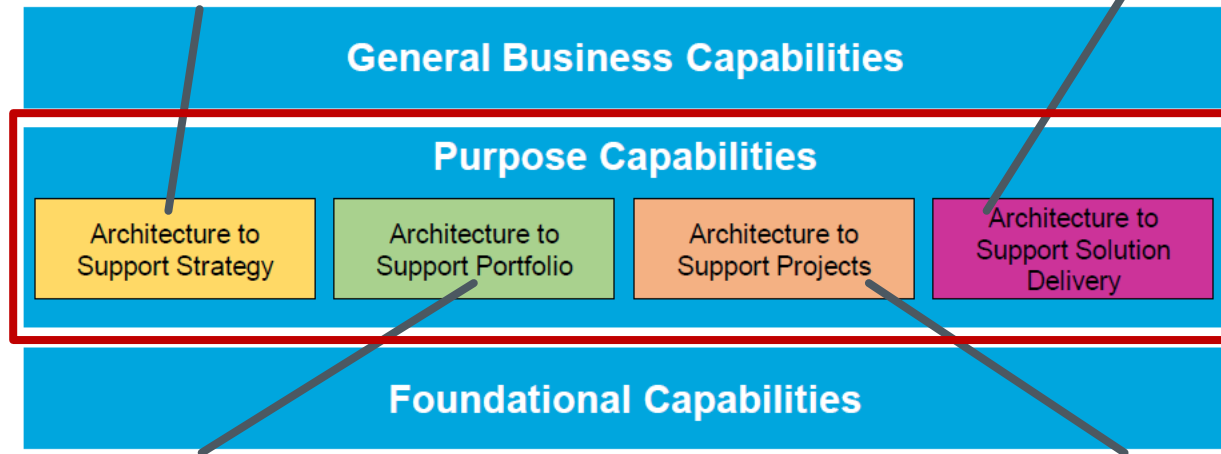


- Posses **organization structures** with roles, responsibilities, skills and processes for EA work
- Treated and run **like any other operational unit within a business**
- **Establishes management capabilities** (e.g., financial, risk & opportunity, performance, supplier, service, communications & stakeholder, resource, quality, configuration, environment)

Architecture Capability – Purposes

Main goal is to develop, use and sustain the architecture to govern change

- **Change & synergies, governance**
- **Multiple programs** or portfolios
- Provide **Target Architecture & roadmaps**
- **Governance framework** for change
- Single **project** or **significant part**
- Support **solution deployment**



- **Alignment, synergies** and **governance**
- **Single portfolio** of projects
- Support **cross-functional & multi-phase**
- **Integration** between projects
- **Single project** (value, compliance)
- Support the **project delivery method**

Architecture Capability – Architecture to support Strategy

The focus lies on an architecture with low depth for the next 3 till 10 years



Purpose of the Architecture

- Deliver EA to **provide an end-to-end Target Architecture**
- **Develop roadmaps** of change over a three to ten-year period

Use of the Architecture

- Identify change initiatives
- Support portfolio and programs
- Set terms of reference
- Identify synergies
- Govern the execution of strategy via portfolio & programs

Enterprise Scope (Breadth)

No pattern - Broad & narrow impact

Level of Detail (Depth)

Low

Time Period (Planning horizon)

3-10 years (Target Architecture)

Architecture Capability – Architecture to support Portfolio

The focus lies on a single subject architecture with low depth for 2 till 5 years



Purpose of the Architecture

- Deliver EA to **support cross-functional, multi-phase and multi-project change initiatives**

Use of the Architecture

- Identify projects
- Set terms of reference for projects
- Align approaches for projects
- Identify project synergies
- Govern project execution

Enterprise Scope (Breadth)

Single subject (the Portfolio)

Level of Detail (Depth)

Typically low but detailed for constraints

Time Period (Planning horizon)

2-5 years

Architecture Capability – Architecture to support Projects

The focus lies on a narrow architecture with detailed depth for less than 2 years



Purpose of the Architecture

- Deliver EA to **support the Enterprise's project delivery method**

Use of the Architecture

- Clarify the purpose and value of the project
- Identify requirements to address synergy and future dependency
- Assure compliance
- Support integration and alignment between projects

Enterprise Scope (Breadth)

Narrow (projects within the portfolio)

Level of Detail (Depth)

Detailed, more constraints than guidance

Time Period (Planning horizon)

<2 years

Architecture Capability – Architecture to support Solution Delivery

The focus lies on a narrow architecture with detailed depth for less than 2 years



Purpose of the Architecture

- Deliver EA that is **used to support the solution deployment.**

Use of the Architecture

- Define how the change will be designed and delivered
- Identify constraints
- Define control and architecture requirements to the design
- Establish a governance framework for change

Enterprise Scope (Breadth)

Very narrow (one project)

Level of Detail (Depth)

Most detailed

Time Period (Planning horizon)

<2 years

Architecture Capability – Development (1/4)

The TOGAF ADM can be used to iteratively develop an EA Capability

Capability topic	ADM	Level	Enterprise deliverables	EA Capability deliverables
Enterprise & EA Context	Phase B	Partial Strategic	- Goals, objectives, initiatives, competitive and tactic analysis - Operating model (partners, suppliers) - Explore what-if scenarios and scorecards	Goals
Business Objectives	Phase A	Capability	None	- Provide initial goals and objectives - Select a reference EA Capability and maturity model - Candidate EA Capability - Candidate operating model - EA Capability gap and priority roadmap
Architecture Governance	Phase B	Partial Segment / Capability	- Enterprise Risk Management Model - Governance Model	- Risk Management Model - Governance Model - Extend candidate operating model to include EA governance - Initial Architecture Partition Model - Trace to EA Capability goals

Architecture Capability – Development (2/4)

The TOGAF ADM can be used to iteratively develop an EA Capability

Capability topic	ADM	Level	Enterprise deliverables	EA Capability deliverables
Alignment with other Frameworks	Phase B & Partial Phase C (Data)	Partial Capability	▪ Reference models for key frameworks ▪ Capability assessment of key frameworks	▪ Framework touch-points ▪ Extend candidate operating model to include other frameworks ▪ Extend EA governance and EA risk management ▪ Initial EA Content Framework aligned to other frameworks and EA governance ▪ Candidate architecture partition model ▪ Trace to EA Capability goals ▪ EA Capability
Customization of Architecture Contents & Metamodel	Phase C (Data)	Capability	None	▪ EA Content Framework ▪ EA Content Metamodel ▪ Viewpoint Library ▪ Architecture Repository Model ▪ Trace to EA Capability goals ▪ Initial EA Content Framework and architecture repository gap

Architecture Capability – Development (3/4)

The TOGAF ADM can be used to iteratively develop an EA Capability

Capability topic	ADM	Level	Enterprise deliverables	EA Capability deliverables
Organization Model	B	Capability	None	- EA organizational model - Select reference EA skills framework - Initial alignment with enterprise job titles and roles - Initial accountability matrix for EA Content Framework and initial architecture repository - Organizational gap and priority roadmap
Process Model	B, C (App) and D	(Partial) Capability	- Process model highlighting touch-points between EA Capability and enterprise processes the EA Capability supports²⁷ - Performance matrix for key processes and organization - Accountability matrix for EA Content Framework and organization	- Process model - Architecture repository application model - Matrix for EA Content Framework and architecture repository application architecture - Process and architecture repository gap and priority roadmap
Roadmap	E	Capability	Create a roadmap highlighting development of the EA Capability by changes in the: - Organizational model - Process model - EA Content Framework - Architecture repository	Trace roadmap to EA Capability goals

Architecture Capability – Development (4/4)



The TOGAF ADM can be used to iteratively develop an EA Capability

Capability topic	ADM	Level	Enterprise deliverables	EA Capability deliverables
Establishing & Evolving	Phase F and G	Capability	Transition the EA Capability Roadmap to an Implementation & Migration Plan	Execute the Implementation & Migration Plan to build the EA Capability your enterprise desires

Architecture Capability – Benefits

The capability supports the execution of a defined & effective governance



Increased transparency of accountability and **informed delegation** of authority



Increased visibility **supporting** internal processes and external parties' **requirements**



Internal value and **shareholder value** creation



Process, concept and component re-use across all organizational business units



Proactive risk and opportunity management through controlling and monitoring

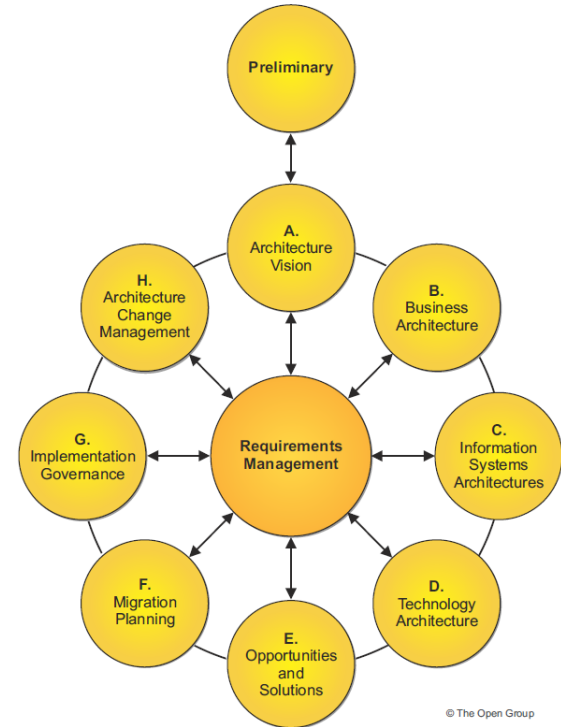


Protection of the existing assets through re-use of existing architectural components

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Study the content – Go through the ADM, its phases, techniques and deliverables. If you know the method, you can get at least 24 correct answers.



Complete exam questions – Test your TOGAF knowhow and familiarize yourself with the structure & characteristics of the exam by answering as many practice questions as possible.



Test under real conditions – Simulate a TOGAF exam with 40 questions in 60 minutes. Learn how to deal with challenging questions in times of low concentration.



Check your exam infrastructure – Test your IT for a TOGAF® proctored exam. Ensure you have a redundant internet connection. Make sure it is quiet in your room.

Time management



Force a constant pace – As a native English speaker, you only have 90s on average to answer a question. Skip difficult questions and look for quick wins instead.



Use the complete time frame – The devil is in the detail, especially with tricky questions. Use any remaining time for a quality check. Start with the answers where you have the most doubts.

Question analysis



Read questions carefully – Go through each question and all the answers. Pay attention to the details. A single word can make all the difference.



Pay attention to key terms – Main TOGAF terms like concern, stakeholder or governance point to relevant phases, techniques or content of the EA framework.



Identify the ADM phase – Many questions & answers define the phase(s) of the Architecture Development Method. Use this context to find the right answer.



Stick to the TOGAF spirit – Select the answer that most closely matches the key concepts of the TOGAF Standard, e.g., ADM phases, deliverables and techniques.



Sort out the distractor – In most cases, one answer has nothing to do with the TOGAF Standard and is very different from other answers. Exclude this incorrect option.



Look for similar answers – For some questions, the best and second-best answers are similar. Choose the one that comes closest to the TOGAF Standard.



Always choose an answer – Only one answer is correct and wrong answers do not give minus points. Therefore, always put a tick in the box. Always.

Definition

Application Architecture

A **description** of the **structure and interaction** of the **applications** that provide key business capabilities and manage the data assets.

Definition

Architecture

1. The **fundamental concepts or properties** of a **system** in its **environment** embodied in its **elements, relationships** and in the principles of its **design and evolution**.
(Source: ISO/IEC/IEEE 42010:2011)
2. The **structure of components**, their **inter-relationships** and the **principles and guidelines governing** their **design and evolution** over time.

Definition

Architecture Building Block (ABB)

An **architectural component** that **specifies** the required **Solution Building Blocks (SBBs)** at a more **logical (or supplier-independent)** level.

Definition

Architecture Continuum

A **categorization mechanism**, with increasing detail and specialization, for the **components and artifacts stored** in the **Architecture Landscape** or **Reference Library (part of the Architecture Repository)**.

This Continuum **begins with foundational definitions** like reference models, core strategies and basic building blocks. From there it **spans to Industry Architectures** and all the way to an **Organization-Specific Architecture**.

Definition

Architecture Development Method (ADM)

The **core** of the TOGAF framework. A **multi-phase, iterative approach** to **develop** and **use** an **Enterprise Architecture** to **shape and govern business transformation**.

Definition

Architecture Domain

The **architectural area being considered**. The TOGAF framework follows the tradition of dividing Enterprise Architecture into **four primary architecture domains: business, data, application and technology**. **Other domains** (motivation, security, governance, etc.) **may span those four primary domains**.

Definition

Architecture Framework

A **conceptual structure** used to **plan, develop, implement, govern and sustain an architecture**.

Definition

Architecture Governance

The **practice of monitoring and directing architecture-related work**. The goal is to **deliver desired outcomes** and **adhere** to relevant **principles, standards and roadmaps**.

Definition

Architecture Landscape

The **architectural representation of assets** in use, or planned, by the enterprise at **particular points in time**.

Definition

Architecture Level

Levels provide a framework for **dividing** the **Architecture Landscape** into **levels of granularity**. Note: Architecture levels are distinct from architecture partitions.

Definition

Architecture Model

A **representation** of a **subject of interest**. An architecture model **provides** a **smaller scale, simplified and/or abstract representation** of the subject matter.

Definition

Architecture Partition

A **subset of architecture** resulting from **dividing** that *architecture* to facilitate its **development and management**.

Definition

Architecture Principle

A **qualitative statement of intent** that **should be met by the architecture**.

Definition

Architecture View

A **representation of a system** from the **perspective** of a related set of **concerns**.

Definition

Architecture Viewpoint

A **specification** of the conventions for a **particular kind of architecture view**. An architecture viewpoint can also be seen as the **definition or schema** for that kind of **architecture view**. It establishes the conventions for constructing, interpreting and using an architecture view to address a specific concern (or set of concerns) about a system-of-interest.

Definition

Architecture Vision

A **succinct description** of the **Target Architecture** that **describes** its **business value** and the **changes** to the enterprise that will result from its successful deployment. It serves as an **aspirational vision** and a **boundary** for detailed architecture development.

Definition

Artifact

An **architectural work product** that **describes an aspect of the architecture**.

Definition

Baseline

A **specification** that has been **formally reviewed** and **agreed upon**, that thereafter serves as the **basis for further development or change** and that can be **changed only** through **formal change control procedures** or a type of procedure such as configuration management.

Building Block

A **potentially re-usable component** that can be combined with other building blocks to **deliver architectures and solutions**.

Definition

Business Architecture

A **representation of holistic, multi-dimensional business views** of: **capabilities**, end-to-end **value delivery**, **information** and **organizational structure**; and the **relationships** among these business views and strategies, products, policies, processes, initiatives and stakeholders.

Relates business elements to business goals and elements of other domains.

Definition

Business Capability

A **particular ability** that a **business** may **possess or exchange** to **achieve** a specific **purpose**.

Definition

Business Model

A model describing the **rationale** for how an **enterprise creates, delivers and captures value**.

Definition

Capability

An **ability** that an **organization, person or system** possesses.

Definition

Capability Architecture

An architecture that describes the **abilities** that an **enterprise** possesses.

Definition

Concern

An **interest in a system** relevant to one or more of its **stakeholders**.

Definition

Communications & Stakeholder Management

The **management of needs of stakeholders** of the Enterprise Architecture practice. It also **manages the execution of communication** between the practice and the stakeholders and the practice and the consumers of its services.

Definition

Data Architecture

A description of the **structure** of the enterprise's **major types** and **sources of data**, **logical data assets**, **physical data assets** and **data management resources**.

Definition

Deliverable

An **architectural work product** that is **contractually specified** and in turn **formally reviewed, agreed and signed off** by the stakeholders.

Represent the **output of projects**. If in documentation form it will typically be archived at completion of a project or **transitioned into an Architecture Repository** as a reference model, standard, or snapshot of the Architecture Landscape at a point in time.

Definition

Digital Architecture

The inclusive architecture focused on a **combination of Enterprise Architecture, data science, telecommunications and Internet of Things (IoT), security, artificial intelligence, cognitive science, neuroscience, robotics and social medias** to deliver operational services.

Definition

Enterprise

The **highest level** (typically) of description of an **organization** and typically **covers all missions and functions**. An enterprise will often span **multiple organizations**.

Definition

Enterprise Continuum

A **categorization mechanism** for **classifying architecture** and **Solution Building Blocks (SBBs)** as they evolve from **generic** to **specific applicability** (or vice versa).

Definition

Gap

A **statement of difference between two states**. Used in the context of **gap analysis**, where the difference between the Baseline and Target Architecture is identified.

Definition

Governance

The **discipline of monitoring & guiding the management of a business** (or Information Systems/Information Technology landscape) to **deliver the business outcomes required**.

Definition

Interoperability

1. The ability to **share information** and **services**.
2. The ability of two or more systems or components to **exchange** and **use information**.
3. The ability of systems to **provide and receive services** from other systems and to use the services so interchanged to enable them to operate effectively together.

Definition

Logical

Implementation-independent. A logical architecture is an implementation-independent definition of the architecture.

Definition

Metamodel

A model that describes the **entities used in building an Architecture Description**, their **characteristics** and the key **relationships** between those entities.

Definition

Modeling

A technique through **construction of models** which enables a **subject** to be represented in a form that **enables reasoning, insight and clarity** concerning the essence of the subject matter.

Definition

Physical

Real-world, tangible. A **logical architecture** is realized through a **physical architecture**.

Definition

Requirement

A **statement of need**, which is **unambiguous, testable or measurable** and **necessary** for acceptability.

Definition

Roadmap

An abstracted plan for business or technology change, typically operating across multiple disciplines over multiple years. Normally used in the phrases Technology Roadmap, Architecture Roadmap, etc.

Definition

Role

1. The usual or **expected behavior of an actor**, or the **part somebody or something** plays in a particular **process or event**. An actor may have a number of roles.
2. The **part an individual** plays in an **organization** and the **contribution** they make through the **application** of their **skills, knowledge, experience and abilities**.

Definition

Segment Architecture

A detailed, formal description of **areas within an enterprise**, used at the **program or portfolio level** to organize and **align change** activity.

Definition

Solution Architecture

A **description** of a **discrete** and **focused business operation** or activity and **how IS/IT** supports that operation.

Definition

Solution Building Block (SBB)

A **physical or implementation-specific component** that **realizes** part or all of one or more **logical Architecture Building Blocks (ABBs)**. There are business, application and technology SBBs.

Definition

Solution Continuum

A **categorization mechanism**, with increasing detail and specialization, for the **components and artifacts stored** in the **Solutions Landscape** or **Reference Library** (part of the **Architecture Repository**).

Definition

Stakeholder

An **individual, team, organization**, or class thereof, having an **interest in a system**.

Definition

Standards Library

A **library of standards** that **can be used to define** the particular services and other **components of an Organization-Specific Architecture**.

Definition

Strategic Architecture

A summary formal **description of the enterprise**, providing an organizing **framework for operational and change** activity, and an **executive-level, long-term view** for direction setting.

Definition

Target Architecture

The **description** of a **future state** of the **architecture** being developed for an organization. There may be **several future states** developed as a roadmap to show the evolution of the architecture to a target state.

Definition

Technology Architecture

A description of the **structure and interaction** of the **technology services** and **technology components**.

Definition

Transition Architecture

A formal description of one **state of the architecture** at an architecturally **significant point in time**.

One or more Transition Architectures may be used to describe the **progression in time** from the Baseline to the Target Architecture.

Definition

Value Stream

A **representation** of an **end-to-end collection of activities** that **create an overall result for a customer, stakeholder, or end user.**

Definition

Viewpoint Library

A **collection** of the **specifications of architecture viewpoints** contained in the **Reference Library** portion of the **Architecture Repository.**

Definition

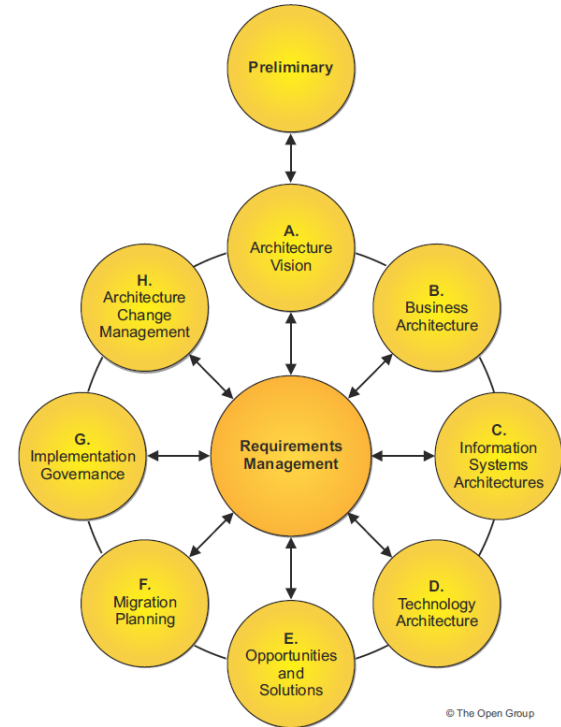
Work Package

A **set of actions** identified to **achieve one or more objectives** for the business.
A work package can be a **part of a project, a complete project, or a program**.

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Document outline

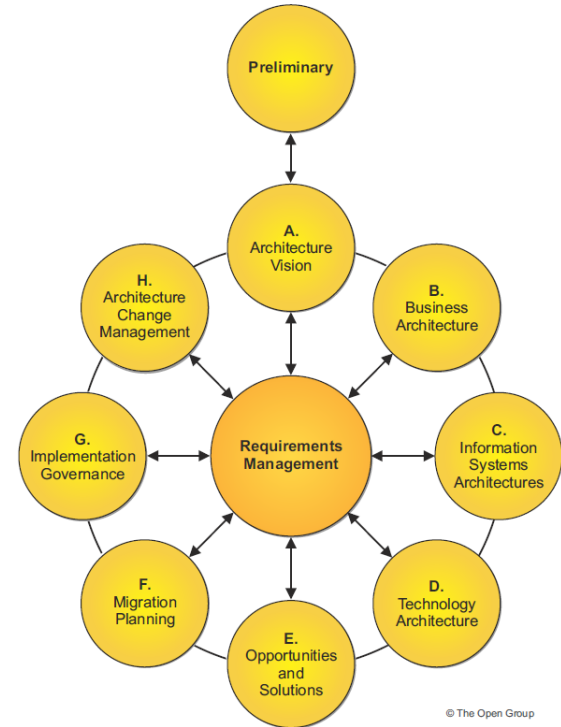
Enterprise Architecture Foundation with the TOGAF Standard, 10th Edition



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Congratulations. You've made it. Thanks for your participation!
Good luck for your TOGAF® Enterprise Architecture Part 1 Exam (Foundation)



Dr. Christopher Schulz
Enterprise Architect

***„Architecture should speak of its time and place,
but yearn for timelessness.”***

- [Frank Gehry](#), Canadian-American Architect



Tobias Smuda
Enterprise Architect

Please give us feedback by writing a review!
Thanks in advance.